COLLEGE OF INFORMATION

Voting & Civic Participation

Alabama Black Belt Counties

Prepared for: Southern Poverty Law Center (SPLC)

Course: INST490 · Section: 0103

Date: May 18, 2026

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About This Report

This report presents the findings of eleven student consulting teams working on behalf of the Southern Poverty Law Center (SPLC) through the INST490 iConsultancy capstone program at the University of Maryland's College of Information. All work was completed January through May during the Spring 2026 semester. The analyses and deliverables in this report were produced by undergraduate students in their final semester of study, working under faculty supervision and in direct consultation with SPLC staff.

UMD iConsultancy Capstone Instructor: Jesse Klein, Ph.D. jrklein@umd.edu
UMD iConsultancy Director: TJ Rainsford tjrains@umd.edu

The Clients

The Southern Poverty Law Center (SPLC) is a nonprofit civil rights organization headquartered in Montgomery, Alabama. Its work spans litigation, policy analysis, data-driven research, and public education aimed at addressing systemic inequities affecting marginalized communities. The teams represented in this report worked with staff from three SPLC units: the Data and Research Team; the Democracy—Voting Rights Litigation Team and Policy Team; and the SPLC Alabama State Office.

Fred McBride - SPLC Senior Supervising Data and Research Analyst | Legal Management Fred.Mcbride@splcetner.org
Jack Slate - SPLC AL State Office jack.slate@splcenter.org
Zach Mahafza - SPLC Senior Research Analyst | Legal Management Zach.Mahafza@splcenter.org

The Project

SPLC commissioned this work to support descriptive, historical analysis of civic participation and voting access across two regions of Alabama: the Black Belt counties (this report) and the Gulf Coast counties (Section 105, a companion report). The Black Belt project focused on 17 counties: Barbour, Bullock, Butler, Choctaw, Crenshaw, Dallas, Greene, Hale, Lowndes, Macon, Marengo, Montgomery, Perry, Pike, Russell, Sumter, and Wilcox. All analyses examined conditions affecting Black and Latinx communities in particular, using publicly available or client-approved data sources. All findings are descriptive and based on publicly available data; they do not constitute legal analysis or advocacy.

How the Teams Worked

The project was structured so that each team addressed a distinct dimension of civic participation access, and the teams' work is designed to be read in relation to one another. BB-DIG built and documented the core data infrastructure that most other teams drew from. BB-VOTE and BB-PAD conducted the primary quantitative analyses of turnout trends and physical access to polling locations. BB-TRAC, BB-AACS, and BB-BAND examined compounding structural barriers: transportation access, ADA compliance at polling locations, and broadband availability. BB-CPRI synthesized indicators from those teams into a composite risk index ranking counties by civic participation risk. BB-VIZ translated data across all teams into an interactive client-facing dashboard. BB-CCP developed community personas grounding the quantitative findings in human context. BB-PORT translated findings into materials suitable for policy and organizational decision-making. BB-CPAD produced public-facing civic process materials designed for community members navigating participation for the first time.

How to Read this Report

Each team's section is self-contained and can be read independently. Every section opens with a one-page brief summarizing what the team studied, its key findings, the deliverables it produced, significant challenges encountered, and connections to other teams. The full report narrative follows, including methods, detailed findings, limitations, and recommendations. Cross-references to other teams are included where teams' findings directly inform one another.

Readers approaching this report for the first time are encouraged to begin with BB-DIG (data infrastructure) and BB-VOTE (turnout trends) before moving to the access-focused sections, as those two sections establish the empirical foundation the rest of the project builds on. A companion report covering the Gulf Coast counties is available separately through the Section 105 iConsultancy team at the University of Maryland's College of Information.

COLLEGE OF INFORMATION

Voting & Civic Participation

Alabama Black Belt Counties

Team 1 BB-DIG: Regional Voting Data Integration & Governance

Prepared for: Southern Poverty Law Center (SPLC)

Team Members: Matthew Ritter (Project Manager), Arnav Patel (Client Communicator), Asmita Brahme (Quality Assurance), Bao Nguyen (Team Communicator), Margaret Hermanto (Recorder), Xuan Zhang (Consistency Checker)

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Team 1: BB-DIG

Team Brief

BB-DIG built and maintained the core data infrastructure for the regional analysis, cleaning, standardizing, and documenting voter turnout, voter registration, and eligible population across 17 Alabama Black Belt counties for the 2020 and 2024 General Elections. This work directly enables SPLC to identify where civic participation gaps are largest by race and county, and provides a reusable, documented data foundation for future research teams. Our team's findings and rates use Citizen Voting Age Population (CVAP), a Census Bureau estimate of the number of citizens age 18+ in a given area, broken down by race, as the eligible population denominator.

Key Findings

- Black voter participation fell in 15 of 17 Alabama Black Belt counties between 2020 and 2024 dropping from 62.8% to 57.2% on average while White participation held nearly flat at 68%, nearly doubling the Black-White participation gap from 5.3 to 10.1 percentage points
- Across the 17 counties in 2024, there are 29,357 inactive Black registrations versus 16,489 inactive White registrations. The average inactive rate for Black voters (12.0%) is 3.1 points higher than White (8.9%). Macon County has the most extreme gap at 17.1 percentage points (25.7% Black vs 8.6% White).
- Average Black turnout fell from 59.0% to 51.0% while average White turnout fell from 69.3% to 66.2% . The Black-White turnout gap widened from 10.3 to 15.2 percentage points. Lowndes saw the steepest Black decline at −13.2%.

Deliverables Produced

Six cleaned datasets Three dataset types across two election years, stored as CSV files one row per county, 17 Black Belt counties each.

File	Description	Link
eligible_population_2020_general.csv	CVAP-based eligible population, ballots, and non-participation rates by race for the 2020 general election	GitHub
eligible_population_2024_general.csv	Same structure as above for the 2024 general election	GitHub
voter_registration_2020_general.csv	ALVR-based active and inactive registration counts and rates by race for the 2020 general election	GitHub
voter_registration_2024_general.csv	Same structure as above for the 2024 general election	GitHub
voter_turnout_2020_general.csv	SoS-based turnout by gender, age, and race with absentee rates for the 2020 general election	GitHub
voter_turnout_2024_general.csv	Same structure as above for the 2024 general election	GitHub

Data dictionaries and documentation) Each dataset folder contains one Excel data dictionary (.xlsx) with one row per column and one markdown documentation file (.md) covering sources, limitations, methodology, and intended use. All twelve files are co-located with their respective CSVs in the repository. GitHub — dataset folders | Google Drive mirror

Standardization document: A standalone reference document defining naming conventions, FIPS code sourcing, data structure rules, race scope decisions, source hierarchy, and methodology decisions that apply across all six datasets. Google Drive

Eight reproducible cleaning and validation scripts Jupyter notebooks (.ipynb) stored in the /scripts/ folder, organized by dataset type. Five cleaning scripts produce the six CSVs; three validation scripts document internal SoS discrepancies and cross-source comparisons. GitHub — scripts folder

Challenges

The two most significant challenges were both source data quality issues. The ALVR file had a broken two-row header that required manually setting column names in the cleaning script, and CVAP estimates produced registration rates above 100% in 15 of 17 counties due to known undercounting in rural majority-minority communities, which required us to switch to ALVR-only calculations mid-semester. Both fixes are documented in the standardization document and data dictionaries for future teams

Connections to Other Teams

BB-DIG serves as the data foundation for the regional analysis. Our six cleaned CSVs with standardized FIPS codes and county names are designed to be used directly by any team conducting county-level analysis across the Black Belt counties. Teams relying on our data include BB-VIZ for visualizations and BB-PAD for spatial and precinct-level analysis.

Final Report and Deliverables

Project Context

BB-DIG serves as the foundational data infrastructure team within the broader 11-team regional analysis. By identifying, cleaning, and standardizing datasets related to voting access and civic participation in the 2020 and 2024 elections, BB-DIG has developed a structured and well-documented data foundation that supports current and future analysis of regional voting patterns. In addition to producing these cleaned datasets, BB-DIG has established a consistent data format, naming conventions, and geographic identifiers, along with standardized documentation of data sources and limitations. This work was largely independent and did not rely on datasets or deliverables from other project teams during the course of the project. However, our dataset deliverables are designed to support a wide range of uses across the 11 different teams, including continued analysis, cross-project comparisons, and research on civic participation in the Alabama Black Belt.

Methods

Our data stems from public annual election records and demographics reports for the 17 counties of interest in the Alabama Black Belt. The Alabama Secretary of State (AlSoS) serves as a main source specifically useful for voter registration and turnout data. This source publishes state-level information regarding election results, including categories such as amount of votes, gender, age, and race. This is especially useful to us as they can be filtered further down to the county level. Data was collected primarily from 2 years, 2020 and 2024. These years both marked general elections for the state.

Another key source of data is the Census Bureau Citizen Voting-Age Population (CVAP). This was used to calculate eligible voting populations for our later analysis. The CVAP program provides 5-year estimates of each county's citizen voting age population. This brings up an important criterion for the basis of our voting information. That is the fact that US citizenship is an enforceable requirement for people to participate in voting. Again, CVAP files for both 2020 and 2024 estimates were used here. One important factor to note here is that this data may have a margin of error since it is an estimated rather than confirmed amount. We expand on this below in our limitations section, and they are written out in our data dictionaries.

A smaller, yet still particularly influential source of data was provided by the Redistricting Data Hub (RDH). We used this data to identify and cross-check Federal Information Processing Standards (FIPS) codes for the regions of interest within the Alabama Black Belt. This is not a key file that composes our final dataset deliverable, but it acts as a checkpoint to ensure our counties are identified and paired with their registration figures appropriately.

Analytical Approaches

After data collection, our goal was to clean, integrate, and standardize the key information of interest. We followed this structure to ensure that our team knew where we were at any given time during the project. Raw data files were collected and noted down from the sources mentioned above. At this point, these files were meticulously reviewed to understand their schema, variables, and relevance to our final deliverable. Later steps, like filtering and separation of data, were only on our radar because of this initial review of the data.

The next step was standardization, and this is where FIPS codes excel. They served as the main identifier for location, and as a result, we could split datasets down into the 17 counties in the region. The obvious choice would have been to standardize data using county names, but the naming here is inconsistent. That is why we opted to use FIPS codes instead of names. This allowed for clean merging of files and a uniform way of interpreting what each code represents. FIPS codes are 5-digit strings that all start with 01 and are followed by their respective 3-digit county code.

Another way we chose to format our datasets is through breakdowns of information by race. The inclusion of racial information acted as categories for comparison and analysis. While there were easily 20 or more available categories for race, it was essential to filter and choose the most important subset of this level of organization. 3 races from this demographic are in focus throughout our data, and those are African American voters, White voters, and Latinx voters. We made this choice based on feedback from SPLC on their work towards voting rights in Alabama. This does not mean that data regarding any/all other races is not important or relevant to voting data. One factor that influenced our choices is the recorded population sizes of these groups of people in the 17 counties, and they make up a very small percentage of the population, to create accurate rate data. Additionally, the margin of error for such rates would be much higher due to the small sample size we are working with. On the other hand, the amount of African Americans, White people, and Latinx people in this region represents a noticeably large portion of the population that can be represented thoroughly through our data.

The next step in our analysis was the manual calculation of participation rates. Some sources did provide such rates, but because of the uneven organization of data prior to standardization, some of these rates may not have been comparable in the end. As a precaution, we chose to use other reliable information to calculate participation rates. This also led to distinct types of datasets being created: eligible population, voter registration, and voter turnout. Eligible population datasets use CVAP counts to determine the number of people who are legally allowed to vote. This was also useful to determine the subset of that group that could vote but chose not to participate. Voter registration datasets calculate turnout rate at elections, comparing inactive registered voters and total registered voters. Voter turnout datasets follow a similar comparison as the previous type of dataset, but total active registered voters are put up against voters from that group who show up to vote.

The point behind all of this is to say that all the datasets we have cannot be directly compared to just any other dataset blindly. Once someone understands what each rate calculated in these datasets means in the context of the bigger project, they may be able to compare datasets, but it is not a 1-to-1 comparison. Regardless, the appropriate data dictionaries for each file are a good place to start to read up on the background of the calculated rates.

Another aspect of rate calculations is the consideration of significant figures in those numbers. As such, rates are calculated and remain as decimals instead of clean, rounded, or truncated percentages. Our thoughts behind this are to uphold precision behind our rates by leaving the numbers as they are. However, not all of our numbers are as messy. Some fields require and are only composed of integers. This includes the count of eligible voting populations, case ballots, registered voters, and absent voters. These are just a few examples of that. This differentiation in the types of numerical data presented is intentional for the reasons above.

One of the last steps of our analysis is quality assurance. This is done after data cleaning and standardization. The values in datasets are spot-checked for overall accuracy, including handling of null/missing values. Cross-validation is also done for each county using other reference files from the RDH. Something to note here is that records with noticeable gaps in their data were not automatically dropped. They were kept in the dataset without placeholders, but they were mentioned in the documentation for that data. We want to uphold transparency for all of our data and not manipulate data at the regional level too much.

Tools Used

We used a few key tools throughout this project to assess, clean, and analyze datasets for our final deliverables. The process by which we worked through our data files remained mostly the same, so the tools used were the same overall. We used the Python programming language when performing data cleaning and transformation. As a language, it has generally easy readability, and the team is comfortable building simple cleaning flows that can be used by anyone with access to our repository. Within Python, we also used the pandas library as a way to manage our raw data files, including the later filtering and merging of datasets. This library would also help us provide organized CSV files for the repository.

The Python scripts were created exclusively through Jupyter Notebooks. These notebooks still contain code split into individual cells. This specific format worked well for us since each step for data cleaning could be properly commented on within the code files as well. That means our scripts can also double as documentation for the code it houses.

Our coding was done in an environment supported by Visual Studio Code (VS Code). Through VS Code, our team wrote, tested, and ran the Python scripts locally. We are also housing all our code and their version in a shared GitHub repository. This is great for promoting collaboration across the team, and it is very useful for informing the rest of the team when updates have been made to our work. Because of the ability to view the commit history of all our code and documentation, it also functions as a reliable tool for assessing accountability of work. The final repository contains data separated by whether it is raw or cleaned data, cleaning scripts, and documentation. This should help an average user navigate through all our files in the event of future work on this topic being conducted.

Challenges and Negotiations

Throughout the project, we encountered many data quality and methodology issues that we as a team needed to decide on. Each challenge brought up a certain limitation in either our source data or how we were presenting our data. However, these challenges and limitations ultimately led to how we structured our final datasets and the metrics that we wanted to present to the SPLC

Our first main challenge was dealing with the ALVR file structure. The Alabama voter registration file from the Secretary of State had a broken two-row header in the raw data sets that we were trying to scrape. This made it a challenge because pandas could not parse the filesets correctly due to this header issue. The first row had important labels as well, like Active and Inactive, with a second row underneath with the race breakdown. To fix this problem, as this was the main data we were working with, we manually set the clean column names in the cleaning script and dropped the first two rows of imported data. We were able to understand the data in the ALVR file to manually fix it, and all of this is documented in the script documentation and our standardization document.

The next challenge we encountered was dealing with the CVAP data. We noticed that when we initially calculated registration rates by dividing ALVR registration counts by CVAP eligible population, there were rates exceeding 100% in most Black Belt counties. Specifically, 15 of the 17 counties showed total registration rates above 100% for black voters. After looking into the data more and reading about CVAP documentation, this is a known limitation of CVAP survey estimates, which commonly undercount eligible voters in rural majority-minority communities. We negotiated as a team a methodology change to remove CVAP-based registration rates entirely and replace them with the active_share columns using ALVR data only. This change helped us avoid the source disagreement while still being able to present counts from CVAP, so if the SPLC wanted to use that data or notice the discrepancy, they could. We have this documented in our data dictionary for the voter_registration files.

We came across a source of disagreement between the Secretary of State and CVAP data when it came to small populations in the 17 counties. When we created our eligible population data set, there were some cells where voted_latinx would exceed the eligible_latinx. We noticed specifically that Greene in 2024 had 4 eligible versus 12 voted, Wilcox 2024 had 1 eligible versus 4 voted, and Hale 2020 had 10 eligible versus 10 voted. We chose to cap the missing voters at 0 to handle this source disagreement rather than give the negative values. We documented the affected counties specifically in the data dictionaries of the eligible population dataset. This goes back to the above limitation, where the CVAP is a Census Bureau survey estimate, so there are margins of error where the SoS counts actual ballots cast. The CVAP survey underestimated smaller populations, so this issue affected two of our cleaned datasets.

One last challenge that stemmed from CVAP data was picking our denominator for race-based turnout. At first, in our voter_turnout cleaning script, we used the CVAP eligible population as the denominator for race-based turnout rates. This was producing unreliable results in counties with small Latinx populations, like Wilcox, for example, which showed a 400% turnout. After we noticed this, we had a team discussion and decided that we should switch to ALVR active plus inactive registered votes as the denominator because it matched how the overall turnout rate is actually calculated (ballots divided by registered), and it produced valid county-level rates. This methodology change resulted in us updating our data dictionary and script for voter_turnout.

The last main limitation we noticed was when we ran internal verification for the Secretary of State data with our verification scripts and found that there were slight discrepancies. The Secretary of State's age, gender, and race breakdown files didn't match the Secretary of State's total file across most of the counties. For example, Butler County stood out with a 418 ballot discrepancy in 2024 between the breakdown files and the total file, while the other counties were within approximately 50 ballots. As a team, we chose to keep the Secretary of State's total file as the authoritative ballot count and document the discrepancies within our data dictionaries. This was a data source issue, not a cleaning issue, so we felt we had to document it and move forward.

Deliverables and Findings

Eligible Population Datasets

Overview

The eligible population datasets represent the broader view through which civic participation in the Alabama Black Belt can be measured. The two datasets we have here cover every citizen of voting age who was legally eligible to cast ballots in the 2020 and 2024 General Elections. This matters a lot to a civil rights organization like SPLC, because it captures the full scope of the civic disengagement, including people who never made it onto the voter poll in the first place.

The datasets cover all 17 Alabama Black Belt counties across both election cycles. Each row represents a single county. The columns are grouped around the main three racial groups that SPLC has identified as central to its voting right work in the region, which being Black, White and Latinx voters. For each group, the dataset provides the number of eligible citizens, the number of who casted, the number of who didn't and the rate of non participation.

Cleaning Script

Script: eligible_pop_2020_2024_cleaning.ipynb

This is the first script that must be run in the project pipeline. The voter registration and voter turnout cleaning scripts both depend on its outputs for eligible population denominators.

The script loads Census Bureau Citizen Voting Age Population (CVAP) files for both election years, aggregates from block group to county level, and filters to the 17 Black Belt counties using FIPS codes as the join key. It then loads the Alabama Secretary of State race turnout files for each year. The 2020 Secretary of State (SoS) race file has embedded line breaks in its column headers, which the script handles by explicitly renaming columns after loading rather than relying on the raw header. County names are standardized to title case and stripped of whitespace before joining.

Missing voter counts are calculated as eligible minus votes, with a floor of zero applied to handle cases where CVAP underestimates eligible voters relative to actual ballots. Missing voter rates are calculated as missing voters divided by eligible voters, rounded to 4 decimal places. The not_voted_gap is calculated as Black missing rate minus White missing rate. All outputs are sorted alphabetically by county and exported with null string values per the standardization document.

Documentation

Sources

- Census Bureau CVAP 2020 file — primary source for the eligible population by race
- Census Bureau CVAP 2024 file — primary source for the eligible population by race
- Alabama Secretary of State 2020 race turnout file — source for ballots cast by race
- Alabama Secretary of State 2024 race turnout file — source for ballots cast by race
- Redistricting Data Hub (RDH) 2024 voter file — source for FIPS codes only; not in final output

Limitations

- CVAP is a survey estimate, not a confirmed count. Margins of error are available in the original Census Bureau files but are not included in these CSVs.
- CVAP is known to undercount eligible voters in rural majority-minority communities. In several counties, this caused vote counts to exceed eligible counts, which is mathematically impossible. The team applied a floor of zero to missing voter counts in those cases. Affected counties: Hale 2020 (10 eligible Latinx, 10 voted), Greene 2024 (4 eligible Latinx, 12 voted), Wilcox 2024 (1 eligible Latinx, 4 voted).
- Latinx rates should be interpreted with particular caution in counties where the eligible Latinx population is under 100. Very small sample sizes make the rates statistically unreliable.
- Race categories are self-reported at voter registration, not at the time of casting a ballot.
- Hispanic was renamed to Latinx in all outputs to align with SPLC language preferences.
- FIPS codes are stored as 4-digit integers. Standard federal practice uses 5-digit codes with a leading zero (01005 rather than 1005 for Barbour County).
- Missing values are stored as the literal string null per the standardization document.

Intended Use

These datasets are intended for SPLC's voting rights litigation, policy work, and community outreach. They support analysis of who could have voted but did not, including people who never registered. The not_voted_gap column directly quantifies the racial equity gap in civic participation at the county level without requiring any additional calculation. These files should be used alongside the voter registration and voter turnout datasets for the complete picture.

Data Dictionary: Both the 2020 and 2024 eligible population files share an identical schema. Example values shown are from Barbour County. The 2024 file uses the same column names, descriptions, and calculation methodology.

[Eligible_population_2020_general_data_dictionary]

[Eligible_population_2024_general_data_dictionary]

Key Findings and Visualization: Black voter participation fell in 15 of 17 Alabama Black Belt counties between 2020 and 2024 dropping from 62.8% to 57.2% on average while White participation held nearly flat at 68%, nearly doubling the Black-White participation gap from 5.3 to 10.1 percentage points

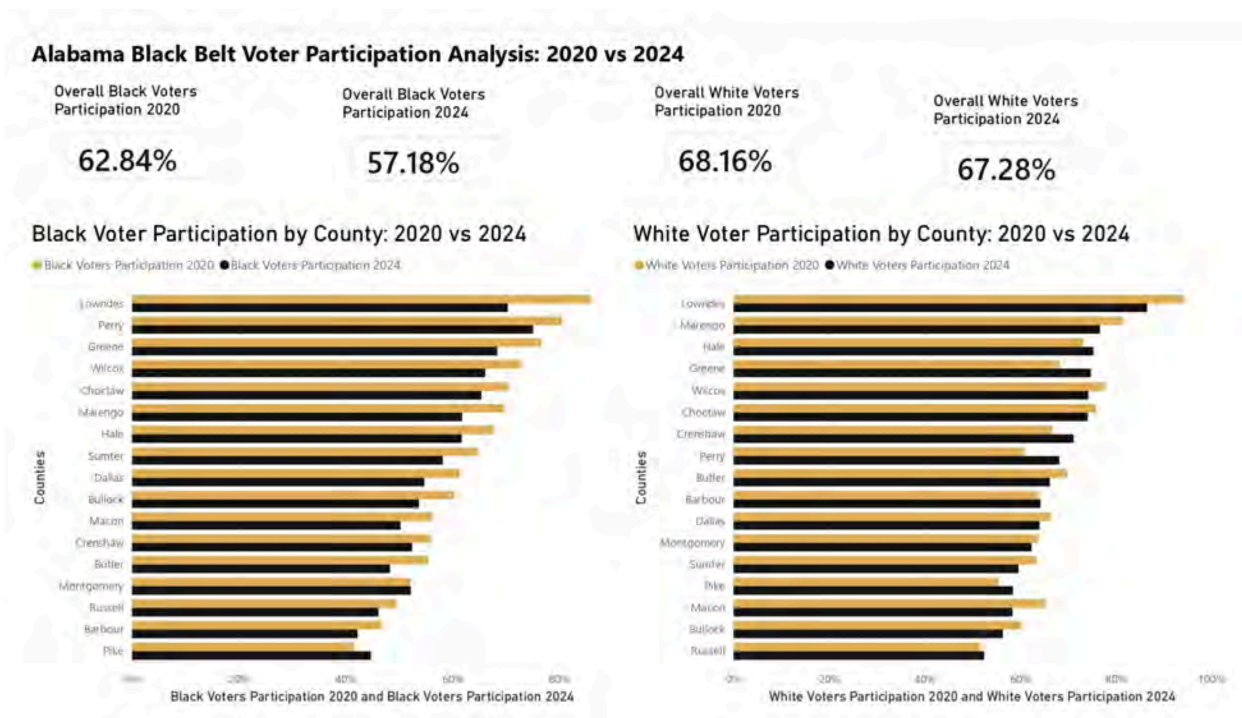

Figure 1. Black and White voter participation rates by county across 17 Alabama Black Belt counties, 2020-2024 general elections Source: Alabama Secretary of State, 2020 and 2024; Census Bureau CVAP, 2020 and 2024.

Voter Registration Datasets

Overview

The voter registration datasets step down from the full eligible population and focus on who successfully registered and how healthy those registrations are. The dataset shows everyone who registered and how many are in active status versus how many are at risk of being removed from the rolls before the next election. An inactive registration means they did not vote in recent elections or did not respond to a confirmation mailing. Inactive voters can still vote, but may need to cast a provisional ballot.

Both files use the Alabama Voter Registration (ALVR) October pre-election snapshot as the primary source, covering active and inactive registration counts broken down by race across all 17 counties. The October snapshot was chosen as the closest available pre-election picture of who was on the rolls. Each row is one county. CVAP eligible population counts are included for reference but are not used as denominators in any calculated columns; that decision was made after testing revealed that CVAP produced registration rates above 100% in 15 of 17 counties.

Cleaning Script

Script: voter_registration_general_2020_cleaning.ipynb / voter_registration_general_2024_cleaning.ipynb

The registration cleaning scripts load the ALVR October sheet for the respective year. The ALVR file has a broken two-row header, the first row contains race category labels, and the second row contains sub-labels, which prevents pandas from parsing column names correctly. The scripts handle this by manually setting clean column names and dropping the first two header rows before any processing.

County names in the ALVR file are in ALL CAPS and are converted to title case before joining. FIPS codes are joined from RDH using the county name as the merge key. The total registered count per race is calculated as active plus inactive from ALVR. The active_share is calculated as active divided by total registered using ALVR data only. The inactive_rate is calculated as inactive divided by total registered. All rates are stored as decimals rounded to 4 decimal places. CVAP eligible population is loaded separately and merged in for reference, but not used as a denominator.

Documentation

Sources

- Alabama Voter Registration (ALVR) October 2020 sheet — primary source for active and inactive registration counts by race
- Alabama Voter Registration (ALVR) October 2024 sheet — primary source for active and inactive registration counts by race
- Census Bureau CVAP 2020 and 2024 files — included for the eligible population reference only; not used as denominators
- Redistricting Data Hub (RDH) — source for FIPS codes only; not in final output

Limitations

- Race columns are limited to Black, White, and Latinx per project scope. Asian, American Indian, Federally Registered, Other, and Not Identified categories are excluded from calculated columns but remain in active_total and inactive_total.
- CVAP registration rates were tested and removed. Dividing ALVR registration counts by CVAP eligible population produced rates above 100% in 15 of 17 counties due to known CVAP undercount in rural majority-minority communities. Active_share columns use ALVR data only. CVAP counts remain in the file for users who want to perform their own rate calculations with a different denominator.
- The ALVR October snapshot may differ slightly from Election Day registration totals as late registrations and Election Day changes are not captured.
- Inactive status does not mean a voter cannot vote — inactive voters can still cast a provisional ballot or update their registration. However, they may face additional barriers compared to active registrants.
- Latinx registration counts are very small in several counties. Wilcox has 9 registered Latinx voters and Sumter has 13; rates derived from these samples are statistically unreliable.
- Hispanic was renamed to Latinx in all outputs. FIPS codes are stored as 4-digit integers. Missing values are stored as null.

Intended Use

These datasets support SPLC's work on voter purge practices and registration access gaps. The inactive_rate_black column is the key field for identifying racial disparities in purge risk at the county level. Macon County's 25.7% Black inactive rate versus 8.6% White inactive rate is the most severe disparity in the dataset. The gap between active_share_black and active_share_white shows where Black registrations are structurally less stable than White registrations heading into an election.

Data Dictionary:

[Voter_registration_2020_general_data_dictionary]

[Voter_registration_2024_general_data_dictionary]

Key Findings and Power BI Visualization:

Across the 17 counties in 2024, there are 29,357 inactive Black registrations versus 16,489 inactive White registrations. The average inactive rate for Black voters (12.0%) is 3.1 points higher than that of White voters (8.9%). Macon County has the most extreme gap at 17.1 percentage points (25.7% Black vs 8.6% White).

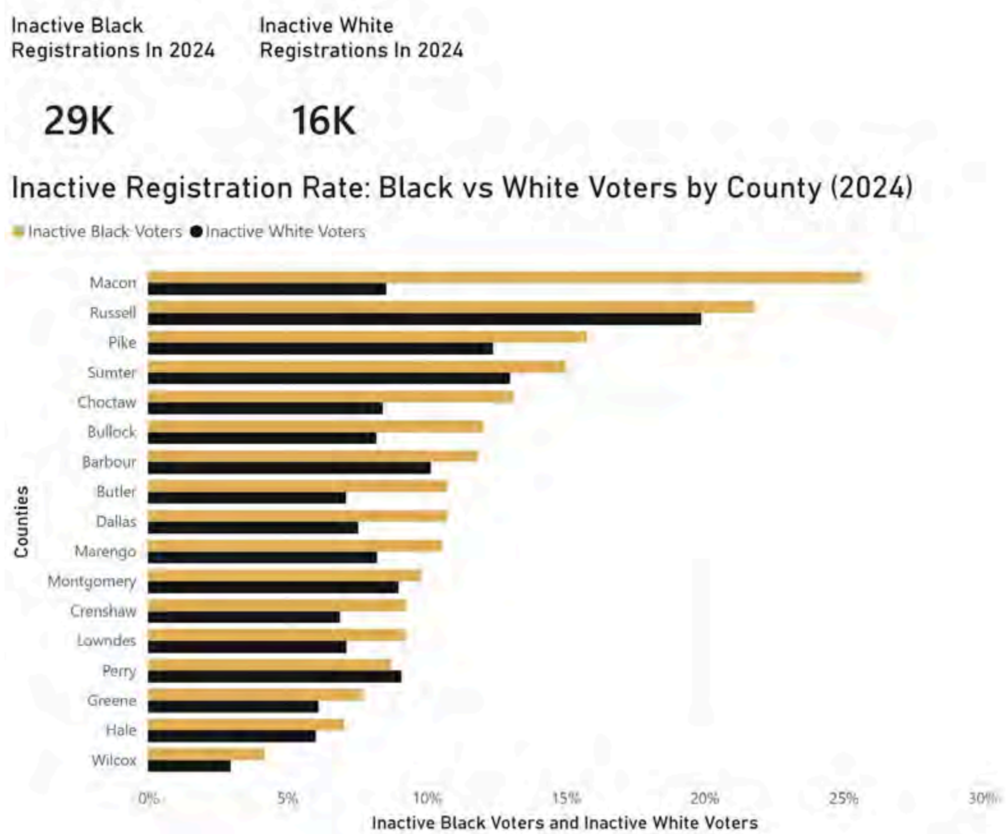

Figure 2. Inactive voter registration rates for Black and White voters by county across 17 Alabama Black Belt counties, 2024 general election   Source: Alabama Voter Registration (ALVR), 2024

Voter Turnout Datasets

Overview

The voter turnout datasets are the most granular of the three types, documenting who actually showed up on Election Day across multiple demographic dimensions. Each file contains 23 columns covering total ballots, registered voters, overall turnout rate, absentee ballot counts, breakdowns by gender, a full age bracket distribution from 18-29 through 80-plus, and race-specific ballot counts and turnout rates for Black, White, and Latinx voters.

The files draw from four Alabama Secretary of State source files merged on county name: the total file, gender file, age file, and race file. All four were validated against each other using dedicated validation scripts before the cleaning scripts were finalized. The overall turnout_rate is calculated as total ballots divided by total registered voters from ALVR. Race-specific turnout rates use ALVR total registered voters (active plus inactive) as the denominator, which matches how the overall turnout rate is calculated.

Cleaning Script

Scripts: voter_turnout_general_2020_cleaning.ipynb / voter_turnout_general_2024_cleaning.ipynb

The turnout cleaning scripts merge four SoS source files: total, gender, age, and race on county name. The 2020 SOS race file has embedded line breaks in column headers that are stripped by the script before processing. County names are standardized to title case and filtered to the 17 Black Belt counties before any calculations.

The ALVR October sheet is loaded separately and used to build total registered voter counts by race, which serve as the denominators for the three race-based turnout rate columns. The age brackets 80-89, 90-99, and 100-plus from the SOS age file are consolidated into a single ballots_age_80_plus column because the 90-99 and 100-plus counts are too small per county to report reliably as separate fields. Absentee rate is calculated as absentee ballots divided by total ballots cast. All rates are stored as decimals rounded to 4 decimal places and exported with null string values.

Two validation scripts — sos_gender_age_race_total_2020_validation.ipynb and sos_gender_age_race_total_2024_validation.ipynb were run before the cleaning scripts to document internal SOS discrepancies. Butler County had a 418-ballot discrepancy in 2024 between the SOS gender/age breakdown files and the SOS total file. The team treated the SOS total file as authoritative and documented the discrepancy in the data dictionary.

Documentation

Sources

- Alabama Secretary of State total, gender, age, and race turnout files for 2020 — primary source for ballot counts and registered voter totals
- Alabama Secretary of State total, gender, age, and race turnout files for 2024 — same
- Alabama Voter Registration (ALVR) October 2020 sheet — source for race-based registered voter denominators
- Alabama Voter Registration (ALVR) October 2024 sheet — same
- Redistricting Data Hub (RDH) — source for FIPS codes only; not in final output

Limitations

- Race columns ballots_black, ballots_white, and ballots_latinx do not sum to total_ballots_cast. Asian, American Indian, Federally Registered, Other, and Not Identified race categories are excluded from the race-specific columns but are included in total_ballots_cast.
- Butler County has a 418-ballot discrepancy between the SOS gender/age breakdown files and the SOS total file in 2024. All other counties have discrepancies of 50 ballots or fewer. The SOS total file is authoritative; the discrepancy is a source data issue, not a cleaning issue.
- Small discrepancies between SOS breakdown files and the total file exist across most counties. These are documented in the verification scripts.
- The ALVR October snapshot may differ slightly from Election Day registration totals. The denominator for race-based turnout rates reflects the October registration snapshot rather than the exact Election Day rolls.
- Race categories are self-reported at voter registration. SOS race categories may not reflect how a voter identifies at the time of casting a ballot.
- Hispanic was renamed to Latinx. FIPS codes are 4-digit integers. Missing values are stored as null.

Intended Use

These datasets support analysis of who voted in the 2020 and 2024 General Elections, broken down by gender, age, and race. The absentee_rate columns are particularly relevant to SPLC's access-oriented work, documenting the sharp collapse in absentee voting between elections. The turnout_rate_black and turnout_rate_white columns are the primary fields for racial turnout comparison and should be used alongside the not_voted_gap from the eligible population files and inactive_rate_black from the registration files for the full three-layer picture of where the civic participation gap originates, compounds, and manifests.

Data Dictionary:

Voter_turnout_2020_general_data_dictionary

Voter_turnout_2024_general_data_dictionary

Key Findings and Visualization:

Average Black turnout fell from 59.0% to 51.0% while average White turnout fell from 69.3% to 66.2%. The Black-White turnout gap widened from 10.3 to 15.2 percentage points. Lowndes saw the steepest Black decline at –13.2%.

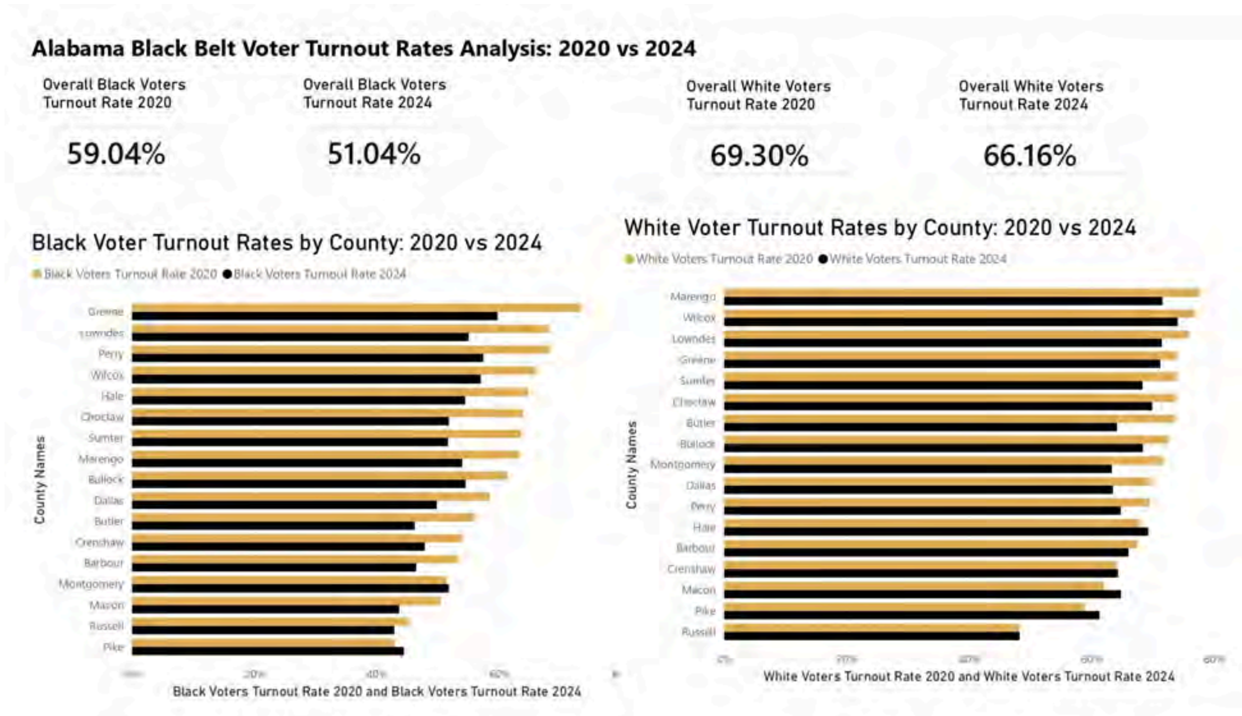

Figure 3. Black and White voter turnout rates by county across 17 Alabama Black Belt counties, 2020-2024 general elections Source: Alabama Secretary of State, 2020 and 2024; Alabama Voter Registration (ALVR), October 2020 and 2024.

Connections to Other Teams

BB-DIG serves as the data foundation for the regional analysis. Our six cleaned CSVs with standardized FIPS codes and county names are designed to be used directly by any team conducting county-level analysis across the Black Belt counties. Teams relying on our data include BB-VIZ for visualizations and BB-PAD for spatial and precinct-level analysis.

Recommendations

Recommendations for the client (SPLC)

The cleaned datasets reveal consistent patterns across the 17 Black Belt counties that warrant target action. SPLC should prioritize voter outreach in counties where Black turnout rates, calculated against total registered voters, are lowest relative to other counties, as these represent the largest gaps between the registration and actual participation. Inactive registration rates among Black voters are also elevated across multiple counties, flagging a population that is registered but at risk of being purged before future elections. SPLC should monitor these figures year over year and invest in re-registration support ahead of each election cycle. For county-to-county comparison, rate-based fields such as turnout_rate_black and inactive_rate_black should be used rather than raw ballot counts, since population size varies considerably across the 17 counties and raw numbers can be misleading. Finally, Latinx voter data should be treated as directional rather than definitive. Population sizes are small in several rural counties, and the AlVR registration figures for Hispanic voters include cases like Wilcox County, where missing values were filled with zero, which affects the reliability of Latinx turnout and registration rates in those specific counties.

Recommendations for future project teams

Future teams should be aware that measurable discrepancies exist between Alabama Secretary of State-reported totals and RHD/L2 voter file data. The comparison script quantifies these differences in both voted counts and registration numbers of 2024, and future teams should run this check against any new election year data before deciding which source to treat as authoritative. The internal SOS validation scripts confirmed that the gender, age, and race demographic breakdowns are internally consistent with SOS total ballots figures for both 2020 and 2024, which gives confidence in using SOS race data for turnout analysis, but this check should be repeated for any new election year added to the dataset.

If staking out, the standardization document should be finalized before any cleaning scripts are written. Column naming conventions, the choice of turnout denominator, and the file naming formats were decisions that required retroactive fixes across multiple scripts, which added significant rework. Aligning on these decisions upfront as a team would have saved considerable time during the licensing phase.

The most natural next step for a future capstone team would be to extend this analysis to the 2020 midterm election, which would allow SPLC to see whether the participation gaps observed in 2020 and 2024 General Election hold in lower turnout cycles, a period when disparities in civic participation tend to widen. Adding a longitudinal view across three election cycles would also make the dataset significantly more useful for SPLC's long-term advocacy and outreach planning.

What methods worked/didn't

- We felt that breaking up the project into three datasets worked to answer a question that the SPLC could use, specifically broken up with a clear denominator. It made our data easy to use and allowed us to lay out guidelines on what each rate measured. This distinction is important, and it's important for future teams to understand what our datasets do, so they don't just combine our data.
- Sticking with the ALVR as the authoritative source helped us produce defensible rates and avoid any source disagreement between RDH and CVAP. It may be important to future teams that deal with conflicting datasets and have to decide which data they want to prioritize. Secretary of State data is reliable when it comes to civic participation across all 50 states, so it should definitely be worth looking into.
- We wrote specifically in each data dictionary how they referenced the other datasets, which really helped us as a team map out how the files all relate to each other. We added this later in the project when we had some coordination issues, so it not only helps us as a team, but also helps users themselves understand the data.
- What didn't work for us was making naming decisions and titles of visualization charts before we had our methodology confirmed. This led to a lot of renaming throughout the semester and labeling certain visualizations incorrectly because we hadn't solidified our standardization document or methodology. In the future, it is important to establish strict naming conventions and ensure that data is being used properly according to its data dictionary.
- As a team, we wish we had noticed some of our data quality issues a little earlier, like the CVAP undercount issue, because this led to a lot of late additions and changes to the project

that affected many different aspects. In-depth data quality checks are important, and it's also important to really understand the data that is pulled from sources. Data from multiple sources is going to have conflicts, so doing that early on is better than realizing it while cleaning the data.

Conclusions

Over the course of 10 weeks, BB-DIG successfully developed a comprehensive data infrastructure that supports analysis of voting access and civic participation across the Alabama Black Belt. Using publicly available datasets and client-approved data sources, BB-DIG collected, cleaned, and integrated datasets covering the 2020 and 2024 general elections. Work was centered on structuring data to capture turnout, non-participation, and demographic patterns across key racial groups, ensuring the datasets could support various analyses. Python, Excel, GitHub, and scripting workflows were utilized to clean, standardize, and document the data.

Key deliverables include three cleaned datasets, a data dictionary, reproducible scripts, and a standardization document, all of which are designed to ensure reusability and transparency in our workflow. Together, these deliverables provide SPLC with a strong foundation for future research, enabling a deeper analysis of gaps in civic participation, which will support decision-making in these regions.

Team contact information is provided below.

Matthew Ritter mritter4@terpmail.umd.edu, Bao Nguyen bnguye26@terpmail.umd.edu, Margaret Hermanto mhermant@terpmail.umd.edu, Asmita Brahme aubrahme@terpmail.umd.edu, Xuan Zhang xzhang37@terpmail.umd.edu, Arnav Patel apatel81@terpmail.umd.edu

Deliverables Checklist

Deliverable 1: Requirements document detailing expected results of the semester

- Link: https://drive.google.com/file/d/17zLI0cbXlU9RcRyULtA6WQw41a0W16yy/view?usp=sharing
- Format: PDF report.

Deliverable 2: Clean and organize datasets for regional civic participation analysis

- There are six clean datasets, each categorized into its own respective folder via the following link: https://github.com/MatthewRitt/BB-DIG/tree/main/data_storage/clean_data
- You can also view them in Google Drive with this link: https://drive.google.com/drive/folders/1sOKmlyDcs5h4hI3lHpYcA6j6lBKG5olF?usp=drive_link
- Type: CSV

Deliverable 3: Data dictionary and documentation describing sources and limitations

- There are data dictionaries and data documentation located in each of the six dataset folders via the following link: https://github.com/MatthewRitt/BB-DIG/tree/main/data_storage/clean_data
- You can also view them in Google Drive with this link: https://drive.google.com/drive/folders/1sOKmlyDcs5h4hI3lHpYcA6j6lBKG5olF?usp=drive_link
- Type: data dictionaries are Excel; data documentation is MD files

Deliverable 4: Documented standards for data structures, naming conventions, and geographic identifiers to enable consistent and reusable regional analysis

- Link: https://drive.google.com/file/d/1AgFQeXsgz9Xp32_JcdoyboKGS9qgw0gU/view?usp=sharing
- Type: PDF

Deliverable 5: Final presentation of the findings of the project

- Link: https://drive.google.com/file/d/1r7yicWQKCz-NKchjHGRC5LbFou344lcq/view?usp=sharing
- Type: PDF

Deliverable 6: Final report, including any recommendations as appropriate

- Link: https://docs.google.com/document/d/1f2Igfui5inakO4U4lgOk1t2BADNSWtChNa3h3C0yCiQ/edit?usp=sharing
- Type: Google Doc

COLLEGE OF INFORMATION

Voting & Civic Participation

Alabama Black Belt Counties

Team 2 BB-VOTE: Voter Turnout and Participation Trends

Prepared for: Southern Poverty Law Center (SPLC)

Team Members: Brayden Hull, Benjamin Donkor, Griffin Biddle, Bryan Cristales Rivas, Evan Carr

Course: INST490 · Section: 0103

Date: May 18, 2026

Table of Contents

Team 2: BB-VOTE

Section Brief

The BB-VOTE team analyzed voter turnout and civic participation trends across 17 Alabama Black Belt counties using data from the 2020 and 2024 general elections and the 2024 Congressional District 2 race. This analysis was conducted for the Southern Poverty Law Center (SPLC) to better understand whether Black and Latinx communities in these counties are participating in elections, supporting SPLC's broader civil rights and racial justice work. Our team's findings and rates use Citizen Voting Age Population (CVAP), a Census Bureau estimate of the number of citizens age 18+ in a given area, broken down by race, as the eligible population denominator.

Key Findings

- Voter turnout declined in all 17 Black Belt counties between 2020 and 2024, with total ballots decreasing by nearly 23,000 votes (raw ballots cast).
- Black voter turnout (voting-eligible population-based) declined from 56.8% in 2020 to 51.9% in 2024, with 16 of 17 counties showing a decrease in Black civic participation.
- Latinx civic participation increased slightly but remains below 1% of total voters in most counties, making county-level statistical conclusions unreliable.
- Black voter percentage was a strong and consistent predictor of Democratic vote share across all three elections analyzed (r = 0.9855, 0.9877, and 0.8951).
- The 2024 CD-2 election results validated the impact of the Allen v. Milligan redistricting decision, with Shomari Figures winning as one of Alabama's first two Black representatives in Congress..

Deliverables Produced

- SQL database and query findings analyzing turnout and racial participation across three election cycles. BB-VOTE Project

- 13 choropleth maps published to ArcGIS Online, visualizing turnout and demographic trends. Maps
- Cleaned and standardized datasets across the 2020 and 2024 general elections and the 2024 CD-2 race, stored as CSV and Excel files in the project data folder. Datasets

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Challenges

The team faced limitations in data availability, as racial participation data from the Alabama Secretary of State is only reported at the county level, making precinct-level or CD-2 specific racial breakdowns impossible. Latinx voter data presented significant reliability issues in most counties due to extremely small population sizes, with some counties reporting CVAP estimates as low as 1–4 individuals.

Connections to Other Teams

BB-VIZ provided the color guide used for the choropleth maps built in ArcGIS Pro, ensuring visual consistency across the final report.

Final Report and Deliverables

Team Members

Project Manager - Brayden Hull

Brayden was responsible for coordinating overall team progress, ensuring project deadlines and deliverables were met, and maintaining collaboration among team members. His role also involved monitoring workflow and ensuring all members had opportunities to contribute throughout the semester.

Team Communicator - Benjamin Donkor

Benjamin managed internal team coordination by organizing meetings and ensuring all members remained informed about project goals/deadlines, and ongoing tasks. His role helped maintain consistency and alignment across all phases of the project.

Client Liaison and Lead Writer - Griffin Biddle

Griffin oversaw formal communication with the Southern Poverty Law Center (SPLC) and served as the primary writer and reviewer for all major deliverables before submission. He was also responsible for ensuring written materials maintained a professional and consistent standard appropriate for client facing reports and documentation.

Technical Lead - Bryan Cristales Rivas

Bryan coordinated the project's analytical and technical components, including data collection, database management, SQL analysis, and statistical evaluation. His responsibilities included organizing technical tasks, supporting team members with programming and analysis work, and ensuring the accuracy of analytical outputs.

Documentation and Visuals Coordinator - Evan Carr

Evan maintained records of meetings, project discussions, and team decisions throughout the semester. His role ensured that project goals and ongoing progress were consistently documented. Additionally, he was a lead in constructing the visualizations for the final deliverables.

Project Context

The Southern Poverty Law Center (SPLC) is a nonprofit civil rights organization established in 1971 and based in Montgomery, Alabama. Its work centers on defending civil rights, fighting hate and extremism, and addressing racial injustice through legal action. For our client, having access to carefully analyzed and reliable data on voter turnout and trends among marginalized populations in Alabama's Black Belt counties is essential. They aim to better understand whether black and latinx communities in these counties are participating in elections.

Although our project will not directly explore the reasons behind why these communities do or do not vote, the data we produce will play a key role in supporting that kind of analysis. We will collect and examine data to compare geographic voter turnout and election outcomes with demographic information in order to uncover patterns and trends. This includes analyzing relationships such as registered voters versus those who actually cast ballots, registered versus unregistered populations, and identifying counties with higher or lower turnout rates. Insights from these analyses will support SPLC's community focused initiatives aimed at serving marginalized groups.

Methods

The study analyzed 17 counties using Alabama Secretary of State data, including total ballots cast, race participation, and precinct results. Data was cleaned and stored in a MySQL database. Pearson correlation was used to measure the relationship between Black voter percentage and Democratic vote shares.

1. Project Approach

The primary objective of this project was to analyze voter turnout and racial participation trends across Alabama's Black Belt counties using data from the 2020 and 2024 election cycles. We found and collected existing data, then cleaned and analyzed it in order to identify turnout trends and relationships between race and election outcomes. The project focused specifically on Black and Latinx civic participation and the electoral impact of Alabama's 2024 Congressional District 2 (CD - 2) following the Supreme Court decision in Allen v. Milligan.

2. Data Sources

The analysis incorporated publicly available election, demographic, and geographic datasets from multiple federal and state sources. Primary election data was obtained from the Alabama Secretary of State, while demographic and Voting Eligible Population (VEP) estimates were sourced from the United States Census Bureau.

Primary Data Sources

- Alabama Secretary of State - Precinct Level General Election Results (2020 and 2024)
- Alabama Secretary of State - General Election Total Ballots Cast Reports
- Alabama Secretary of State - General Election Participation by Race Reports
- Alabama Secretary of State - 2024 Congressional District 2 Election Results
- United States Census Bureau - American Community Survey (ACS) demographic estimates
- United States Census Bureau - Citizen Voting Age Population (CVAP) Special Tabulations (2016–2020 ACS 5 Year)

- CVAP Special Tabulations (2016–2020 ACS 5 Year) — County and Congressional District level
- Redistricting and county level demographic reference data used for geographic validation and comparison

Additional Project Datasets

The team also incorporated supporting datasets and project files stored within the shared project repository and client data archive, including:

- County level turnout datasets
- Precinct level election result files
- Race participation summaries
- CVAP demographic tables
- Congressional district boundary reference data
- Supplemental geographic and turnout comparison datasets used during SQL analysis and visualization development

These datasets were reviewed and standardized prior to analysis to ensure consistent county naming conventions, formatting, and election year alignment across all files.

3. Geographic Scope

The project analyzed 17 Alabama counties:

Black Belt Counties (17)
Barbour, Bullock, Butler, Choctaw, Crenshaw, Dallas, Greene, Hale, Lowndes, Macon, Marengo, Montgomery, Perry, Pike, Russell, Sumter, and Wilcox.

Additional analysis was conducted for the 2024 Congressional District 2 election, which included the subset of counties located within the newly redrawn district boundaries following Allen v. Milligan.

Deliverables and Findings

Voter turnout declined in all Black Belt counties from 2020 to 2024. Total ballots decreased significantly across the region. Black civic participation declined in most studied counties, with only Montgomery showing a small increase. VEP-based Black voter turnout declined from 56.8% in 2020 to 51.9% in 2024, while Latinx voter turnout remained consistently low at approximately 19% in both election cycles. Latinx civic participation increased in most counties, but it remains less than 1 percent of total voters in most areas, making reliable conclusions difficult. Race was the most consistent factor in election outcomes. Counties with higher percentages of Black voters tended to show higher Democratic vote shares.

1. Findings Overview

This document summarizes the findings from the SQL analysis of voter turnout and participation data across three election cycles in Alabama's Black Belt counties. The analysis focuses on trends in Black and Latinx civic participation.

2. Data Sources & Methodology

2.1 Election Cycles Analyzed

- 2020 General Election (Presidential) - Biden vs. Trump
- 2024 General Election (Presidential) - Harris vs. Trump
- 2024 General Election (2nd Congressional District) - Figures vs. Dobson

2.2 Geographic Scope

Black Belt Counties (17):

Barbour, Bullock, Butler, Choctaw, Crenshaw, Dallas, Greene, Hale, Lowndes, Macon, Marengo, Montgomery, Perry, Pike, Russell, Sumter, Wilcox

Note: While this report focuses exclusively on the 17 Black Belt counties as assigned, the project presentation included Gulf Coast counties (Baldwin, Clarke, Conecuh, Escambia, Mobile, Monroe, Washington) for comparative purposes. Analyzing both regions provided the client with a broader view of voter turnout trends across southern Alabama and helped contextualize Black Belt findings within the wider regional landscape. The Gulf Coast comparison also supported the analysis of Alabama's 2nd Congressional District, which spans both regions.

2.3 Data Sources

- Alabama Secretary of State - Precinct Level General Election Results (2020, 2024)
- Alabama Secretary of State - General Election Total Ballots Cast (2020, 2024)
- Alabama Secretary of State - General Election Participation by Race (2020, 2024)
- Alabama Secretary of State - 2024 General Election CD - 2 Precinct Results
- U.S. Census Bureau - CVAP Special Tabulation 2016 - 2020 ACS 5 Year (County and CD level)
- U.S. Census Bureau - CVAP Special Tabulation 2020 - 2024 ACS 5 Year (County and CD level)
- U.S. Census Bureau - ACS 1 Year 2024 (S2901: Citizens 18 Years and Over by Race - Alabama statewide)

3. Methodology Notes & Assumptions

3.1 VEP Based Turnout Rate Calculation

In addition to the registered voter based turnout rates reported by the Alabama Secretary of State, this project calculated VEP based (Voting Eligible Population) turnout rates for black and latinx voters, using CVAP data as the denominator. This is the academically preferred method as it measures participation among all eligible citizens, not just those registered to vote.

Formula: Black/Latinx Turnout Rate = (Race-Specific Ballots Cast ÷ Race-Specific CVAP) × 100

3.2 Total Federally Registered Exclusion

The 'Total Federally Registered (may be of any race)' column was excluded from all racial participation analyses. This field represents an administrative registration category in Alabama's voter file, not a self identified race category. It is a separate recordkeeping pathway that may include voters of any race and, therefore, cannot be used as a demographic indicator.

3.3 CD - 2 County Scope

Not all Black Belt counties fall within the 2024 2nd Congressional District boundaries. CD-2 analysis was limited to the following Black Belt counties within the district:

Barbour, Bullock, Butler, Crenshaw, Macon, Montgomery, Pike, Russell

4. Key Findings

4.1 Overall Voter Turnout Declined from 2020 to 2024

Turnout decreased in all 17 counties between the 2020 and 2024 general elections. This is consistent with national trends, because turnout declines between presidential election cycles are common and may reflect differences in campaign intensity and voter mobilization.

Region	2020 Avg Turnout	2024 Avg Turnout	Change	2020 Total Ballots	2024 Total Ballots
Black Belt	61.78%	56.86%	-4.92%	248,335	225,537

Table 1. Overall voter turnout by region, 2020 and 2024. Source: Alabama Secretary of State, 2020, 2024.

- Black Belt total ballots decreased by nearly 23,000 votes, which was a significant drop
- Counties with the largest turnout declines: Sumter (-7.76%), Lowndes (-7.66%), Greene (-6.58%)
- Counties with the smallest declines: Macon (-2.59%) and Crenshaw (-2.65%)

4.2 Black Voter Turnout Rate (VEP Based)

Using CVAP as the VEP denominator, Black voter turnout rates were calculated for all 17 study counties. This is the preferred method for measuring true participation among eligible Black voters.

Metric	2020	2024	Change
Statewide Black Ballots Cast	562,519	511,534	-50,985
Statewide Black CVAP (VEP)	989,960	984,877	-5,083
Black Turnout Rate (VEP Based)	56.8%	51.9%	-4.9%

Table 2. Black voter turnout rate (VEP-based) across Alabama's Black Belt counties, 2020 and 2024.Source: Alabama Secretary of State, 2020, 2024; U.S. Census Bureau CVAP Special Tabulation 2016-2020 and 2020-2024.

- Black voter turnout declined by 4.9 percentage points from 2020 to 2024

- Highest black turnout 2020: Lowndes County (86.17%)
- Highest black turnout 2024: Perry County (75.11%)

Black Civic Participation Declined in Most Counties:

County	Black Voters 2020	Black Voters 2024	Change
Dallas	12,221	10,418	-1,803
Macon	7,043	6,097	-946
Lowndes	4,860	3,916	-944
Montgomery	51,522	51,925	+403
Russell	9,746	9,217	-529

Table 3. Black voter participation by selected Black Belt county, 2020 and 2024. Source: Alabama Secretary of State, 2020, 2024.

- Dallas had the largest absolute drop in Black Belt Black civic participation (-1,803)
- Montgomery was the only major Black Belt county where black civic participation increased (+403)
- Black civic participation declined in 15 out of 17 counties from 2020 to 2024

4.3 Latinx Civic Participation

Latinx civic participation is documented in this project, but meaningful analysis is severely constrained by extremely small population sizes across all 17 study counties. VEP based Latinx turnout rates were calculated but show significant reliability issues in most counties.

Metric	2020	2024	Change
Statewide Latinx Ballots Cast	20,478	23,430	+2,952
Statewide Latinx CVAP (VEP)	106,827	123,709	+16,882
Latinx Turnout Rate (VEP Based)	19.2%	18.9%	-0.3%

Table 4. Latinx voter turnout rate (VEP-based) across Alabama's Black Belt counties, 2020 and 2024. Source: Alabama Secretary of State, 2020, 2024; U.S. Census Bureau CVAP Special Tabulation 2016-2020 and 2020-2024.

- Latinx turnout rate remained consistently low at approximately 19% in both elections

Latinx Civic Participation by County (Selected Counties):

County	Latinx Voters 2020	Latinx Voters 2024	Change
Russell	287	329	+42
Montgomery	604	715	+111
Pike	47	59	+12

Table 5. Latinx voter participation by selected Black Belt county, 2020 and 2024. Source: Alabama Secretary of State, 2020, 2024.

- Latinx civic participation actually increased in most counties from 2020 to 2024
- Montgomery County showed the largest increase in Latinx participation among Black Belt counties (604 to 715 voters)
- Latinx civic participation remained very low across all counties
- Despite increases, latinx voters remain a very small percentage of total voters, less than 1% in most counties
- The small sample size makes it statistically impossible to draw meaningful conclusions about latinx voting behavior at the county level

4.4 Race is a Consistent and Significant Factor in Voting

Analysis across all three election cycles reveals a clear and consistent pattern: the racial composition of a county is strongly correlated with candidate support. This trend appeared in the 2020, 2024 Presidential, and 2024 CD - 2 races.

Election	High Black % Winner	Low Black % Winner	Tipping Point
2020 Presidential	Biden (D)	Trump (R)	~40 - 45% Black
2024 Presidential	Harris (D)	Trump (R)	~40 - 46% Black
2024 CD-2	Figures (D)	Dobson (R)	~40 - 44% Black

Table 6. Relationship between Black voter percentage and election outcomes across Alabama's Black Belt counties, 2020 and 2024. Source: Alabama Secretary of State, 2020, 2024; U.S. Census Bureau CVAP Special Tabulation 2016-2020 and 2020-2024.

- Counties with Black voter percentages above ~40 - 45% consistently voted for Democratic candidates
- Counties with Black voter percentages below ~40% consistently voted for Republican candidates
- Russell County was consistently the most competitive, narrowly voting for the Democratic candidate in all three elections
- Macon (77% black) and Bullock (72% black) were the strongest Democratic counties across all elections
- Crenshaw (20% black) and Washington (20% black) were the strongest Republican counties in the Black Belt counties

Latinx Voter Analysis:

Election	Latinx Correlation	Strength	Interpretation
2020 Presidential	-0.2622	Very Weak	No meaningful relationship
2024 Presidential	-0.1844	Very Weak	No meaningful relationship
2024 CD-2	0.2301	Very Weak	No meaningful relationship

Table 7. Pearson Correlation Coefficient results for Latinx voter percentage and Democratic vote share across Alabama's Black Belt counties, 2020 and 2024. Source: Alabama Secretary of State, 2020, 2024; U.S. Census Bureau CVAP Special Tabulation 2016-2020 and 2020-2024.

- Latinx voter percentage shows no meaningful correlation with candidate support across all three elections
- Although several counties had larger Latinx voter populations, many counties in the study area still contained very small Latinx voting populations relative to the total electorate.
- Despite increases in Latinx participation from 2020 to 2024, the numbers remain too small for statistically reliable conclusions

4.5 2024 CD - 2 Results Validate Allen v. Milligan

The 2024 CD - 2 election directly demonstrates the impact of the Allen v. Milligan Supreme Court decision and the subsequent redistricting that created a near majority black congressional district:

County	Figures (D)	Dobson (R)	% Black	% Latinx	Winner
Macon	6,134 (78.67%)	1,663 (21.33%)	77.14%	0.33%	Figures
Bullock	2,996 (73.06%)	1,105 (26.94%)	72.25%	0.44%	Figures

Montgomery	58,668 (65.56%)	30,813 (34.44%)	57.63%	0.79%	Figures
Russell	10,397 (50.76%)	10,085 (49.24%)	44.23%	1.58%	Figures
Crenshaw	1,488 (23.07%)	4,962 (76.93%)	19.59%	0.26%	Dobson
Barbour	4,119 (42.38%)	5,601 (57.62%)	38.11%	0.50%	Dobson
Butler	3,301 (39.17%)	5,127 (60.83%)	36.96%	0.22%	Dobson
Pike	4,911 (37.34%)	8,240 (62.66%)	31.75%	0.44%	Dobson

Table 8. 2024 CD-2 election results by Black Belt county, including Black and Latinx voter percentages. Source: Alabama Secretary of State, 2024; U.S. Census Bureau CVAP Special Tabulation 2020-2024.

- Shomari Figures (D) won the district, becoming one of Alabama's first two black representatives in Congress.
- The redistricting successfully gave black voters the opportunity to elect a candidate of their choice
- Latinx voters remained a small percentage of total voters in most counties.
- Russell County (1.58% latinx) had the highest latinx percentage in CD - 2 and was the most competitive county
- Without the court ordered redistricting, these voters would have been split across multiple majority white districts

5. Statistical Analysis

5.1 Method: Pearson Correlation Coefficient

The Pearson Correlation Coefficient (Pearson's r) was used to measure the strength and direction of the linear relationship between the percentage of Black voters in a county and the Democratic candidate's vote share. This method was selected because:

- The data are quantitative, both variables are continuous (percentages and vote counts)
- It measures the strength and direction of the relationship between racial composition and Democratic vote share.
- Pearson correlation was appropriate because both variables were continuous percentage measures and the analysis aimed to evaluate the linear relationship between racial composition and Democratic vote share.
- Results range from -1 to +1, where +1 indicates a perfect positive relationship

5.2 Variables Analyzed

- Independent Variable (X): Percentage of black voters in each county
- Dependent Variable (Y): Democratic candidate vote share in each county
- Unit of Analysis: County (13 counties for CD - 2, 24 counties for Presidential races)

5.3 Results: Black Voter Analysis

Election	Correlation (r)	Strength	Democratic Candidate	Interpretation
2020 Presidential	0.9855	Very Strong	Joseph R. Biden	Black voter percentage has a strong positive correlation with Biden vote share.
2024 Presidential	0.9877	Very Strong	Kamala D. Harris	Race showed the strongest statistical relationship with Democratic vote share across all elections analyzed.
2024 CD - 2	0.8951	Strong	Shomari Figures	Black voter percentage was strongly associated with Figures support.

Table 9. Pearson Correlation Coefficient results for Black voter percentage and Democratic vote share across Alabama's Black Belt counties and CD-2, 2020 and 2024. Source: Alabama Secretary of State, 2020, 2024.

5.4 Results: Latinx Voter Analysis

Election	Correlation (r)	Strength	Democratic Candidate	Interpretation
2020 Presidential	-0.2622	Very Weak	Joseph R. Biden	No meaningful relationship between Latinx % and Biden support
2024 Presidential	-0.1844	Very Weak	Kamala D. Harris	No meaningful relationship between Latinx % and Harris support
2024 CD - 2	0.2301	Very Weak	Shomari Figures	No meaningful relationship between Latinx % and Figures support

Table 10. Pearson Correlation Coefficient results for Latinx voter percentage and Democratic vote share across Alabama's Black Belt counties and CD-2, 2020 and 2024. Source: Alabama Secretary of State, 2020, 2024.

5.5 Interpretation

Correlation Strength Scale:

Correlation Value	Strength
0.9 - 1.0	Very Strong
0.7 - 0.9	Strong
0.5 - 0.7	Moderate
0.3 - 0.5	Weak
-0.3 - 0.3	Very Weak / Negligible

Table 11. Pearson Correlation Coefficient strength scale used for interpreting statistical results. Source: Evans, J. D. (1996). Straightforward Statistics for the Behavioral Sciences. Brooks/Cole Publishing.

Black Voter Findings:

- All three elections show strong to very strong correlations, confirming Black voter percentage is a statistically significant factor in voting
- The 2024 Presidential correlation (0.9877) is the strongest, meaning the relationship between Black voter percentage and Harris support was nearly perfect
- The 2020 and 2024 Presidential correlations are nearly identical (0.9855 vs 0.9877), confirming this is a consistent structural pattern, not a one time event
- The CD - 2 correlation (0.8951) is slightly lower than the Presidential races, this is expected because CD - 2 was a newly drawn competitive district
- Counties with higher Black voter percentages consistently favored Democratic candidates.

Latinx Voter Findings:

- All three elections show very weak correlations, Latinx voter percentage is not a reliable predictor of candidate support

- Negative correlations in Presidential races (-0.2622 and -0.1844) are likely because counties with higher l=Latinx populations also tend to be more Republican leaning overall
- This does not indicate Latinx voters prefer Republicans, the sample sizes are too small for meaningful conclusions
- Latinx voters make up less than 1% of voters in most counties studied, making statistical analysis unreliable
- The contrast between black (strong) and Latinx (very weak) correlations highlights the importance of sample size in statistical analysis

5.6 Statistical Conclusion

The Pearson Correlation Coefficient analysis provides strong statistical evidence that Black voter percentage is a significant and consistent factor in voting behavior across all three election cycles studied. The near perfect correlations in the Presidential races ($r > 0.98$) and the strong correlation in the CD - 2 race ($r = 0.8951$) confirm that the percentage of Black voters in a county is a highly consistent indicator of Democratic candidate support in Alabama's Black Belt counties.

In contrast, Latinx voter percentage shows no meaningful correlation with candidate support across all three elections. This is primarily due to the very small Latinx population in the study area, most counties have fewer than 50 Latinx voters, making it statistically impossible to draw reliable conclusions about Latinx voting behavior at the county level.

Data Visualizations

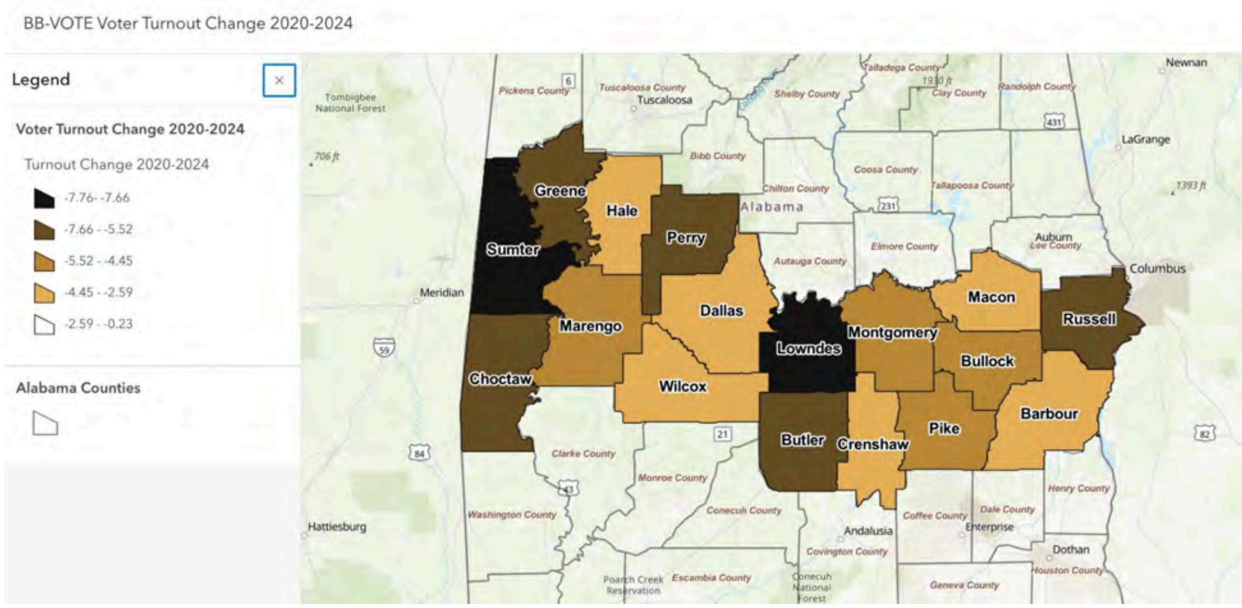

Figure 1. Overall voter turnout change by county across Alabama's Black Belt counties, 2020-2024. Source: Alabama Secretary of State, 2020, 2024.

Interactive map available at:

[*https://uofmd.maps.arcgis.com/apps/mapviewer/index.html?webmap=738e45573c7d4b94935c4c1d719392dd&center=-86.409663%2C32.150606&scale=2311162.217155*](https://uofmd.maps.arcgis.com/apps/mapviewer/index.html?webmap=738e45573c7d4b94935c4c1d719392dd&center=-86.409663%2C32.150606&scale=2311162.217155)

This map shows the change in overall voter turnout rates between the 2020 and 2024 general elections across the 17 Alabama counties included in the study. Counties are visualized based on whether turnout increased or decreased and by how much, allowing for clear geographic comparison in the Black Belt counties. This visualization supports the project goal of identifying geographic turnout trends over time. It directly highlights that turnout declined across all counties. This helps contextualize the regional disparities identified in the statistical analysis.

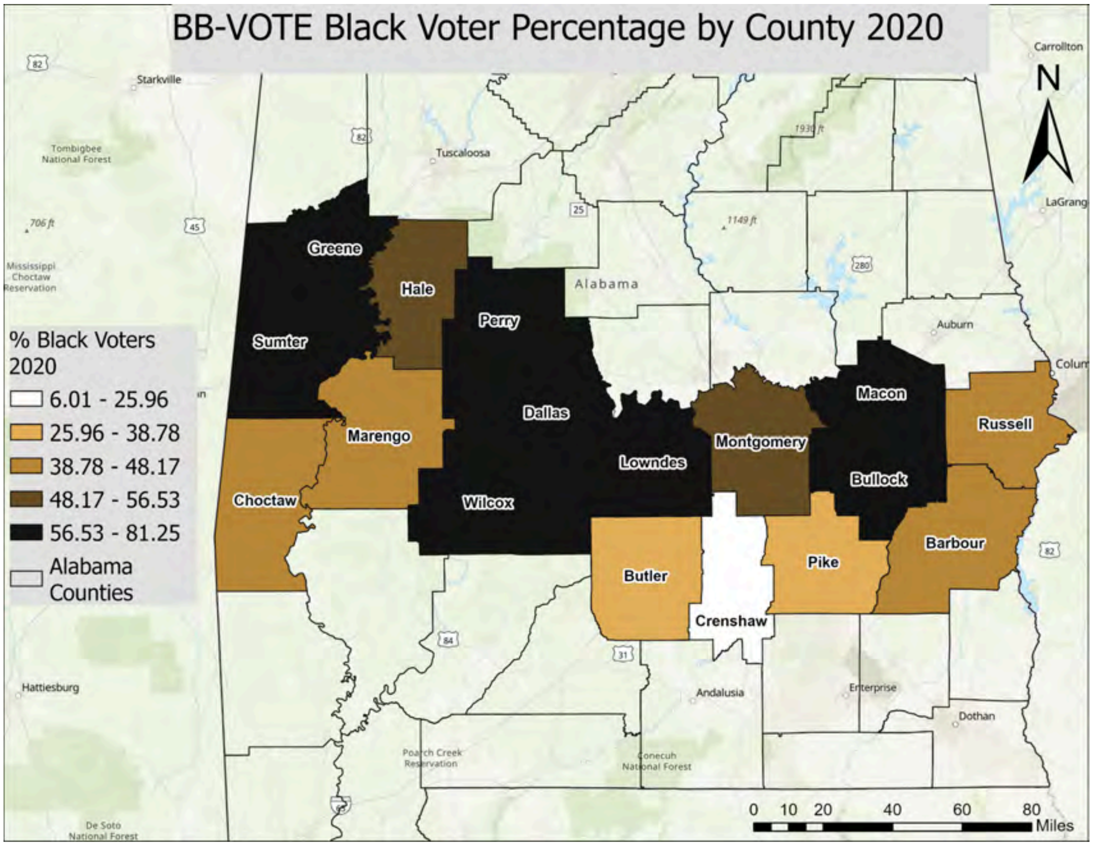

Figure 2. Black voter percentage by county across Alabama's Black Belt counties, 2020. Source: Alabama Secretary of State, 2020.

Interactive map available at:

https://uofmd.maps.arcgis.com/apps/mapviewer/index.html?webmap=f43b01bf3ecc46ba8f4c7111e40d2a2c&center=-87.794092%2C32.017724&scale=9244648.868618

This map displays the percentage of Black voters (or Black voting eligible population proxy) across the 17 counties in 2020. It provides a baseline snapshot of racial composition before the 2024 election cycle. This visualization supports the project's objective of examining the relationship between racial composition and voter behavior. It is used as a baseline to compare how racial geography aligns with voting patterns in 2020.

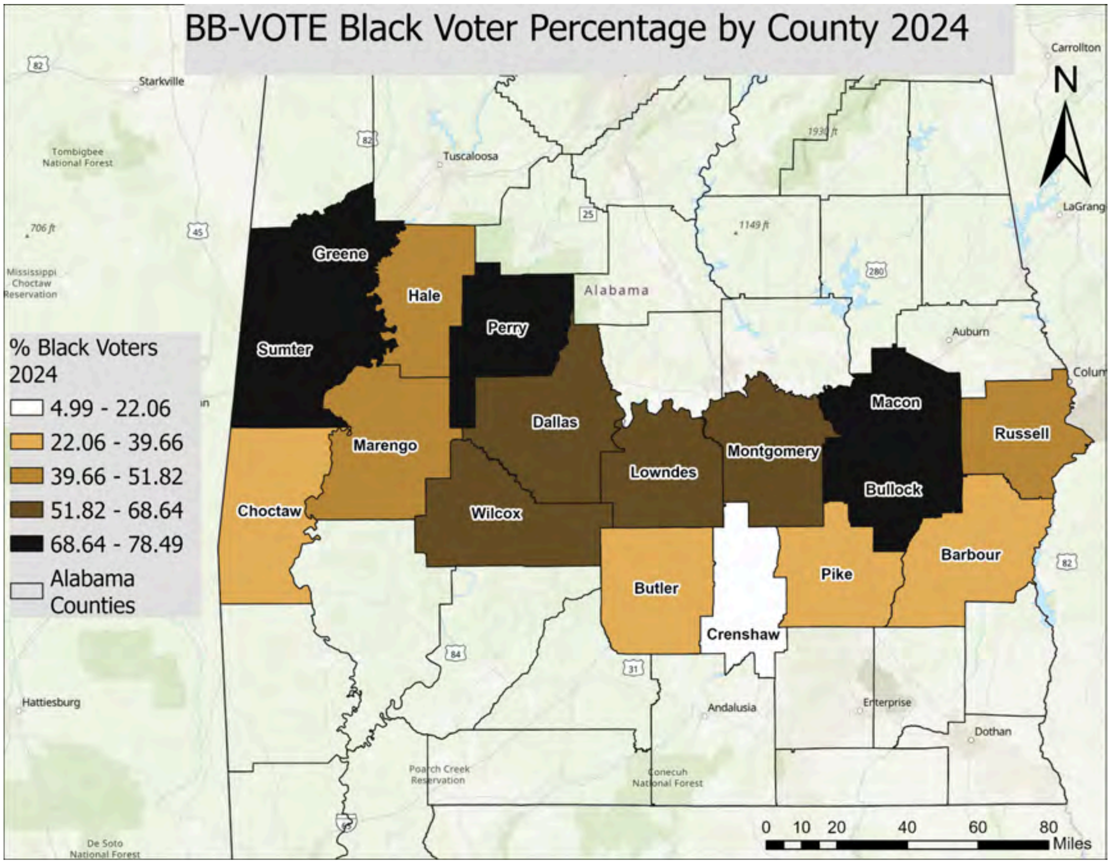

Figure 3. Black voter percentage by county across Alabama's Black Belt counties, 2024. Source: Alabama Secretary of State, 2024.

Interactive map available at:

https://uofmd.maps.arcgis.com/apps/mapviewer/index.html?webmap=2c6dad1120604af2a83a3864180fd609&center=-88.398663%2C31.857347&scale=9244648.868618

This map shows the distribution of Black voter percentage across the same counties in 2024. It allows for comparison with 2020 to identify whether the racial composition changed or remained stable across the study period. This supports the project goal of analyzing whether shifts in demographic composition correspond with changes in turnout or voting outcomes. It provides the updated context used in correlation and turnout analysis.

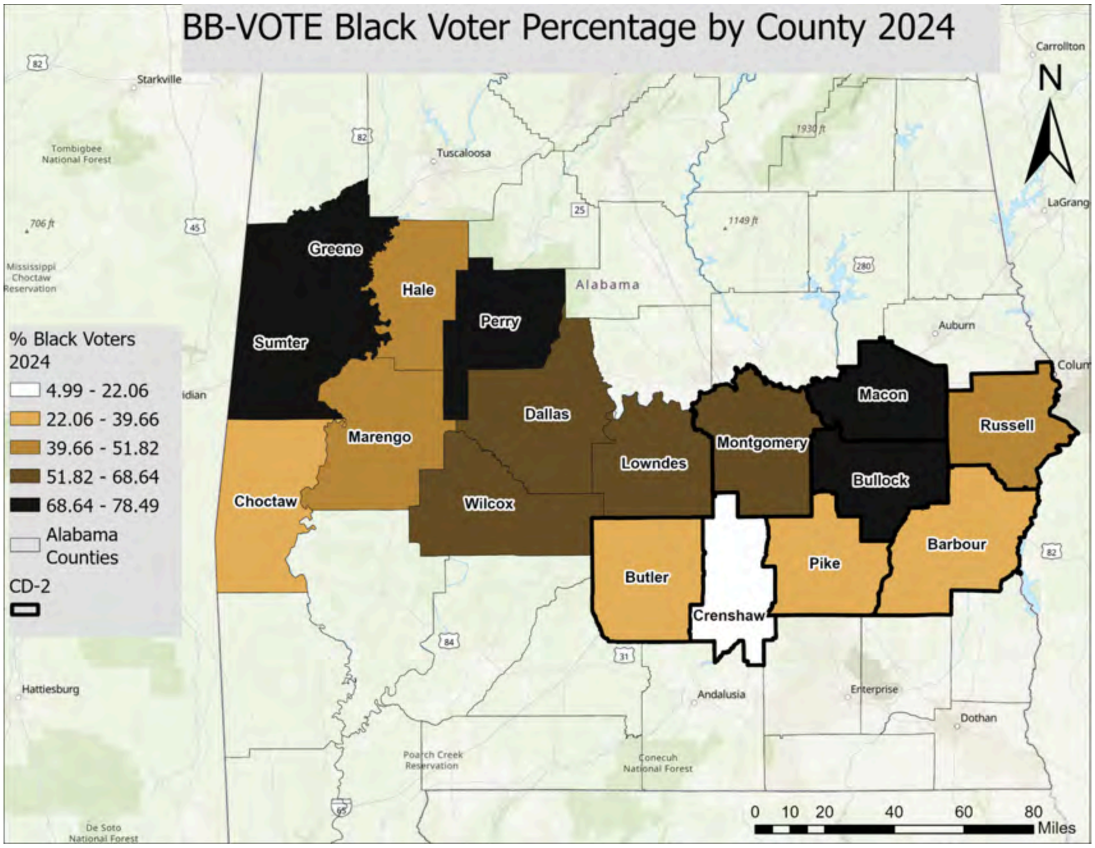

Figure 4. Black voter percentage by county across Alabama's Black Belt counties with 2024 CD-2 district boundary overlay. Source: Alabama Secretary of State, 2024; U.S. Census Bureau CVAP Special Tabulation 2020-2024.

Interactive map available at:

https://uofmd.maps.arcgis.com/apps/mapviewer/index.html?webmap=7d362532fea64d22bd1ae44db0321821&center=-85.799154%2C31.851175&scale=4622324.434309

This map overlays Black voter percentage with the 2024 Congressional District 2 boundary. It shows how the district lines intersect with areas of higher black population concentration. This visualization directly supports the project's focus on the impact of Allen v. Milligan and redistricting. It helps illustrate how CD - 2 was structured around majority black population areas and provides geographic context for the 2024 election outcome analysis.

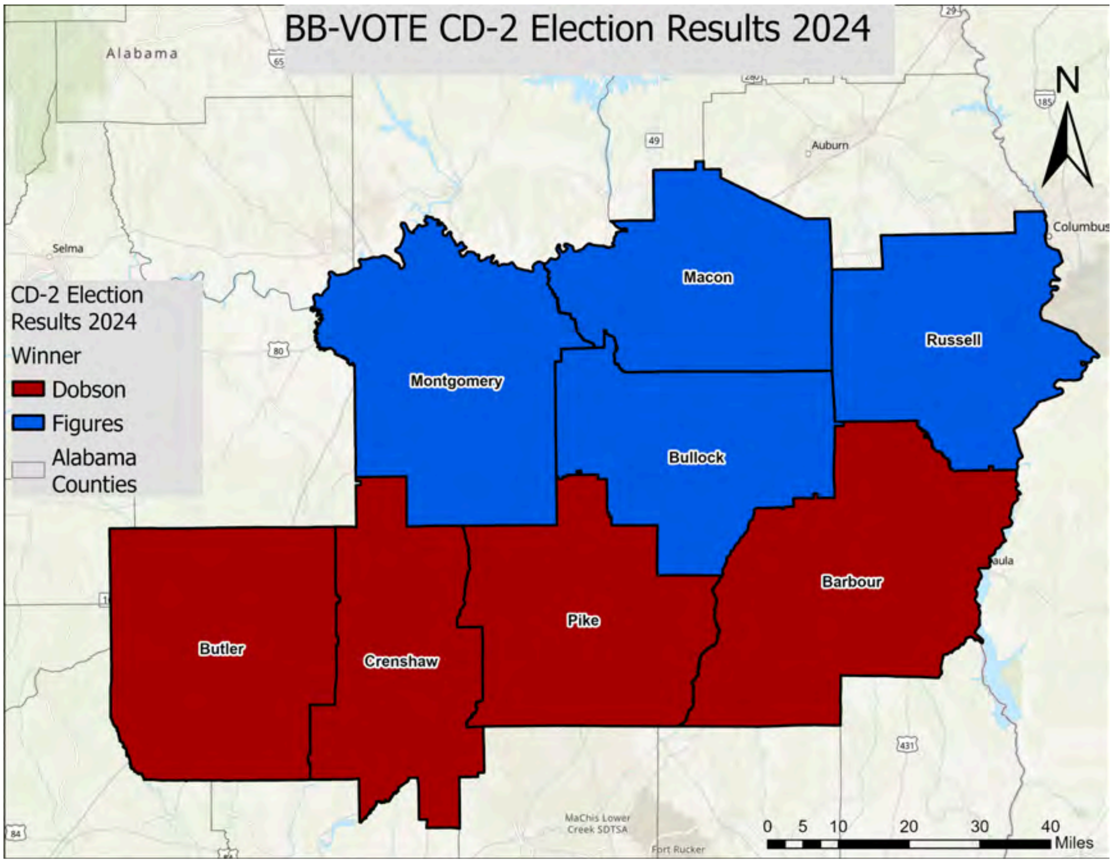

Figure 5. 2024 CD-2 election results by Black Belt county showing vote share for Shomari Figures (D) and Caroleene Dobson (R). Source: Alabama Secretary of State, 2024.

Interactive map available at:

https://uofmd.maps.arcgis.com/apps/mapviewer/index.html?webmap=87e9d0574ce3469ea46cff9c3b4cf0c5&center=-85.706853%2C32.651459&scale=4622324.434309

This map displays the 2024 Congressional District 2 election results by county, showing vote share differences between Democratic and Republican candidates. This visualization connects racial composition and electoral outcomes within the newly drawn district. It supports the project goal of evaluating whether the redistricted district enabled black voters to elect a preferred candidate, as discussed in the findings section.

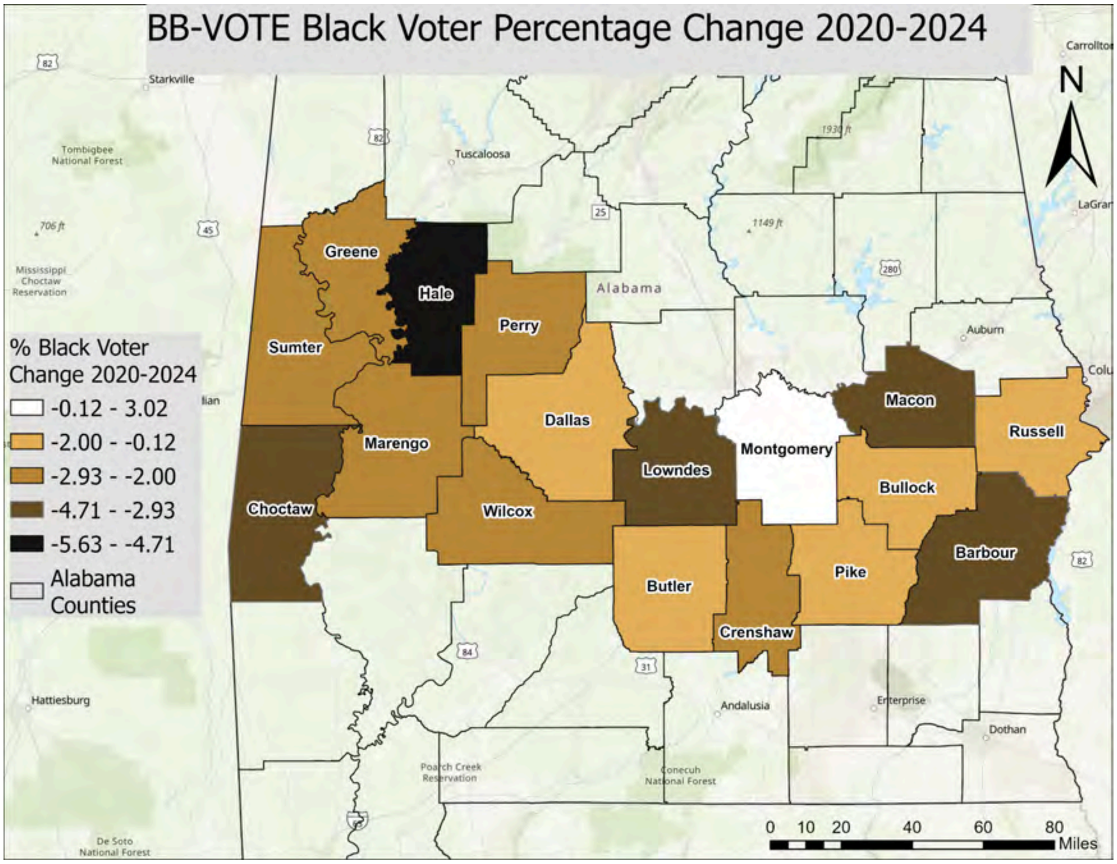

Figure 6. Change in Black voter percentage by county across Alabama's Black Belt counties, 2020-2024. Source: Alabama Secretary of State, 2020, 2024.

Interactive map available at:

https://uofmd.maps.arcgis.com/apps/mapviewer/index.html?webmap=f47c15bc12894863b5b2dc0fe7de838a&center=-86.835727%2C32.574331&scale=4622324.434309

This map shows the change in Black voter percentage between 2020 and 2024 at the county level. This visualization supports the project's objective of identifying whether racial composition changed meaningfully over time. It helps confirm whether observed turnout and voting patterns are driven by demographic change or behavioral trends rather than population shifts.

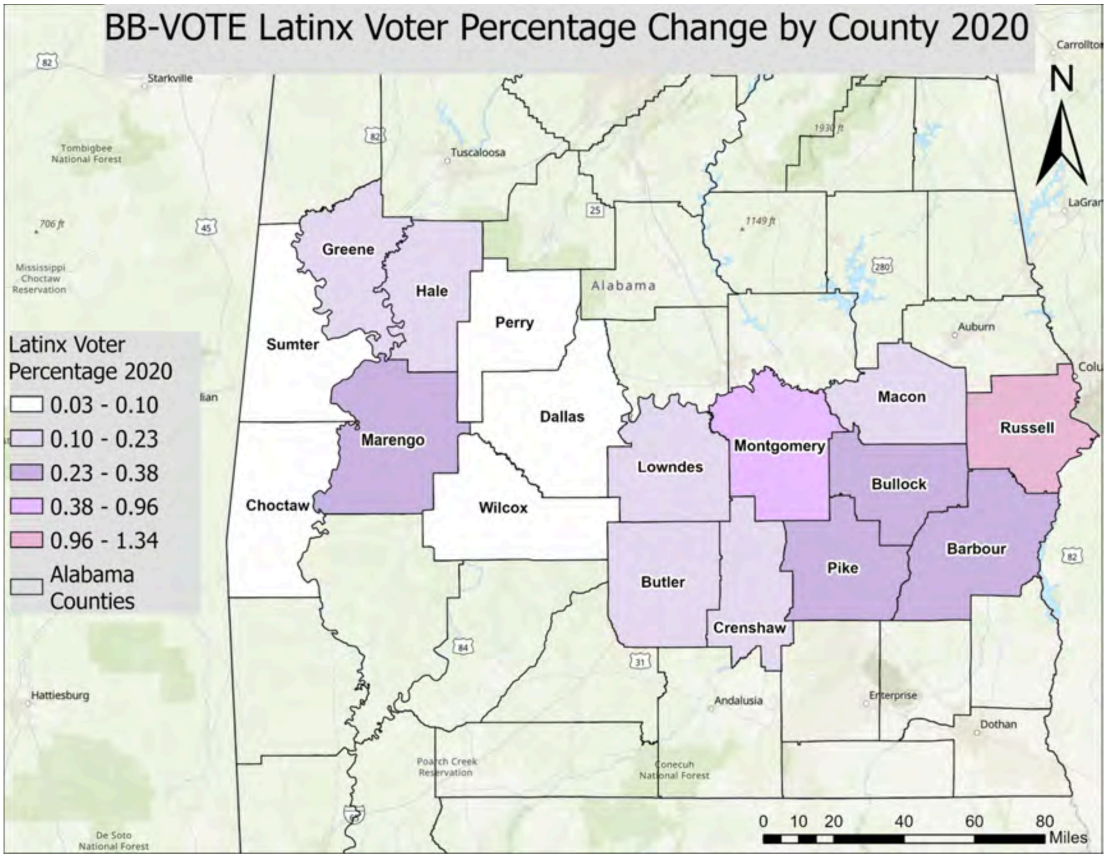

Figure 7. Latinx voter percentage by county across Alabama's Black Belt counties, 2020. Source: Alabama Secretary of State, 2020.

Interactive map available at:

https://uofmd.maps.arcgis.com/apps/mapviewer/index.html?webmap=00b020184adc4ac4958765958072cf5d&center=-86.752503%2C32.622558&scale=4622324.434309

This map shows the percentage of Latinx voters across counties in 2020. It highlights the distribution of a relatively small population group across the study area. This supports the project's goal of evaluating minority participation trends, while also providing context for the statistical limitation that latinx population sizes are extremely small at the county level.

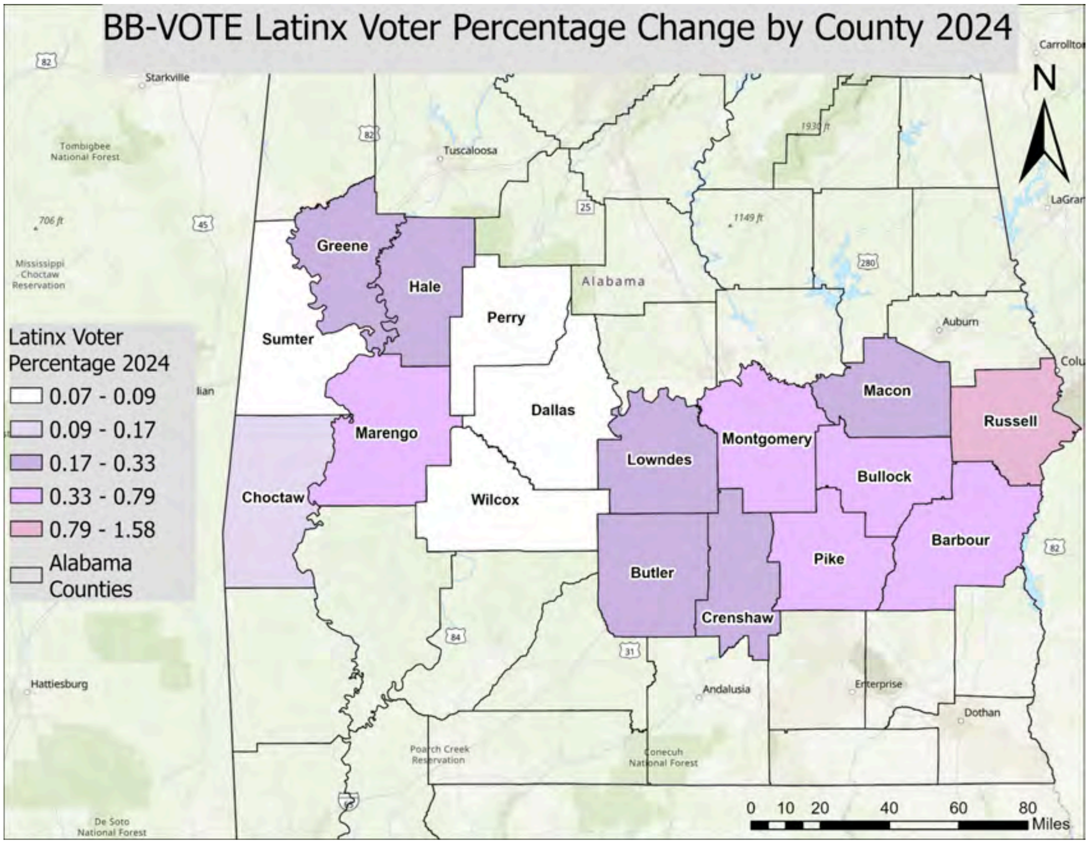

Figure 8. Latinx voter percentage by county across Alabama's Black Belt counties, 2024. Source: Alabama Secretary of State, 2024.

Interactive map available at:

https://uofmd.maps.arcgis.com/apps/mapviewer/index.html?webmap=b137dc7eba33415492b46a6d48d98c3e&center=-86.820181%2C32.601779&scale=4622324.434309

This map shows Latinx voter percentage in 2024, allowing comparison with 2020 to identify changes in distribution or population growth. This visualization supports the analysis of Latinx civic participation over time. It helps reinforce the finding that while participation increased slightly, the population remains too small for strong statistical inference.

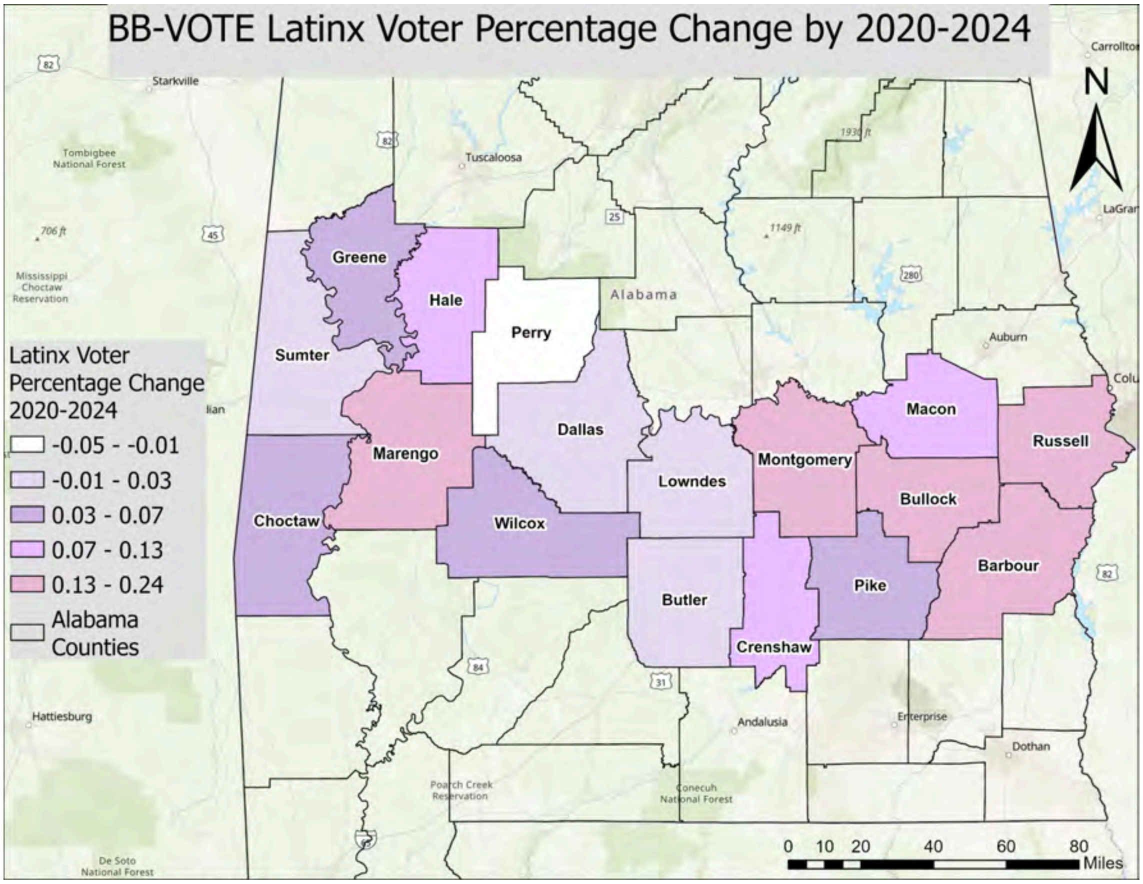

Figure 9. Change in Latinx voter percentage by county across Alabama's Black Belt counties, 2020-2024. Source: Alabama Secretary of State, 2020, 2024.

Interactive map available at:

https://uofmd.maps.arcgis.com/apps/mapviewer/index.html?webmap=a444f7e727244f559e318be9ff04650b&center=-86.424429%2C32.618316&scale=4622324.434309

This map visualizes the change in Latinx voter percentage between 2020 and 2024 across counties. This supports the project goal of identifying demographic shifts over time, but also reinforces a key limitation: even when percentage changes appear large, absolute population sizes remain extremely small, limiting interpretability.

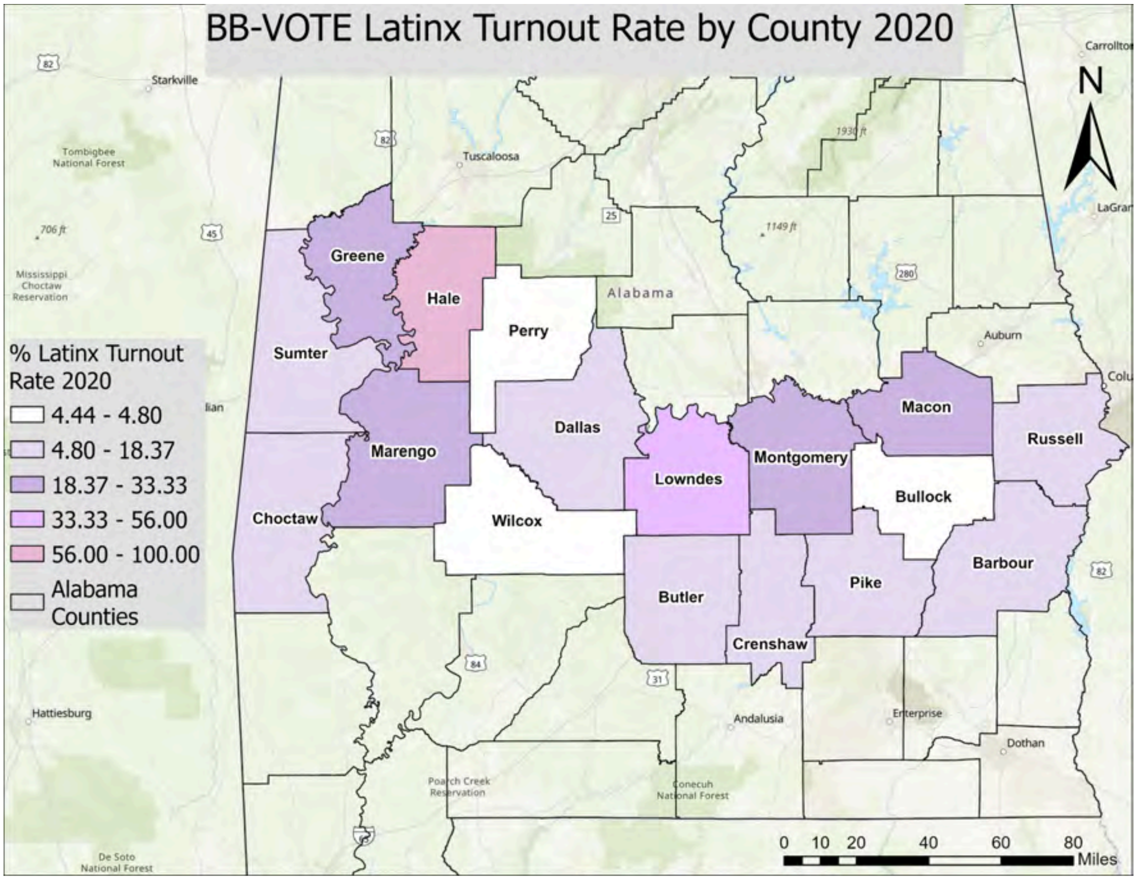

Figure 10. Latinx voter turnout rate (VEP-based) by county across Alabama's Black Belt counties, 2020. Source: Alabama Secretary of State, 2020; U.S. Census Bureau CVAP Special Tabulation 2016-2020.

Interactive map available at:

https://uofmd.maps.arcgis.com/apps/mapviewer/index.html?webmap=0a25d561486740409442c6c64600d1d18&center=-86.616088%2C32.641426&scale=4622324.434309

This map shows Latinx turnout rates by county in the 2020 election cycle. This visualization is used to assess baseline Latinx civic participation prior to 2024. It supports the methodological section by illustrating how turnout rates can appear unstable in counties with very small CVAP estimates.

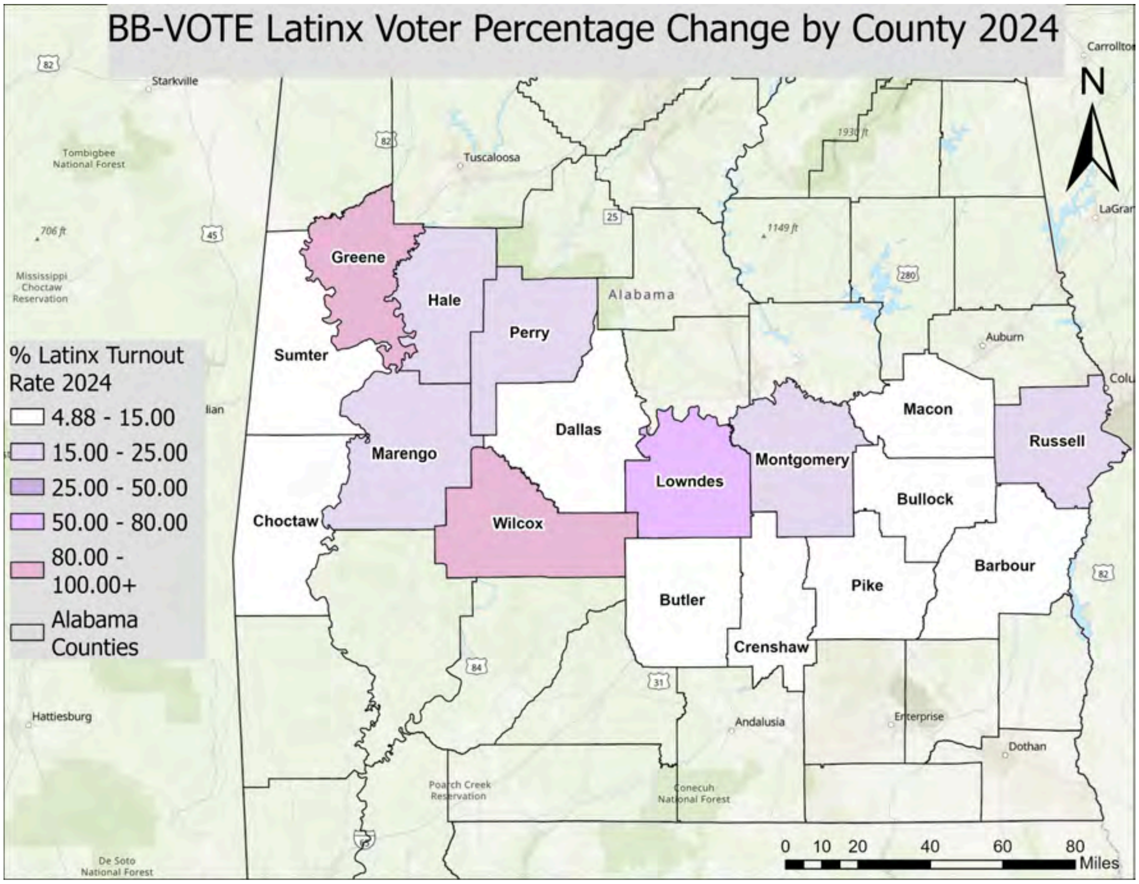

Figure 11. Latinx voter turnout rate (VEP-based) by county across Alabama's Black Belt counties, 2024. Note: Wilcox and Greene counties show rates exceeding 100% due to extremely small Latinx CVAP estimates and should be interpreted with caution. Source: Alabama Secretary of State, 2024; U.S. Census Bureau CVAP Special Tabulation 2020-2024.

Interactive map available at:

[*https://uofmd.maps.arcgis.com/apps/mapviewer/index.html?webmap=ee69f293f90d4f38b184686fa286966f&center=-86.642479%2C32.186292&scale=4622324.434309*](https://uofmd.maps.arcgis.com/apps/mapviewer/index.html?webmap=ee69f293f90d4f38b184686fa286966f&center=-86.642479%2C32.186292&scale=4622324.434309)

This map showcases Latinx turnout rates for 2024 across the same counties. This supports comparison across election cycles and reinforces the project's conclusion that Latinx turnout rates are not statistically reliable at the county level due to small population sizes.

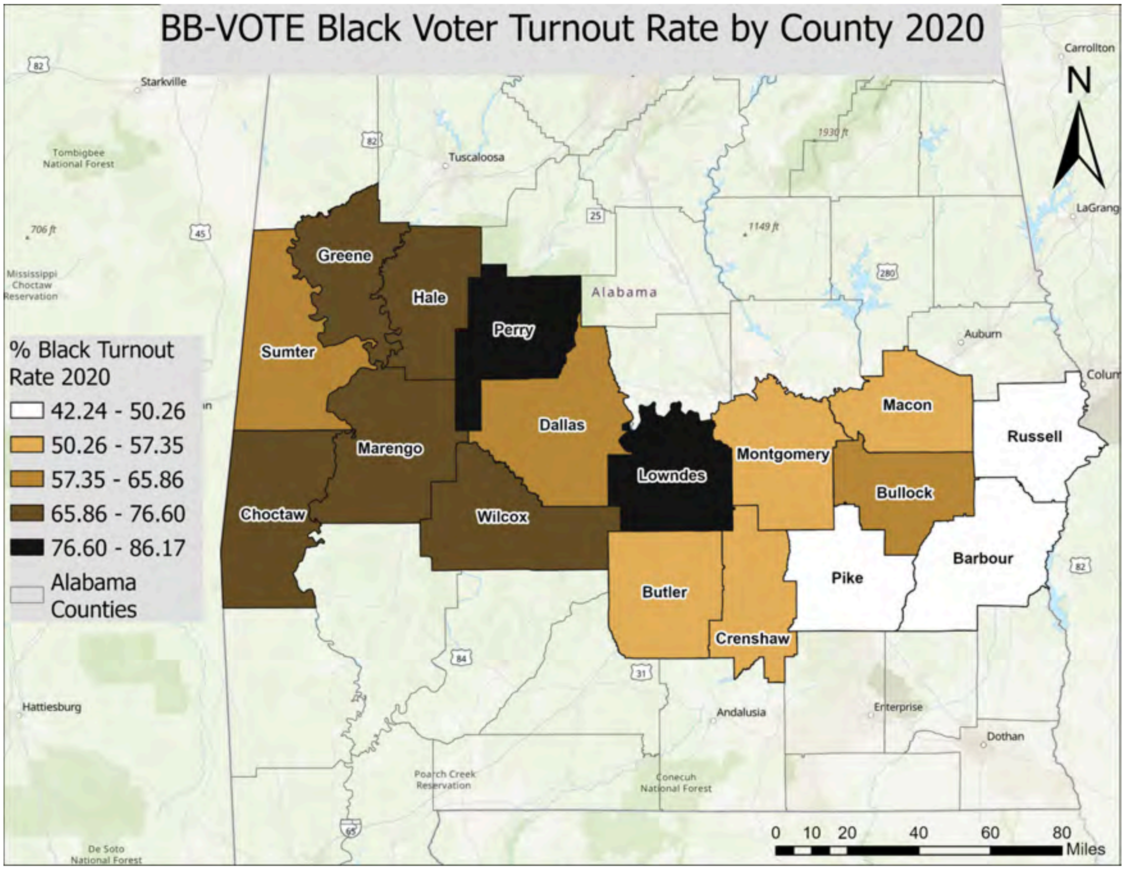

Figure 12. Black voter turnout rate (VEP-based) by county across Alabama's Black Belt counties, 2020. Source: Alabama Secretary of State, 2020; U.S. Census Bureau CVAP Special Tabulation 2016-2020.

Interactive map available at:

https://uofmd.maps.arcgis.com/apps/mapviewer/index.html?webmap=5fea6be6203a47c387fc8b67be0a6ebe&center=-86.794643%2C32.425409&scale=4622324.434309

This map shows off Black turnout rates across counties in the 2020 election. This visualization directly supports the core analysis of the project by showing baseline Black civic participation across the study region. It provides the comparison point for evaluating turnout decline in 2024.

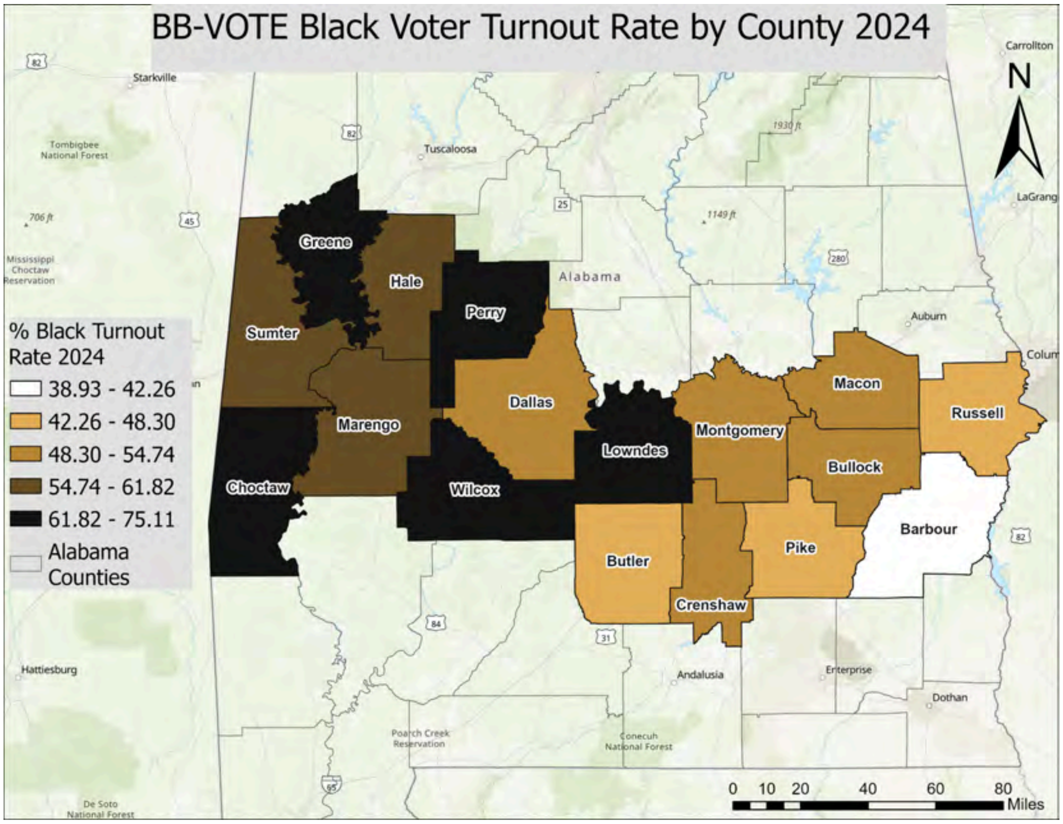

Figure 13. Black voter turnout rate (VEP-based) by county across Alabama's Black Belt counties, 2024. Source: Alabama Secretary of State, 2024; U.S. Census Bureau CVAP Special Tabulation 2020-2024.

Interactive map available at:

https://uofmd.maps.arcgis.com/apps/mapviewer/index.html?webmap=eccdcbbbe8f14d8ab8450723991056b8&center=-86.60678%2C32.363107&scale=4622324.434309

This map shows off Black turnout rates in 2024 across all counties. This visualization is central to the project's findings on declining black civic participation. It supports the statistical analysis showing a statewide decline and reinforces the regional pattern of stronger declines in Black Belt counties.

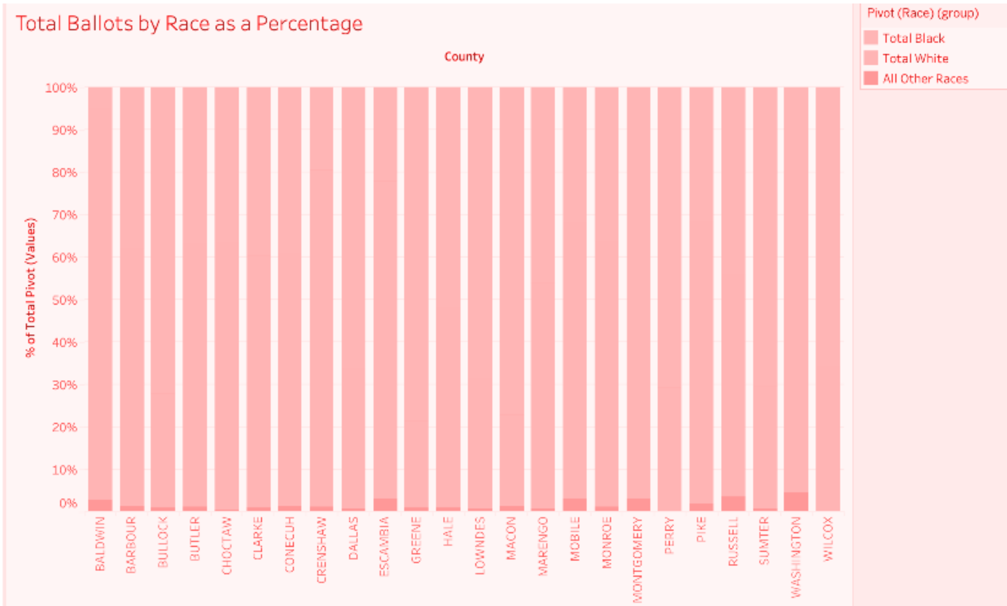

Figure 14. Total ballots cast by race as a percentage of total voters across Alabama's Black Belt counties, 2020 and 2024. Source: Alabama Secretary of State, 2020, 2024.

This graph breaks down total ballots cast by race as a percentage of overall turnout across the study counties. This visualization supports the project goal of understanding racial composition of turnout and how different groups contribute to total voter participation.

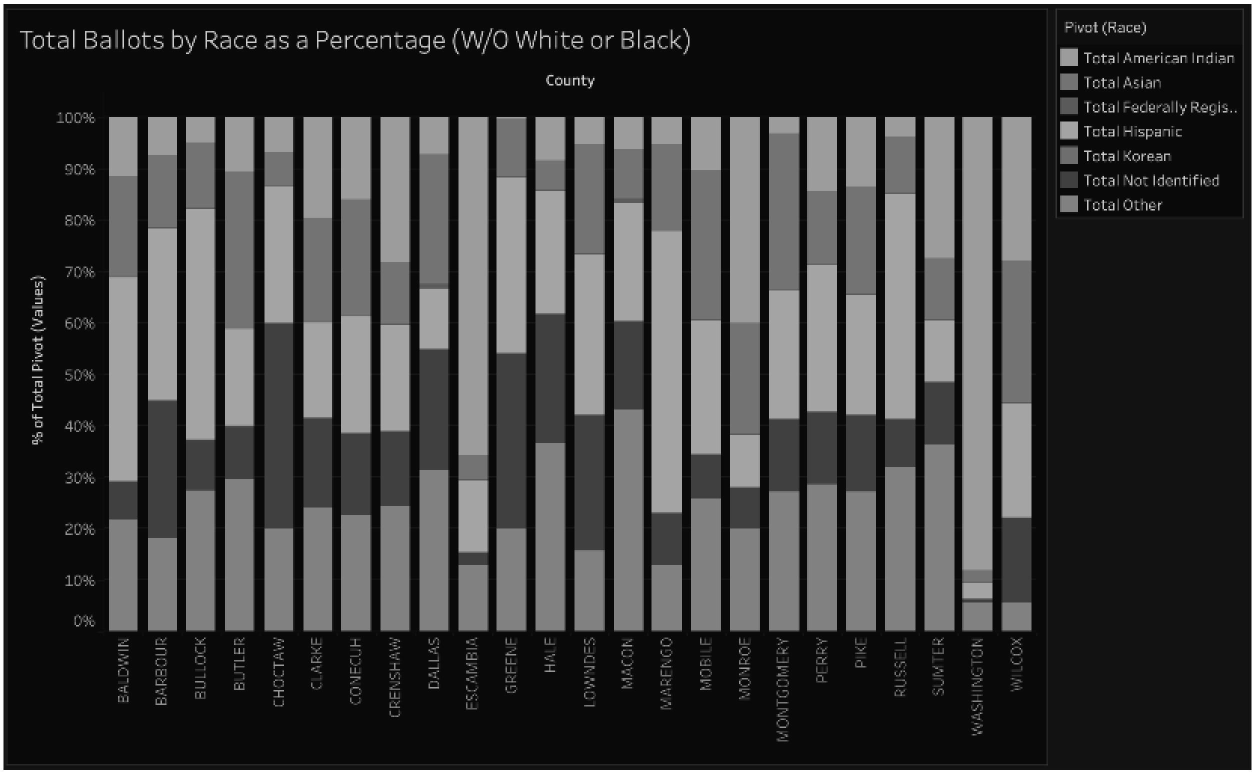

Figure 15. Total ballots cast by race as a percentage of total voters, excluding White and Black voters, across Alabama's Black Belt counties, 2020 and 2024. Source: Alabama Secretary of State, 2020, 2024.

This graph removes white and Black voters from the breakdown to highlight smaller racial and ethnic groups more clearly, including Latinx participation. This supports the project's goal of examining minority participation trends while acknowledging that small population sizes can distort interpretation when included in full scale comparisons.

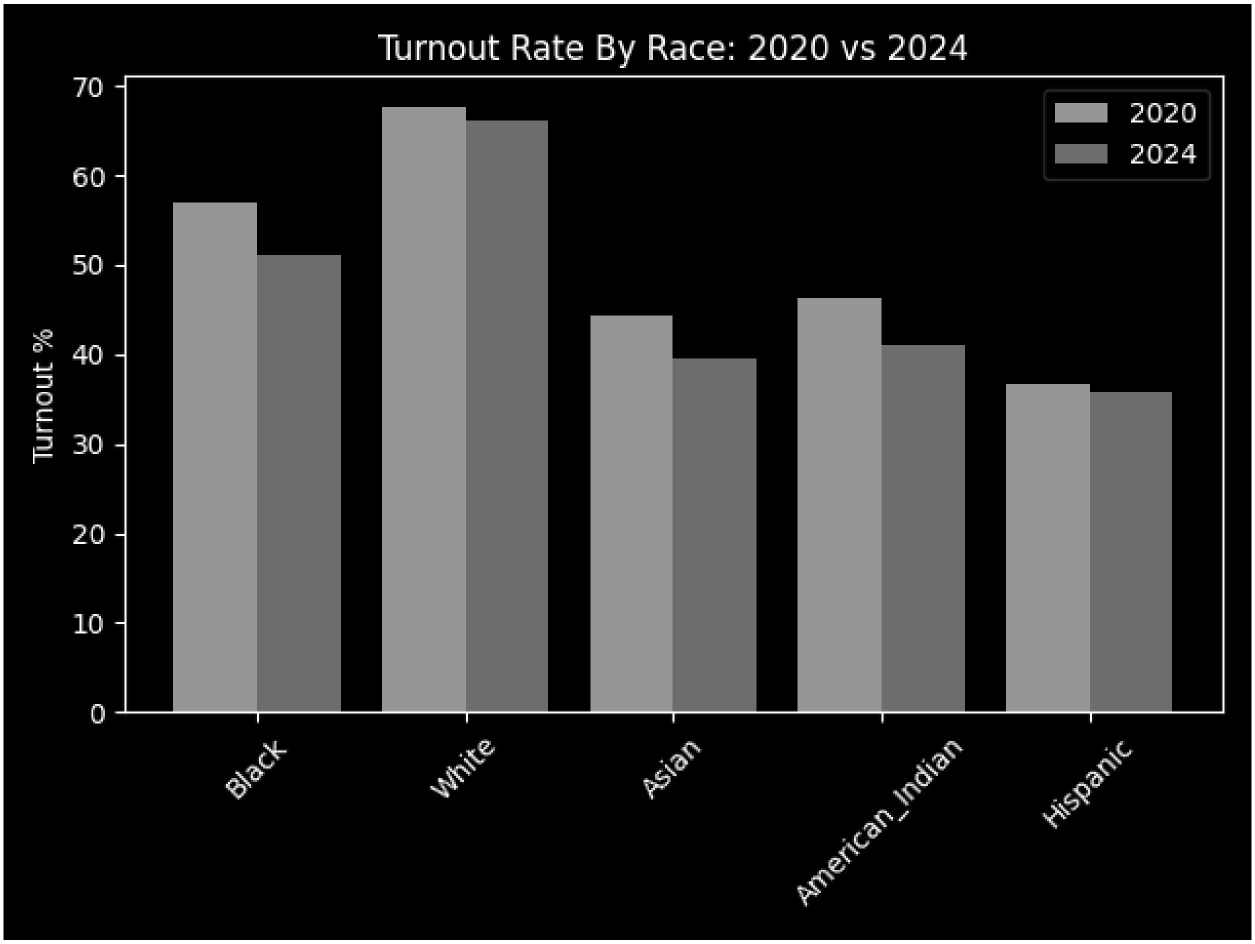

Figure 16. Voter turnout rate by race comparing 2020 and 2024 general elections across Alabama's Black Belt counties. Source: Alabama Secretary of State, 2020, 2024; U.S. Census Bureau CVAP Special Tabulation 2016-2020 and 2020-2024.

This graph compares turnout rates by race between the 2020 and 2024 elections. This visualization directly supports the project's central analysis of changes in civic participation over time. It reinforces the finding that Black turnout declined across cycles while Latinx turnout remained relatively low and unstable.

Limitations

1. General Limitations

The largest limitation throughout this project was time. Our aspirations to learn about the counties were high, but we only had so much time throughout the semester to make progress. Additionally, we all still had other classes and things in our lives that limited our time. We had to stay coordinated throughout to keep aligned on our goals and avoid tangents in specific pieces of work. Our group needed to meet the timing requirements of both the course and the clients while maintaining our own individualized timeline. We used a task plan throughout the project to track our progress and keep track of our due dates.

Data availability was another major constraint for this project. Election reporting varied across counties, limiting the level of geographic detail available for analysis. We were able to make it work and ultimately find data for each county, the reliability of our findings varies by county. It is very possible that certain census datasets undercounted certain populations, such as individuals experiencing housing instability or in institutional settings. Although data accessibility was initially a concern, all required datasets were ultimately obtainable through publicly available sources.

While working with this data, it was important for us to handle the data ethically and responsibly. We respected data privacy by using only public federal data, so we used only demographic and county data. We did not share any personal information in our project, no names, addresses, etc. We ensured our data was clearly organized to avoid misinterpretation. The insights are realistic, nothing outlandish, and no findings that feel stretched. All of our analysis insights were within our projected scope.

2. Data Collection Limitations

The Alabama Secretary of State reports race participation data only at the county level. No precinct level or congressional district level racial breakdown is available. This made it impossible to calculate a CD - 2 specific Black or Latinx turnout rate. Clarke and Mobile counties are split between congressional districts, making precise CD - 2 ballot level calculations methodologically unreliable. The 2020 ACS 1 Year estimates were not published by the Census Bureau due to covid data collection issues. The 2016 - 2020 ACS 5 Year CVAP Special Tabulation was used as the standard methodological substitute. Race participation data are at the county level. Precinct level racial breakdowns are not available. Voter file data may not capture all demographic nuances such as multiracial voters.

3. Latinx Voter Data Limitations

Latinx voter numbers are too small across most counties for statistically meaningful trend analysis. Wilcox County 2024 shows a latinx turnout rate of 400% due to a CVAP estimate of only 1, which is mathematically correct but statistically meaningless. Greene County 2024 shows a latinx turnout rate of 300% due to a CVAP estimate of only 4. Hale County 2020 shows a latinx turnout rate of 100% due to a CVAP estimate of only 10. Counties with latinx CVAP under 50 should be interpreted with caution. Latinx voters represent only 2.1% of eligible voters in CD - 2.

4. Methodological Limitations

Total Federally Registered voters excluded from racial participation analysis. This administrative category cannot be classified by race. Precinct level data contains minor discrepancies vs. certified totals (< 1% difference). Voter file data may not capture all demographic nuances such as multiracial voters. VEP turnout rates use CVAP as the denominator. CVAP estimates are based on ACS surveys, subject to sampling error, particularly in small populations.

Conclusion and Recommendations

1. Recommendations for the Southern Poverty Law Center (SPLC)

The findings from this project provide SPLC with organized and standardized turnout data that can support future voting rights advocacy, outreach efforts, and regional analysis. Several patterns identified in the analysis suggest areas that may warrant additional attention and investigation. First, SPLC should prioritize further outreach and engagement efforts in Black Belt counties that experienced the largest declines in voter turnout and black civic participation between 2020 and 2024. Counties such as Sumter, Lowndes, Greene, and Dallas showed some of the sharpest turnout declines in the study region. Because race was shown to be a highly significant predictor of voting behavior across all three elections analyzed, declining participation in these counties may have substantial electoral implications.

Second, SPLC may benefit from continued monitoring of Alabama's 2nd Congressional District following the Allen v. Milligan redistricting decision. Counties such as Russell and Mobile demonstrated relatively competitive election outcomes compared to other counties in the district, suggesting that turnout fluctuations in these areas could meaningfully influence future elections.

Third, the project identified important limitations in publicly available demographic voting data. The lack of precinct level racial participation data restricts more detailed geographic analysis. SPLC may benefit from pursuing or developing additional datasets related to polling location access, voter registration trends, transportation accessibility, or multilingual outreach efforts to support more comprehensive future analyses.

Finally, the cleaned datasets, SQL database structure, and turnout calculations developed during this project provide SPLC with a centralized foundation for future election analysis. These resources reduce the need for repeated data collection and preprocessing and allow future research efforts to focus more directly on interpretation, outreach strategy, and policy evaluation.

2. Recommendations for Future Project Teams

Future teams expanding on this work should prioritize extending the dataset to additional election cycles, particularly the 2016 Presidential Election and the 2022 Midterm Election. Including additional election years would provide a longer historical baseline for identifying turnout and participation trends over time. The team also recommends incorporating additional demographic variables such as age, income, education level, and polling location accessibility metrics. Combining turnout data with geographic accessibility or socioeconomic indicators could support more comprehensive multi factor analyses of civic participation.

Several methodological lessons may help future teams work more efficiently. First, county naming conventions and election reporting formats varied significantly across datasets and required substantial cleaning before analysis could begin. Establishing standardized formatting procedures early in the project would reduce preprocessing time considerably. Second, MySQL proved effective for organizing and querying large election datasets, particularly when working with multiple election years and county level demographic data. Future teams should continue using relational database structures for scalability and consistency.

Future teams should also interpret latinx voter turnout data cautiously at the county level. In many counties, latinx CVAP estimates were extremely small, producing statistically unreliable turnout percentages. These findings are still valuable descriptively but should not be treated as strong predictive indicators without larger sample sizes or broader geographic aggregation. Finally, this project demonstrated the importance of maintaining detailed documentation throughout the semester. Query logs, dataset descriptions, methodology notes, and cleaning procedures significantly improved workflow consistency and would allow future capstone teams to begin analysis more efficiently rather than repeating early stage organizational work.

Our contact information is attached here for any questions and interest in further developments:

Brayden Hull (bhull@terpmail.umd.edu, 240-409-3102)

Griffin Biddle (gbiddle2@terpmail.umd.edu, 240-483-8017)

Bryan Cristales Rivas (bcrist10@terpmail.umd.edu, 240-883-1060)

Benjamin Donkor (bdonkor@terpmail.umd.edu, 301-401-4966)

Evan Carr (ecarr1@terpmail.umd.edu, 443-934-1015)

Glossary

VEP (Voting Eligible Population): The total number of individuals in each geographic area who are legally eligible to vote, regardless of registration status. Used as the preferred denominator for turnout calculations in this project.

CVAP (Citizen Voting Age Population): A U.S. Census Bureau measure of citizens 18 years and older in a given geographic area, broken down by race and ethnicity. Used as the VEP proxy in this project, sourced from the Census CVAP Special Tabulation.

RDH (Redistricting Data Hub): A nonprofit data provider that aggregates and standardizes precinct level election results and demographic data. Used as a primary data source for this project.

Turnout Rate (VEP Based): The percentage of the Voting Eligible Population that cast a ballot in each election, calculated as (Ballots Cast by Race / CVAP by Race) x 100. This is the preferred academic measure of civic participation.

ACS (American Community Survey): An ongoing demographic survey conducted by the U.S. Census Bureau that provides estimates of population characteristics, including race, ethnicity, and income, at the county and sub county level.

Allen v. Milligan: A 2023 U.S. Supreme Court decision upholding the Voting Rights Act and requiring Alabama to redraw congressional district boundaries to create a second majority black district. This directly resulted in the creation of the new CD - 2, in which Shomari Figures was elected in 2024.

Pearson Correlation Coefficient (r): A statistical measure of the strength and direction of the linear relationship between two variables, ranging from -1 (perfect negative) to +1 (perfect positive). Used in this project to measure the relationship between the black voter percentage and the Democratic vote share.

SPLC (Southern Poverty Law Center): The client organization for this project. A nonprofit civil rights organization based in Montgomery, Alabama, focused on defending civil rights and advancing racial justice.

Deliverables Checklist

BB_VOTE-Voter Turnout and Participation Trends-SQL Findings Zip Folder-vFinal

BB_VOTE-Voter Turnout and Participation Trends-Final Report-vFinal

BB_VOTE-Voter Turnout and Participation Trends-Maps Visualizations-vFinal

BB_Vote-Voter Turnout and Participation Trends-README

Voting & Civic Participation

Alabama Black Belt Counties

Team 3 BB-PAD: Polling Place Access & Distance Analysis

Prepared for: Southern Poverty Law Center (SPLC)

Team Members:

Data Analysis: Myles Sartor, Mawa Clarke

Design: Janice Imuze, Jacy Dickstein, Jayden Ervin

Course: INST490 · Section: 0103

Date: May 18, 2026

Table of Contents

Team 3: BB-PAD

Team Brief

Client and Project Overview

This project is intended for the Southern Poverty Law Center (SPLC), specifically the Alabama office and associated research and voting rights-focused teams. SPLC functions as a nonprofit legal advocacy organization focused on civil rights, public interest litigation, and social justice initiatives across the Southern United States. For this project, the organization was interested in better understanding how polling place access, transportation conditions, and geographic distance may influence voting accessibility and civic participation within Alabama communities.

The project additionally supported broader SPLC interests related to identifying potential accessibility disparities, understanding infrastructural barriers affecting voter access, and exploring how geographic and transportation-related conditions may impact civic engagement among different populations and regions.

Key Findings

- Polling location accessibility across Montgomery County was shorter than expected, with an average walking distance ranging from 1.02 to 1.58 miles depending on the routing method.
- The areas in Montgomery County with higher percentages of Black residents had shorter walking distances to polling places due to their concentration in urban areas. Poverty and transportation limitations still pose accessibility concerns.
- We found that Traditional GIS maps are difficult to be interpreted by the general public which led the team to create an accessible VoteAccess prototype.

Deliverables Produced

- Montgomery County Voter Access Dashboard: An interactive ArcGIS Dashboard application providing real-time monitoring of polling place accessibility across Montgomery County, Alabama. The dashboard integrates block group-level accessibility scores, demographic indicators, polling place coverage statistics/finder, directional routing capabilities, and walking isochrone polygons into a unified client-facing interface.
- GitHub: Comprehensive geospatial analysis project on GitHub with documentation that examines polling place accessibility in Montgomery County, Alabama. The documentation shows how we calculate walking distances from population centers to polling locations, integrate demographic vulnerability indicators, analyze voter turnout patterns, and produce actionable insights to improve electoral access. Multiple routing engines (Euclidean, Manhattan, OSRM network, OSRM pedestrian, Google Routes API) are employed alongside isochrone generation to provide a multi-faceted accessibility assessment.
- VoterAccess Website Prototypes: A civic-oriented website prototype (VoteAccess) displaying accessibility indicators and functioning as a preparatory platform designed to communicate polling place access, travel conditions, transportation availability, and voting-related accessibility information through a user-centered interface.

Challenges

- The design team had trouble with finding effective GIS and advanced visualization, which caused the project to shift towards a UX-focused accessibility product. The team had limited understanding on design systems outside of figma due to school curriculum.

- Time constraints and lack of a finalized prototype interfered with user-testing, refinement and development leaving parts of the prototype strictly conceptual. This causes our final prototype to not be as in-depth as intended

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Connections to Other Teams

- Team 5: ADA Accessibility & Compliance at Polling locations helped guide our projects accessibility feature within the VoteAccess prototype

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Final Report and Deliverables

Team Members

Design Team— Jacy Dickstein

Jacy was responsible for taking on a client-facing role during the project, where she led the first client interview and coordinated questions with her team members. In addition, she contributed to the design and development of the Figma prototype. She also played a role in selecting the features within the prototype, ensuring they aligned with both SPLC's goals and BB-PAD's project scope.

Project Manager/ Figma Design Lead — Janice Imuze

Janice was responsible for overseeing team coordination, project organization, project management platform maintenance, and workflow management across the project. Responsibilities included coordinating communication between teams, managing deliverable organization and development times lines. With the Design Team, Janice led the VoteAccess prototype through interface design.

Design Team — Jayden Ervin

Jayden was responsible for conducting meetings with clients and coordinated questions that informed our design process and data analysis moving forward. Jayden contributed to testing different design platforms in the research process and contributed to the development of the Figma prototype.

Data Team — Mawa Clarke

Data Team/API Integration Focus — Mawa was mainly responsible for developing the Python-based geospatial polling accessibility API designed to support interactive civic accessibility analysis and future frontend integration. My contributions included spatial data processing, nearing polling location calculations, demographic integration, accessibility classification systems, API response generation, and geospatial visualizations using polling place, census tract, and hospital infrastructure datasets for Montgomery County, Alabama. She also hosted both Data Team focuses on GitHub, including updated versions of code and constructed a README for future references.

Data Team — Myles Sartor

Myles was responsible for the end-to-end data processing pipeline, designing and implementing all Python scripts for multi method distance calculations, building the PostgreSQL/PostGIS database with fifteen analytical views, and leading the cloud migration to Supabase with automated Google Apps Script reporting. He configured all three ArcGIS interactive applications within the Voter Access Dashboard (Nearby Polling Location Finder, Walkability Isochrones Viewer, and Directional/Routing Widget). He also conducted the R statistical analysis including regression modeling, correlation matrices, and demographic comparisons. He authored the comprehensive project README and managed GitHub version control and deployment.

Section I: Project Context

Our team contributed to the broader BB-PAD capstone initiative by focusing on polling place accessibility and voting access patterns for communities within Montgomery, Alabama, as identified by the Southern Poverty Law Center (SPLC). The project scope included identifying polling locations, examining potential accessibility barriers and closures using available geographic and polling-related data, and exploring ways to visually communicate voting access patterns across the Montgomery and broader Black Belt region. By the conclusion of the 15-week semester, the team produced a detailed analysis of calculated polling place distances for selected communities alongside a conceptual civic-focused prototype designed to communicate accessibility-related information in a more user-centered format. The project additionally included summaries of potential access challenges, documentation of analytical methods, and broader geographic assumptions explored throughout the research process. In terms of interdisciplinary collaboration across capstone teams, our team primarily relied on the design guidelines developed by Project 11: Civic Participation Process & Access Design (BB-CPAD), which helped maintain consistency in visual standards and alignment with SPLC's preferred branding, color palette, and overall design language across final deliverables.

Section II: Methods

2.1 Data Analysis

The analysis integrated within these categories of data required distinct means for acquiring, cleaning, and validating certain procedures.

Election Data

2024 precinct-level general election results were obtained from the Redistricting Data Hub (RDH) as a shapefile containing precinct boundaries joined with vote counts for presidential, judicial, and congressional races. The RDH documentation specifies a column naming convention following a specific pattern. In G[YEAR] [OFFICE] [PARTY] [CANDIDATE], the party codes are D (Democrat), R (Republican), I (Independent), and O (Other/Write-In). The G24PREDHAR decodes as 2024 General, President, Democrat, Harris. Office codes include PRE (President), CON## (US House District ##), CFJ (Chief Justice), AJ# (Associate Justice), CV# (Court of Civil Appeals), CR# (Court of Criminal Appeals), PSC (Public Service Commission), and A## (Amendment). The RDH documentation notes that for Montgomery County specifically, the shapefile was sourced directly from the county GIS website, ensuring higher boundary accuracy than counties using Census VTD files.

2020 precinct-level results were sourced from the University of Florida Election Lab, which compiled data from the Alabama Secretary of State. The UFL documentation contains an important methodological note on absentee and provisional ballots that were reported countywide and allocated to precincts based on precinct-level vote share. This allocation method means that precinct level 2020 totals are approximations rather than exact counts, which is a limitation that is documented transparently in all deliverables. The 2020 shapefile uses the same column naming convention (G20PREDBID for Biden Democratic votes, G20PRERTRU for Trump Republican votes, G20PRELJOR for Jorgensen Libertarian votes).

A county-level registration and turnout report was used (2020_General_Total_Ballots_Cast_Report.csv) from the Alabama Secretary of State provided registered voter counts (165,228) and total ballots cast (99,286) for Montgomery County. This served as the denominator for accurate turnout rate calculations (60.1% in 2020).

Demographic Data

American Community Survey 5-year estimates (2020-2024) were retrieved programmatically using the tidycensus R package, which provides an R interface to the U.S. Census Bureau's APIs. Data was pulled at two geographic levels. At the tract level (71 units in Montgomery County), the complete demographic data was utilized. This includes total population, race/ethnicity counts (White alone, Black or African American alone, Asian alone, Hispanic/Latino total), median household income, poverty status (total population for whom poverty status is determined, population below poverty level), vehicle access (total occupied housing units, units with no vehicle available), and educational attainment (population 25 years and over, bachelor's degree or higher). At the block group level (203 units in Montgomery County), the same variables were used, but with a critical limitation. Poverty estimates at the block group level are suppressed by the Census Bureau due to small sample sizes and high margins of error. All 203 block groups returned NULL values for poverty-related variables.

Census 2020 population-weighted centroids (CenPop2020_Mean_BG01.txt) provided the coordinate reference points for all distance calculations. These centroids are scientifically superior to geometric centroids because they account for where people actually live within each geographic unit rather than simply using the mathematical center. The file contains 12-digit GEOIDs, latitude, longitude, and population for every block group in the United States and was filtered to Montgomery County using the FIPS code prefix 01101 (state 01 + county 101).

Polling Place Data

A client-provided statewide Alabama polling place shapefile (Al_Polls_Flood_SLED.shp) was the foundation for all distance calculations. The shapefile bundle contained the standard ESRI components: .shp (geometry points representing each polling location), .dbf (attribute table with polling place name, county, address, and precinct assignment), .shx (shape index linking geometry with attributes), .prj (coordinate reference system definition), .cpg (character encoding, UTF-8), .sbn and .sbx (spatial indices for faster querying), and .shp.xml (metadata). Filtering the statewide to Montgomery County yielded 49 active polling locations, matching the expected norm for the year 2020.

Infrastructure Data

OpenStreetMap road network data for Montgomery County was downloaded programmatically using the osmnx Python package, which queries the Overpass API for features matching specified tags. The query retrieved all highway features within Montgomery County's boundary and tagged each segment for sidewalk presence based on two OSM attributes. There was sidewalk (values: yes, both, right, left, separate, no) and footway (values: sidewalk, yes, both). We developed a helper function that classified each road segment as having a sidewalk if either attribute indicated pedestrian infrastructure. The resulting dataset contained the full county road network with highway classifications, sidewalk presence indicators, footway information, and street names.

The City of Montgomery Open Data Portal provided paving project information for road quality assessment. The paving data includes road classification (arterial, collector, local), width in feet, and construction status (complete, in progress). We developed a scoring algorithm that adjusts baseline score of 50 based on road class (arterial/highway: -20, collector: -10, local +10), width (under 25 feet: +10, over 40 feet: -10), and status (complete +10, in progress: +5), producing a road quality score from 0-100 for each segment intersecting a route corridor.

Routing Engine Data

OpenStreetMap Alabama extracts were processed through two routing engines.

OSRM (Open Source Routing Machine): The Alabama OSM extract was processed with OSRM's osrm-extract and osrm-contract tools to build optimized routing graphs for both driving (car profile, port 5001) and walking (foot profile, port 5002). The driving profile accounts for one-way streets, turn restrictions, and speed limits. The walking profile uses pedestrian-accessible pathways including sidewalks, footpaths, and pedestrian crossings. Two separate instances were required because the routing profiles differ fundamentally in which network edges are traversable.

Valhalla: The Alabama OSM extract was processed through Valhalla's tile-building pipeline within a Docker container to generate routing tiles optimized for pedestrian isochrone generation. The denoise parameter (0.11) was applied to smooth jagged edges from network analysis, producing cleaner polygon boundaries.

Analytical Approaches

Distance Calculation

Five distinct distance metrics were computed between all 203 population centers and all 49 polling places, producing 9,947 origin-destination pairs for each method.

Euclidean Distance (Python/scipy): Straight-line "as the crow flies" distance using scipy.spatial.distance.cdist with the Euclidean metric. Distances were converted from degrees to miles using the approximation 1° ≈ 69 miles. While this introduces minor distortion, it is appropriate for the small geographic area of a single county. The formula used was $\text{sqrt}((x_2 - x_1)^2 + (y_2 - y_1)^2)$.

Manhattan Distance (Python/scipy): Grid-based taxicab distance using the cityblock metric, calculated as $|x_2 - x_1| + |y_2 - y_1|$. This was more appropriate for urban areas with rectilinear street networks. Manhattan distance provides a middle ground between Euclidean approximation and full network routing.

OSRM Driving Distance (local OSRM, port 5001): Actual road network driving distance computed through the OSRM routing engine using the car profile. Each origin-destination pair was requested individually with a 50ms delay between requests to avoid overwhelming the local server. Distances were returned in meters and converted to miles.

OSRM Walking Distance (local OSRM, port 5002): Pedestrian network distance using the foot profile. Same request pattern as driving distances but using pedestrian-accessible pathways.

Google Walking Distance (Google Routes API): Google's pedestrian routing algorithm accessed through the Routes API Compute Route Matrix endpoint. Processing was batched with 6 origins per request (294 elements, safely under the 625-element API limit) and 2.5-second delays between batches to respect free-tier rate limits. Exponential backoff (2s, 4s, 8s, 16s, 32s, 64s, 128s, 256s, 512s) was implemented for rate-limited requests.

For each population center, the minimum distance to any polling place was recorded alongside the index of the nearest facility. Full pairwise distance matrices (203 x 49) were preserved for each method for detailed spatial analysis.

Accessibility Scoring

A composite accessibility score (0-1) was created by normalizing and weighting three distance components using min-max scaling.

- Euclidean distance to nearest polling place (40% weight)
- Manhattan distance to nearest polling place (30% weight)
- Distance to nearest civic infrastructure reference point (30% weight)

Six reference points were used throughout this process. Montgomery City Hall and five community centers (Loveless, Houston Hill, Highland Gardens, Hayneville Road, Hunter Station) that served as neighborhood hubs. The minimum distance to any reference point was used as the civic infrastructure metric.

Categories were assigned as Excellent (0-0.25), Good (0.25-0.50), Fair (0.50-0.75), and Poor (0.75-1.0).

Walkability Enhancement

The enhanced walkability analysis incorporated OpenStreetMap sidewalk data and Montgomery County paving quality through a buffer corridor approach. For each population center's route to its nearest polling place, a 0.05-mile buffer corridor was created along the straight-line path. All OSM road segments and paving project segments intersecting this corridor were analyzed for sidewalk presence and road quality.

A composite walkability score (0-100) combined three weighted components:

- Distance score (0-40 points): Based on walking distance thresholds (<0.5 mi: 40, <1.0 mi: 32, <2.0 mi: 24, <3.0 mi: 16, <5.0 mi: 8, >5.0 mi: 0)
- Road quality score (0-30 points): Paving quality score normalized to 0-30 (supplemented by construction status and type of road/road width)
- Sidewalk score (0-30 points): 20-30 points if sidewalks present (scaled by coverage percentage), partial credit if no sidewalks but some coverage, 0 if no sidewalk infrastructure

Categories were assigned as Excellent (80-100), Good (60-79), Fair (40-59), Poor (20-39), and Very Poor (0-19).

Isochrone Generation

Valhalla's pedestrian routing engine generated walkable area polygons for 5, 10, 15, 20, and 30-minute walking thresholds around each polling place. For each of the 49 polling places, five isochrone requests were made (245 total polygons). Area calculations used UTM Zone 16N projection (EPSG:32616) for accurate square mileage measurements rather than degree-based approximations. Population served within each isochrone was computed through spatial join with population center point geometries, summing the population column for all enclosed points.

Vulnerability Index

A composite vulnerability score was calculated using z-score normalization of three indicators identified by the SPLC as markers of at-risk communities:

- Percent Black or African American population
- Percent population below poverty level
- Percent households without vehicle access

Each component was standardized using z-score transformation ($z = (x - \mu) / \sigma$). The three z-scores were averaged with equal weights to produce a composite vulnerability index. Communities were categorized as High (index > 1.0), Moderate (0.0 to 1.0), Low (-1.0 to 0.0), or Very Low (< -1.0) vulnerability.

Statistical Analysis

All statistical analysis was performed in R using tidyverse for data manipulation and broom for model output formatting. Three linear regression models examined relationships between demographic variables and accessibility metrics:

- **Model 1:** Walkability Score ~ % Black + % Poverty ($R^2 = 0.046$)
- **Model 2:** Walking Distance ~ % Black + % Poverty + % No Vehicle ($R^2 = 0.105$)
- **Model 3:** Walking Distance ~ Vulnerability Index ($R^2 = 0.095$)

Model 1 found that poverty was a significant predictor of walkability score ($p = 0.038$, $\alpha = 0.05$), but the model explained only 4.6% of variance. Model 2 found that percent Black was a significant predictor of walking distance ($p = 0.008$, $\alpha = 0.01$), with each percentage point increase in Black population associated with a 0.0017-mile decrease in walking distance. Model 3 found that the vulnerability index was a highly significant predictor ($p = 8.98 \times 10^{-6}$, $\alpha = 0.001$), with higher vulnerability associated shorter walking distances.

Pearson correlation coefficients were calculated between walking distance and all numeric variables (walkability score, Euclidean distance, driving distance, demographic percentages, vulnerability index), with p-values and strength classifications (strong: $|r| > 0.5$, moderate: $|r| > 0.3$, weak: $|r| \leq 0.3$). The strongest correlations with walking distance were driving distance ($r = 0.644$), walking OSRM ($r = 0.640$), and Euclidean distance ($r = 0.609$). Demographic correlations were weaker for percent Black ($r = -0.307$), median income ($r = 0.320$), and vulnerability index ($r = -0.308$).

Independent samples t-tests compared walking distances between high (>50%) and low (<50%) percent Black areas, finding a statistically significant difference of 1.30 miles ($t = 3.58$, $p < 0.001$), with high percent-Black areas having shorter distances (Mean = 1.22, Standard Deviation = 0.98) than low percent Black areas (Mean = 2.52, Standard Deviation = 2.93).

Tools Used

Python (≥ 3.8): pandas (tabular data manipulation), geopandas (spatial data operations), numpy (numerical computation), shapely (geometric operations including Point, LineString, and buffer creation), scipy (distance matrix computations via cdist), requests (HTTP API calls to OSRM, Google Routes, and Valhalla), matplotlib (static visualization), contextily (basemap tiles), osmnx (OpenStreetMap data download). All Python scripts were developed in a conda environment with package versions pinned for reproducibility.

R (≥ 4.0): tidycensus (Census ACS data retrieval via API with user-provided key), tidyverse (data manipulation and visualization pipeline), sf (spatial data handling with simple features), tigris (Census geographic boundary data), broom (tidy model output formatting), ggplot2 (static visualization and choropleth maps), ggspatial (basemap tiles for maps via annotation_map_tile), patchwork (combining multiple ggplot figures), pheatmap (correlation heatmap generation), rosm (OpenStreetMap tile downloading and caching for offline use).

SQL: PostgreSQL 15 with PostGIS 3 extension enabled for spatial data handling. The PostGIS extension provides spatial data types (GEOMETRY, GEOGRAPHY), spatial indexes (GiST), and spatial functions (ST_DWithin, ST_Contains, ST_AsText, ST_SetSRID, ST_MakePoint). The database schema was normalized to Third Normal Form (3NF) with foreign key relationships between tables.

Routing Engines: OSRM 5.27 (local installation, ports 5001/5002 for driving and walking profiles respectively), Valhalla 3.4 (Docker container, port 8002 for pedestrian isochrone generation), Google Routes API (Compute Route Matrix endpoint for pedestrian distances).

Cloud Infrastructure: Supabase (cloud PostgreSQL hosting with Edge functions for REST API access), Google Apps Script (JavaScript-based automation with time-driven triggers for hourly data refresh), Google Sheets (intermediary data cache for Looker Studio integration)

GIS & Visualization: ArcGIS Online (Dashboards for monitoring, Instant Apps for public-facing tools, Experience Builder for interactive applications), ArcGIS Pro (desktop GIS for database connection and advanced spatial analysis).

Addressing Challenges

Poverty Data Suppression at Block Group Level

The Census Bureau suppresses block group-level poverty estimates when sample sizes are small or margins of error are high, returning NULL values. For Montgomery County, all 203 block groups had NULL poverty data. This was discovered during the R statistical analysis when correlation and regression functions failed with "no complete element pairs" errors. The initial debugging process involved verifying data integrity at each pipeline stage while checking the ACS pull script output, the CSV export, the PostgreSQL import, and the R data loading before identifying the root cause as Census suppression rather than a processing error.

The solution was to shift demographic analysis to the tract level (71 units), where poverty data is complete with 0 NULL values. Accessibility scores calculated at the block group level (203 units) were joined with tract-level demographics using GEOID substring matching. This meant the extraction of the first 11 characters of the 12-digit block group GEOID to obtain the tract GEOID, then left-joining tract demographics. This approach preserves the fine-grained spatial resolution of distance calculations (block group level) while enabling demographic analysis with complete data (tract level). The trade-off is that all block groups within the same tract receive identical demographic values, which reduces variance in the regression models.

Precinct Boundary Changes Between Election Cycles

Redistricting between 2020 and 2024 caused precinct name and boundary changes that prevented direct year-over-year comparison. The 2024 RDH shapefile contained 51 precincts with names like "101 WILSON COMM & ATHLETIC CTR", while the 2020 UFL shapefile contained 49 precincts with names like "AFED Conference Ctr". Automated name matching failed due to inconsistent abbreviations, punctuation differences, and facility name changes.

A manual mapping dictionary was created through careful comparison of precinct locations using geographic coordinates, polling facility addresses from the client-provided shapefile, and facility name cross-referencing. The mapping required understanding the precinct numbering system (100-series for central/west Montgomery, 200-series for south/southwest, 300-series for east, 400-series for north/northeast, 500-series for rural south county) and identifying which 2020 precincts had been consolidated, renamed, or split.

Of 51 precincts in 2024, 41 were successfully matched to 2020 equivalents. The remaining 10 precincts represent new construction (e.g, "204 REBIRTH CHRISTIAN MINISTRY", "511 AUM

OUTREACH CTR") or precinct consolidation that occurred during redistricting. These unmatched precincts are excluded from comparative turnout analysis and documented in the election processing script output. The matched pairs are stored in the precinct_name_mapping table in PostgreSQL and the precinct_mapping dictionary in election_data_processing.py.

Network Port Blocking

Direct PostgreSQL connections from cloud services (Looker Studio, ArcGIS Online) to the local database were blocked by university network firewall rules. Testing confirmed that ports 5432 (direct PostgreSQL) and 6543 (connection pooler) were both unreachable from external services. Multiple workarounds were attempted: ngrok TCP tunneling (failed due to verification requirements), Cloudflare tunneling (failed due to configuration complexity), and mobile hotspot (failed due to the same port blocking on the cellular network).

The solution that worked was a complete cloud migration. The local PostgreSQL database was exported using pg_dump, converted to a plain SQL file, and imported into a Supabase cloud PostgreSQL instance. A Supabase Edge Function (Deno-based serverless function) was deployed to serve as a REST API wrapper, accepting POST requests with view names and returning JSON query results over standard HTTPS (port 443), which is universally accessible. A Google Apps Script was created to fetch data from the Edge Function on an hourly schedule and populate Google Sheets as an intermediary cache that Looker Studio can reliably reach

This architecture introduces a one-hour latency between database updates and dashboard refreshes, but it completely bypasses the network restriction problem and provides a sustainable model for long-term client access.

Google API Rate Limiting

The Google Routes API Compute Route Matrix endpoint imposes a limit of 625 elements per request (origins x destinations). With 203 origins and 49 destinations (9,947 elements total), the calculation required batching into 34 separate requests of 6 origins each (294 elements, safely under the limit). The free tier imposes additional rate limits that necessitate 2.5 second delays between batches, extending total processing time to approximately 5-10 minutes.

Six origin rows failed during the initial batch processing run (centers 157-162), likely due to transient API errors or rate limit enforcement. Rather than reprocessing the entire batch, a secondary fill script was developed that identifies missing values in the output CSV, extracts only those GEOIDs, fetches their coordinates from the population centers file, and requests distances for just the missing rows. This targeted approach avoided redundant API calls and completed in under a minute.

GEOID Format Inconsistency

The accessibility scores CSV stored GEOIDs in a short format missing the leading zero (e.g., 11010026002 instead of 011010026002). This occurred because pandas reads numeric-looking strings as integers by default, stripping leading zeros during CSV export. When these GEOIDs were later joined with census data (which uses the full 12-digit string format), the merge failed entirely, producing 0 matched rows and breaking the R statistical analysis pipeline.

The fix required format conversion logic in multiple places. The Python fill scripts convert short format GEOIDs to full format before coordinate lookup, the R modeling pipeline pads TRACTCE values to 6 digits before constructing tract GEOIDs, and the PostgreSQL views use substring matching to extract tract GEOIDs from block group GEOIDs. The root cause of this pandas' automatic type inference was documented for future teams to prevent recurrence by explicitly specifying dtype = {'GEOID': str} at the point of CSV implementation.

ArcGIS Data Deduplication

The GeoJSON export for ArcGIS Online contained seven duplicate block group records, inflating the feature count from 203 to 210. This occurred because the SQL export query used SELECT DISTINCT, but the duplicate rows had subtle differences in floating-point precision for distance columns, causing the database to treat them as distinct. ArcGIS dashboard indicators displayed inflated counts and slightly skewed averages as a result.

The issue was resolved through going into the ArcGIS content section for that specific feature layer and manually deleting the duplicate records because there were only 7 of them. This created an array of unique GEOIDs by iterating through variables and only having the originals despite the previously uploaded layer falsely creating duplicates.

2.2 Interactive Polling Accessibility API Development

Introduction and System Overview

A Python-based geospatial polling accessibility API system was developed to support interactive civic accessibility and demographic analysis, along with frontend integration (utilizing Figma) for public-facing applications. While the broader project focused on spatial accessibility modeling and vulnerability analysis at the county level, this component focused and emphasized real-time user interaction and geospatial analyzing. This system was designed as a prototype civic technology framework capable of receiving user-generated location inputs, identifying nearby polling infrastructure, calculating accessibility metrics, and returning demographic and spatial information in structured API-style responses.

The API system functioned as a bridge between backend geospatial analysis and frontend visualization platforms such as Figma-based interface prototypes. The framework integrated polling place data, census tract demographic information, hospital infrastructure data, and geospatial distance calculations into a unified querying system capable of producing structured JSON outputs suitable for future dashboard deployment, web applications, or mobile civic accessibility tools.

The system architecture emphasized flexibility and adaptability. Although the initial implementation focused on Montgomery County, Alabama, the workflow was designed to support future expansion to additional Black Belt counties and broader statewide analyses. The API framework also supports future integration or additional civic infrastructure layers including schools, public transit stops, emergency shelters, broadband access points, and flood vulnerability indicators.

Polling Place Data

As previously mentioned, polling place data originated from the statewide Alabama polling location shapefile titled *Al_Polls_Flood_SLED.shp*. The shapefile contained geospatial point representations of active polling locations across Alabama alongside extensive attribute metadata. Relevant attributes extracted for analysis included precinct names, street addresses, county identifiers, municipality information, ZIP codes, flood hazard metadata, and geographic coordinates.

The original shapefile utilized the EPSG:4269 coordinate reference system (NAD83 geographic coordinates). Because accurate distance calculations cannot reliably be performed using angular degree units, all polling location geometries were transformed into EPSG:5070 (NAD83 / Conus Albers Equal Area). This projected coordinate system represents measurements in meters and is commonly used for regional-scale spatial analysis within the United States. The CRS transformation ensured that Euclidean distance calculations between polling locations and census tract centroids were spatially accurate and computationally consistent.

The statewide dataset was then filtered to Montgomery County to align with the project's geographic focus and reduce unnecessary computational overhead during distance calculations and API querying operations. Fintering produced a subset of 49 active polling locations within Montgomery County. Polling location geometries were validated using GeoPandas geometry validation procedures to remove null or invalid spatial records prior to analysis.

Additional metadata from the polling place shapefile included FEMA flood hazard labels such as flood zone classifications and zone subtype categories. Although not fully operationalized in the current API implementation, these environmental vulnerability indicators were stored within the dataset to support future expansions involving disaster resilience, flood exposure, and environmental justice analyses related to civic accessibility.

Demographic Data

Demographic information was received from the Alabama 2020 Census Tracts shapefile, which contained tract-level population and race variables collected through the Census. Variables considered for analysis included total population counts and racial/ethnic composition variables such as White alone, Black or African American alone, Asian alone, American Indian and Alaska Native alone, Native Hawaiian and Pacific Islander alone, Hispanic/Latino population, and multiracial population totals.

The tract dataset also included Census geographic identifiers (GEOIDs), tract names, land area measurements, water area measurements, and polygon geometries representing official census tract boundaries. Montgomery County tracts were isolated using the county FIPS code 101, producing a localized demographic dataset containing 71 census tracts.

To support point-based accessibility calculations, polygon tract geometries were converted into centroid representations using GeoPandas and Shapely geometry operations. Centroids were used as population reference points for nearest distance calculations because they provide a simplified approximation of tract-level population locations while substantially reducing computational complexity.

The centroid generation process required careful geometry handling to avoid conflicts between polygon and point geometries. Original polygon geometries were removed after centroid creation, and centroid geometries were reassigned as the primary geometry column within a new GeoDataFrame structure.

GEOID fields from multiple datasets were standardized as string data types to ensure compatibility during tabular and spatial joins. Duplicate geometry columns and duplicated GEOID fields were programmatically removed during merge operations to avoid conflicts generated by Pandas and GeoPandas suffix assignment behavior.

Demographic percentages were computed within the API system by dividing racial subground counts by total tract population counts. These values were rounded to two decimal places and returned within API responses as tract-level demographic summaries associated with user-input locations.

Hospital Infrastructure Data

Hospital infrastructure data was incorporated into the file as an additional civic accessibility reference layer to allow for future context in polling place accessibility relative to broader public service infrastructure. Hospital point locations were loaded from a statewide hospital shapefile containing geographic coordinates and healthcare facility locations throughout Alabama.

The hospital dataset was filtered to Montgomery County. Hospital geometries were transformed into EPSG:5070 to maintain coordinate system consistency with polling place and census tract datasets.

Nearest neighbor distance calculations were then performed between census tract centroids and nearby hospital facilities. Distances were converted from meters to miles using the standard conversion factor of 1609.34 meters per mile. These calculations allowed for comparative analysis between polling accessibility and healthcare accessibility patterns throughout Montgomery County. The integration of hospital infrastructure data expanded the project beyond election accessibility alone and allowed the API system to function as a broader civic infrastructure accessibility framework. This comparative approach provided additional analytical value by identifying whether communities with limited polling accessibility also experienced reduced proximity to other essential public services.

The architecture of the API system was intentionally designed to support future expansion to additional infrastructure datasets including schools, libraries, emergency response facilities, transportation areas, and broadband access networks.

Spatial Data Processing and Coordinate Systems

Because the project integrated multiple geospatial datasets originating from different coordinate systems and spatial standards, preprocessing and CRS standardization represented a critical stage of the workflow. Polling locations, census tract geometries, and hospital infrastructure layers initially utilized differing coordinate systems and projection definitions.

All datasets were validated for missing CRS metadata prior to transformation. Datasets lacking explicitly defined coordinate systems were assigned EPSG:4326 before reprojection. All spatial layers were then transformed into EPSG:5070 (NAD83 / Conus Albers Equal Area), which provides meter-based planar coordinates suitable for regional accessibility analysis and distance computation.

Spatial validity checks were performed to remove null geometries and invalid spatial records before analytical processing. Bounding box comparisons were additionally conducted to confirm geographic alignment between polling locations and tract boundaries after reprojection.

Spatial indexing structures generated by GeoPandas were used to improve nearest neighbor querying performance during repeated distance calculations. These indexing operations reduced computational overhead associated with iterative geometry comparisons between tract centroids and polling place locations.

Analytical Approaches

The API system utilized nearest-neighbor geospatial analysis techniques to evaluate polling accessibility across Montgomery County. Census tract centroids were treated as origin base points representing population centers, while polling place geometries functioned as destination points representing voting infrastructure locations.

For each tract centroid, Euclidean distance calculations were performed against all polling locations using GeoPandas geometric distance operations. The nearest polling location was identified by locating the minimum distance value across all candidate polling place geometries. Distances were initially calculated in meters within the projected coordinate system and subsequently converted into miles for interpretability within user-facing outputs.

Additional nearest-neighbor calculations were conducted against hospital infrastructure layers to support comparative civic accessibility analysis. These calculations produced tract-level measurements for both polling access and healthcare access.

The system also incorporated dynamic geocoding functionality through the Geopy Nominatim geocoder service. User-generated inputs were processed through a classification workflow that first attempted direct precinct matching against polling location names. Inputs failing precinct matching were subsequently treated as addresses and geocoded into geographic coordinates.

Geocoded coordinates were transformed into projected geometries compatible with the EPSG:5070 analytical environment. Spatial containment operations were then used to identify the census tract containing the user input location. Once the tract was identified, demographic attributes associated with that tract were extracted and formatted into API-ready outputs.

The API framework generated structured query responses containing:

- User input classification
- Nearest polling location
- Polling place address
- Polling accessibility distance
- Accessibility classification
- Associated census tract identifier
- Tract-level demographic information and composition

The modular architecture of the system allows for future integration with live REST APIs, cloud database systems, and frontend dashboard frameworks.

Accessibility Classification Framework

Polling accessibility categories were assigned using threshold-based classification rules derived from distance-to-polling-place calculations. These classifications were designed to simplify interpretation of accessibility conditions for frontend visualization systems and public-facing interfaces.

Accessibility categories were defined as:

- "Very Close" for polling distances under 1 mile

- “Moderate” for distances between 1 and 3 miles
- “Limited” for distances between 3 and 5 miles
- “Poor Access” for distances greater than 5 miles

These categories provided a qualitative interpretation layer on top of continuous spatial distance measurements. The classification system was designed to support rapid visual interpretation within dashboards, choropleth maps, and API-driven civic applications.

The threshold system additionally supports future expansion toward weighted accessibility scoring systems incorporating transportation infrastructure, sidewalk availability, flood vulnerability, and public transit accessibility.

Polling Place Data Interactive Query and API Generation System

A major component of the project involved developing an interactive querying framework capable of generating structured API-style outputs from user-provided inputs. The system was designed to simulate backend logic for future civic accessibility applications and frontend prototypes.

The query system accepted multiple forms of user input including:

- Street addresses
- ZIP codes
- Polling precinct names
- Public institutions
- Hospitals
- Civic landmards

Inputs were categorized based on matching and geocoding results. Precinct names were matched directly against polling place records, while other inputs were processed through geocoding services to obtain geographic coordinates.

Outputs were returned in structured JSON format to support interoperability with frontend visualization systems and future REST API deployment. JSON responses contained nested objects representing polling location information, demographic summaries, and accessibility classifications.

The API design emphasized flexibility. Additional infrastructure layers, routing engines, or demographic variables can be integrated into future versions without major restructuring of the system architecture.

The API framework additionally served as the backend conceptual foundation for future Figma-based civic accessibility interface prototypes intended to improve public interaction with polling accessibility information.

Visualization Methods

Exploratory visualizations were generated using Matplotlib and GeoPandas to analyze spatial accessibility patterns and demographic relationships throughout Montgomery County.

Generalized visualizations included:

- Choropleth maps of polling accessibility distances
- Census tract racial composition maps
- Polling accessibility classification maps
- Histograms showing distance distributions
- Bar charts displaying accessibility category frequencies
- Scatterplots comparing racial composition against polling distance
- Comparative scatterplots of hospital accessibility versus polling accessibility

Scatterplot analyses were used to examine potential relationships between demographic composition and polling accessibility outcomes.

Visual outputs were exported as high-resolution PNG figures suitable for inclusion within reports, dashboards, and presentation materials.

Python Tools and Development Environment

Python 3.11 served as the primary programming language environment for API development, geospatial analysis, and data preprocessing operations. The system was implemented within a virtual environment to ensure package isolation and reproducibility across development sessions.

Primary Python libraries included:

- Pandas for tabular data manipulation and merging operations.
- GeoPandas for geospatial analysis and spatial joins.
- Shapely for geometric operations and centroid generation.
- Matplotlib for static visualization and exploratory graphics.
- Geopy for geocoding and address-to-coordinate conversion.
- JSON utilities for structure API response generation.

GeoPandas spatial indexing functionality was used to optimize repeated geometry comparison operations during nearest-neighbor calculations. CRS transformations and geometry validation procedures were heavily utilized throughout preprocessing workflows to maintain analytical consistency between datasets.

The API system was designed as a scalable prototype capable of future deployment within Flash or FastAPI backend environments connected to cloud-hosted spatial databases and interactive frontend applications.

2.3 Design

Introduction

One of the primary stakeholders identified throughout the project was everyday voters, particularly individuals living in areas affected by polling place and transportation accessibility challenges. As the project evolved, the Design Team recognized that while raw datasets, visualizations, and analytical findings may be useful for stakeholders such as election administrators, local agencies, and SPLC research teams, these forms of information were often less accessible to the general public. Within the Design Team, there was a shift from initially exploring data visualization and geographic mapping approaches to communicate polling place access and distance analysis findings toward exploring how these findings could instead be translated into a more user-centered product. The team's goal was to create an easy-to-parse interface that allowed voters to better understand factors that may affect their ability to reach polling places, including travel distance, roadway conditions, transportation limitations, and polling location accessibility.

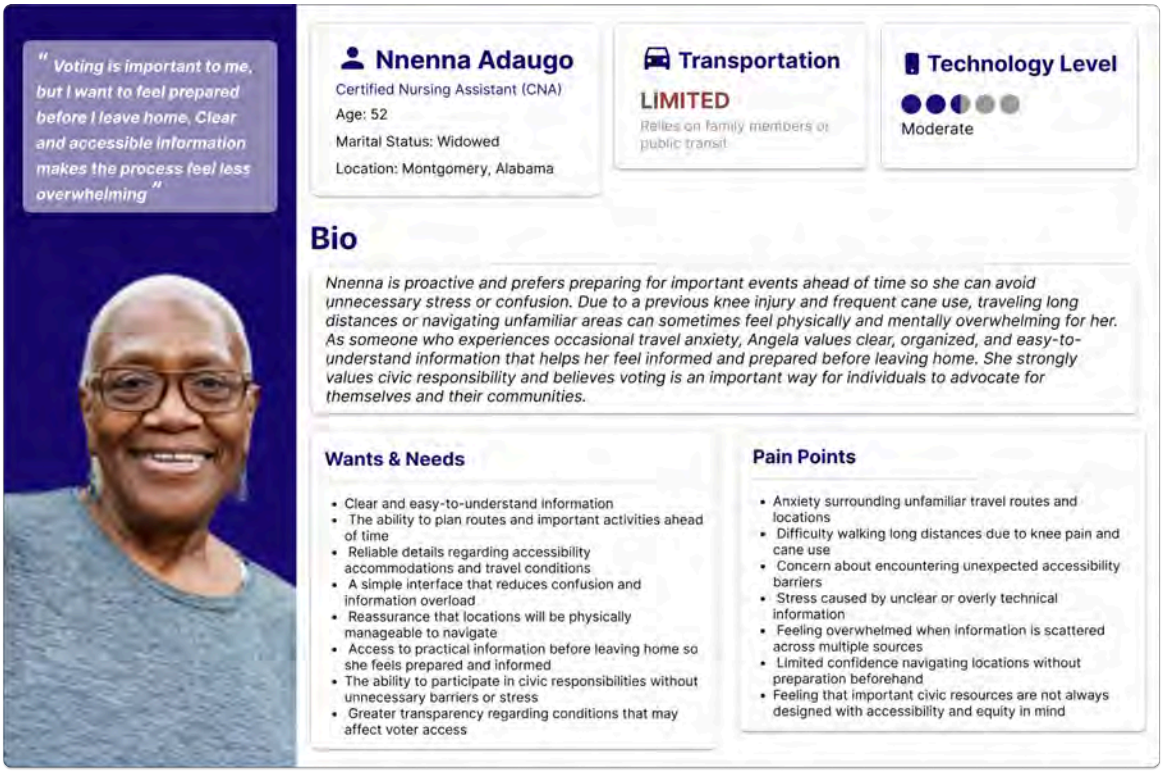
Rather than functioning solely as a means of visualizing data, the prototype evolved into a simplified civic-focused accessibility interface influenced by findings and trends identified by the Data Analysis Team and intended to translate otherwise technical information into a more practical and understandable user experience. However, before developing the interface itself, the Design Team first needed to better understand who the product was intended for.

Persona Development

Due to time constraints, direct user research and usability testing following prototype development were outside the scope of the project. Instead, the Design Team developed an insight-informed persona, Nnenna Adaugo, based on broader project findings and recurring accessibility-related themes identified throughout the research process.

Before developing the final interface and organizing features into a cohesive and navigable experience, the Design Team explored potential user scenarios and voter accessibility concerns through persona development. The final persona was created to better understand how individuals may be affected by geographical, transportation-related, and accessibility barriers to voting access.

The persona also helped contextualize how factors such as transportation limitations, travel conditions, mobility concerns, and accessibility accommodations may affect voter experience and interaction with the product.



User persona detailing the demographic background, transportation access, technology familiarity, wants and needs, and voting-related pain points of Nnenna Adaugo, a 52-year-old immigrant living in Montgomery, Alabama.

Product Concept and Beginnings

VoteAccess is a civic-focused accessibility and preparatory resource developed as one component of the broader BB-PAD (Polling Place Access & Distance Analysis) project. Throughout the project, the Design Team recognized that while the broader team produced multiple data-driven visualizations and analytical findings, many of these outputs were more beneficial to technically inclined stakeholders than everyday voters. The team therefore explored how polling place access and distance-related findings could instead be translated into a more user-centered experience that provided voters with clearer, more practical insight into the broader issue of voting accessibility and how it may affect them directly.

VoteAccess was conceptualized as a simple and easily navigable two-screen platform intended to help users prepare for Election Day regardless of election type. Users begin by entering their address or zip code, after which the platform generates information curated to their location and assigned polling area. The resulting interface includes polling location details, route-based map visualizations, estimated travel conditions, transportation availability, accessibility accommodations, election information, and voting support resources.

The platform additionally explores the use of simplified indicators and conceptual scoring systems intended to communicate otherwise complex polling place access and distance-related conditions in a more understandable and user-friendly format. Many of these interface features and informational categories were directly influenced by broader project findings produced by the Data Analysis Team as well as insight-informed persona development explored throughout the design process.

The overall product concept was intended to function as both an informational and preparatory resource while additionally serving as an accessibility-oriented "equity measurer" designed to help users better understand the environmental, infrastructural, and transportation-related factors that may influence voter access within their area.

Design Choices

Several design choices throughout VoteAccess were directly influenced by insights and broader accessibility patterns identified by the Data Analysis Team. Findings related to polling place accessibility, transportation conditions, and distance measurements revealed that while average polling distances across analyzed areas appeared relatively short (generally under two miles), distance alone was insufficient for understanding broader voting accessibility conditions. Walkability disparities, inconsistent roadway maintenance, transportation limitations, and varying environmental conditions all contributed to accessibility concerns that may affect voter experience and turnout.

To address these findings, the VoteAccess interface was intentionally structured around multiple accessibility-focused categories rather than relying solely on polling distance measurements such as maps or straight-line distance calculations. Sections including "Ease of Travel," "Road Quality," "Polling Place Accessibility," and "Polling Information" were developed to communicate how infrastructural, environmental, transportation-related, and accessibility-focused factors may collectively influence polling place access beyond geographic distance alone.

This organizational structure additionally supported the broader goal of VoteAccess as a preparatory civic resource by helping users better anticipate potential accessibility barriers, understand travel conditions, and identify available accommodations prior to Election Day.

Scoring Indicators:

- Section: "Your Voting Access"
    - Indicator: Polling Place Accessibility gauge — 72 out of 100 (Moderate Access)
    - Explanation:
        - The conceptual polling place accessibility gauge was partially influenced by scoring exploration conducted by Data Analysis Team member Mawa, who developed accessibility-related scoring concepts based on distances between centralized areas and polling locations. However, within VoteAccess, the indicator was further adapted into a broader conceptual accessibility gauge intended to reinforce the idea that polling place accessibility is influenced by multiple interconnected transportation and environmental conditions rather than distance alone.

          Rather than presenting users with isolated statistics, the gauge explored how broader accessibility conditions could be summarized into a more understandable and preparatory indicator system aligned with the civic-focused goals of VoteAccess. A "Moderate Access" score, for example, was intended to communicate that voting accessibility conditions may contain both supportive and limiting factors depending on transportation, roadway conditions, infrastructure, and polling-site accessibility.

        - Gauge Levels
            - Very Limited: 1-25
            - Mild: 25-6-500
            - Moderate: 51-75
            - High: 76-100
- Section: "Travel Conditions"
    - Indicator: Road Quality Score – 54.6/100 (Moderate)
    - Explanation:
        - This indicator was influenced by findings related to roadway quality, documented route maintenance, and walkability variability identified through analysis conducted by team member Myles Sartor. The intention of this feature was for the indicator to be informed by historical road quality data and continuously adapt as roadway and infrastructure conditions change over time.
- Section: "Polling Place Accessibility"
    - Indicator: High
    - Explanation:
        - The "Polling Place Accessibility" section was conceptually intended to incorporate findings related to ADA accessibility and polling-site accommodations explored by Team 5: ADA Accessibility & Compliance at Polling Locations. Due to project timeframe limitations, the section remained conceptual during prototype development. However, future iterations of

VoteAccess could potentially integrate more detailed accessibility and compliance findings in order to transform the section into a more data-informed and functional accessibility resource.

Tools & Platforms Used

1. Figma
    a. Primary platform for interface design, prototyping, organization, component creation, and mild interactive screen development throughout the VoteAccess prototype
2. Iconify Plugin (Figma):
    a. Used to source consistent iconography related to transportation, navigation, and voting-related information indicators throughout the website.
3. ChatGPT:
    a. Used to assist in generating the map visual and placeholder geographic representation intended to demonstrate distance.

Visual Identity & Design System

The visual identity of the prototype was designed to establish VoteAccess as a civic-focused product by balancing the professionalism commonly associated with civic platforms with simplicity, readability, and accessibility, while still maintaining subtle visual associations with the Southern Poverty Law Center (SPLC). Because the prototype represents only one focus area within SPLC's broader scope of work, the Design Team intentionally avoided directly replicating SPLC's primary branding and design system. Instead, VoteAccess was given its own independent visual identity and presentation, allowing the platform to function as a standalone public-facing resource while still acknowledging SPLC as the supporting organization behind the project.

Typography:

Inter was chosen as the primary typeface throughout the VoteAccess interface due to its readability, clarity, and reliability across multiple digital environments. Its familiar and highly legible structure makes it well-suited for a civic-focused platform centered around accessibility and ease of use. Variations in weight and scale were used throughout the interface to establish visual hierarchy while still maintaining consistency across the platform.

Typography Role	Typeface	Font
Main Headers	Inter	Bold, 64px
Section Titles	Inter	Bold, 48px
Section Subtitles	Inter	Regular, 24px
Card Labels/ Titles	Inter	Semi Bold, 40px
Card Body Text	Inter	Regular, 30px

Color:

The color palette was designed to reflect the nature of VoteAccess as a reliable civic platform that evokes calmness, professionalism, and consistency. Color combinations and contrast levels were tested for readability and accessibility using Web Content Accessibility Guidelines (WCAG) scoring tools. Additionally, the low saturation of the palette was intentionally used to reduce potential cognitive overload, as the interface contains multiple layers of informational content within a single experience. Subtle gradient treatments were also incorporated throughout select cards to introduce depth, soften visual contrast, and maintain visual cohesiveness across the platform.

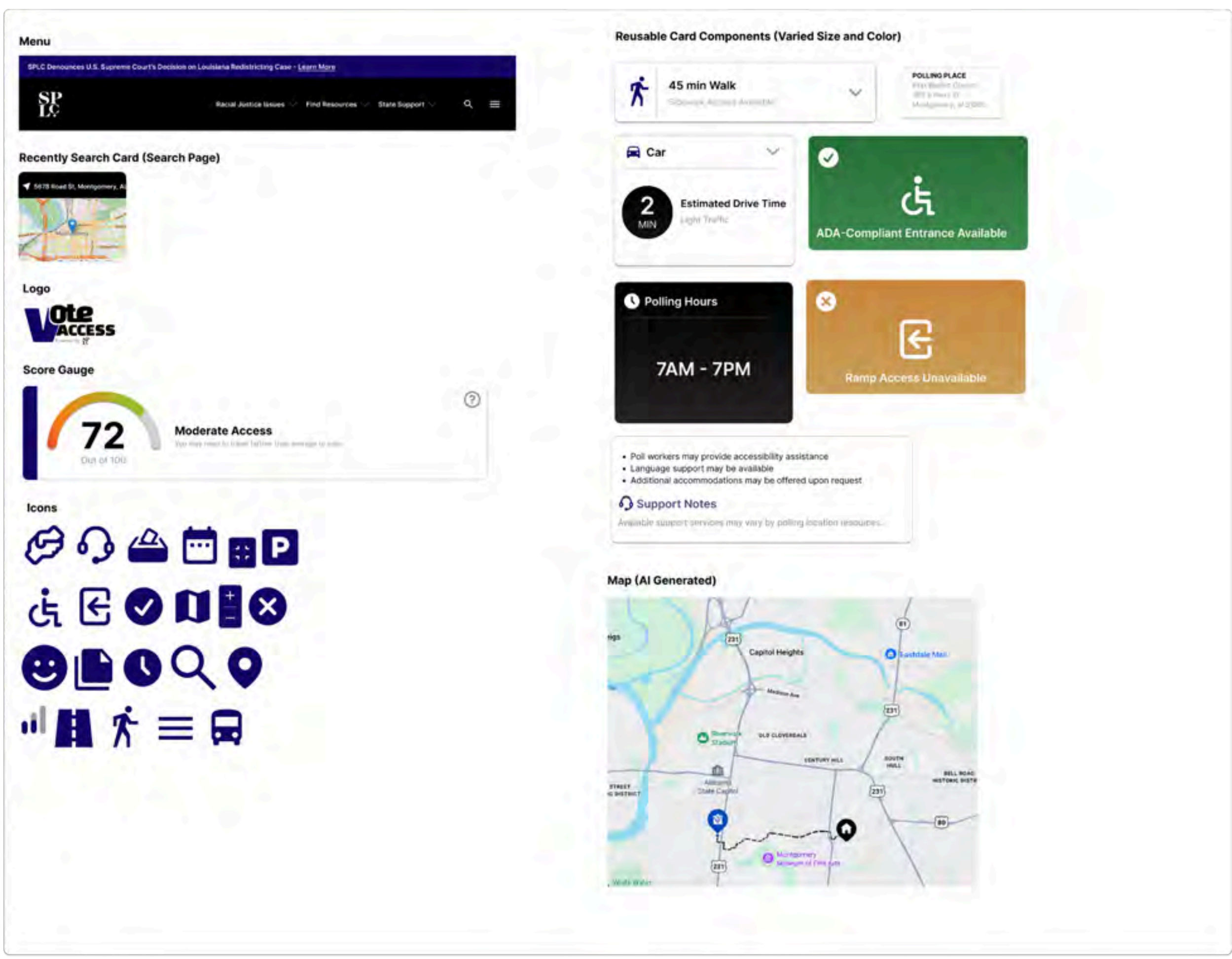
Color	Hex Code	Opacity	Usage/ Role

	#150765	100%	• Primary Color • Top Navigation • Icons • Cards • Logo Branding • Footer Accents • Accents • Titles
	#51467A	100%	• Support (Travel Notes)
	#000000	100%	• Primary Text Color on Light Backgrounds • Navigation/ Header
	#FFFFFF	100%	• Primary Text Color on Dark Backgrounds • Page Color • Cards
	#656161	73%	• Subtitles/ Microcopies • Interaction Buttons • Secondary Hierarchy
	#656161	61%	• Dividers • Card Strokes
	#287039	100%	• Positive Accessibility Indicators • High Accessibility • Icons

	#418B52	100%	• Text support to #287039
	#C8873E	100%	• Communicates Limitation

Design System:

A reusable component system was built and implemented throughout the interface in order to maintain visual hierarchy and consistency while organizing large amounts of information into more manageable and digestible sections. Repeated card structures, indicators, score gauges, icons, and interface elements were incorporated to simplify navigation while reinforcing hierarchy, usability, and consistency across the platform.



Design system components displaying reusable interface elements and iconography incorporated throughout the VoteAccess prototype.

Design Challenges & Limitations

One of the primary challenges throughout the project involved determining how polling place access and distance-related findings could be visually communicated in an understandable way while still meeting the broader goals of the project. The Design Team initially intended to incorporate more traditional data visualization and geographic mapping approaches. However, as non-Information Science majors, the team had limited experience working with large-scale datasets and geographic visualization tools in a manner that could effectively support the broader work of the Data Analysis Team.

Several platforms, including Flourish and Genially, were explored during early development stages, but the team ultimately struggled to create visualizations that effectively communicated the broader accessibility concerns and user experience goals of the BB-PAD project. As a result, the project direction gradually shifted from a primarily visualization-focused approach toward a more user-centered concept such as VoteAccess.

Time constraints also became a major limitation throughout the design process. Because a substantial portion of the project timeline was spent experimenting with visualization platforms and determining a workable project direction, the team had limited time for deeper conceptualization, iteration, and refinement while developing VoteAccess within Figma. Rather than progressing through extensive low-fidelity ideation phases, the prototype development process moved relatively quickly into mid-fidelity interface design.

Additionally, direct user research, usability testing, and iterative user feedback were outside the scope and timeframe of the project. While insight-informed persona development and broader project findings helped guide design decisions, the prototype was not validated through formal user testing or real-world implementation. As a result, VoteAccess remains conceptual and could be further expanded or refined in future iterations of the product.

The prototype's accessibility indicators, scoring systems, and informational categories were also developed primarily as conceptual communication tools rather than finalized analytical measurements. While broader themes and findings identified by the Data Analysis Team directly influenced the structure of the interface and several accessibility-related categories, many numerical values, visual indicators, and displayed figures included throughout the prototype were created to demonstrate how polling place access information could potentially be communicated within a real-world product environment. For example, while generalized accessibility labels and scoring concepts were informed by broader project analysis, many displayed prototype figures were not generated through live or finalized analytical systems.

Lastly, the map visualization and geographic representation within the prototype remain static. Due to limited experience with GIS platforms and interactive mapping systems, the map displayed within the prototype was generated with the assistance of generative AI tools, specifically ChatGPT. While conceptual, the map was intended to demonstrate the potential for route-centered interaction and accessibility-focused interface design rather than function as a live geographic system.

Overall, the prototype should be understood primarily as a conceptual exploration of an accessibility-focused civic product rather than a fully functional polling access platform.

Project Adjustments

Challenges	Negotiated Adjustment/ Resolution
Difficulty developing advanced	The project direction shifted toward a UX-focused

geographic visualizations	accessibility and communication-based product rather than highly technical visualization systems.
Limited GIS and a large-scale data visualization	Responsibilities became more specialized between teams, with the Data Analysis Team focusing on analytical dashboards and findings while the Design Team focused on accessibility communication and user experience design.
Limited project timeframe	The team narrowed the feature scope and transitioned more quickly into mid-fidelity prototyping within Figma in order to maximize development time.
Lack of direct user research and usability testing	Used broader project findings and recurring accessibility-related themes to develop the insight-informed persona, Nnenna Adaugo, which helped guide interface organization and feature prioritization throughout the design process.
No live geographic mapping system	Used a conceptual static map prototype to demonstrate route visualization and potential user interaction without relying on a live geographic mapping system.
No finalized analytical scoring model	Used conceptual indicators and simplified scoring systems to demonstrate how polling place access conditions could potentially be communicated within the platform.

Deliverables and Findings

Data Analysis Team

Montgomery County Voter Access Dashboard

Description: An interactive ArcGIS Dashboard application providing real-time monitoring of polling place accessibility across Montgomery County, Alabama. The dashboard integrates block group-level accessibility scores, demographic indicators, polling place coverage statistics/finder, directional routing capabilities, and walking isochrone polygons into a unified client-facing interface.

Rationale: This deliverable addresses the SPLC's need for a practical, accessible tool that non-technical stakeholders can use to explore voting access patterns without requiring GIS expertise or programming knowledge. The dashboard transforms complex spatial analysis into an intuitive visual format with filtering capabilities, pop-up information, applications, and layer toggles. It meets the project objective of creating "solution-based products" by providing a reusable template that can be extended to additional counties by updating data sources and map extents.

Dashboard Elements: The dashboard header displays four live indicator elements showing key summary statistics at a glance. The Total Block Groups indicator reads 203, representing all Montgomery County census block groups analyzed. The Total Polling Places indicator reads 49, representing all 2020 Montgomery County polling places. The legend shows that polling places labels are colored blue, with the colors of the circles getting darker as average distances (Miles) increase. The size of the circle labels grow larger as the population the polling place serves grows larger, and each block group (population center) surrounding these polling place is colored-coded on their general accessibility/walkability categories (Dark Green – Excellent, Green – Good, Yellow – Fair, Poor – Orange, Very Poor – Red). Clicking any point opens a detailed pop-up displaying the block group's GEOID, geographic coordinates, walking distance to a nearest polling place (Miles), nearest polling place name, Black population percentage, no-vehicle percentage, poverty rate, total population, vulnerability level, and walkability category. This level of detail enables stakeholders to investigate specific communities of concern. The pop-ups for polling places show the facility name, street address, average walking distance from assigned population centers (Miles), total population served, and number of population centers assigned to that location. This enables rapid identification of polling places that may be over or under-serving their designated areas. The walkability isochrones application embedded within the dashboard visualizes 245 time-sorted walkable area polygons across five distinct time intervals. It also requires no login or authentication beforehand, making it accessible to those who need it. The 5-minute isochrones (darkest blue) show the most immediate walkable area before moving onto 10, 15, 20, and 30-minute intervals (lightest blue). These isochrones show the maximum reasonable walking distance, with visual progression enabling stakeholders to understand how pedestrian coverage expands around each polling place. All five time intervals are visible simultaneously, but layer visibility controls enable toggling individual time intervals on and off for focused analysis. SPLC staff and researchers can understand spatial patterns of specific coverage around the area. As an intuitive visual representation of walking patterns, the isochrones complement distance metrics well. Rather than just the specific numbers, each isochrone shows the reachable areas with given walking times over straight-line approximations. Any gaps or errors in judgement during the distance calculations from before can be filled in when you're able to see actual coverage and overlapping service areas. Dropdown selection filters within the dashboard allow for dynamic data exploration. There are distance and area measures, basemap selections, layer enabling/disabling, and poor/very poor access area displays. All of these features allow stakeholders to focus on specific data elements without visual clutter. The polling location finder application embedded within the dashboard is a tool built with the Nearby template that enables people to enter any Montgomery County address and find the closest polling places with walking or driving directions. The application requires no login or authentication, making it accessible to all voters. It helps serve the SPLC's constituency by providing an accessible, no-login required tool for voters to locate their nearest polling place and understand the walking or driving distance involved. The practical utility behind the 0-5 mile search radius distance slider(2.5-mile default) aligns with a focus on walkable access, and directions that leverage the distance analysis work. The search bar accepts any street address, intersection, or place name. Upon searching, the application queries the polling places and population center feature layers and returns results stored by distance from the entered address. Results display the earlier pop-up details regarding each polling place and population center within the given radius, as well as exact distances from the specified address. You can also interactively click anywhere within the given map layers instead of typing in a specific address, allowing the same searching or querying process to take place. The embedded directional routing widget application provides the same address search and directions functionality as the Nearby app, but has slightly more features. You can create routes by entering in addresses or clicking on the map, as well as provide stopping points along the way alongside departure times. The modes you can use are driving time, driving distance, trucking time, trucking distance, walking time, walking distance, rural driving time, rural driving distance.

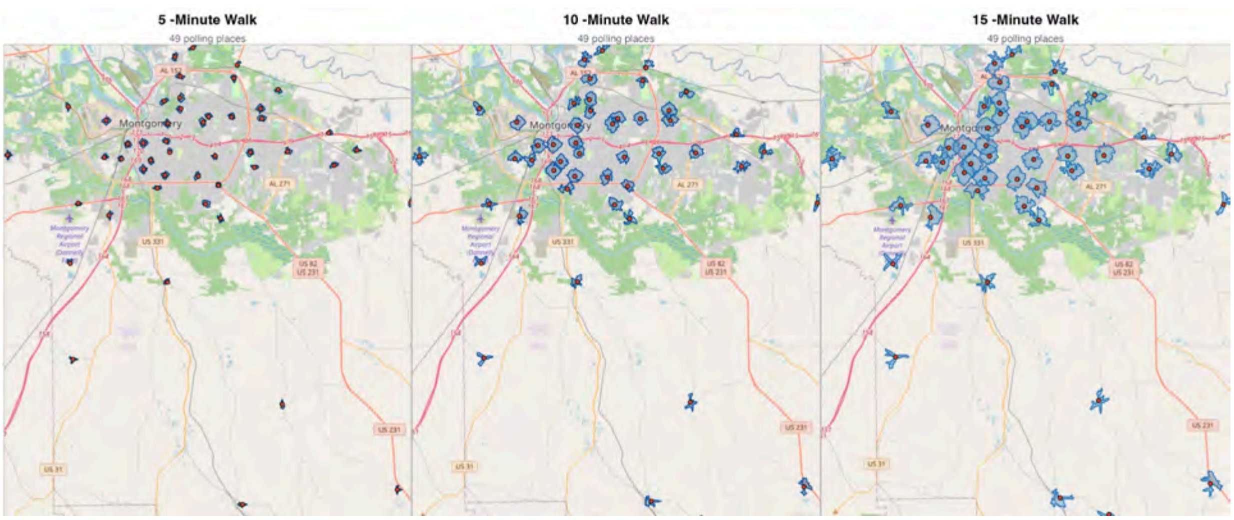
Statistical Analysis and Key Findings

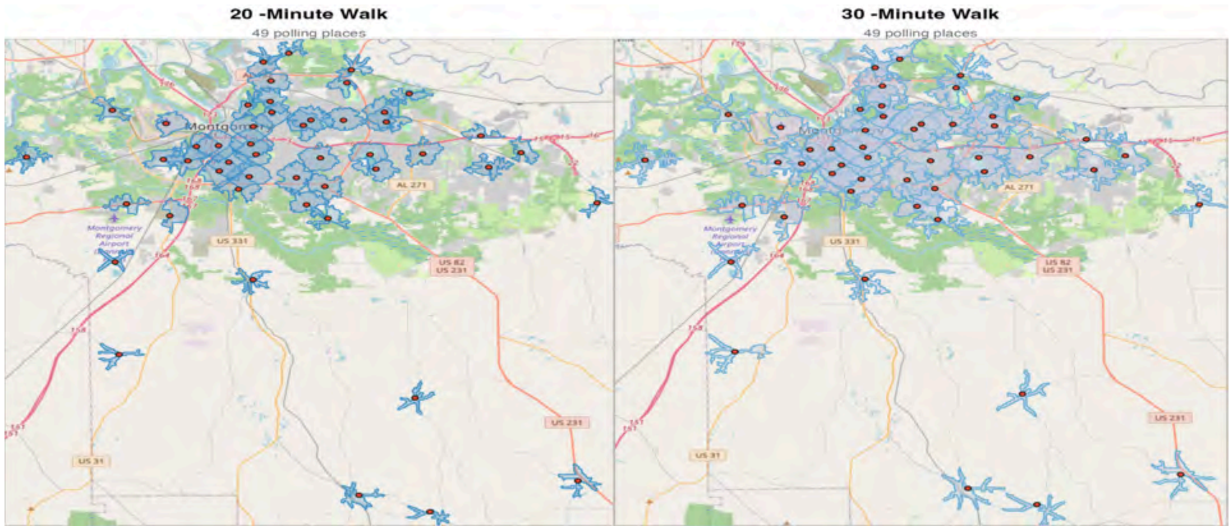
Comprehensive statistical analysis of voter turnout, demographic disparities, and polling place accessibility including regression models, correlation analysis, and descriptive statistics are present throughout our findings. The core analytical nature of our project included transforming raw distance calculations and demographic data into evidence-based findings and recommendations. The five distance calculation methods produced consistent results across Montgomery County. Euclidean straight-line distance averaged 1.02 miles per population center to its nearest polling place. Manhattan grid-based distance averaged 1.27 miles. OSRM driving distance, which reflects actual road networks, averaged 1.44 miles. OSRM walking distance averaged 1.40 miles. Google walking distance, which uses a more sophisticated pedestrian routing algorithm, averaged 1.58 miles.

The cross-comparison ratios between methods reveal important patterns. The driving-to-Euclidean ratio of 1.41x indicates that actual road routes are typically much shorter than straight-line distance, which is a counterintuitive finding explained by the fact that Euclidean distance crosses straight through buildings, waterways, and other barriers, while road networks follow efficient transportation corridors (actual pathways to destination). The OSRM walking-to-Euclidean ratio of 1.37x similarly indicates that pedestrian pathways are more direct than straight-line approximations. However, the Google-to-OSRM walking ratio of 1.13x suggests that Google's routing algorithm accounts for pedestrian barriers and realistic walking paths that OSRM's simpler foot profile may miss.

The 15-minute walkable coverage analysis revealed that the average polling place in Montgomery County has a walkable area of 0.74 square miles, serving approximately 1,408 people. The total walkable area across all 49 polling places is 36.16 square miles. Coverage varies substantially between polling places, with the best-covered location serving dramatically more residents than the least-covered locations in rural areas.

Walkability category distribution showed that 8.4% of areas Excellent walkability, 31.5% Good, 32.5% Fair, 22.2% Poor, and 5.4% Very Poor. This means 27.6% of Montgomery County block groups fall into the Poor or Very Poor categories, which is over a quarter of the county's population centers. The Montgomery road quality score of 54.6/100 indicates that, on average, walking routes are in fair condition, with room for improvement in road classification, width, and construction status.





Faceted Isochrone Map showing 5/10/15/20/30-minute walking coverage with polling places on OpenStreetMap basemap. Each panel displays walkable area polygons for a specific time threshold, with polling places marked as red dots (Generated by isochrone_final.R).

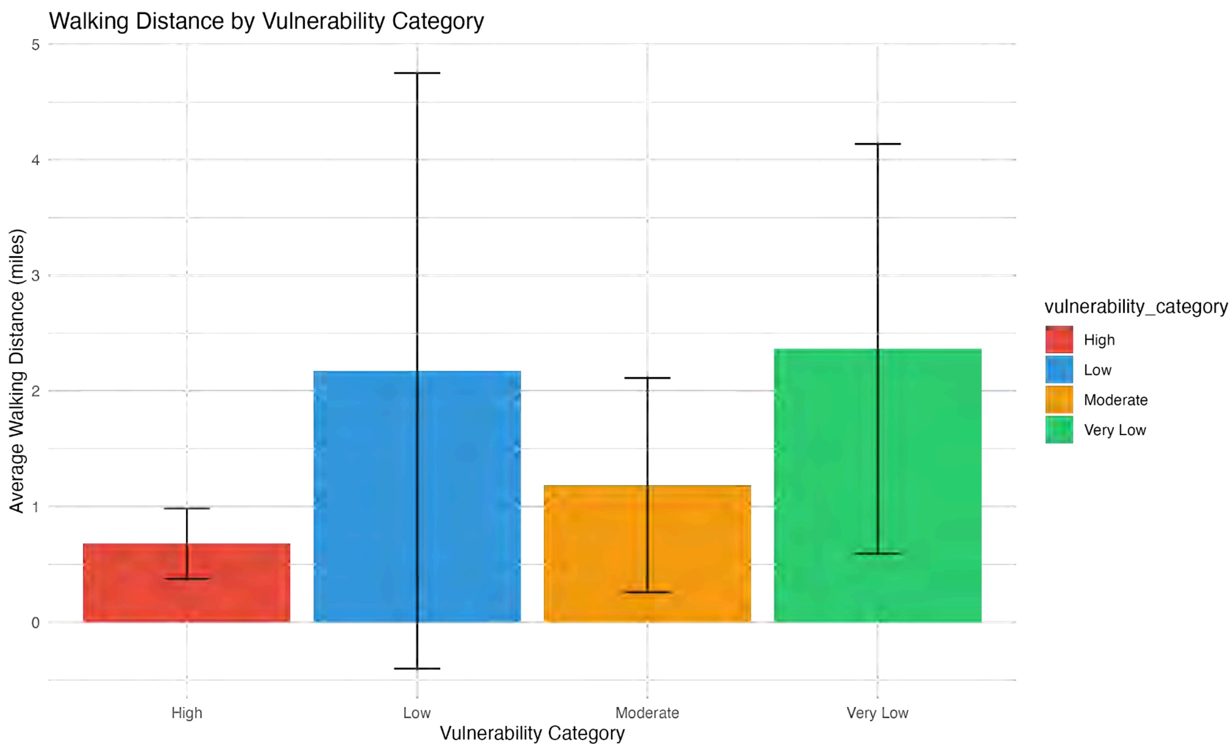
Demographic Disparities

The most significant finding of this analysis is the paradoxical relationship between vulnerability and walking distance. High-vulnerability areas are defined as communities with above-average percentages of Black residents, poverty, and households without vehicles. They have significantly shorter walking distances to polling places than low-vulnerability areas. High percent Black (>50%) areas average 1.22 miles of walking distance, compared to 2.52 miles for low percent Black areas, which is a difference of 1.30 miles that is statistically significant ($t = 3.58$, $p < 0.0001$).

This finding is confirmed by the regression models. Model 1 found that walkability scores are not significantly predicted by demographic variables. Model 2 found that percent Black is a significant predictor of walking distance, with each percentage point increase in Black population associated with a 0.0017-mile decrease in walking distance. Model 3 found that the vulnerability index is a highly significant predictor, with higher vulnerability associated with shorter walking distances. All three models point in the same direction, meaning that physical access to polling places is equitable or even favorable to high-vulnerability communities in Montgomery County.

The correlation analysis reinforces this finding. Walking distance is negatively correlated with percent Black ($r = -0.307$, $p < 0.001$), percent poverty ($r = -0.258$, $p < 0.001$), and vulnerability index ($r = -0.308$, $p < 0.001$). It is positively correlated with median income ($r = 0.320$, $p < 0.001$). In other words, wealthier, whiter areas tend to have longer walking distances to polling places. This is the opposite of what would be expected if polling place distribution were inequitable.

The priority areas analysis confirmed this pattern from a different angle. A query for block groups that are simultaneously high vulnerability and poor/very poor walkability returned zero results. No community in Montgomery County meets both criteria simultaneously. High-vulnerability communities have adequate physical access. Communities with poor walkability tend to be lower-vulnerability.

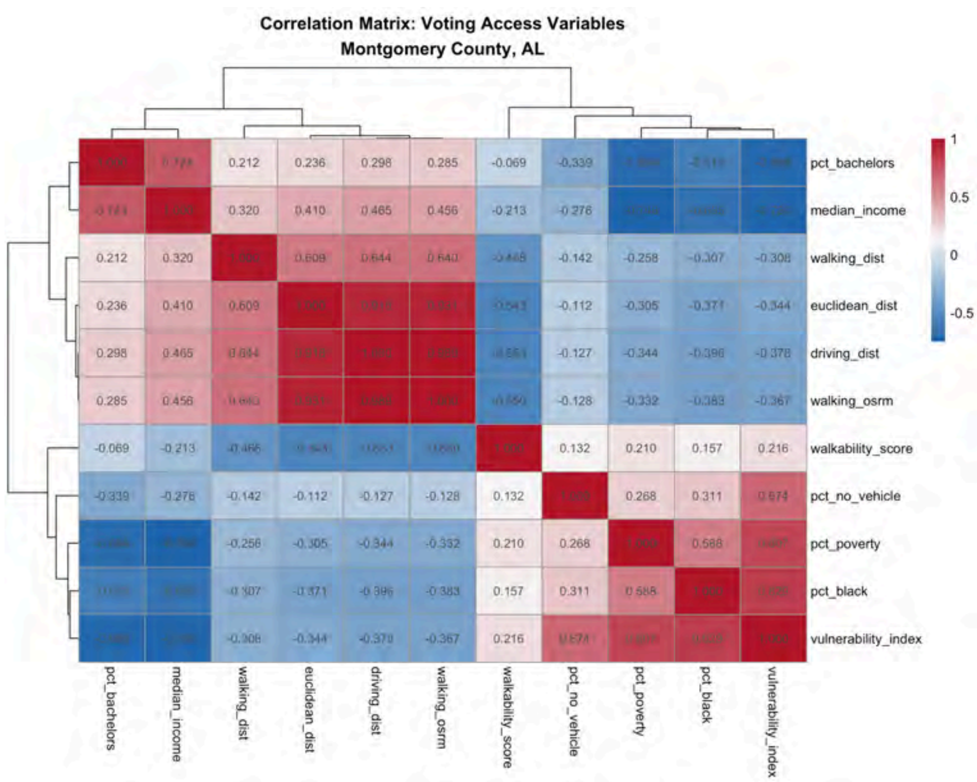


Vulnerability Bar Chart comparing walking distances across four vulnerability categories. High vulnerability areas (n=13) show the shortest average walking distance at 0.68 miles, while Very Low vulnerability areas (n=10) show the longest at 2.37 miles. Error bars represent one standard deviation (Generated by vulnerability_bar_chart.R).

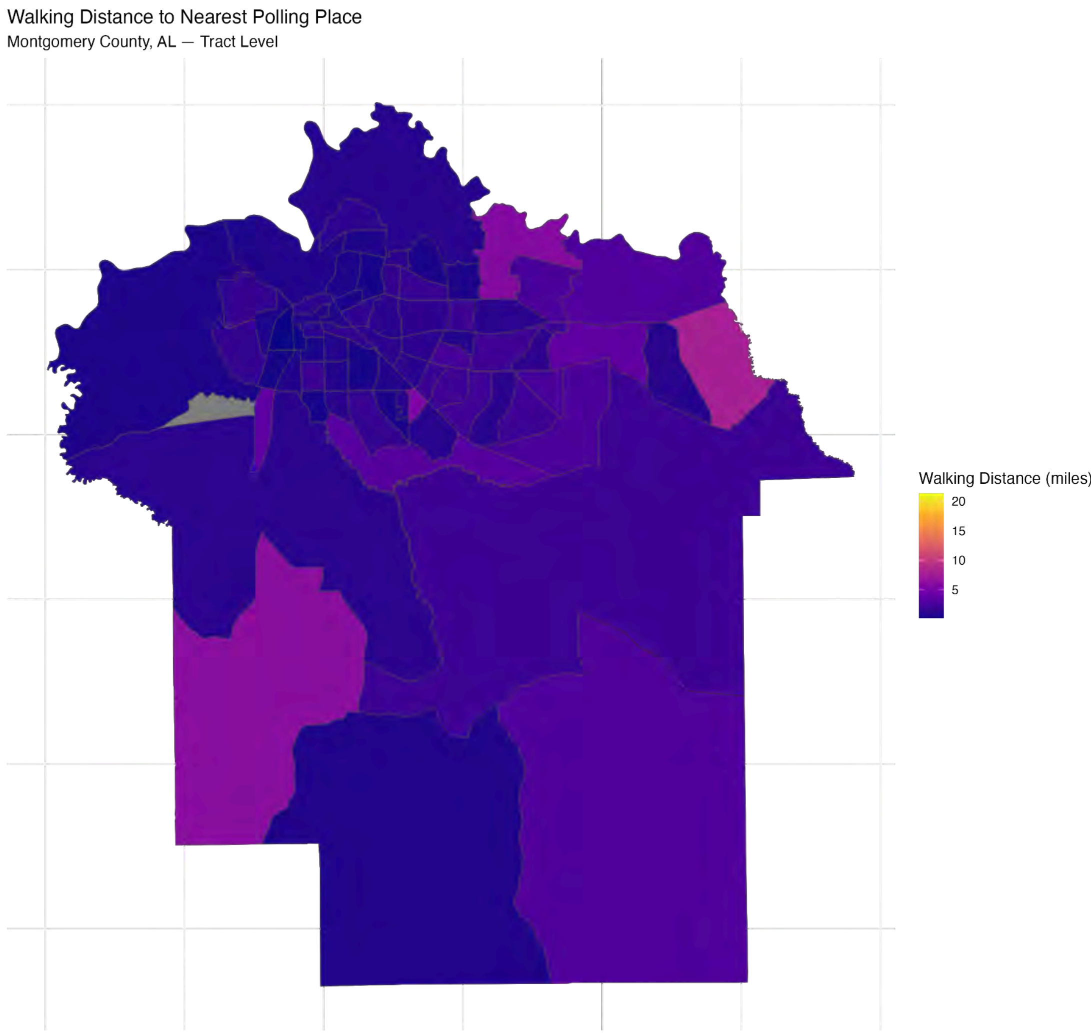
Infrastructure Equity

While physical distance does not disadvantage vulnerable communities, infrastructure quality does. The sidewalk equity analysis revealed that lower-vulnerability areas have more sidewalk coverage than higher-vulnerability areas. The average sidewalk coverage percentage varies systematically by vulnerability category. This creates a situation where high-vulnerability communities, despite being physically closer to polling places, may face more dangerous or less comfortable walking conditions due to missing sidewalks, poor road quality, and inadequate pedestrian infrastructure.

The routing outliers analysis identified multiple block groups where actual Google walking distances are more than 1.5 times their Euclidean distances, indicating significant pedestrian detours. These detours are likely caused by highway barriers, missing crosswalks, disconnected street grids, or other infrastructure obstacles that are available in the routing_outliers database view and represent concrete intervention points for infrastructure improvement.



Correlation Heatmap displaying Pearson correlation coefficients between walking distance, walkability score, Euclidean distance, driving distance, walking OSRM distance, and demographic variables (percent Black, percent poverty, percent no vehicle, percent bachelor's degree, median income, vulnerability index, Generated by correlation_heatmap.R).



Choropleth Map of Montgomery County census tracts colored by average Google walking distance to nearest polling place. Darker purple indicates longer walking distances, concentrated in rural areas in the south and east of the county. Lighter shades indicating shorter distances cluster in the urban core (Generated by choropleth_map.R).

Turnout Patterns

Voter turnout analysis between 2020 and 2024 revealed concerning trends. Only 35% of matched precincts saw increased turnout from 2020 to 2024, while 65% experienced declines. The average turnout per precinct was −350 votes, and the median change was –598 votes. The average percentage change was +54.9%, but this figure is misleading because it is skewed by a few precincts with very small 2020 turnout that saw large proportional increases. The median percentage change, which is less sensitive to outliers, tells a more reliable story.

Critically, accessibility metrics show negligible correlation with turnout change. The correlation between vote change and average walking distance is r = −0.142, and the correlation between vote change and walkability score is r = 0.089. Neither is statistically significant. This means that precincts with better physical access to polling places did not have meaningfully different turnout changes than precincts with worse access. Physical distance, while important for individual voter experience, does not appear to be a primary driver of aggregate turnout patterns in Montgomery County.

This finding does not diminish the importance of polling place accessibility. Rather, it contextualizes accessibility as one component of a multidimensional voting access landscape. Transportation availability, voter education, registration processes, time-off-work policies, and other socioeconomic factors likely play larger roles in determining whether and how people vote. The weak correlation between distance and turnout suggests that improving pedestrian infrastructure alone, while valuable for community quality of life, may not significantly impact voter participation without complementary investments in other access dimensions.

Cloud Database and Live Reporting Pipeline

A fully automated cloud database infrastructure enabling live querying of analytical views from any internet connected tool. The pipeline migrates the local PostgreSQL database to Supabase cloud hosting and establishes hourly automated ETL processes through Google Apps Script.

This deliverable addresses the SPLC's potential need for ongoing monitoring beyond the capstone timeline. By establishing cloud infrastructure, the analytical database becomes a living resource rather than a static snapshot. The pipeline is designed to be replicable for additional counties with minimal modification, such as changing county FIPS codes in the processing scripts, running the same SQL schema creation, and adding new view names to the Apps Script array.

The pipeline consists of four layers. The Supabase cloud PostgreSQL database serves as the authoritative data store, containing all 15 analytical views plus the underlying tables. A Supabase Edge Function, written in TypeScript and running online, provides a REST API endpoint that accepts POST requests with view names and returns JSON query results. The function authenticates using the service role key, which has full database access but is only exposed server-side.

A Google Apps Script, triggered hourly by a time-driven installable trigger, iterates through an array of 11 view names. For each view, it sends a POST request to the Edge Function endpoint with the view name in the request body, receives the JSON response, and writes the data to a dedicated sheet within a Google Sheets spreadsheet. Each view gets its own sheet, with column headers automatically generated from the JSON response keys and nested objects (like GeoJSON geometry) flattened into string representations.

The Google Sheets serve as an intermediary cache that any data visualization tool can reliably reach. Google Looker Studio, planned for future development, will connect directly to these sheets as data sources, enabling live-updating dashboards without requiring direct database access.

All four pipeline layers are deployed and operational. The Supabase database was migrated from the local PostgreSQL instance and contains all tables, views, and data. The Edge Function was tested with curl commands and returns valid JSON for all 11 views. The Apps Script is deployed as a web app with an hourly time trigger and successfully populates all sheets. The Google Sheets file contains one sheet per view with current data. This and the Looker Studio integration is documented as planned future work.

Comprehensive Project Documentation

A detailed GitHub repository containing all processing scripts (Python, R, SQL), database schemas, analytical queries, visualization outputs, and a comprehensive README documenting every component of the project was drafted. The repo is organized to enable replication by the client and future teams.

The scripts/ directory contains 16 Python scripts (distance calculations, election processing, polling place standardization, Google API batch processing and fill scripts, OSRM network calculations, Valhalla isochrone generation, walkability enhancement, distance merging, ArcGIS export, and visualization generation), 6 R scripts (ACS data pull, turnout modeling, R visualizations), 5 SQL scripts (table creation, data import,, precinct mappings, view creation, analytical queries), and the Google Apps Script code.

The data/ directory organizes raw and processed data by category. There's census demographics, election results, polling places, shapefiles (precincts, boundaries, infrastructure), and analytical outputs (distance metrics, accessibility scores, isochrone polygons).

The outputs/ directory contains all visualization files. There's maps (isochrone panel, walking distance choropleth), figures (correlation heatmap, vulnerability bar chart), tables (turnout change, walkability summary, merged analysis data, regression results), and cached basemap tiles for offline rendering.

Data Analysis – Interactive Polling Accessibility API Development

Interactive API Query Findings

The interactive polling accessibility API successfully processed both address-based and precinct-based user queries across Montgomery County. Query testing demonstrated that the system was capable of utilizing user-provided addresses in order to identify the nearest polling facility, calculating accessibility distances, and associating the queried location with its corresponding census tract and demographic information and profile.

Address-based queries produced varying accessibility classifications across Montgomery County ZIP codes and civic institutions. For example, queries associated with central Montgomery locations such as "36104 Montgomery AL" and "Alabama State University" returned polling distances of approximately 0.1 miles, categorized as "Very Close" accessibility. In contrast, other ZIP-code-based queries such as "36116 Montgomery AL" and "36106 Montgomery AL" returned distances exceeding one mile, classified as "Moderate" accessibility. These variations demonstrate measurable differences in polling accessibility across spatially distinct areas of Montgomery County.

The API additionally demonstrated successful integration of demographic analysis with polling accessibility metrics. Returned census tract demographic profiles revealed substantial variation in racial composition across queried locations. For example, the census tract associated with "Carver High School" contained a population that was approximately 93.69% Black or African American, while the tract associated with "Dalraida Elementary School" contained a substantially lower Black population percentage of approximately 23.84%. Despite these demographic differences, both locations exhibited relatively short polling distances under one mile, illustrating how API can support comparative civic accessibility analysis across demographically distinct communities.

Healthcare-related queries further demonstrated the flexibility of the system's infrastructure integration capabilities. Queries involving "Baptist Medical Center East," "Baptist Medical Center South," and "Jackson Hospital" successfully returned nearby polling facilities and accessibility classifications, illustrating the feasibility of comparing election accessibility alongside broader civic infrastructure systems such as healthcare access.

The API also demonstrated operational resilience through basic error-handling functionality. Invalid or unmatched user inputs generated structured error responses ("Input not found") rather than causing system failure, improving reliability for future frontend deployment and user-facing applications.

Collectively, the API query outputs demonstrated the viability of an interactive civic accessibility platform capable of integrating geospatial analysis, demographic context, polling infrastructure, and dynamic user interaction into a unified backend-ready system.

Design Team:

VoteAccess Website Prototype

1. Description: A civic-oriented website prototype (VoteAccess) displaying accessibility indicators and functioning as a preparatory platform designed to communicate polling place access, travel conditions, transportation availability, and voting-related accessibility information through a user-centered interface.
2. Link to prototype: BB-PAD Prototype (has some functionality)
3. How to Navigate:

   The VoteAccess prototype contains several interactive elements intended to demonstrate how users may navigate and interact with the platform.

   - On the Search Page (Home), click the "See My Score" button to generate results.
   - Users will then be directed to the results page displaying polling place access information related to their location.
   - Hover over the question mark icon within the "Your Voting Access" section to view additional information regarding the accessibility score.
   - The results page is vertically scrollable and organized into multiple informational sections.

- Within the map section, users may click the fullscreen icon located on the right side of the map to expand the map view. Clicking the collapse icon returns the map to its original size.
- To return to the Search Page, users may scroll back to the top of the interface and click the SPLC button located within the navigation area.

Recommendations

Recommendations for Client (SPLC) – Design

If SPLC chooses to continue building our project into a functional design, our team has some recommendations to support the client during this process. Our first recommendation would be to conduct usability research to improve the platform's scanability and visual design. Specifically, we would recommend SPLC tested the prototype with voters across the Black Belt region. For these results to guide future design decisions, SPLC should test the interface with voters from diverse backgrounds with differing levels of access to transportation. Doing so would inform SPLC about design changes that could improve user experience for real-world users. SPLC should also consider integrating Google Maps into our prototype. Adding a feature that allows users to quickly transition from our platform to an interactive map with clear directions to their nearest polling location could further improve usability. Additionally, this integration might ensure more accurate travel times, directions, and real-time updates to the route. Furthermore, SPLC could prioritize ADA labeling and quick-access shortcuts within the visual design of our platform. Accessibility tags such as "Ramp

Availability" or "Wheelchair Accessible" would allow users with disabilities to filter out nearby polling locations that fail to meet their needs. If possible, SPLC could also work to include real-time updates for each polling location. This could notify users via email or in-app notifications about location closures, current wait times, and any temporary accessibility barriers.

Recommendations for Future Teams

For future teams continuing to further the work on polling accessibility and civic participation research, our group has recommendations and suggestions we would like researchers to consider while continuing our progress. Our first recommendation for future team members is to begin narrowing the scope for the data visual portion of the project. While working on the project, our team explored multiple mapping and visualization platforms before working with Figma and ARC-GIS to finalize our designs. While this process shaped the outcome of our work, it consumed a significant amount of our project timeline. Establishing a clear project direction within the first week would allow future teams to dedicate time to refinement and user testing as soon as possible.

Our second recommendation for teams is to assess technical capabilities and experience amongst team members at the start of the project. While doing our project, we found that it required knowledge and experience with GIS analysis, data visualization, UI/UX design, database management, and accessibility communication. With our team split amongst Information Science majors and Information Designers, our lack of overlap in our backgrounds creates growing pains to further our progress. Future teams need to establish individual strengths and establish realistic goals that can be achieved moving forward.

Our Third Recommendation is to possibly explore live geographic mapping systems and real API integration to implement into the project. The VoteAccess prototype is currently static and more conceptual to demonstrate how polling accessibility information could potentially be communicated to voters. However, implementing API and mapping systems would improve the practicality and accuracy of the project. It would be able to implement real time routing, travel conditions, and updates on polling information that would make the prototype more reliable for the user and investors.

Recommendations for Clients (SPLC) – Data Analysis

1. **Shift advocacy focus from polling placement to pedestrian infrastructure**

The central finding of this analysis is that physical distance to polling places is not the primary access barrier for vulnerable communities in Montgomery County. High-vulnerability areas (predominantly Black, higher poverty) have shorter walking distances to polling places than low-vulnerability areas (1.22 miles vs. 2.52 miles, $p < 0.001$). Polling places are already equitably distributed across the county, and no block group simultaneously meets criteria for both high vulnerability and poor walkability. The access gap is not about where polling places are located, it is about the quality of the pedestrian infrastructure connecting residents to those locations.

The sidewalk equity analysis reveals that high-vulnerability communities have less sidewalk coverage than lower-vulnerability communities, despite being physically closer to polling places. The routing outlier analysis identifies specific block groups where actual walking routes are more than 1.5 times longer than straight-line distance due to highway barriers, missing crosswalks, and disconnected street grids. These are not problems that can be solved by moving polling places. They require infrastructure investment.

Specific recommendations include having the SPLC prioritize sidewalk installation, crosswalk improvements, and pedestrian connectivity in the specific block groups identified in the routing_outliers database view. These represent concrete, data-backed intervention points where infrastructure investment would most directly improve voting access. The GEOIDs of affected block groups are available for direct use in advocacy materials and funding applications.

2. **Investigate non-distance barriers to voting as complementary advocacy priority**

The weak correlation between physical accessibility and voter turnout change ($r = -0.142$ for distance, $r = 0.089$ for walkability score) indicates that something other than polling place location is driving participation patterns in Montgomery County. Only 35% of precincts saw increased turnout from 2020 to 2024, despite polling places being equitably distributed. If distance were the primary barrier, we would expect precincts with better access to show higher turnout growth, but this pattern does not appear in the data.

This finding suggests that the SPLC should investigate other potential barriers, such as transportation availability (while only 2-5% of county households lack vehicle access, this figure may mask geographic concentration in specific communities), voter education and registration processes, time-off-work policies, polling place hours, wait times, and other factors that this project's data cannot directly measure. The accessibility analysis provides a foundation for this work by establishing that physical access is not the bottleneck in Montgomery, allowing resources to be directed toward identifying what actually is.

3. Invest in cloud hosting for long-term monitoring capability

The project's Supabase cloud database and automated Google Sheets pipeline are fully operational but require ongoing hosting ($15-25/month for a small cloud PostgreSQL instance) to remain accessible beyond the current capstone timeline. Without this investment, the database reverts to limited potential in which static snapshots frozen in time at project completion replace dynamic querying updates. With this, the SPLC gains a living monitoring tool that can be updated annually with new election results and census data at a minimal marginal cost.

The dashboard templates, analytical views, and processing scripts are already built. Updating the data requires running the ACS pull script with the new census year, processing new election results through the existing Python pipeline, importing the updated CSVs to the cloud database, and refreshing the dashboard data sources. This annual maintenance could be performed by anyone with basic SQL and Python skills, or a team looking to develop it further.

4. Deploy targeted interventions for Very Poor walkability areas

Eleven block groups (5.4% of Montgomery County) fall into the Very Poor walkability category, with composite scores below 20/100 and walking distances exceeding 5 miles in many cases. These represent the most severe access gaps in the county, with areas where walking to a polling place is effectively impossible, regardless of pedestrian infrastructure quality. While these areas tend to be lower-vulnerability by demographic measures, their residents still face a fundamental access barrier that distance alone creates.

It would be in SPLC's best interest to deploy mobile polling units or establish satellite early-voting locations to serve the Very Poor walkability block groups during election periods. The specific GEOIDs are available in the accessibility_scores_enhanced_complete.csv file, filterable by walkability_category = 'Very Poor'. These locations can be mapped and integrated into the Nearby Polling Location Finder application so that voters in affected areas can identify alternative voting options.

5. **Extend the analysis to additional Black Belt counties for regional perspective**

The entire processing pipeline includes Python scripts, SQL schema, R analysis, ArcGIS templates, and cloud infrastructure. However, it's county agnostic and designed for replication. Extending this analysis to Dallas, Lowndes, Perry, Wilcox, and other Black Belt counties would provide regional comparative insights that are impossible to obtain from a single-county study.

Key questions that multi-county analysis could answer include: Does the paradoxical finding (high-vulnerability areas having shorter walking distances) hold across the Black Belt, or is

Montgomery County an outlier? Are there counties where distance is the primary barrier, and if so, what distinguishes them from counties where it is not? Do the sidewalk equity findings replicate across the region, suggesting a systematic infrastructure pattern?

The marginal cost per county is primarily processing time, which is roughly 2-3 hours for OSRM routing, plus a few hours of dashboard configuration and paving project data collection, rather than new development. The pipeline is built and it only needs new input data.

Recommendations for Clients (SPLC) – Data Analysis – API System

1. API Integration & Frontend Deployment

Future development teams should prioritize full integration of the geospatial polling accessibility API into the existing Figma-based frontend prototype and future production web platforms. The current API infrastructure successfully demonstrates backend geospatial querying capabilities, dynamic polling place lookup functionality, demographic integration, and structured JSON response generation. However, full frontend integration would significantly enhance usability by allowing users to interact with polling accessibility information through a public-facing interface.

Integrating the API into a production based frontend application would allow users to:

- Search polling accessibility information by address, ZIP code, institution, or precinct.
- Visualize nearby polling locations dynamically on interactive maps.
- Receive accessibility classifications in real time.
- View demographic and civic infrastructure context for queried areas.

- Improve public awareness of election accessibility conditions.

This API was intentionally designed to support future deployment within civic technology platforms.

2. Expansion of Civic Infrastructure Datasets

Future teams should continue expanding the API's integrated civic infrastructure datasets beyond polling locations and hospitals. Additional infrastructure layers could include:

- Public transportation routes and bus stops.
- Sidewalks and ADA accessibility infrastructure.
- Early voting locations.
- Mail-in ballot drop boxes.
- Schools and community centers.
- Broadband and internet accessibility.
- Flood-risk and environmental hazard overlays.
- Multilingual election assistance locations.

Expanding these datasets would allow SPLC and future researchers to analyze civic accessibility more holistically and identify overlapping infrastructure issues affecting vulnerable communities.

3. Automated Data Updating & Live Infrastructure Monitoring

The current implementation primarily utilizes static geospatial datasets. Future development efforts should incorporate automated data pipelines and scheduled updates through APIs, cloud databases, or government open-data portals.

Potential improvements include:

- Live polling place updates.
- Automated precinct boundary synchronization.
- Real-time road closure integration.
- Election cycle updates.
- Dynamic geocoding services.
- Cloud-hosted spatial databases.

Automated infrastructure updating would improve long-term sustainability and ensure that accessibility analyses remain accurate across election cycles.

4. Enhanced Spatial Analytics & Accessibility Modeling

Future teams should build upon the existing accessibility framework by incorporating additional transportation and mobility metrics. While the current implementation focuses primarily on nearest-neighbor spatial accessibility, future analyses could integrate:

- Network-based walking routes.
- Public transit travel times.
- Vehicle ownership constraints.
- Disability accessibility considerations.
- Pedestrian safety analysis.
- Traffic modeling.
- Transportation accessibility.

These additions would produce more realistic civic accessibility measurements and strengthen future election accessibility research.

5. Public-Facing Civic Technology Applications

The API infrastructure demonstrates a high potential for deployment as a public-facing civic technology platform supporting election transparency and voter accessibility awareness. SPLC may consider developing:

- Interactive public dashboards.
- Mobile-friendly polling accessibility applications.
- Community accessibility monitoring tools.
- Election resource locators.
- Civic infrastructure equity dashboards.

These tools have the potential to improve public engagement, increase awareness of polling accessibility disparities, and support community advocacy efforts related to equitable election infrastructure.

6. Future Machine Learning & Predictive Analytics Opportunities

Future research teams may additionally explore predictive analytics and spatial modeling and machine learning applications using the existing geospatial infrastructure. Potential directions include:

- Identifying areas at elected risk of civic inaccessibility.
- Forecasting infrastructure strain during elections.

- Clustering underserved communities.
- Predicting accessibility disparities under polling location closures.
- Optimizing future polling location placement.

The API’s modular geospatial architecture provides a strong foundation for these future analytical extensions.

Recommendations for Future Project Teams – Data Analysis

Begin with the database not the scripts

The PostgreSQL/PostGIS schema and analytical views in create_views.sql represent the most valuable intellectual artifact of this project. The views encode the key analytical questions, such as which block groups have the worst access, how does walkability relate to demographics, where are the routing outliers (pre-built), and reusable SQL. a future team should start by reading the views, running useful_queries.sql to see what data is available, and understanding the table relationships before writing any new code.

This approach would save approximately three weeks of work. The current team spent significant time discovering that block group data is suppressed by the Census Bureau, that GEOID format inconsistencies break merges, and that precinct boundaries changed between 2020 and 2024. All of these lessons are documented here so the next team does not repeat the investigation into the issues.

Use OSRM as the primary distance engine, reserve Google for validation

OSRM is free, runs locally with no API keys or rate limits, and processes 9,947 origin-destination pairs in 5-20 minutes depending on the routing profile. Google's Routes API provides more sophisticated pedestrian routing, accounting for sidewalk quality, pedestrian crossings, and terrain, but hits free-tier rate limits after 10,000 requests and requires a paid billing account for production use.

The approach that worked was using OSRM as the primary distance metric for all population centers, then validating results against Google Walking on a sample of 10-20% of routes. For multi-county expansion, OSRM is the only practical option without a paid Google API account. The OSRM setup documentation in the README provides complete instructions for installation, Alabama OSM data download, graph extraction, and server startup.

Enforce GEOID as a string type from the moment of data incorporation

Pandas' default behavior of reading numeric-looking strings as integers will strip leading zeros from GEOIDs, creating a cascade of merge failures downstream. This occurred in multiple scripts across the project, such as distance calculations, Google API batch processing, R statistical analysis, and ArcGIS export. Each instance required separate debugging and format conversion logic.

The solution is simple but must be applied at the first point of data contact. Explicitly specify dtype = {'GEOID': str} when reading any CSV containing GEOIDs with pd.read_csv(). This single line prevents the entire class of GEOID-related bugs. The pop_centers_conversion.py script demonstrates the correct pattern and should be used as template for new county processing.

Budget for cloud hosting from project's start

The project spent approximately 2 days working through network restriction issues, with ngrok tunneling, Cloudflare alternatives, VPN connections, and mobile hotspots that would've been avoided entirely if the database had been cloud-hosted from the beginning. University networks block outbound PostgreSQL connections on ports 5432 and 6543 as a security measure, and these restrictions are unlikely to change.

Supabase provides a free tier (500MB database, sufficient for approximately 5-10 counties) and requires about 30 minutes to set up. For production use beyond the free tier, budget $15-25/month. The Edge Function REST API wrapper that bridges the cloud database to external services requires about an hour to deploy and test. Building this infrastructure at project start, rather than retrofitting it at the end would save significant troubleshooting time and enable live client demos throughout the development process.

Natural continuation of this work includes multiple considerations

- **Multi-county comparative analysis across the Black Belt region:** The pipeline is built, but it needs county-specific input data and processing time. Start with Dallas County (Selma), which has similar demographics to Montgomery but different geography.
- **Public transit data integration:** The Montgomery Area Transit System (MATS) provides bus service throughout the country. GTFS (General Transit Feed Specification) data, if available,

would enable transit accessibility scoring, measuring distance to nearest bus stop plus walking distance from destination stop to polling place. This would add a critical dimension to the accessibility analysis that currently assumes walking or driving as the only modes.

- **Elevation and terrain analysis:** The current routing engines account for road networks but not all types of topography. USGS 3DEP elevation data could be integrated to calculate route elevation gain, identifying polling places that are technically within walking distance but unreachable for pedestrians with mobility limitations due to steep terrain.
- **Looker Studio dashboard deployment:** The Google Sheets pipeline is populating with fresh data hourly. A Looker Studio report connected to these sheets would provide the SPLC with live-updating dashboards without requiring ArcGIS Online accounts or GIS expertise.
- **Public transit accessibility scoring as an additional metric alongside walking distance:** Enables a more complete picture of how voters without personal vehicles can reach polling places.

Recommendations for Future Project Teams - Data Analysis - API System

1. Transition from Euclidean Distance to True Network Routing

This project currently relies primarily on centroid-based Euclidean distance calculations for accessibility analysis. Future teams should integrate routing engines as briefly used and demonstrated in the ARCGIS portion of this project such as:

- Open Source Routing Machine (OSRM)
- Google Routes API

- Valhalla
- OpenRouteService

This would allow:

- Real walking/driving distance
- Travel time estimation
- Turn restrictions
- More realistic accessibility modeling

Network routing would significantly improve the accuracy of voter accessibility measurements.

2. Transition from Euclidean Distance to True Network Routing

This script already includes filtering for multiple Alabama Black Belt counties. Future teams should expand the analysis regionally to include:

- Dallas County
- Lowndes County
- Perry County
- Wilcox County

This would allow:

- Regional equity comparisons
- Rural vs urban accessibility analysis
- Identification of broader infrastructure disparities across the Black Belt

3. Include Additional Infrastructure Variables

Future work should move beyond distance-only accessibility by integrating broader infrastructure quality measurements such as:

- Sidewalk availability
- Road quality
- Street lightning
- Public transportation access
- ADA accessibility

This would create a more comprehensive “walkability” or “voting accessibility” index rather than simply distances.

4. Develop a Web Application

The interactive lookup system developed in this project can serve as a foundation for a larger public-facing platform.

Future improvements could include:

- Flash/FastAPI backend
- Real–time routing and navigation
- Mobile-friendly design

This would make the analysis more actionable and accessible for public users.

Conclusions

1.1 Data Analysis Team

The BB-PAD project successfully delivered a comprehensive polling place accessibility analysis for Montgomery County, Alabama, integrating five distinct distance calculation methods, demographic vulnerability indicators, sidewalk infrastructure data, and voter turnout patterns into a unified analytical framework. Three interactive ArcGIS applications provide the SPLC with practical tools for exploring voting access patterns at different levels of technical sophistication. From the public facing Nearby app to the researcher-oriented isochrone viewer. Furthermore, a cloud database infrastructure with automated hourly refresh establishes the foundation for long-term monitoring and multi-county expansion in the future.

The central finding challenges a common assumption in voting access advocacy that states physical distance to polling places is the primary barrier for vulnerable communities. In Montgomery County, high-vulnerability areas (predominantly Black, higher poverty, less vehicle access) have significantly shorter walking distances to polling places than low-vulnerability areas. Polling places are already somewhat equitably distributed in urban areas. No block group simultaneously meets criteria for both high vulnerability and poor walkability. The access gap is not about where polling places are located. It is about the quality of pedestrian infrastructure connecting residents to those locations.

This finding carries important implications for advocacy strategy. Resources directed toward polling place relocation are unlikely to meaningfully improve voting access in Montgomery County because the current distribution is already favorable to vulnerable communities. Resources directed toward sidewalk installation, crosswalk improvement, pedestrian bridge construction, and street connectivity would address the actual barrier identified by this analysis. The routing_outliers database view provides a specific, data-backed list of intervention points where infrastructure investment would most directly improve pedestrian access to polling places.

The project also demonstrates that equitable physical access does not automatically translate to equitable voter participation. Only 35% of precincts saw increased turnout from 2020 to 2024, and accessibility metrics show negligible correlation with turnout change. This finding does not diminish the importance of polling place accessibility where safe walkable routes to polling places are a public good regardless of their impact on turnout statistics. However, it contextualizes accessibility as one component of a multidimensional voting access landscape. Transportation availability, voter education, registration processes, time-off-work policies, and other socioeconomic factors likely play larger roles in determining aggregate participation patterns in this specific region.

The tools, scripts, and infrastructure built for this project are designed for replication. Every Python processing script accepts a county FIPS code as its primary parameter. Every SQL view can be created in a new schema for a new county. Every ArcGIS template can be pointed at new feature layers and map extents. The cloud database pipeline provides a model for sustainable, long-term monitoring that can be extended to the broader Black Belt region. The investment required for additional counties is primarily processing time and configuration effort, not new development.

In addition to the broader routing and infrastructure analysis, this project also demonstrated the importance of integrating spatial data processing directly into Python-based analytical projects. Working extensively with shapefiles, GeoJSON exports, coordinate reference systems, centroid generation, and demographic joins provided a deeper understanding of how GIS methodologies can support public policy and accessibility research at scale. The project reinforced how geographical information systems is not something that can only be used through traditional GIS software, but can also be integrated into computational pipelines and methods that support automation, statistical analysis, visualization, and interactive application development.

The addition and integration of GeoPandas, Shapely, geocoding services and spatial visualization tools allowed for the transformation of these raw geographical datasets into powerful accessibility measure tools capable of being interpreted and used by the public. With this, a specific connection has been identified between spatial computing, civic infrastructure analysis, and real-world challenges seen in community accessibility.

Overall, the BB-PAD project provides both a technical framework and a research foundation for future civic accessibility studies. By combining GIS analysis, database engineering, routing systems, demographic modeling, and interactive visualization, the project demonstrates how computational approaches can support more data-driven decision making on voting access and public infrastructure planning.

1.2 Design Team

The Design Team concluded that analytical findings and traditional data visualizations alone are often too technical to effectively communicate civic accessibility concerns to general public audiences. Translating otherwise complex polling place access and distance-related information into a more understandable and user-centered product became essential to making broader project findings feel more actionable, preparatory, and personally relevant to voters.

The project additionally revealed that accessibility extends beyond physical polling place conditions alone. Readability, informational hierarchy, ease of navigation, and simplified communication systems all play significant roles in how users understand and interact with civic information and accessibility-related resources.

VoteAccess also demonstrated the potential for accessibility-focused civic products to function not only as informational and preparatory resources, but as tools capable of contextualizing broader environmental, infrastructural, and transportation-related barriers that may affect voter access and turnout. Through a more user-centered approach, platforms such as VoteAccess may additionally help users feel more informed, prepared, and aware of how accessibility conditions can shape their voting experience.

Contacts

Team Member	Contact Information
Jacy Dickstein	Email: jacyjdickstein@gmail.com Phone: 609-922-6230
Janice Imuze	Email: imuzejanice@gmail.com Phone: 443-787-2723
Jayden Ervin	Email: jayden.ervin03@gmail.com Phone: 240-443-1993
Mawa Clarke	Email: Mawa515@yahoo.com Phone: 240-688-9812
Myles Sartor	Email: msartor@terpmail.umd.edu Phone: 732-947-1044

Deliverables Checklist

Deliverables Folder: 3 BB-PAD - Polling Place Access and Distance Analysis

- ☑ Deliverable 1 - VoteAccess Figma Prototype: BB-PAD VoteAccess Figma Prototype (With minor interactivity)
- ☑ Deliverable 2 - Montgomery County Voter Access ArcGIS Dashboard
- ☑ Deliverable 3 - GitHub
- ☑ Deliverable 4 - BB-PAD Sheet Cumulative Link Sheet (Containing Links Listed Above)
- ☑ Deliverable 5 - Visualizations

Voting & Civic Participation

Alabama Black Belt Counties

Team 4 BB-TRAC: Transportation Barriers to Civic Participation

Prepared for: Southern Poverty Law Center (SPLC)

Team Members: Don Brazelton, Yeongjae Roh, Oswalt Vasquez, Setareh Mahjoob, Mateen Vafa

Course: INST490 · Section: 0103

Date: May 18, 2026

Team 4: BB-TRAC

Team Brief

Team Members:

Don Brazelton

- Led transportation analysis, liaising shuttle availability, income, and transit data to identify compounding barriers to voting participation. Developed Tableau visualization to communicate findings to clients. Additionally served as the team's primary communicator with Alabama's Dept. of Transportation. Furthermore, drew key inferences from the full data findings to help shape conclusions and recommendations.

Oswalt Vasquez

- Provided data on the ages of voters in different counties. The data was used to create visualizations in Tableau. Ensure that all the visualizations were accessible and merged all the visualizations into one workbook on Tableau.

Mateen Vafa

- I helped with finding and sourcing data and identifying polling stations across the Alabama Black Belt region alongside the rest of my group members to support our Tableau visualizations. I also helped review vehicle ownership and polling place data to explore how transportation access can affect voting participation. I also reviewed our map to check for quality and review new additions to the map like highways and geographic reference points.

Yeongjae Roh

- Manage collecting and processing the datasets utilized for the team's geographic analysis. I gathered and cleaned vehicle ownership percentage data and polling station data, structuring them into a format for examination. By combining the collected data, I contributed to providing meaningful analytical materials for the client. Based on those data, I created maps of the Black Belt region in Tableau, visualizing both the percentage of household vehicle ownership and the number of polling stations.

Setareh Mahjoob

- Sourced, cleaned, and analyzed data on the age demographics of voters and their relation to total ballots cast by county for the 2020 and 2024 general elections, using Tableau to create visualizations reflected in the final workbook. As project manager, I was primarily responsible for coordinating communications with SPLC representatives, particularly in instructor-facilitated meetings. Additionally, I assisted significantly in organizing team deliverables and drafting report sections for consistency and accuracy with our data analyses.

Clients:

- Fred McBride: *SPLCs Senior Supervising Data and Research Analyst*
  - fred.mcbride@splcenter.org
- Jack Slate: *SPLC AL State Office*
  - jack.slate@splcenter.org
- Zack Mahafza: *SPLC Senior Research Analyst*
  - zach.mahafza@splcenter.org
- Kristen Akey: *Contractor assisting SPLC with data analysis in iConsultancy Partnership*
  - kristen.k.akey@gmail.com

Key Findings

- Bullock, Pike, and Crenshaw have no recorded transit service, yet all three rank in the middle ⅓ for income. Transit absence is not exclusively a poverty problem, but an infrastructural one.
- ⅚ of the lowest-income counties are served by only demand-response transit, with no fixed-route options.
- Dallas County has the highest car-free household rate in the region at 15.1%, a median income of $36,810 and only demand response transit. Roughly 1/7 households have no vehicle and no reliable transportation to reach polls.

Deliverables Produced

- Income & Shuttle Map which interactively shows: county, median household income, transportation agency, service type, hours of operations, & phone number.
- Percentage of Households Without Vehicles Map which visually displays the percentage of car ownership status by all 17 counties in Black Belt region.
- Polling places by County in Alabama Black Belt, 2000 which indicates the number of polling locations per county in Black Belt region.

Challenges

- Identifying a fixed-bus route service across each of the Black Belt Counties remains unresolved, as the vast majority of transit options within the region function as demand-response shuttles with no publicly available route data. As a suggestion from Alabama's Department of Transportation, our team decided to focus on shuttle data to supplement the lack of available public bus information.

- When merging data on Tableau to add points that had information about the county chair on the map, we ran into some issues. We would get an error on Tableau when trying to merge the data together. We decided that we were going to add the information manually instead of merging the data.
- We encountered difficulties in obtaining a consistent and unified dataset of polling place numbers across all 17 Black Belt counties. While some counties have data for recent election years, others counties were missing. It was challenging to collect all 17 counties' dataset based on a specific year

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Final Report and Deliverables

Project Context

BB-Trac team focused on identifying transportation barriers to civic participation across counties in the Alabama Black Belt region. Within the broader SPLC regional analysis, the project centered around how access and lack of access alike to transportation services, vehicle ownership, household income, and polling site distribution intersect to affect civic participation and voting accessibility. Our team heavily relied on data from the U.S. Census Bureau, resources from the Alabama Department of Transportation, and transit information from Alabama Dept. of Transportation & ALTrans.org. Our team's findings support those from teams examining polling place access and distance, as well as voter turnout and participation trends.

Methods

1. Transportation and Income Analysis

County median household income was sourced from 2024 U.S. Census for all 17 Black Belt Counties. Transportation data was acquired through direct outreach to ALTrans.org (Alabama's) statewide rural transit support organization, which provided shuttle and transit information for each county. Information included: transit agency names, service types, contact information, and hours of operation. Both income and transit information were cleaned, then combined in Microsoft Excel, where counties were ranked 1-17 by median household income.

Cleaned income and transit data was imported into tableau, to produce an interactive map of the Black Belt Region. Counties are shaded on a blue color gradient which reflects median household income, with county seats plotted as an additional reference point to users. Hovering over any county surfaces will open the full dataset for that county including income figures, transit agency in said county, service type, hours of operation, etc, allowing SPLC to visualize county level disparities quickly.

1.1 Challenges with transportation Data

Locating credible transportation data for each of the 17 counties proved to be our group's most significant obstacle. Our initial efforts to find public bus route data through conventional public databases came up largely empty. Following a recommendation from our SPLC client team, direct outreach happened with the Alabama Department of Transportation (ALDOT). This spearheaded our team to communicate with ALTrans.org. This communication yielded the transit data necessary to complete our analysis.

2. Vehicle Availability Analysis

Vehicle ownership data was collected from Census Reporter (Table B08201) of the U.S. Census Bureau's ACS 5-Year Estimates (2019-2023). This dataset provides a different level of current household vehicle ownership statistics across all 17 Black Belt counties: no vehicles, one vehicle, two vehicles, three vehicles, and four or more vehicles. Our team was refined and focused on the 17 Black Belt counties in Alabama and broke down into the county level.

The Percentage of vehicle ownership statuses were calculated for each county and supported as a key factor of transportation issues. The refined dataset was imported into Tableau to create a map that shows different conditions in each county in Black Belt region. Each area is represented in a blue color based on their percentage of vehicle ownership. Users can select different levels of vehicle ownership categories: no vehicles, one, two, three, four or more, to view the proportion of each category within 17 counties on a separate map. By referring to this interactive map, the SPLC can explore vehicle ownership patterns and more easily identify which counties are facing the most transportation disadvantages.

Deliverables and Findings

4.1 Deliverable 1: Percentage of Households Without vehicles Map

Description

This map illustrates the percentage of households without a vehicle among the 17 Alabama Black Belt counties. Each county reflects the proportion of households having no vehicles based on the color, with a darker section indicating the higher rate of households without vehicles. This map was created based on the data from the U.S. Census Bureau's American Community Survey (ACS), Census Reporter (Table B08201).

Rationale

Since rural regions such as the Black Belt counties have public transit limitations, vehicle availability is a key indicator of transportation access. Local residents without a personal vehicle face significant barriers to reaching polling locations. By visualizing these gaps county by county, this map allows the Southern Poverty Law Center (SPLC) will able to identify which county has the most transportation limitations.

Content

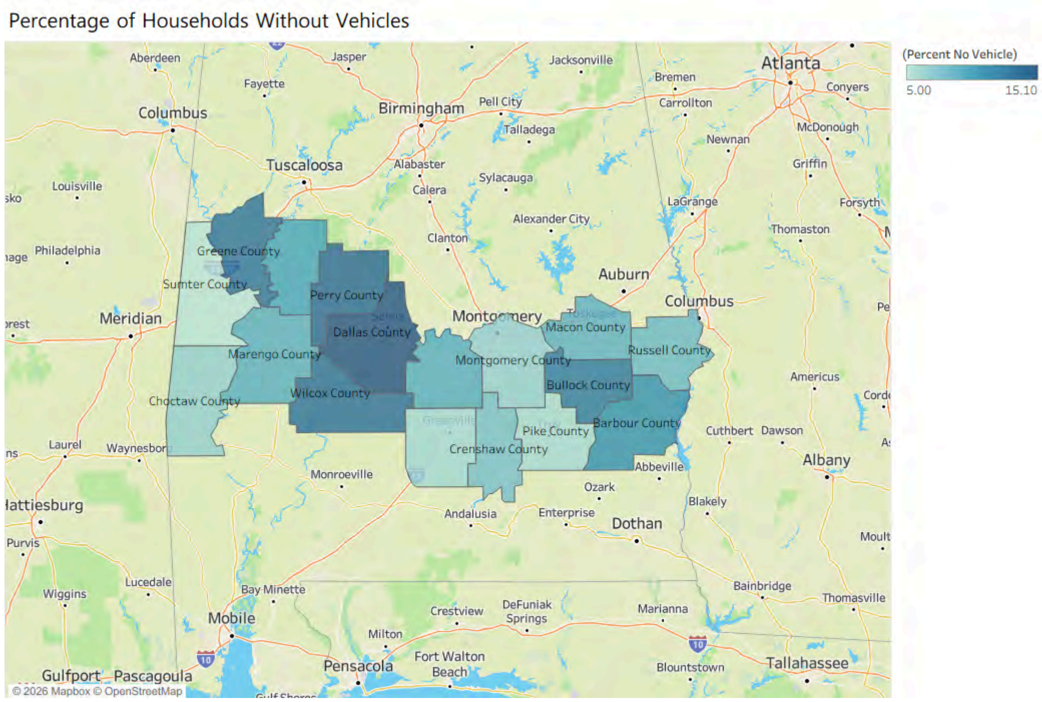

Figure1. County-level percentage of households without a vehicle, Alabama Black Belt counties. Darker colors show higher rates of zero-vehicle households. Source from U.S. Census Bureau, American Community Survey (ACS), 2019-2023. The map was created in Tableau.

Finding

This map shows different levels of vehicle accessibility between each Black Belt counties. In Dallas County, 2327 out of a total of 15,416 households reported no vehicle available which has the highest rate 15.1% in Black Belt regions. Perry (13.7%), Wilcox(13.3%), Greene(13.1%), and Bullock (12.6%) also recorded more than 12% for households without vehicles. These five counties correspond to lower income areas, demonstrating a lack of vehicle ownership and economic have a close relationship. However, Sumter (5.0%), Pike (5.8%), and Butler (5.9%) reported the lowest rates of households without a vehicle.

Connection to Other Team

Vehicle ownership data is considered as one of the most direct information for measuring transportation barriers by each county. It can be used in the development of a comprehensive civic engagement risk index. In addition, teams dealing with community context and persona can reference this data to reflect residents daily the transportation issues by the county.

4.2 Deliverable 2: Polling Places by County in Alabama Black Belt, 2000

Description

This map displays the total number of polling stations across the 17 counties of Alabama's Black Belt region in 2000. Darker shades have a larger number of polling stations in that county, while lighter shades mean that fewer polling stations are available to the residents. This map was created using data from the Office of the Alabama Secretary of State.

Rationale

The number of polling places within a county directly impacts on local residents for voting. In counties with fewer polling stations, people without their private transportation face multiple barriers because of absence of a vehicle and a lack of accessible polling sites. By mapping the number of polling places along with transportation data, SPLC can identify counties requiring additional attention regarding both the number of polling stations and transportation support.

Content

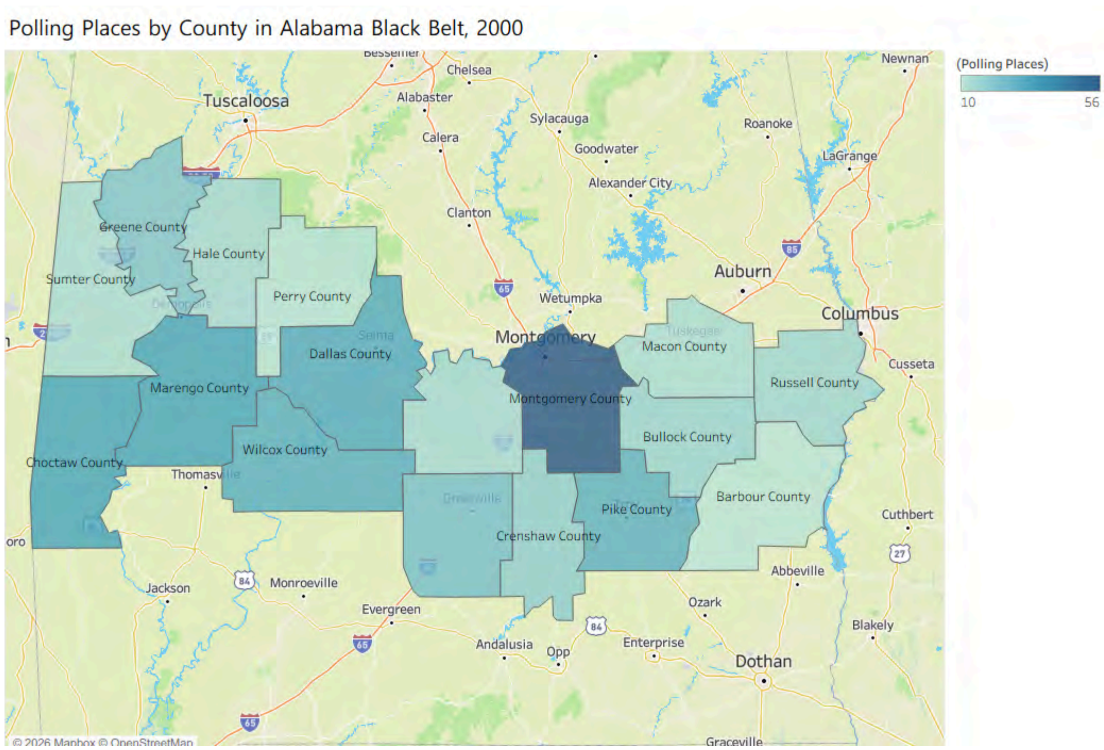

Figure2. Polling Places by County in Alabama Black Belt, 2000. Darker color areas have more polling places. Source from the Office of the Alabama Secretary of State. The map created in Tableau.

Finding

This map demonstrates the variation in the number of polling places across the Black Belt counties. Montgomery County has 56 polling places which is the highest number in the region, reflecting the county's population density and urbanization. On the other side, Barbour County has the fewest polling stations with 10, Perry (11), Macon (13), and Sumter (13). Especially some counties with the fewer number of polling places such as Perry and Bullock, they also encounter high rates of no vehicle ownerships. This indicates that residents in these areas face a dual challenge; limited access to polling places and lack of transportation.

Connection to Other Team

With this data, other teams can conduct analyses of distance and accessibility issues by associating which countries have the fewest polling locations relative to their geographic characteristics and population size. If counties have both low number of polling locations and high percentage of car-free represent areas considered as a place where transportation barriers and polling access barriers reinforce each other. Furthermore, this data can be used as a clue for recognizing which counties are most vulnerable to voters because of structural barriers.

4.3 Deliverable 3: Median Household Income & Transportation Data

Description

This map displays an interactive choropleth of the 17 Black Belt Counties shaded blue by median household income, with county seats plotted as geographic reference points. Hovering over any county will reveal its current transit agency (as listed on ALTrans.org) service type, hours of operations and median household income.

Rationale

Income and transit availability are two direct structural barriers to voter participation. Income, transit availability and vehicle ownership are all directly correlated with civic participation. Counties where all three coverage unfavorably show the highest barriers to reaching polling locations.

Darker hues represent median household income ranging from $31,495 to $58,153. The lowest-income area is located in the western portion of the region and several of these same counties rely solely on demand-response transit or have no transit service at all.

Content

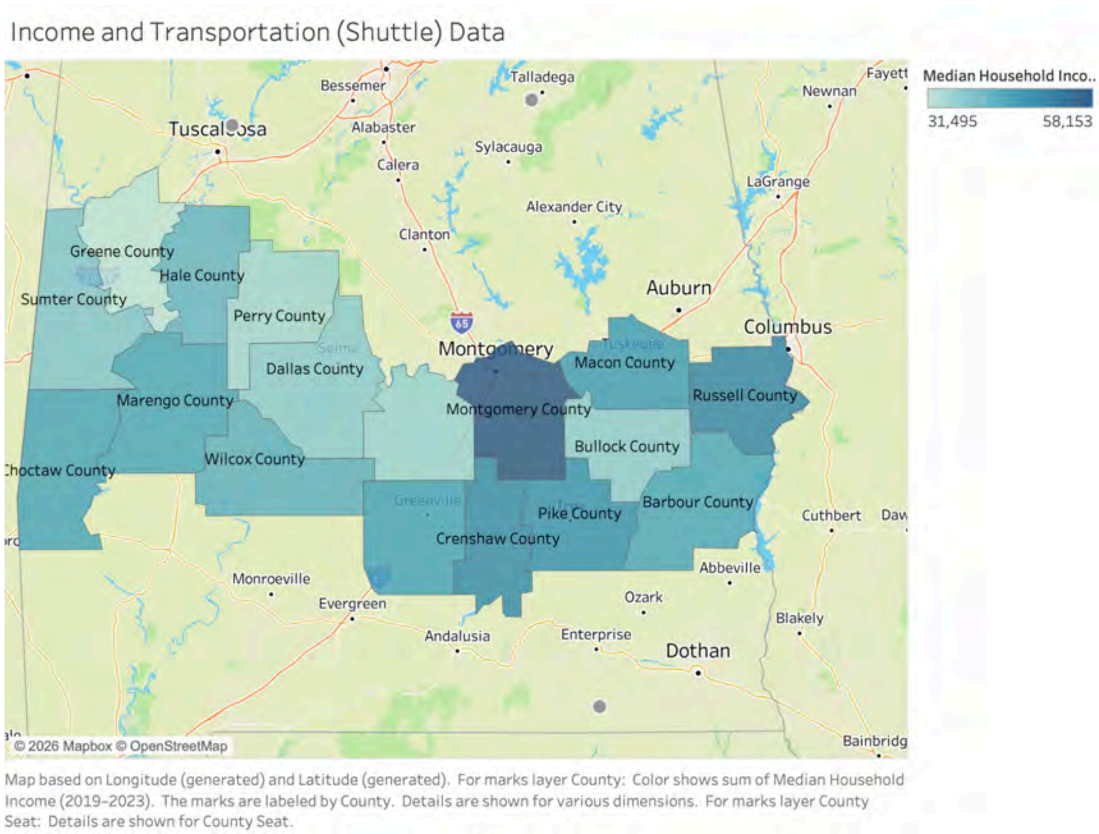

Figure 3. Black Belt counties shaded by median household income. Darker hues indicate higher income levels. Income data sourced from the U.S. Census Bureau (2024). Hovering over a county reveals its transit agency, service type, hours of operation, median household income. The map was made with Tableau.

Findings

Montgomery County has the highest median household income at $58,153, while Green County has the lowest at $31,495. A median income gap of over $26,000 between neighboring counties reflects disparities in local ability to access and participate in voting within the same region.

The lowest-income counties are located in the Western Black Belt, where transit coverage is also weakest. This suggests that poverty and transit absence, or the lack thereof reinforce each other geographically.

Several counties that rank in the middle ⅓ of income still lack fixed-route transit entirely, indicating that transit infrastructure gaps or not only limited to the poorer counties.

Dallas County stands as the most acute case. It ranks among the lowest ⅓ in income, while also having the highest car-free household rate in the region, only being served by demand-response shuttle transit.

Connection with other Teams

This map directly supports other teams' analyses. The Polling Place & Distance analysis team can use income and transit availability as overlapping barriers when assessing whether the local population can realistically reach polling locations. The Risk Index team can incorporate our income and transport data to create a risk index, e.g., vulnerability scoring. The ADA Accessibility team may find income data relevant, as lower-income counties are less likely to have resources to maintain equitable polling place infrastructure.

Recommendations

Recommendations for the client (SPLC)

Based on our findings, we believe that the SPLC should consider prioritizing counties that demonstrate high percentages of both households without vehicles and limited polling place accessibility. Some examples of counties include Dallas, Perry, and Bullock counties. These counties demonstrate overlapping barriers to transportation and voting access, which may disproportionately affect civic participation. Additionally, we believe the SPLC could benefit from continued partnerships with organizations such as ALTrans.org. Improving the documentation of transportation limitations throughout rural counties in the Alabama Black Belt would also be of significant benefit.

Recommendations for future project teams

For future teams working on this topic, the next team should know that there is no accessible fixed-route bus data. Meaning that there are no designated buses that have set schedules picking up people and dropping them off at certain locations. Instead, there are shuttles that people call for to set a trip, but this information is not collected from these shuttle organizations. For any type of transit information, teams should reach out to ALTrans.org. For any type of information regarding income and households, future teams should go to the U.S. Census Bureau, a reliable source to get data from. Future teams should not heavily rely on organizations to get back to them with the information they are looking for. Something we as a team would do differently if we had more time is to make the Tableau maps more interactive by being able to click on a county, and only that county's information will display, nothing else. Doing this, the income and ownership of cars would have to be merged. At the moment, we have the income data and the car ownership in two different workbooks.

Conclusions

The primary objective of this research is to examine voting barriers throughout Alabama's Black Belt counties. The team designed an interactive Tableau map that revealed county-level disparities when it came to income, car accessibility, and transit availability by incorporating data from the U.S Census and transportation data that was collected by contacting ALTrans.

The results indicated that transportation obstacles are a significant problem throughout the Black Belt counties. There are no buses that have designated routes that people can use to get to places. People's main transportation was shuttles with no fixed-route alternatives. Counties like Dallas stand out due to a significant number of people not having access to a car, low income, making transportation making it difficult to get to polling locations.

Don Brazelton: donb@terpmail.umd.edu

Yeongjae Roh: yroh@terpmail.umd.edu

Oswalt Vasquez: ovasqu11@terpmail.umd.edu

Setareh Mahjoob: smahjoob@terpmail.umd.edu

Mateen Vafa: mvafa@terpmail.umd.edu

Deliverables Checklist

- [x] Visualization (Maps + Graphs)
- [x] Data
- [x] Transport and Income Tableau Map
- [x] Vehicle Ownership Tableau Map
- [x] Polling Place Tableau Map
- [ ] Packaged File Tableau Income/Transit

Voting & Civic Participation

Alabama Black Belt Counties

Team 5 BB-AACS: ADA Accessibility & Compliance at Polling Locations

Prepared for: Southern Poverty Law Center (SPLC)

Team Members: Samantha An (Project Manager), Jessica Barke (Documentation Lead), Katherine Delaney (Client Facing Review Lead), Megan Otchet (Software Developer), Catalina Herrera (Integration/Consistency Checker), Derik Luc (Research Lead)

Course: INST490 · Section: 0103

Date: May 18, 2026

Team 5: BB-AACS

Team Brief

Section Brief

BB-AACS built the methodological blueprint and data framework by analyzing the 49 polling sites in Montgomery County. By pairing high-resolution satellite imagery and GIS mapping with boolean logic, we turned federal mandates, such as the ADA and the Voting Rights Act, into practical, measurable checklist data. This matters because identifying and removing physical barriers directly helps achieve higher voter turnout, driving the SPLC's mission to boost participation among Black, Latinx, and disabled communities who are historically left behind. Because our evaluation protocols are fully standardized, future research teams can easily export this process to the remaining 16 Black Belt counties. Ultimately, this work allows the broader project to see whether polling place hurdles are just isolated local issues or part of a systemic, region-wide trend suppressing voter turnout.

Key Findings

- Finding 1: Physical entry barriers are more prevalent than site access issues, as only 55% of locations (27 of 49) could be confirmed to have fully accessible entrances, while 75% provided compliant parking and exterior routes.
- Finding 2: Infrastructure gaps disproportionately affect smaller, less-developed sites, which frequently lack both designated parking lots and formal passenger drop-off

areas, forcing voters with mobility impairments to navigate street-level accommodations without assistance.

- Finding 3: A compliance gap exists between site arrival and building entry, where 20% of polling locations provide accessible parking and exterior routes but fail to provide a confirmed accessible entrance due to non-compliant door hardware or high thresholds.

Deliverables Produced

- [ ] Deliverable 1: Dataset
    - [ ] Dataset
- [ ] Deliverable 2: Accessibility Analysis Report
    - [ ] Accessibility Analysis Report
- [ ] Deliverable 3: Accessibility Visualization Summary
    - [ ] Can be emailed to client- cannot be directly linked

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Challenges

A major obstacle we faced is not having a dataset that was already available to us. We overcame this by creating our own dataset that included all of the polling locations in Montgomery County, AL and looking them up on google earth to assess what accessibility standards they met.

Connections to Other Teams

We shared our standardized ADA guidelines with Team 7, helping them integrate physical accessibility metrics into their own analysis and recommendations.

Final Report and Deliverables

Project Context

As part of a comprehensive 11-team regional analysis of the Alabama Black Belt, this Montgomery County assessment serves as a critical benchmark, providing a cohesive dataset that addresses the broader study of voting barriers. While our team's immediate focus remained localized to the 49 polling sites within Montgomery, the structured dataset and metadata protocols we developed can be designed to be exported and scaled across the remaining study areas. Because we established the baseline for using high-resolution satellite imagery and boolean logic to track ADA compliance, future work can consist of other teams relying on our work as the methodological blueprint for their own county-level research. Ultimately, our findings in Montgomery act as the foundation that allows the broader project to determine if voting barriers are specific to this county or part of a wider, systemic trend throughout the Alabama Black Belt.

Methods

To fulfill the project goals, our team executed a rigorous virtual audit of Montgomery County's voting infrastructure. We transformed raw precinct lists into a structured, evidence-based dataset designed to identify systemic physical barriers to the ballot. We created our dataset using Google Sheets to ensure our information is well organized and easy to analyze.

Site Identification and Sampling:

We assessed all 49 active polling locations in Montgomery County, Alabama. The sampling frame was established using the official Montgomery County Election Center precinct list.

Data Collection and Virtual Inspections:

Data was collected through a systematic process using high-resolution satellite imagery and street-level photography via Google Earth and Google Maps Street View.

To maintain transparency and allow for future verification, the team recorded the following metadata for each site:

- Image Capture Date: The specific date the Google imagery was acquired was noted to account for any infrastructure changes since the photo was taken.
- Image Links: A direct Google Earth URL was archived for each location, pinning the exact street-level view used during the audit.

Evaluation Criteria and Checklist:

The assessment was guided by the U.S. Department of Justice ADA Checklist for Polling Places. This standardized federal tool was used to evaluate whether the physical infrastructure met minimum accessibility requirements.

The dataset uses a boolean (True/False or Check Mark) logic to indicate the presence of ADA-compliant features. In your final report, the coding is interpreted as follows:

- TRUE (or ✓): Indicates the feature was clearly present and met the visual criteria of the ADA Checklist for Polling Places

- FALSE: Indicates the feature was either missing or did not meet the visible requirements (e.g., a designated parking area existed, but lacked the required "Van Accessible" vertical signage).
- Contextual Notes: In cases where specific infrastructure was not applicable, for example we noted "No ramp needed" to ensure the "False" mark was not interpreted as a failure of accessibility.

The evaluation was conducted through team members who were assigned to evaluate specific data columns across all 49 sites to ensure consistent application of the ADA criteria. The specific categories analyzed included:

- Accessible Parking: Availability of designated van-accessible spaces, access aisles, and proper vertical signage.
- Passenger Drop-Off Area: Presence of a level area for unloading passengers near the entrance.
- Exterior Route to Accessible Entrance: Analysis of the path from the arrival point to the building (checking for curb cuts, stable surfaces, and lack of obstructions).
- Polling Place Entrance: Evaluation of door widths, maneuvering clearance, and threshold heights.
- Ramps: Inspection of slope steepness, handrail presence, and level landings.
- Elevators/One Floor: Verification that the voting area was located on the ground floor or was accessible via a compliant elevator.
- Additional Notes: Qualitative documentation of temporary barriers or unique site characteristics

Overall, we were able to create a comprehensive dataset ready for analysis.

Maps

Mapline Heat Map

In Mapline, a new map was created by importing the dataset and specifying the address field using the appropriate address data type. Several addresses required additional clarification and standardization, as duplicates existed across the state and some entries were initially mapped outside of Montgomery County, AL. These discrepancies were corrected to ensure accurate geographic placement.

After validating and refining all address data, the corresponding number of accommodations met was added to the dataset. This allowed the application to categorize each location based on the relative number of accommodations satisfied, forming the basis for visual differentiation in the resulting heat map.

Tableau Interactive Map

To create a geographic map in Tableau, the dataset must first include latitude and longitude values for each location. These coordinates were added to the original dataset before importing it into Tableau. Once prepared, the dataset was downloaded as a .csv file and imported as a text file.

After importing the data, a new worksheet was created in Tableau. The map is generated by dragging Longitude to the Columns shelf and Latitude to the Rows shelf. Tableau automatically recognizes this configuration and a map view is displayed.

To represent individual locations, the Marks card is used. The mark type is set to Map, and additional attributes are added to enhance the visualization. For example, Address should be added to the Detail field to distinguish each point, while other attributes can, and should, be included in the Marks Tooltip to provide additional information when hovering over a location on the map.

Filters are then applied to allow interactive control of the map. Each attribute was dragged into the Filters card, and the "Show Filter" option is enabled to allow users to dynamically select which data appears on the map.

In order to emphasize locations with more accommodations met, point size can be adjusted or driven by the "Number of Accommodations Met" location attribute to visually emphasize differences between locations.

ADA Legal Research

To evaluate the legal history of the polling locations in Montgomery County, Alabama, we conducted a review of ADA-related litigation:

General Web Research: We performed broad digital searches using keywords (e.g., "Montgomery County," "ADA compliance," and "voting accessibility") to identify public news reports, DOJ settlements, or civil rights complaints.

Address-Specific Audits: We cross-referenced each polling location's physical address with public records to identify site-specific building code violations or property-related legal actions.

Judicial Record Verification: If there were findings we planned to use AlaCourt (pa.alacourt.com), the Alabama Unified Judicial System's official portal. This would have allowed us to read the specific court cases if we found any.

No active or historical cases were found. This suggests a lack of judicial oversight or legal pressure regarding ADA compliance at these specific locations.

Deliverables and Findings

Deliverable #1: Dataset

Our primary dataset is a structured dataset documenting ADA accessibility features across polling locations in Montgomery County, Alabama. The dataset was compiled using observations gathered through Google Earth and the Americans With Disabilities Act Polling Place Accessibility checklist, which provided a standardized framework for evaluating accessibility conditions by each location. Using the ADA checklist as our guide, we were able to organize the data into measurable accessibility indicators rather than relying on general assumptions about whether a polling location was accessible or not.

The dataset includes variables related to multiple areas of accessibility, such as accessible parking, exterior routes to accessible entrance, polling place entrance, ramps, and whether the polling location is located on one floor or requires elevator access. This approach allows the dataset to serve as both a research tool and a decision making resource. Instead of only identifying that accessibility issues exist overall, the dataset pinpoints which specific polling locations fail to meet particular ADA related standards.

	B1 County									
1	County	Location Site/Name	address	Latitude	Longitude	Accessible Parking	Passenger Drop-Off Area	Exterior Route to Accessible Entrance	Polling Place Entrance	Ramps
2	Montgomery	Ramer Public Library	5444 State Hwy. 94 [Ran	32.0576	-86.2533	☐	☑	☑	☐	☑
3	Montgomery	Snowdoun Volunteer Fire Department	219 Hobbie Rd.	32.2897	-86.3102	☑	☑	☑	☐	☑
4	Montgomery	409 Better Covenant Ministries	4050 Fairgrounds Road	32.3886	-86.2622	☑	☑	☑	☑	☑
5	Montgomery	211 Rufus Lewis Library	3095 Mobile Hwy	32.3301	-86.3548	☑	☑	☑	☑	☐
6	Montgomery	210 Pintlala Fire Department	250 Federal Rd	32.1935	-86.379	☐	☑	☐	☑	☐
7	Montgomery	Cleveland Ave YMCA	1201 Rosa L. Parks Ave	32.3304	-86.3052	☑	☑	☑	☑	☑
8	Montgomery	Wilson Comm. & Athletic Ctr. - Huntingdon College	1155 E. Fairview Ave.	32.3547	-86.2856	☑	☑	☑	☐	☑
9	Montgomery	Alabama State University / Acadome	1595 Robert C. Hatch Dr	32.3578	-86.2781	☑	☑	☑	☑	☑
10	Montgomery	Trenholm State (Patterson Campus, Bldg. D)	3920 Troy Highway	32.3026	-86.2479	☑	☐	☐	☐	☐
11	Montgomery	City of Refugee church	3820 Woodley Rd.	32.3117	-86.2502	☐	☐	☑	☑	☐
12	Montgomery	St. Paul AME Church	706 E. Patton Ave.	32.3379	-86.289	☐	☐	☑	☑	☐
13	Montgomery	Beulah Baptist Church	3703 Rosa L. Parks Ave	32.3196	-86.3038	☑	☑	☑	☑	☐
14	Montgomery	Hayneville Road Community Center	3315 Hayneville Rd.	32.3075	-86.3332	☐	☑	☐	☐	☐
15	Montgomery	Catholic Social Services	4455 Narrow Lane Rd.	32.331	-86.2687	☐	☐	☑	☑	☐
16	Montgomery	Southlawn Baptist Church	5340 Mobile Hwy	32.3148	-86.3732	☑	☐	☑	☑	☐
17	Montgomery	New Home Baptist Church	11015 Jones Rd	32.4382	-86.1375	☑	☐	☐	☑	☐
18	Montgomery	Montgomery Museum of Fine Arts	1 Museum Dr	32.406	-86.2095	☑	☑	☑	☑	☑
19	Montgomery	Vaughn Park Church of Christ	3800 Vaughn Rd.	32.3565	-86.2332	☑	☑	☑	☐	☐
20	Montgomery	104 Midtown YMCA	3455 Carter Hill Rd	32.3423	-86.2891	☑	☑	☑	☑	☑
21	Montgomery	Weeping Willow Missionary Baptist Church	2925 Forbes Dr.	32.4012	-86.2793	☑	☑	☑	☐	☑
22	Montgomery	McIntyre Community Center	1240 Hugh St.	32.3113	-86.3202	☐	☐	☐	☑	☑
23	Montgomery	Vaughn Forrest Church	8660 Vaughn Rd.	32.3925	-86.1502	☑	☑	☑	☑	☑
24	Montgomery	Frazer Church	6000 Atlanta Hwy.	32.4028	-86.1903	☑	☑	☑	☑	☑
25	Montgomery	Macedonia Baptist Church	3835 Macedonia Dr.	32.3005	-86.262	☑	☑	☑	☑	☐
26	Montgomery	Dalraida Baptist Church	3838 Wares Ferry Rd.	32.3712	-86.2296	☑	☑	☑	☐	☐
27	Montgomery	Eastern Hills Baptist Church	3604 Pleasant Ridge Rd.	32.3798	-86.245	☑	☑	☑	☑	☐
28	Montgomery	Eastmonth Baptist Church	4505 Atlanta Hwy.	32.3825	-86.2459	☑	☐	☑	☑	☑
29	Montgomery	Old Elam Baptist Church	2526 Cong. W L Dickinson Dr	32.3923	-86.2615	☑	☐	☑	☑	☐

Montgomery Locations | 3 dataset | ADA info

	L1 Elevators/One floor						
1	Elevators/One floor	Additional Notes	Number of Accomodations Met	Date from image capture	Image Links	Existing/Past Court Case for this l	Location Website (if any)
2	☑	No ramp needed for entrance. No interior images to tell if the inside is accessible	4	April 2024	https://maps.app.goo.gl/NfJiHBGXVK8GY	check eelc and doj disability section	https://mccpl.lib.al.us/ramer-l
3	☑	No ramp needed for entrance. No interior images to tell if the inside is accessible	5	May 2024	https://maps.app.goo.gl/nsddEUqC9XB4W		https://usfiredept.com/snowd
4	☐	No ramp needed for entrance, also no interior images to tell if the inside is accessible	5	January 2026	https://earth.app.goo.gl/?apn=com.google		https://bcm4him.org
5	☑	No Ramp needed for entrance, the interior looks very accessible	5	January 2026	https://earth.app.goo.gl/?apn=com.google		https://mccpl.lib.al.us/rufus-a
6	☑	Gravel parking lot	3	May 2024	https://earth.google.com/web/search/250		https://digitalfirehouse.com/r
7	☐		5	January 2025	https://earth.google.com/web/search/120		https://ymcamontgomery.org/
8	☐	no ramp needed, buildings rooms on ground floor	4	January 2026	https://earth.app.goo.gl/?apn=com.google		https://www.huntingdonhawks
9	☑	Passenger drop off is better since from parking to door is a longer distance	6	April 2009	https://earth.app.goo.gl/?apn=com.google		https://bamastatesports.com
10	☐		1	April 2017	https://earth.app.goo.gl/?apn=com.google		https://www.trenholmstate.ed
11	☑	Only one floor so no need for ramp or elevator	3	Mar 2025	https://earth.app.goo.gl/?apn=com.google		https://www.corlive.org/
12	☑	No designated lot, there is parking across the street, sidewalk seems to be unstable for	3	Jan 2026	https://earth.app.goo.gl/?apn=com.google		https://www.facebook.com/P
13	☐	There are steps for the drop off exit but there is no ramp and roundabout way of getting inside is n	4	Jan 2026	https://earth.app.goo.gl/?apn=com.google		https://www.bbcmontg.org/
14	☐		1	Jan 2025	https://earth.app.goo.gl/?apn=com.google		https://www.fusionmontgomer
15	☐		2	Mar 2025	https://earth.app.goo.gl/?apn=com.google		https://catholicsocialservices
16	☑	One floor so no ramp needed or elevator	4	Jan 2026	https://earth.app.goo.gl/?apn=com.google		https://mgmbaptists.org/welc
17	☑	Possible second floor however there are no steps on the exterior so a ramp is not needed	3	Feb 2025	https://earth.app.goo.gl/?apn=com.google		https://www.newhomemtmisc
18	☑	Website has accessiblity outline	6	February 2023	https://earth.app.goo.gl/?apn=com.google		https://mmfa.org/visit/plan-y
19	☑	Website mentions wheelchair accessibility, no visible ramp or automatic doors.	4	March 2025	https://earth.app.goo.gl/?apn=com.google		https://vaughnpark.com/new
20	☑	All accessiblity features are met, there are a few images of what it looks like inside and from what	6	January 2026	https://earth.app.goo.gl/?apn=com.google		https://www.ymcasenc.org/lo
21	☐	Steps inside the church with no interior ramp shown.	4	March 2025	https://maps.app.goo.gl/C5yVt2s67zXERb		N/A (facebook page)
22	☐	Next to school that does appear to have accommodations	2	January 2025	https://earth.google.com/web/search/124		https://www.fusionmontgomer
23	☐	voting on ground floor, looks like larger facility	5	February 2025	https://earth.app.goo.gl/?apn=com.google		https://vaughnforest.com/
24	☑	Very large facility, many entrances	6	February 2025	https://earth.google.com/web/search/600		https://www.frazer.church/
25	☑	There is only one entrance with steps, but it does not appear that there is a ramp near this entranc	5	March 2025	https://earth.app.goo.gl/?apn=com.google		Facebook: https://www.faceb
26	☐	Several entrances, main one appears to have stairs but there is an alternative side door next to the	3	February 2025	https://earth.app.goo.gl/?apn=com.google		https://www.google.com/sear
27	☑	No ramps on the exterior, but does appear that all entrances are accessible.	5	February 2025	https://earth.app.goo.gl/?apn=com.google		https://easternhills.church
28	☐	No ramps, not super accessible, looks like one possible entrance that doesn't require stairs	4	February 2025	https://earth.app.goo.gl/?apn=com.google		https://eastmont.org
29	☑	No ramps or drop off area, but super small and only one floor.	4	March 2014	https://earth.app.goo.gl/?apn=com.google		N/A

Montgomery Locations | 3 dataset | ADA info

Dataset

Rationale:

This dataset directly supports the project goal of identifying specific ADA compliance gaps at polling locations. Rather than relying on general assumptions about accessibility, the dataset provides detailed, location specific, evidence about where barriers exist. The ADA Polling Place

Accessibility checklist guided our selection of variables by focusing on important accessibility components such as parking, exterior pathways, entrances, and voting areas access. As a result, stakeholders like the SPLC can use the data to better understand which locations present the greatest accessibility challenges and make more informed decisions about where advocacy, resources, or improvements are most needed.

Deliverable 2: Accessibility Analysis Report

Accessible Parking

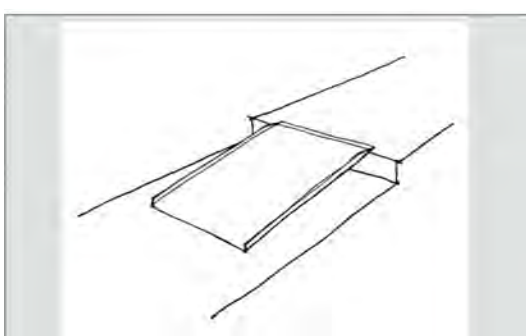
For this category, we looked at whether each polling location had accessible parking and if those spaces were actually usable for individuals who need them. This included checking for designated handicap parking, how close those spots were to the entrance, whether the sames were clearly marked with proper signage, and if the parking lines were visible and organized. We also paid attention to the overall layout of the parking area, since even if a space exists, it may not be accessible if it is too far away or difficult to navigate. Using the ADA Polling Place Accessibility Checklist as a reference, we know that accessible parking should include things like clearly marked spaces, proper signage, and an adjacent access aisle to allow individuals to safely get in and out of their vehicles.While going through the dataset, we noticed that some locations did not have parking lots at all, which already creates a barrier for many voters. In other cases, parking technically existed, but it lacked clear markings or was positioned in a way that made access difficult. In the end there were 37 out of 49 polling locations that had accessible parking, but the quality and usability of those spaces varied a lot, which is important to consider beyond whether they exist.

Van accessible parking signs should be used to designate van accessible parking locations.

Directional signage should be used to show direction to the accessible route, accessible entrance, and voting area.

Exterior Route to Accessible Entrance

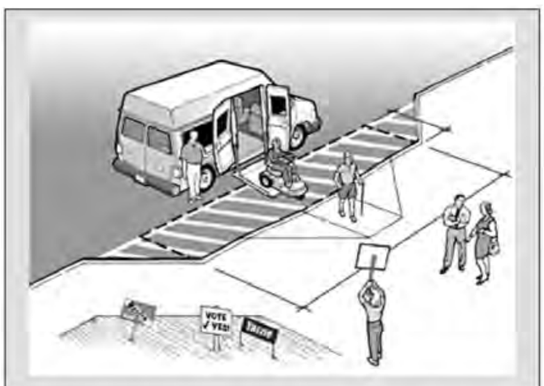
For this category, we focused on whether each polling location had an accessible exterior route leading up to the entrance. We specifically looked for features like curb cuts, smooth and continuous walkways, and pathways that were wide and clear enough for individuals using wheelchairs, walkers, or other mobility aids. We also considered whether there were any obstacles along the route, such as uneven pavement, stairs without ramps, or clutter that could block access. Based on the ADA Checklist, an accessible route should connect parking or drop-off areas directly to the entrance and be free of barriers that would prevent someone with mobility disability from entering independently. Based on our observations, 37 out of the 49 locations had an exterior route that met these criteria. While this shows that a majority of locations do provide some level of accessibility, there is still a significant number that do not, meaning they cannot be considered fully accessible. We also noticed that many of the locations lacking these features tended to be smaller or less developed sites, which may indicate a gap in resources or infrastructure that needs to be addressed.



Portable ramps without handrails can only be used for heights six inches or less and can provide access at a curb or low step. Portable ramps also can be placed flat to cover holes or gaps in a sidewalk.

Passenger drop-off area

For this category, we analyzed whether there is a designated space that allows a voter to transition safely from a vehicle to the accessible route leading to the building. This includes a curb cut or specific road that leads to the front of the building, both allowing a smooth transition for voters to be dropped off. Of the 49 buildings analyzed, 31 sites provided a designated passenger drop-off area or curb cut, allowing voters to enter safely into the building. However, the 17 buildings that do not have a passenger drop-off area face accessibility issues. From our dataset, it showed that the accessibility gaps are most prevalent at smaller locations with limited parking lot size, which led to usually no need for a drop off area. While some of the smaller locations have small enough parking lots for voters, some locations do not have parking lots at all, forcing voters to be dropped off or park on the street. For SPLC's litigation and policy teams, these findings highlight a lack of designated passenger drop off areas , which may present a barrier to voters with mobility impairments and necessitate targeted advocacy for temporary accessibility remediations on Election Day.



Accessible passenger drop-off and loading area

Elevators

For this category, we checked whether or not elevators were available for individuals to navigate to the voting area. The ADA polling location checklist states that an elevator or lift must be maneuverable by individuals using mobility devices. These elevators must be clearly marked and independently operable. It is also important to note that an elevator is only required if the voting area is not on the same level as the entrance to the building. We found that 30 out of the 49 locations we analyzed had an elevator or lift within the building. A significant portion of the buildings we assessed were single story and did not need elevators. For buildings with multiple stories, we would search Google for floor plans or accessibility information. The ADA checklist suggests that polling locations should ideally remain on the same floor as the building entrance in order to avoid the need for an elevator at all.

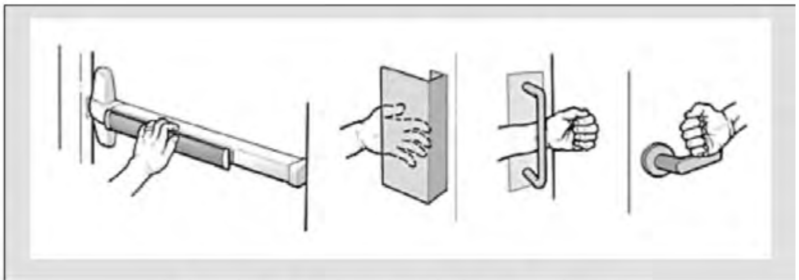
Polling Place Entrance

A polling place must have at least one entrance that is accessible to individuals with disabilities, including sufficiently wide doorways, easy-to-use handles, and the absence of barriers such as high thresholds.

The primary issue identified in our analysis was the presence of inaccessible door handles. In many cases, we were also unable to accurately measure door widths. While some entrances appeared accessible, such as those with clearly marked or double doors, limited visibility made it difficult to confirm certain features. As a result, some "no" responses in this section may reflect insufficient visual information rather than confirmed inaccessibility.

For this category, we assessed whether entrances included door handles operable with one hand, without tight grasping, pinching, or twisting of the wrist, as well as whether doorways appeared wide enough to accommodate a wheelchair and had low thresholds. These features, however, were not always verifiable through online observation.

Overall, 27 out of 49 polling places were identified as having an accessible entrance.

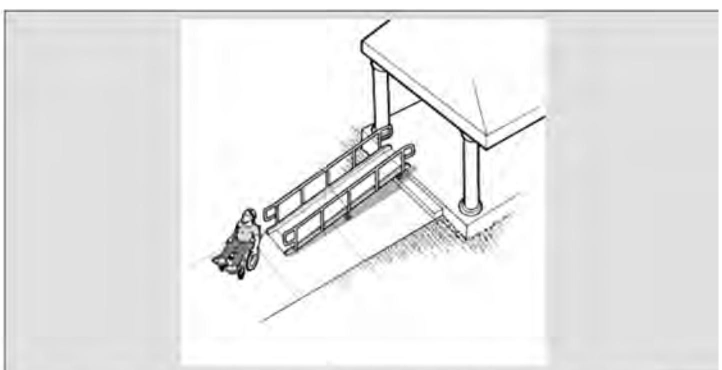


Examples of accessible door hardware

Ramp

In this section we looked at whether each polling location had a ramp that a person using a wheelchair, walker, or another mobility aid could use to enter the building. More specifically, we checked whether there was a ramp present when stairs, curbs, or raised entrances created a barrier to access. We also looked for whether the ramp appeared usable and connected to the entrance in a way that would realistically allow someone with limited mobility to enter the polling place safely. Issues we identified while analyzing the poling locations were that

Out of 48 buildings, we only identified 30 to have visible ramps; however, a portion of the buildings who did not have ramps had no need for them since the building was one floor and level with the parking lot.



A portable ramp with edge protection and handrails is placed over stairs to provide an accessible route on Election Day.

Limitations

Overall Limitations: This analysis relied mostly on Google Earth and street view images, which were not always clear, up to date, or detailed enough to fully evaluate accessibility. Because of this, some locations may have been marked as inaccessible due to limited visibility rather than actual barriers, and in-person assessments would be needed to confirm these findings.

Broken Down Limitations:

Accessible Parking: It was sometimes difficult to determine whether parking spaces were properly marked or usable due to unclear or limited imagery.

Exterior Route to Accessible Entrance: Online images made it challenging to fully assess surface conditions, slopes, or smaller obstacles that may impact accessibility.

Passenger Drop-Off Area: It was not always clear whether informal or temporary drop-off areas existed, especially at smaller locations without defined parking lots.

Elevators: Information on elevators or lifts was inconsistent, as it often required external research beyond visual data, such as floor plans or online sources. This created gaps in data accuracy and made it difficult to confirm accessibility for multi-level buildings or steps inside that were not known.

Polling Place Entrance: Key features such as door width, handle type, and thresholds were often not clearly visible in images, making it difficult to verify accessibility. Some entrances may have been marked as inaccessible due to lack of visual confirmation rather than actual barriers.

Ramp: It was sometimes unclear whether ramps were present or necessary, especially for buildings that appeared level with the ground. Limited angles and image quality may have caused ramps to be missed or incorrectly assessed.

Court Cases for Polling Locations: We researched active and historical court cases associated with these polling location addresses, specifically looking for litigation against Montgomery County regarding ADA compliance. Our search produced no results. This suggests a lack of significant legal action or federal oversight aimed at enforcing ADA standards at these specific sites.

Overall Findings:

Overall, our analysis shows that many polling locations have some accessible features, although it's not consistent across all sites. For example, 37 out of 49 locations had accessible parking and exterior routes, but the quality of those features varied, and in some cases they were difficult to use in practice. Other important features, like accessible entrances (27/49) and passenger drop-off areas (31/49), were missing more often and may create barriers for voters with mobility challenges. We also found that ramps and elevators were present at some locations, but in many cases they were either not clearly visible or not needed due to the building being level. A common pattern we noticed is that smaller or less-developed locations were more likely to be missing key accessibility features, especially when they did not have parking lots or designated drop-off areas. Overall, even though a majority of locations meet some accessibility criteria, there are still clear gaps that could make it harder for some voters to access polling places safely and independently.

Deliverable 3: Accessibility Visualization Summary

Heat Map:

This map (Figure 1) shows the areas with the highest and lowest levels of accessibility. Locations that meet more of the accessibility criteria are deeper shades of red, while locations with insufficient ADA accommodations are more blue and green. The marks on the map cover a one mile diameter around each location.

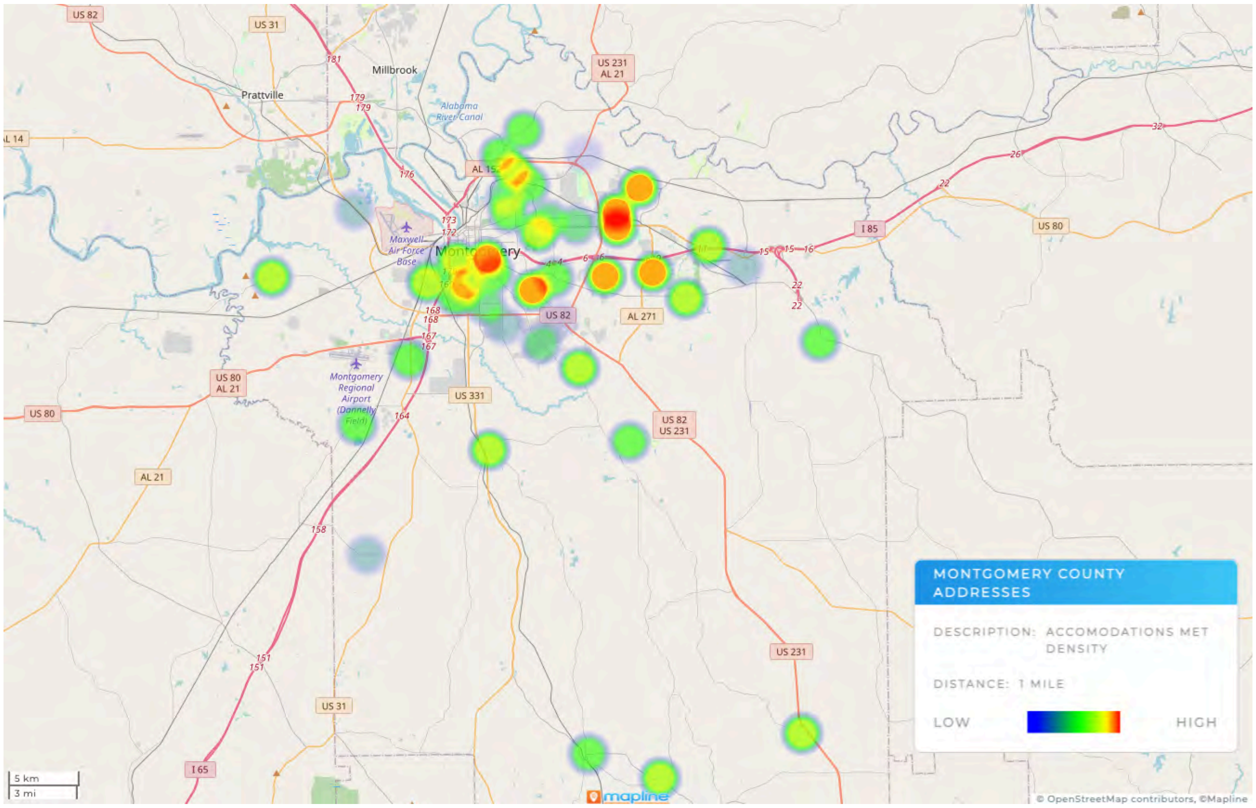


Figure 1. Built using the Deliverable 1 dataset and Mapline (https://app.mapline.com). Refer to methodology

Interactive Accommodations Map:

This interactive map (Figure 2) in Tableau is a simple interface that allows for users to find locations that meet select accommodations. People don't necessarily require all of the same resources, but it is important to know ahead of time, if a polling location meets certain needs. If a user wants to find the closest location with a proper entrance, but doesn't need handicap parking, they can use the interactive map in order to select filters that apply to them.

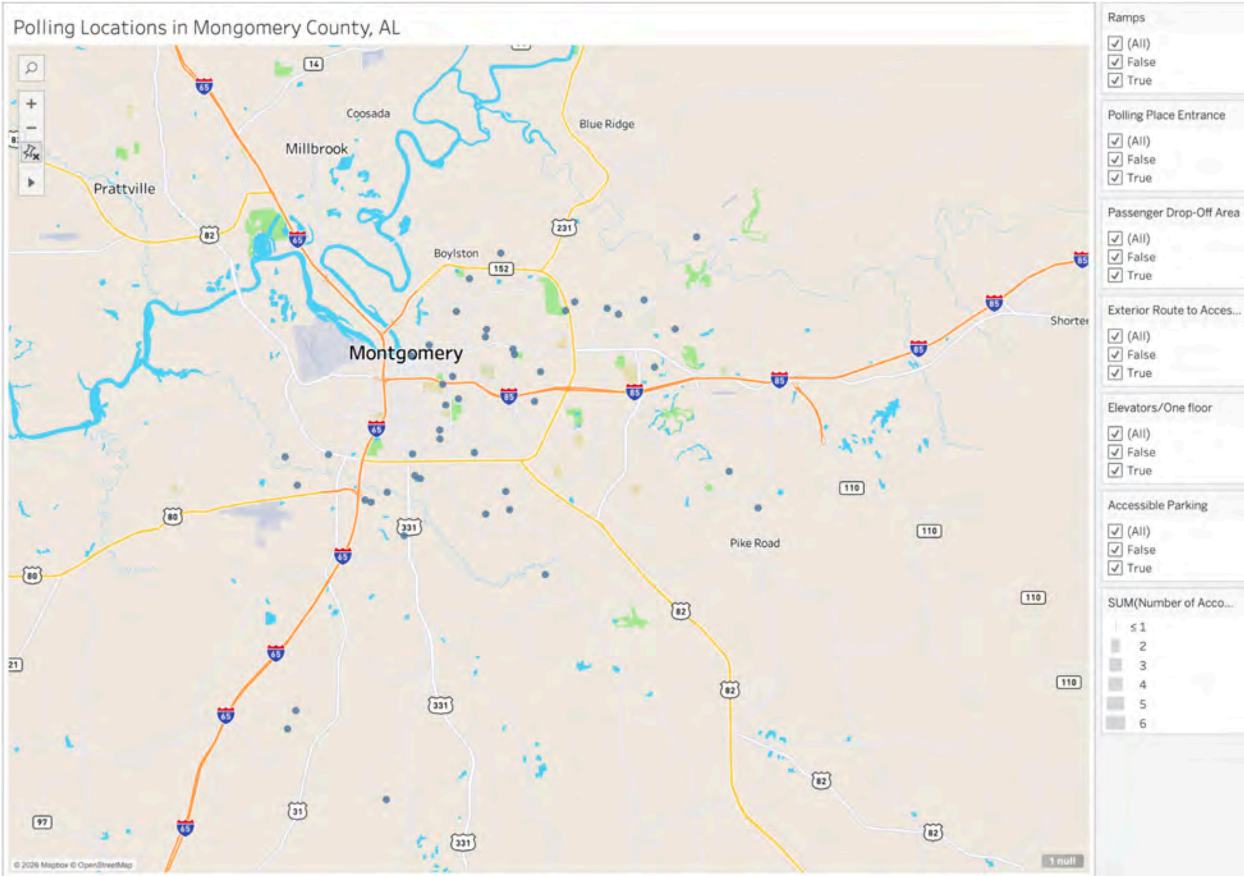


Figure 2. Built using the Deliverable 1 dataset and Tableau.

An additional aspect of being able to see specific location data was added. If the user hovers over any polling place on the map (Figure 2), they are able to see details related to that location. This includes the address, Google Earth image link, location site/name, whether each accommodation is true or false, longitude, latitude, and the total number of accommodations met.

Address:	3604 Pleasant Ridge Rd.
Image Links:	https://earth.app.goo.gl/?apn=com.google.earth&isi=293622097&ius=googleearth&link=https%3a%2f%2fearth.google.com%2fweb%2fsearch%2f3604%2bPleasant%2bRidge%2bRd,%2bMontgomery,%2bAL%2b36109%2f%4032.37864213,-86.25544552,97.38750442a,0d,60y,252.97153767h,86.25861164t,0r%2fdata%3dCpsBGm0SZwolMHg4ODhjMmEwODIyNDQ5ODM3OjB4NGJkMGM3ZWEwMmQ4MTExOBnkYrHycDBAQCE-fV-3Y5BVwCosMzYwNCBQbGVhc2FudCBSaWRnZSBSZCwgTW9udGdvbWVyeSwgQUwgMzYxMDkYASABIiYKJAlHI_1e6TFAQBEqV1n0GzFAQBnutR_Hz49VwCFt7FRAEJBVwEICCAEiGgoWVTdLcjRhSTZqcVpaTElya2dnVHdHdxACOgMKATBCAggAS
Location Site/Name:	Eastern Hills Baptist Church
Accessible Parking:	True
Elevators/One floor:	True
Exterior Route to Accessible Entrance:	True
Passenger Drop-Off Area:	True
Polling Place Entrance:	True
Ramps:	False
Latitude:	32.3798
Longitude:	-86.2450
Number of Accomodations Met:	5

Figure 3. Result of hovering over one example polling address - 3604 Pleasant Ridge Rd.

Recommendations

Recommendations for the client (SPLC)

Prior to this project, there were no existing datasets regarding polling place ADA compliance in Montgomery County, AL. We hope that SPLC can use the data we collected in order to address accessibility issues at the local level. They should be able to use the dataset and visualizations to pinpoint specific locations in Montgomery County that fail to comply with ADA requirements. The data can help inform community decisions or be used to lobby for accessibility improvements. Furthermore, SPLC can use the data collected to raise awareness about accessibility challenges to community members and educate voters. We hope this leads to advocacy efforts on the local level, as the most important part of advocacy is to highlight the voices of those who are personally affected.

Recommendations for future project teams

We used a standardized checklist to create our dataset, which future project teams could use to expand on this work. The logical continuation is to survey other counties besides Montgomery or reevaluate and update the existing locations in the dataset. However, there are a few key points to keep in mind. It is important to be aware that Google Earth is not always accurate (as discussed in Limitations). Teams could consider calling the locations to ask about accessibility features. Some polling locations may also have websites with more information. We found that smaller locations would sometimes have local Facebook pages with clearer images of both the exterior and interior of buildings. You may also find floor plans of certain locations. These resources could be used to determine the presence of accessibility features such as external automatic doors or elevators which might not be visible from Google street view. An approach we considered in the early stages of our project was getting in contact with people in Montgomery County to interview them about what support is needed within their communities. Being able to directly interview community members could help in identifying what accessibility gaps exist and the specific needs that should be advocated for.

Conclusions

In conclusion, this project successfully created a structured and evidence based assessment of ADA accessibility across 49 polling locations in Montgomery County, Alabama. By using the ADA Polling Place Accessibility Checklist alongside Google Earth and Street View, our team was able to identify patterns in accessibility barriers related to parking, entrances, exterior routes, passenger drop-off areas, ramps, and elevators. The project not only produced a working dataset, but also visual tools such as heat maps and Tableau dashboards that made the findings easier to interpret and apply in future advocacy and research efforts.

Overall, our findings showed that although many polling locations include some accessible features, accessibility is not consistently implemented across all sites. Several polling locations lacked important accommodations that could impact a voter's ability to independently and safely access that polling location. Smaller or less- developed polling locations were more likely to have accessibility gaps, especially regarding accessible entrances and passenger drop-off areas. These findings highlight the importance of continued evaluation and improvement of voting infrastructure in order to support equal access to the voting process for individuals with disabilities.

In addition to supporting SPLC's broader research and advocacy goals, this project also established a standardized methodology that future teams can expand across other counties in the Alabama Black Belt. The dataset structure, evaluation criteria, and visualization methods developed through this project provide a foundation for continued accessibility research and future policy recommendations. Although limitations existed due to reliance on virtual imagery, the project demonstrates how publicly available tools and structured data collection can still provide meaningful insight into accessibility barriers at polling locations.

Most importantly, this project highlighted the broader significance of accessibility in the voting process. Access to polling locations is an essential part of civic participation, and barriers related to parking, entrances, or navigation can directly impact an individual's ability to vote independently and safely. By identifying these gaps and organizing the findings into a usable dataset and visual reports, this project provides information that can support future advocacy, research, and accessibility improvements within Montgomery County and beyond.

Contact Information:

Samantha An

Email - san10@terpmail.umd.edu

Jessica Barke

Email - jbarke1@terpmail.umd.edu

Katherine Delaney

Email - kdelane1@terpmail.umd.edu

Megan Otchet

Email - motchet@terpmail.umd.edu

Catalina Herrera

Email - cherrer4@terpmail.umd.edu

Derik Luc

Email - dluc@terpmail.umd.edu

Deliverables Checklist

- [ ] Deliverable 1: Dataset
    - [ ] Dataset
- [ ] Deliverable 2: Accessibility Analysis Report
    - [ ] Accessibility Analysis Report
- [ ] Deliverable 3: Accessibility Visualization Summary
    - [ ] Can be emailed to client- cannot be directly linked

Voting & Civic Participation

Alabama Black Belt Counties

Team 6 BB-BAND: Broadband Access

Prepared for: Southern Poverty Law Center (SPLC)

Team Members: Ronja Markoff (Project Manager), Dominic Balzano (Team Communicator), James Droter (Quality Assurance), Eliott Perez (Documentation Lead), Isa Ansari (Integration/Consistency Checker)

Course: INST490 · Section: 0103

Date: May 18, 2026

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Team 6 BB-BAND

Team Brief

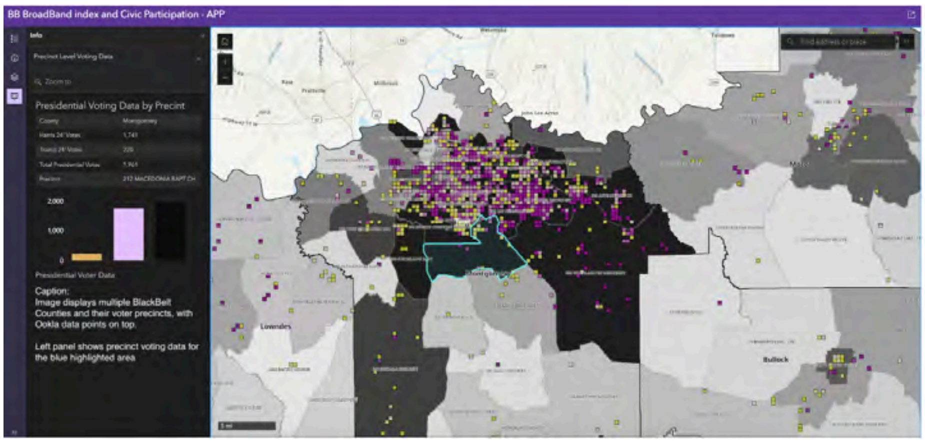
Section Brief

The BB-BAND team was tasked with identifying and compiling broadband access data relevant to civic participation for the Southern Poverty Law Center (SPLC). Using 2024 presidential election data alongside Ookla Speedtest data, the team synthesized an ArcGIS visualization of Alabama's Black Belt Counties. This visualization reveals key insights, highlighting areas for further research and action regarding broadband access for the SPLC.

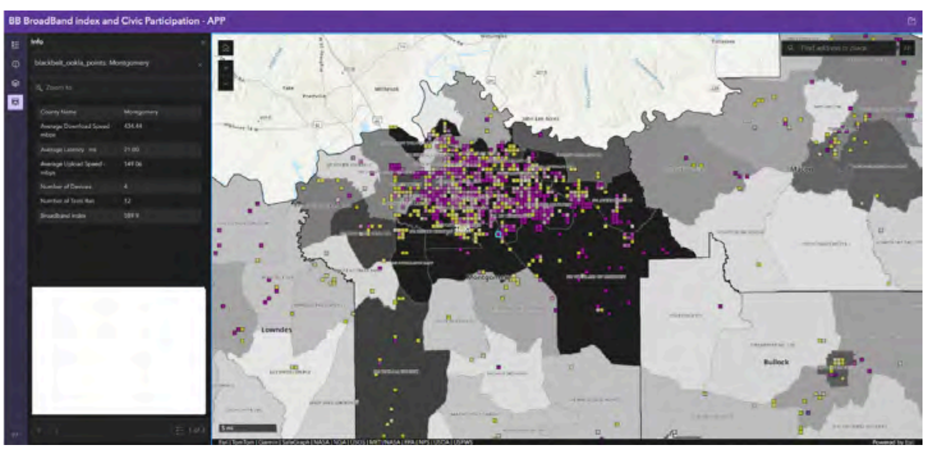
Key Findings

- Finding 1: Voting precincts in Alabama's Black Belt counties with stronger broadband performance, indicated by higher internet speed test volume and broadband index scores above 100, showed higher levels of civic participation in terms of voter turnout in the 2024 presidential election.
- Finding 2: Our correlation is not foolproof. Outliers do exist; certain areas may show a high volume of tests with low broadband index scores, yet still demonstrate strong civic participation, while other areas may show a low volume of tests with high broadband index scores and still demonstrate strong civic participation.
- Finding 3: There is a key difference when breaking civic participation into varying levels, as viewing the Alabama Black Belt from a precinct perspective will result in different findings than viewing counties. In certain precincts, there is strong broadband access in an otherwise scarce county; upon further investigation, this is because resources are not evenly spread throughout a county. Breaking geographic regions into more digestible segments will yield better results.

Deliverables Produced



This image shows the overall appearance of the Alabama Black Belt counties, divided into their voter precincts, with Ookla data points overlaid. The left panel shows precinct voting data for the highlighted area.



This image portrays the overall appearance and a representation of an individual data point in Montgomery County. As shown, users are able to see the number of devices that ran an Ookla speed test, the number of tests run total, and the score results.

Main Map Visualization:

https://www.arcgis.com/apps/instant/sidebar/index.html?appid=62ba6624d6f344d6ab0e1a395d90f8df

Presidential Voter Count in BlackBelt Counties:

https://www.arcgis.com/apps/mapviewer/index.html?webmap=e31daca2a90245f58a2a1685b1a5e41d&center=-86.212725%2C31.872134&scale=1155581.108577

Challenges

While broadband and internet access are commonly acknowledged as significant barriers to civic participation, the Southern Poverty Law Center has not previously analyzed it through a dashboard or interactive map. While this was intimidating at first, it allowed our team to be more exploratory and creative without specific bounds. By working to address this gap, we learned about new interfaces while considering accessibility, long-term effectiveness, and real-world application.

Connections to Other Teams

Team 2: BB-VOTE - In the interest of reliable and consistent polling data, our team utilized a recommendation by the BB-VOTE team, who directed us to the Alabama Secretary of State website. While multiple datasets are available on this site, the data collection process will be consistent regardless of the exact voting data selected. We then used data from the Redistricting Hub, which reviews and publishes data from the Alabama website.

Visualization working group: The visualization team assisted us in constructing a consistent ArcGIS map using their uniform colors guide. We shifted the colors of our map to the purple and yellow hues identified for conformity.

Final Report and Deliverables

Project Context

The strongest cross-team coordination occurred in identifying datasets. In the interest of reliable and consistent polling data, our team utilized a recommendation by Team Two (BB-VOTE), who directed us to the Alabama Secretary of State website. While there were multiple datasets available on this site, the data collection process will be consistent regardless of the exact voting data selected. In order to apply this data, we utilized the Redistricting Hub, which is a platform that pulls voting data from websites such as the Alabama Secretary of State for an in-depth analysis. We further utilized working groups that helped in formatting and presenting deliverables to our client. The visualization team provided a uniform color guide to make our ArcGIS map consistent. With help from our professor, we utilized lessons discovered from teams in a different class section to prepare our ArcGIS programming better.

Methods

To best address the broadband access and civic participation relation and potential gap, we utilized sources such as Ookla, the State of Alabama's data, the Redistricting Hub data, ArcGIS, and diagnostic interpretation. Because SPLC had previously told us that "civic participation" in their context was defined as anything related to casting a ballot, we decided to focus on the 2024 election data. This would work in tandem with the Ookla Speedtest data, where in our map, we cross-reference votes with broadband data to synthesize an evidence-driven conclusion.

As the team focused on broadband access and internet availability, it was vital to identify a reliable data source for speed data. Throughout this process, data from individual carriers, such as AT&T and Verizon, were considered and assessed for applicability. Utilizing private carriers, however, introduces an objectivity risk and data limited to personal clients. To address this concern, Ookla's platform "Speedtest" provides data on "internet speed, quality, and performance for mobile and broadband networks," as stated on their company website. As a team, we concluded that choosing Ookla for broadband data would be best. By not limiting data to individual carriers (Verizon, AT&T, etc.), Ookla provides a strong foundation for the most objective data.

We chose ArcGIS as the primary tool for this project because it offered the most in-depth way to display broadband coverage across the Black Belt counties. ArcGIS was able to geographically visualize counties with the weakest connectivity, how speed around precincts was performing, or where gaps could be better identified. By turning location data into intelligence and interoperable points, the platform will allow additional layers when combined with various teams. Easily allowing new maps and quantitative data to be integrated, ArcGIS is a reliable and consistent tool across capstone teams.

Our team utilized the Redistricting Hub for precinct-level voting data. This website utilizes the Alabama Secretary of State's publicized voting data at a more specified level, as it is divided by precinct rather than county. In particular, we focused on the 2024 presidential election voting data to narrow our scope and provide working information regarding large campaigns. Due to the presidential election having the highest turnout rate, this data best suited our needs for assessing information access via broadband relevant to civic participation. On the Secretary of State website, this data is presented as a PDF file by county, as seen below, meaning it is non-translatable by ArcGIS nor directly applicable to precincts. As a result, we utilized the Redistricting Hub, which features the data by precincts and in an ArcGIS-compatible format.

General Election - November 5, 2024

Official Total Ballots Cast

County	Registered Voters	Total Ballots Cast	Democrat Ballots	Republican Ballots	Absentee Ballots	Turnout as a Percentage of Registered Voters	Absentee as Percentage of Total Ballots
Autauga	46,291	28,388	4,960	13,269	1,670	61.33%	5.88%
Baldwin	207,643	122,542	14,334	69,308	7,771	59.02%	6.34%
Barbour	17,666	9,919	3,409	2,419	593	56.15%	5.98%
Bibb	15,152	9,278	1,245	4,628	351	61.23%	3.78%
Blount	43,601	28,138	1,534	18,524	965	64.54%	3.43%
Bullock	7,181	4,144	2,573	601	406	57.71%	9.80%
Butler	14,530	8,530	2,661	3,087	464	58.71%	5.44%
Calhoun	84,074	48,819	9,152	23,826	2,415	58.07%	4.95%
Chambers	26,794	14,380	4,063	5,509	774	53.67%	5.38%
Cherokee	21,383	13,030	984	8,417	522	60.94%	4.01%
Chilton	31,677	19,824	1,500	8,601	682	62.58%	3.44%
Choctaw	10,767	6,692	2,067	2,414	452	62.15%	6.75%
Clarke	19,143	11,993	4,219	4,898	724	62.65%	6.04%
Clay	10,603	6,759	670	3,855	324	63.75%	4.79%
Cleburne	12,193	7,860	341	5,127	412	64.46%	5.24%
Coffee	39,771	22,367	3,215	12,013	1,331	56.24%	5.95%
Colbert	45,322	27,239	5,064	13,060	1,316	60.10%	4.83%
Conecuh	9,820	6,105	2,005	1,533	479	62.17%	7.85%
Coosa	8,354	5,276	1,227	2,710	210	63.16%	3.98%
Covington	29,306	17,165	1,643	9,756	1,025	58.57%	5.97%
Crenshaw	10,851	6,516	1,126	3,210	314	60.05%	4.82%
Cullman	66,582	43,311	2,281	27,545	1,858	65.05%	4.29%
Dale	37,101	19,207	3,240	9,970	1,043	51.77%	5.43%
Dallas	29,534	15,631	8,526	2,644	1,038	52.93%	6.64%
DeKalb	48,071	29,756	2,393	18,730	1,268	61.90%	4.26%
Elmore	63,582	41,729	6,439	21,891	1,943	65.63%	4.66%
Escambia	28,268	15,009	3,196	7,874	716	53.10%	4.77%
Etowah	76,956	46,281	7,255	25,475	2,605	60.14%	5.63%
Fayette	12,897	8,417	735	4,928	293	65.26%	3.48%

Challenges

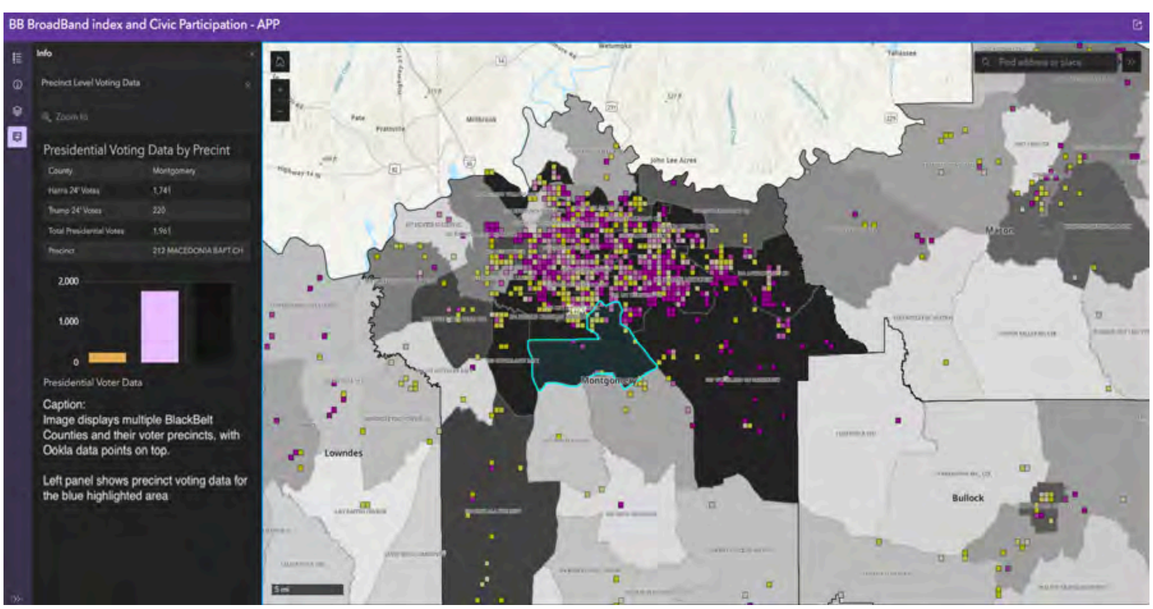
While broadband and internet access are commonly acknowledged as significant barriers to civic participation, the Southern Poverty Law Center has not previously analyzed either through a dashboard or interactive map. While this was intimidating at first, it allowed our team to be more exploratory and creative without specific bounds. By working to find a way to address this gap, we were able to learn new interfaces while considering accessibility, long-term effectiveness, and real-world application.

A significant challenge was identifying the best dataset to use within our program. We relied on publicly available quantitative data from government agencies, as other data collection sources introduced an unnecessary risk of bias. To mitigate this, we communicated with other teams to select a set from the Alabama Secretary of State. While this voting data was corroborated, broadband access data has a strong possibility of biased collection, as it is largely self-reported. Using platforms of specific cell phone and internet carriers would negate outside carriers that do not have the same depth of reporting. This caused a challenge in gathering data, which led us to use Ookla, a third-party platform commonly used for speed tests.

Additionally, FCC data has historically overstated broadband availability because providers report coverage by census block, meaning that if one household in a block is served, the entire block may be labeled as covered. There may also be gaps in rural coverage data, especially in counties with sparse populations. Certain populations, particularly low-income households, elderly residents, undocumented individuals, and those without stable housing, may be underrepresented in survey-based datasets. Additionally, broadband "availability" does not necessarily equal affordability, reliability, or digital literacy, which introduces interpretive limitations.

Deliverables

Our primary deliverable is an interactive multilayer map built in ArcGIS Online displaying broadband performance and civic participation data across 17 Alabama Black Belt counties. The map consists of three layers that work together to allow viewers to visually compare broadband infrastructure quality against precinct-level voter participation patterns.



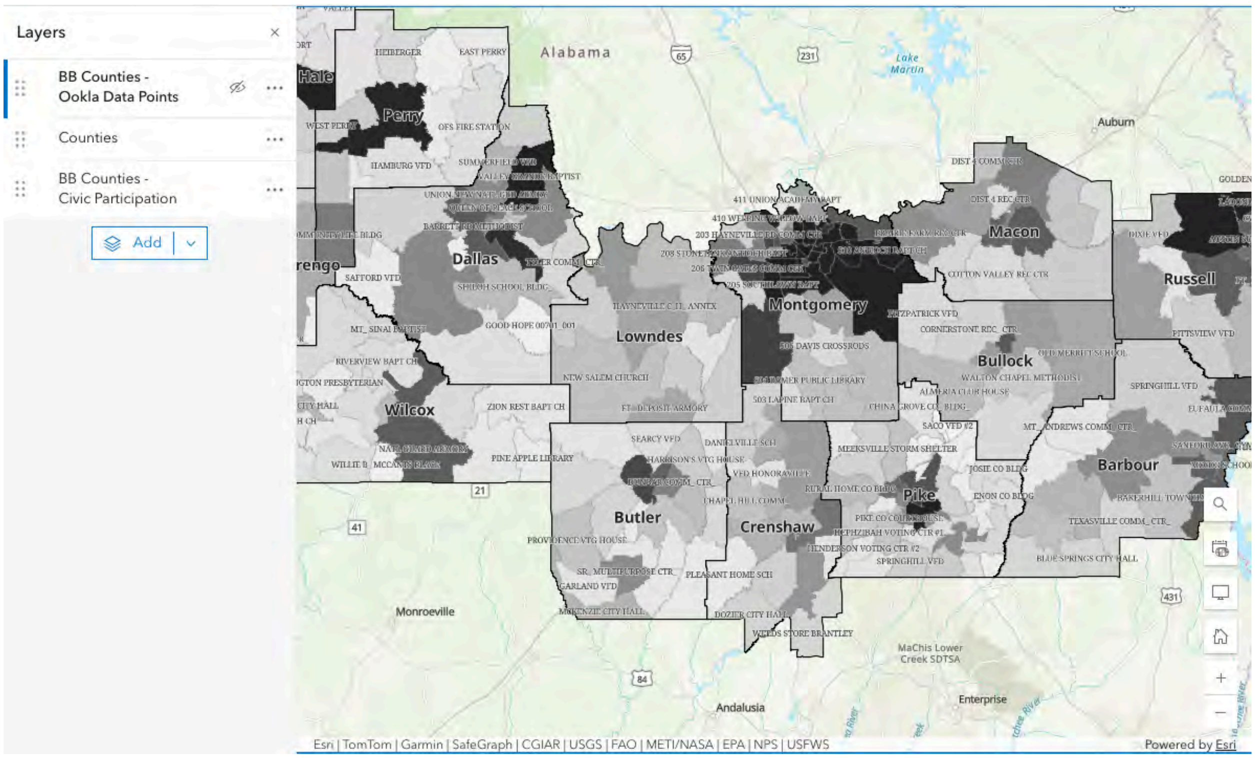
Our team's ArcGIS Visualization

Map Layer 1: Precinct Voter Participation

The base layer of the map displays individual precinct boundary polygons across the 17 Black Belt counties, choropleth shaded by total presidential votes cast in the 2024 general election. This metric, combining votes cast for Kamala D. Harris and Donald J. Trump, serves as the primary measure of civic participation, representing overall voter engagement.

Data source	Redistricting Data Hub, Alabama 2024 General Election Precinct Boundaries and Election Results
URL	https://redistrictingdatahub.org/dataset/alabama-2024-general-election-precinct-level-results-and-boundaries/
Participation Metric	Total presidential votes cast per precinct (G24PREDHAR + G24PRERTRU)
Visualization	Choropleth shading
Output file	blackbelt_precincts_final.geojson

The precinct boundaries and election results were sourced from the Redistricting Data Hub, which compiled 2024 General Election data from the Alabama Secretary of State and verified totals against the official state certification of election results. Precinct boundaries were originally sourced from individual county offices, county websites, and public records requests by the Redistricting Data Hub team.



Processing Steps:

1. Downloaded the shapefile zip from the Redistricting Data Hub
2. Unzipped and loaded the .shp file into Python using GeoPandas in Jupyter Lab
3. Filtered to 17 Black Belt counties using the County field
4. Calculated a total_pres_votes field by summing G24PREDHAR and G24PRERTRU
5. Reprojected to WGS84 (EPSG:4326) and exported as a GeoJSON file
6. Uploaded to ArcGIS Online via Map Viewer and styled as a choropleth using the total_pres_votes field with a purple color ramp.

Python Script Used:

```
import geopandas as gpd

precincts = gpd.read_file('al_2024_gen_all_prec.shp')

black_belt = ['Barbour','Bullock','Butler','Choctaw','Crenshaw',
    'Dallas','Greene','Hale','Lowndes','Macon','Marengo',
    'Montgomery','Perry','Pike','Russell','Sumter','Wilcox']

precincts_filtered = precincts[precincts['County'].isin(black_belt)]
precincts_filtered['total_pres_votes'] = (
    precincts_filtered['G24PREDHAR'] + precincts_filtered['G24PRERTRU'])
precincts_filtered.to_crs(4326).to_file(
    'blackbelt_precincts_final.geojson', driver='GeoJSON')
```

Map Layer 2: County Outline Reference

The second layer provides bold county boundary outlines displayed over the precinct choropleth, allowing viewers to clearly identify which county each precinct belongs to. This layer carries no fill color and serves purely as a visual reference. County names are displayed as labels positioned at the center of each county polygon.

Data Source	USA Counties, ArcGIS Living Atlas (US Census Bureau TIGER/Line 2023 boundaries)
Purpose	Visual reference only for County outlines

This layer was added directly from the ArcGIS Living Atlas within Map Viewer and filtered using attribute expressions to display only the 17 Black Belt counties. Popups were disabled on this layer so that clicking anywhere on the map triggers only the precinct layer popup.

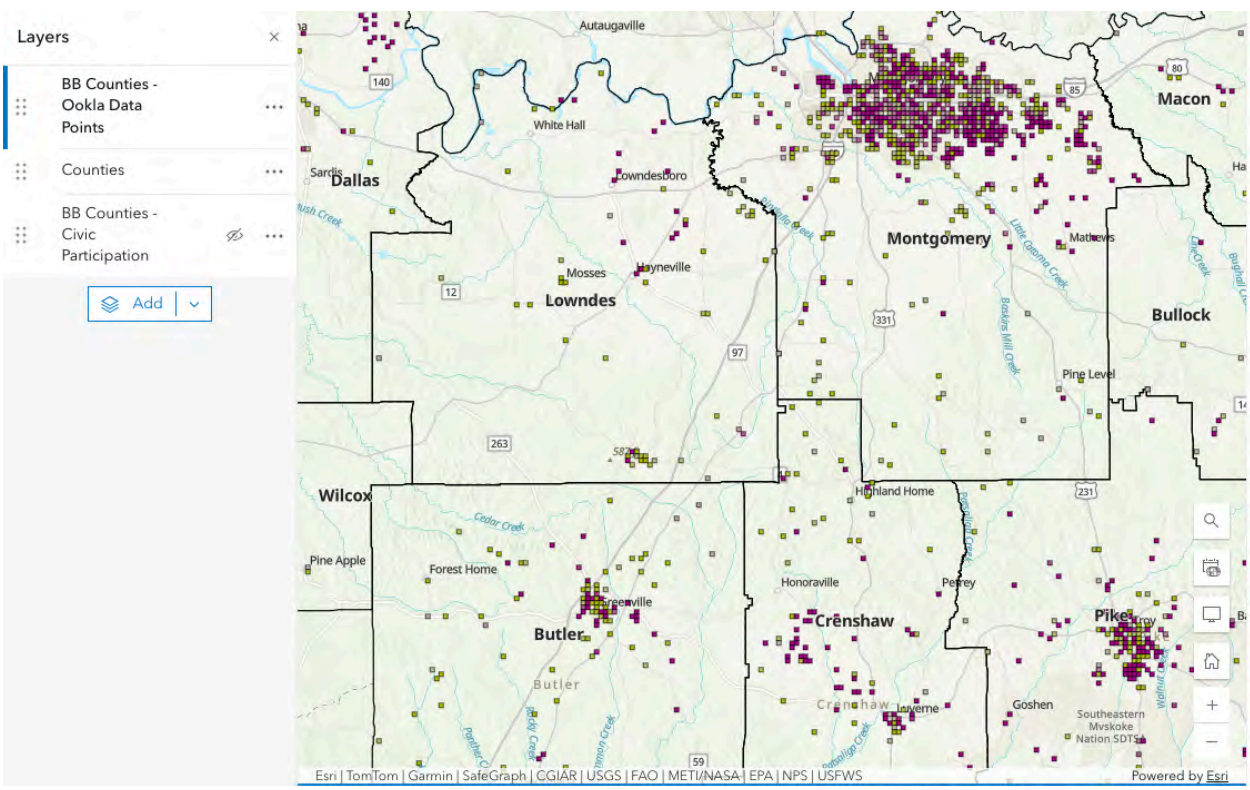
Map Layer 3: Ookla Fixed Broadband Index

The third layer displays individual Ookla speed test tile points across the 17 Black Belt counties, symbolized using a broadband index score that combines average download and upload speeds into a single performance metric. This index, developed by Ookla and Esri, uses the United States federal broadband standard of 100 Mbps download and 20 Mbps upload as its reference point, assigning a score of 100 to tiles that meet this threshold.

Data Source	Ookla open data, fixed broadband performance tiles, Q3 2024
URL	github.com/teamookla/ookla-open-data
Broadband Metric	Broadband index score calculated via Arcade expression
Visualization	Counts and Amounts (Color), purple = above 100, gold = below 100
Output File	blackbelt_ookla_points.geojson

The broadband index is calculated using the following Arcade expression applied directly within ArcGIS Online, based on the official Ookla and Esri methodology:

```
var AverageDownScore = $feature.avg_d_mbps / 100 * 100;
var AverageUpScore = $feature.avg_u_mbps / 20 * 100;

var BroadbandScore = when(
    AverageDownScore >= 100 && AverageUpScore >= 100,
        (AverageDownScore + AverageUpScore) / 2,
    AverageDownScore < 100 && AverageUpScore >= 100,
        AverageDownScore,
    AverageDownScore >= 100 && AverageUpScore < 100,
        AverageUpScore,
    AverageDownScore < 100 && AverageUpScore < 100 && AverageDownScore <= AverageUpScore,
        AverageDownScore,
    AverageDownScore < 100 && AverageUpScore < 100 && AverageUpScore <= AverageDownScore,
        AverageUpScore,
    100);

return BroadbandScore
```



Processing Steps:

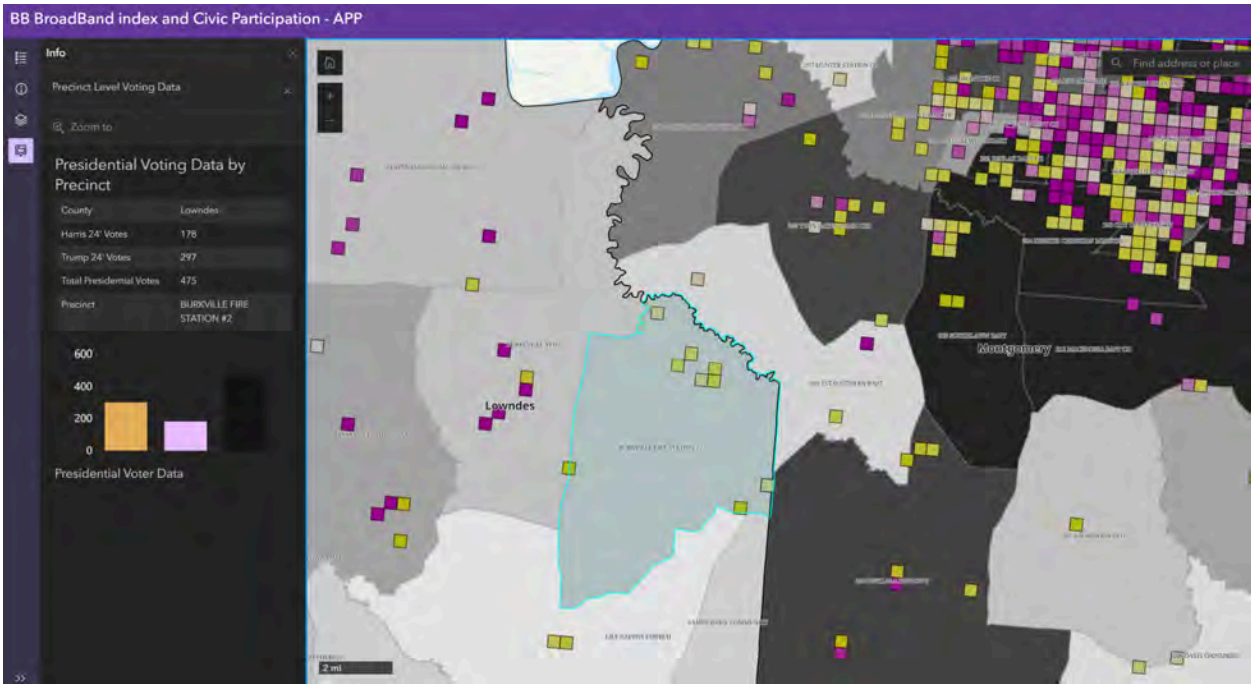
1. Downloaded Q3 2024 fixed broadband tiles from Ookla's open data S3 bucket via Python
2. Downloaded Alabama county boundaries from the US Census Bureau TIGER/Line 2023 via Python
3. Filtered county boundaries to 17 Black Belt counties using Alabama's FIPS code (STATEFP == '01')
4. Performed a spatial join to clip Ookla tiles to only those intersecting the Black Belt county boundaries
5. Converted average speed values from Kbps to Mbps
6. Exported as GeoJSON and uploaded to ArcGIS Online as an individual point layer
7. Applied broadband index Arcade expression for styling using Counts and Amounts (Color)

```
import geopandas as gpd
from datetime import datetime

def get_tile_url(service_type, year, q):
    dt = datetime(year, [1,4,7,10][q-1], 1)
    base = 'https://ookla-open-data.s3-us-west-2.amazonaws.com/shapefiles/performance'
    return
    f'{base}/type%3D{service_type}/year%3D{dt:%Y}/quarter%3D{q}/{dt:%Y-%m-%d}_performance_{service_type}_tiles.zip'

tiles = gpd.read_file(get_tile_url('fixed', 2024, 3))
counties = gpd.read_file('https://www2.census.gov/geo/tiger/TIGER2023/COUNTY/tl_2023_us_county.zip')

al_counties = counties.loc[
    (counties['STATEFP'] == '01') &
    (counties['NAME'].isin(black_belt))
].to_crs(4326)

tiles_clipped = gpd.sjoin(tiles, al_counties[['NAME','geometry']],
    how='inner', predicate='intersects')
tiles_clipped['avg_d_mbps'] = tiles_clipped['avg_d_kbps'] / 1000
tiles_clipped['avg_u_mbps'] = tiles_clipped['avg_u_kbps'] / 1000
tiles_clipped.to_file('blackbelt_ookla_points.geojson', driver='GeoJSON')
```

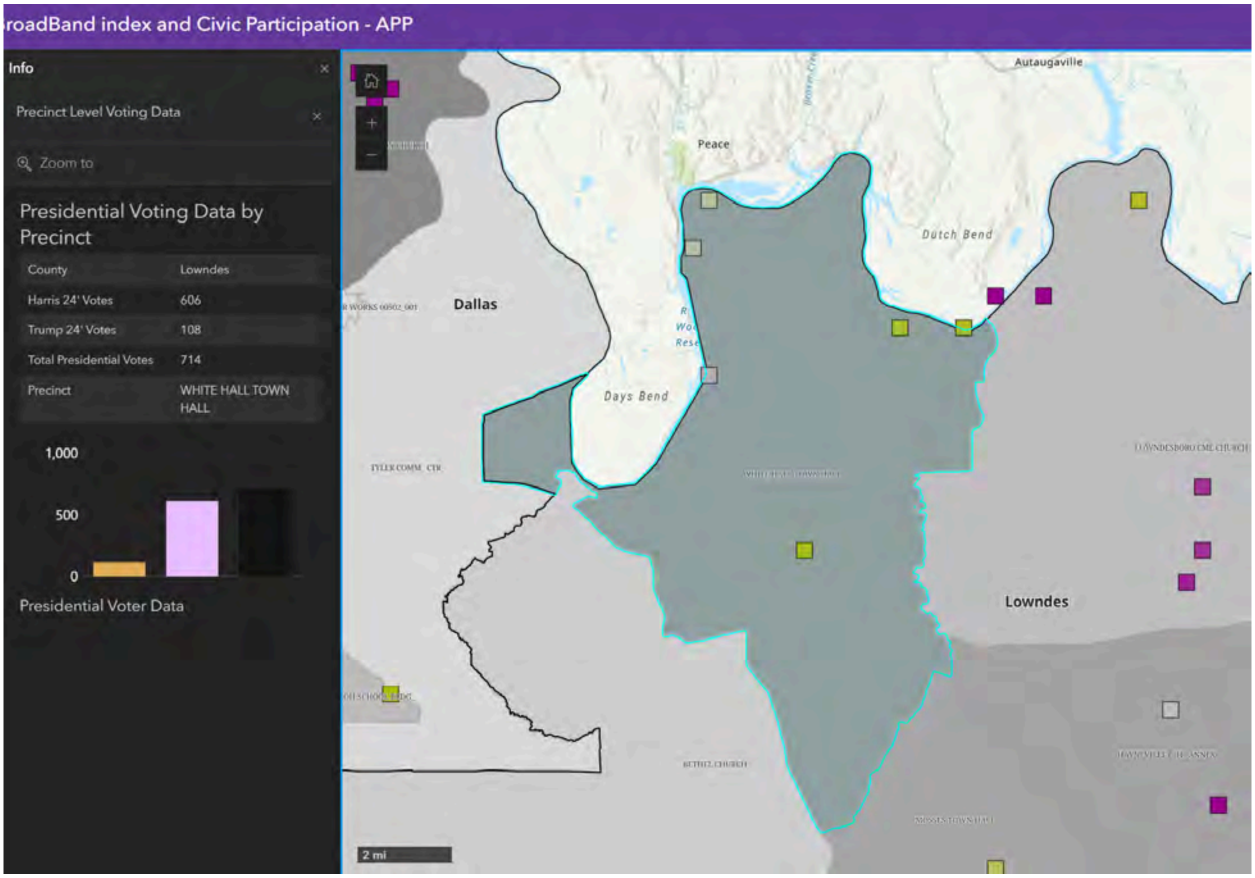
Findings

Voting precincts in Alabama's Black Belt counties with stronger broadband performance, indicated by higher test volume and broadband index scores above 100, tended to show higher levels of civic participation in terms of voter turnout in the 2024 presidential election. Broadband performance here refers to the quality and prevalence of fixed internet connections as measured by Ookla speed tests, specifically the broadband index score, which combines average download and upload speeds relative to the federal standard of 100 Mbps download and 20 Mbps upload. Precincts located in areas where more speed tests were conducted and where those tests returned broadband index scores above 100 corresponded with higher total presidential vote counts, suggesting an identifiable relationship between internet infrastructure quality and voter engagement across the region.

Our correlation, however, is not absolute. Outliers do exist; certain areas show a high volume of tests with low broadband index scores, yet still demonstrate strong civic participation, while other areas may show a low volume of tests with high broadband index scores and still demonstrate strong civic participation. Many examples exist in the map, but one particular case is White Hall Town Hall and Burkville Fire Station #2. Both precincts, relative to other locations, have low reported Ookla Speedtest data. Burkville, however, clearly has more than White Hall, with broadband indexes closer to 100 (but none above).



Burkville Visualization. As observed, multiple speed tests close to the broadband index of 100 were conducted in this precinct (light yellow/close to white shade indicates this, and can be expanded on by clicking each point).



White Hall Town Hall Visualization. Large precinct with few speed testing data points, with only a few that are close to the broadband index of 100.

As seen in the visualizations of Burkville and White Hall, despite having more speed testing data, Burkville did not have more total votes cast during the 2024 presidential election than White Hall. For this reason, the correlation identified between civic participation and broadband data is stated not to be absolute. While the overall visualization suggests it, there are cases such as these two precincts that prompt inquiry. As we will further discuss in our recommendations section, the best way to address this is by attempting to close the gap in the data itself; the low frequency of speed tests reported in these areas is a factor to be taken into consideration.

The most appropriate synthesis that can be accurately summarized is that there is an identifiable correlation between available and reliable broadband access and civic participation. However, correlation does not imply a static relationship. Due to gaps in broader regions across the Black Belt in tandem with cases such as Burkville and White Hall (and this was NOT the only case of this discrepancy), there is still more research and inquiry to be done. Simply stated, while there is an identifiable correlation between broadband access and civic engagement, it is only one of multiple factors that influences civic engagement; the correlation is not absolute, nor does it imply causation. Rather, it demonstrates a pattern of civic engagement, but there are multiple cases across the Black Belt counties showcasing 1) there is not always large amounts of reported broadband data and 2) it is not the sole indicator of a precinct's civic engagement.

Recommendations

Recommendations for the client (SPLC)

We strongly recommend that the Southern Poverty Law Center encourage more of the target population to complete speed testing to gain more comprehensive data. While we were able to pull what is accessible and published currently, there are strong gaps in speed test data and population estimates. This can be completed through flyer campaigns and posting materials at third spaces, such as city halls, libraries, and schools. For deeper exploration, SPLC should overlay other maps, such as population density maps, to assess the severity of this discrepancy. Other teams that have collected and aggregated data, particularly using our election cycle, with ArcGIS, are strong contenders for map integration.

Our map also highlights particular precincts that warrant further study (Burkville and White Hall from earlier being good examples). The Green's Crossroads precinct in Barbour

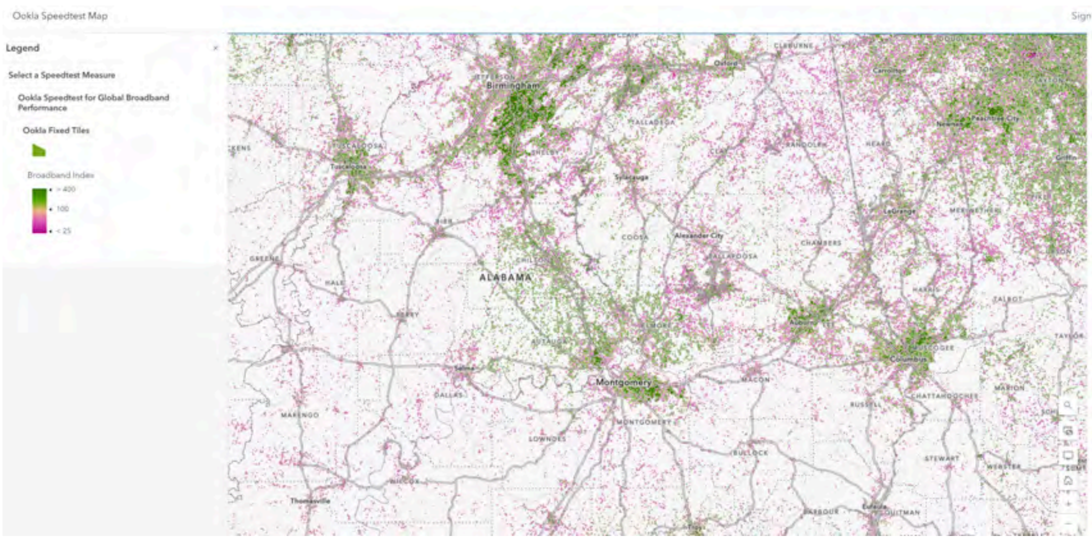
County has a particularly high speed test result despite having a total of 139 votes cast. This low number of ballots would generally be expected to be linked with low internet speed, as it is generally more rural. This answer may result in beneficial insight into whether it is a strong outlier or if broadband access does not have an identifiable correlation with civic participation, as expected, and should be investigated.

We recommend future capstone teams utilize the relationships with the SPLC and Alabama Black Belt community leaders, educational institutions, and other advocacy groups to gain a stronger reasoning into how broadband is affecting civic participation. Local community leaders witness how broadband can have an effect on everyday life and the civic infrastructure. These leaders can help teams identify which voices are most underrepresented and facilitate introductions to residents who would otherwise be difficult to reach. Colleges have faculty and students who may be researching broadband access and can share findings. Advocacy groups have a wide range of connections that can help in getting diverse opinions.

Recommendations for future project teams

The best approach to beginning this research is to identify reliable sources for civic participation. Our group found that the Alabama State Secretary's website provided sufficient information for election engagement data. While for the map visualization, we'd pivoted to using the redistricting hub data (which pulls from the State Secretary data regardless), any group looking to continue our work should begin by canvassing this data/identifying a reliable source. Without this information, there would be a considerable gap in our deliverables.

Additionally, various maps and resources already existed for us to build on and draw inspiration from to create our final deliverable (as mentioned in the deliverables and findings). An Ookla speed test visualization has already been created on ArcGIS. This influenced our group's direction in creating the main deliverable**.**



The full visualization can be viewed here (SDGS Today, 2026).

As observed in our interactive map, when expanding one of these points, the user can see civic participation data (which, for the scope of this project, focuses on casting a ballot), as well as whatever data the Ookla speed test has on the location. This retains the same idea as the previously mentioned SDGS Today visualization on ArcGIS.

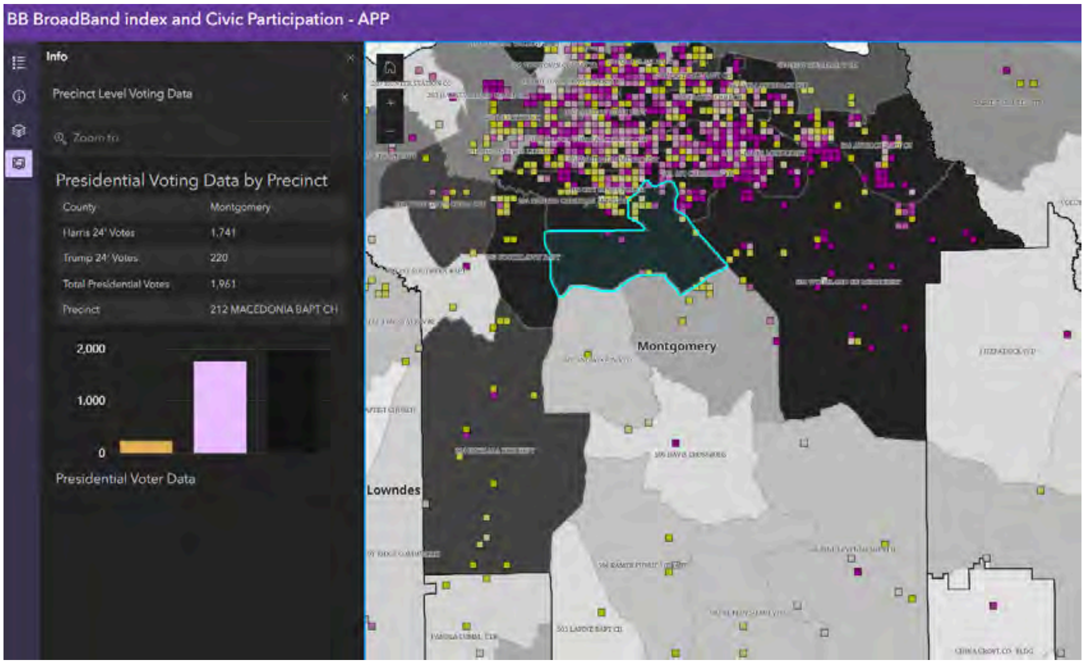


Figure 1.1 - Viewing Precinct data in Montgomery County. Bar graph visualization appears for Presidential Voter Data

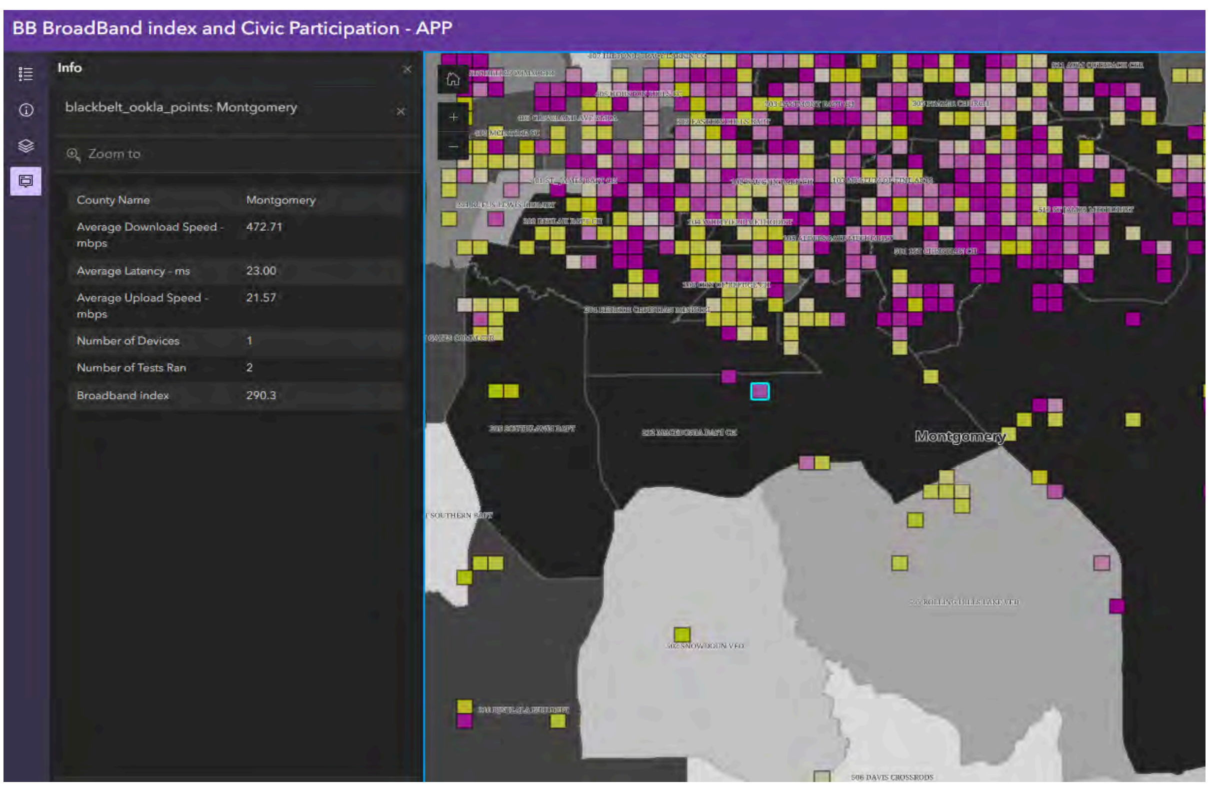


Figure 1.2 - Viewing a single data point in the precinct of interest.

A large barrier for our group, due to the lack of study and research in this field, was figuring out what we would produce. It is for that reason that any future research groups conduct should retain this format for a cohesive, legible presentation of broadband and civic data. We found that this visual presentation works best as users can reasonably scan for relevant data on the county and precinct level, while the visualization is easy to understand and navigate. In addition, SPLC approved of this visualization as it followed the constraints they asked us to abide by. These constraints only asked us to deliver a product that was not abstract or dense, and easy for the community to navigate and understand. It is for these reasons that any future groups looking to continue this research/work should maintain this structure of data presentation, not simply out of presentational demands, but to move past the barrier of identifying a format for the deliverable faster than we did.

When adding additional information regarding this field, there are a few considerations future groups must keep in mind. Broadband data can be very broad in scope, and must be cleanly defined before any insight gathering. Further work on this topic can warrant an expanded scope of what is considered for broadband access. For example, while our group landed on general Ookla speed tests, which evaluated internet speed, other areas of consideration include, but are not limited to:

- Access: Do citizens in these areas have reasonable access to reliable broadband internet?
- Digital Literacy: Broadband access being available is not significant to civic participation if large portions of the population are not themselves familiar with these technologies that influence civic engagement

These are just a few of the possible places teams can expand the initial scope of evaluating internet speeds. It is strongly recommended that whatever data future groups canvas is layered upon/expands on the pre-existing format of the map visualization. For example, teams may find it easier to add additional information in the sidebar expanded section of our deliverable. This, again, goes back to what future teams will define as relevant to their scope of broadband access. In tandem with the format of presentation, this will make any future teams' work more thorough, in-depth, and all while maintaining an intuitive layout.

Generally, teams looking to pick up where our work left off have a few immediate choices for continuing this work:

1. Looking to bridge the gaps in speed test data: There are still considerable gaps in speed test data in various precincts/counties. Groups can work with SPLC or relevant faculty conducting research in this domain to identify more insight into broadband access, where civic participation is concerned
2. Expanding the scope of broadband access: Teams can look to supplement the information our map already provides. This can be done by adding captions, layering other maps, and so forth to demonstrate any relevant broadband access data. As mentioned previously, this can mean shared spaces, digital divide, costs, and so on. Identifying these issues where they may pose challenges in the relationship of broadband access to civic participation are all strong places to continue our group's work.
3. Identifying between correlation and causation: Our group did not have enough data to accurately determine causation, let alone correlation. As mentioned, locations like Green's Crossroads in Barbour County were very interesting to look at due to high connection tests being run, yet simultaneously having a lower civic engagement than

other areas. This can be an interesting point of interrogation for future groups looking to pick up our work.

This list of recommendations, while giving strong direction, is not meant to be exhaustive. There may be other areas of interrogation that groups can continue based on our research where broadband access and civic participation intersect. The premise, however, remains the same. Groups should look to identify areas of concern and supplement that on the pre-existing map index of broadband data for an intuitive presentation that is also complete and thorough.

Conclusions

We set out to examine the effect of broadband access and civic participation across the Alabama Black Belt Counties. We developed an ArcGIS map integrating Ookla internet speed test data and 2024 Presidential election precinct locations that visually showed the regional difference. Our findings revealed that voting precincts located in areas with higher broadband speeds corresponded with stronger levels of voter turnout during the 2024 presidential election. While outliers exist and broadband access is not the sole determinant of civic participation, we found this to have some correlation. This map should best be applied for further integration with other ArcGIS maps to overlay key information regarding voting and information access.

The aim with our deliverable was to create a product/deliverable for SPLC that was actionable; something that not only a team of analysts could look at, but the community could observe and also evaluate themselves. Broadband access in relation to civic engagement has typically been an uncharted territory in this field. As a result, our group's goal was not just to create a coherent deliverable to handoff to SPLC, but also lay the framework for future study and interrogation in this field. This data and analysis already sets the precedent that broadband access clearly holds relation to civic engagement, the next steps are in acting on this reported data while simultaneously conducting more research.

In addition, this project demonstrates the importance of continuing to investigate how digital infrastructure affects communities beyond simple internet availability. Broadband access influences how individuals receive information, communicate with public institutions, participate in elections, and engage with civic resources in their everyday lives. By presenting the information through an accessible ArcGIS interface, our team was able to highlight disparities that may otherwise remain difficult to identify through traditional datasets alone. Ultimately, our work contributes to a broader understanding of how technology access and civic participation intersect in historically underserved communities throughout the Alabama Black Belt.

Team Members:

Ronja Markoff: rmarkoff@terpmail.umd.edu

Dominic Balzano: dbalzano@terpmail.umd.edu

James Droter: jdroter@terpmail.umd.edu

Elliott Perez: eperez11@terpmail.umd.edu

Isa Ansari: ansari99@terpmail.umd.edu

Deliverables Checklist

Deliverables Checklist:

Interactive ArcGIS Online Map - APP	https://www.arcgis.com/apps/instant/sidebar/index.html?appid=62ba6624d6f344d6ab0e1a395d90f8df
Interactive ArcGIS Online Map - Map viewer	https://www.arcgis.com/apps/mapviewer/index.html?webmap=5cd51b914e0a4146b11fa4b5a031530d
blackbelt_ookla_points.geojson	Ookla fixed broadband tile points clipped to Black Belt counties
blackbelt_precincts_final.geojson	Precinct boundary layer with total presidential votes
Counties.json	Outlines and labels For Counties only

Voting & Civic Participation

Alabama Black Belt Counties

Team 7 BB-CPRI: Civic Participation Risk Index

Prepared for: Southern Poverty Law Center (SPLC)

Team Members: Michelle Akem (Project Manager), Eli Jean (Team Member), Noah Lee (Team Member), Vonn Savaya (Team Member), Ethan Schwartzberg (Team Member)

Course: INST490 · Section: 0103

Date: May 18, 2026

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Team 7: BB-CPRI

Team Brief

Section Brief

BB-CPRI built a composite civic participation risk index that took into account 15 variables encompassing infrastructure access, demographic variables, geographic isolation, and individual capability for the 17 Black Belt counties during the 2020 General Election. This team aims to synthesize the findings of teams 2-6 into a single viewpoint, quantifying the likelihood of the Black Belt counties to vote during that period to give SPLC a basis for targeting advocacy and resource allocation efforts within that region.

Key Findings

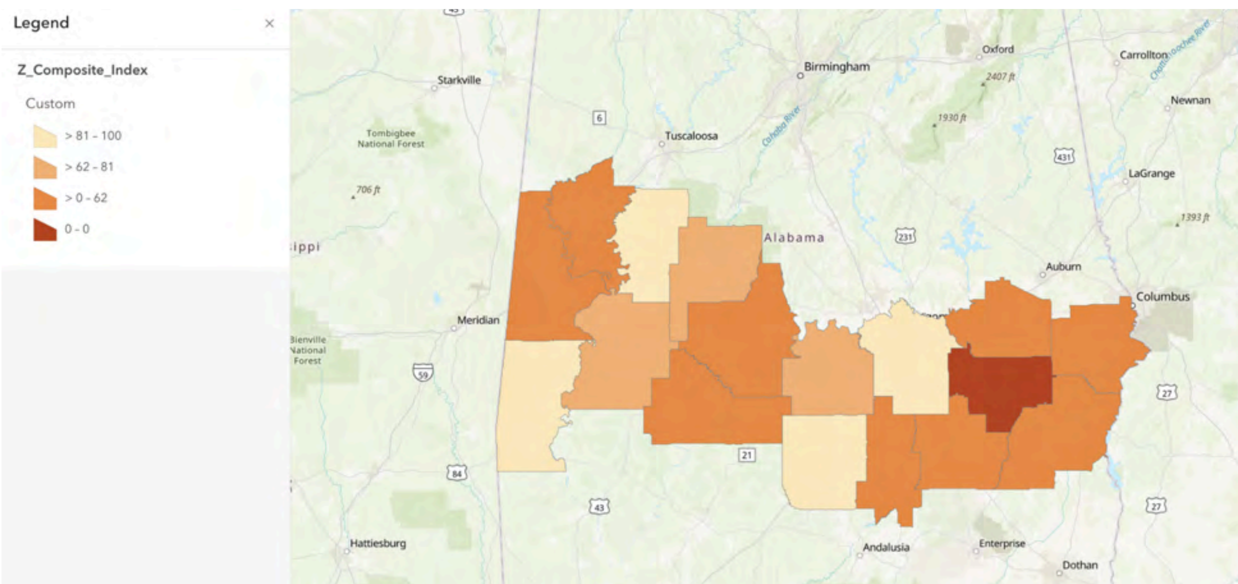
- From the overall composite index, the standout counties are Bullock, with a low index score of 0.00, signifying the highest risk of not participating in voting. Barbour has the second-lowest score of 46.31. Butler has a score of 100, meaning the least risk, and Montgomery is the second highest with 94.42.
- Montgomery County, the largest county by population and home to Alabama's capital, ranked second in civic participation access despite a low subindex 4 ranking, which correlated to individual capability factors such as language access, mobility access through personal vehicles, and health status with self-reported physically unhealthy days. This suggests that while Montgomery's external resources, such as transportation and broadband access, provide good conditions for voting, individual citizens may be hindered by personal issues.
- Within the geographic isolation subindex that measured factors such as the proportion of community colleges in the county, the proportion of congregations in the county, polling place distance, and the rurality of the county, over half the counties were low risk, while a segment of counties in the southeast corner of the map were 4 out of the 5 highest risk

counties in that category. This implies a possible gap in the public resources provided in that region, requiring more hands-on work to connect citizens to information.

Deliverables Produced

- Civic Participation Factors Dataset - A spreadsheet table containing all the data we used to create the composite index, comprising 15 variables across 17 counties.
  - Link: [Civic Participation Factors Dataset]

- Composite Index - A map showing the Black Belt counties and their risk in civic participation, ranging from 0 (high risk) to 100 (low risk).
  - Picture:



- Web Application - A front-facing interface that provides an accessible experience for users to interact with the risk data.
  - **Link:** https://github.com/vsayasa/risk-index

Challenges

The main challenges our team encountered were with the data and ArcGIS. Because the data we included were static numbers, we could only assess their credibility through the data provider, such as the United States Census, and clearly note the limitations in the report, as we weren't editing or imputing the data. Additionally, ArcGIS is not collaboration-friendly for synchronous work, so we had two of the five members working on that platform, and any changes to the index had to be recalculated manually by them. The index uses 2020 as its reference year because the majority of structural input variables, including rurality, vehicle access, and disability rates, are available consistently across 2020 data sources. Extending the index to 2024 would require updating all input variables, which is recommended as a next step for future teams.

Connections to Other Teams

> Team 3 - BB-PAD/Team 4 - BB-TRAC/Team 5 - BB-AACS/Team 6 - BB-BAND: These teams' topics helped us consider the inclusion of factors such as polling distance, public transit, disability percentages, and broadband access within our index to capture a more comprehensive picture of the region's voting landscape.

Final Report and Deliverables

Project Context

This project contributes to a broader 11-team regional analysis examining barriers to civic participation across Alabama counties. We were in charge of building the composite civic participation risk index (BB-CPRI) to synthesize multiple access-related indicators into unified measures of civic participation risk. The construction of the index relied on several inputs, including public datasets, internal analysis, and input from SPLC. As a result, the composite index serves as a summary tool that other teams can use to identify geographic areas with varied access-related risk factors. By integrating multiple factors into a single framework, this project reveals a realistic perspective on civic participation access in the Southern Alabama county area.

Methods

There were a number of helpful data sources that allowed us to create our own dataset to aid in the development of the composite risk index.

Public datasets we resourced included:

Source	Description
Census County Population Counts 2020-2025	Used to obtain county populations in 2020 for proportion variables
Alabama Community College System	A list of the community colleges in Alabama.
ALTRANS	Map of transit providers per county in Alabama.
County Health Rankings & Roadmaps	Used to obtain estimates on the average self-reported unhealthy days for each county in the Black Belt in 2020
L2 Data	Used to obtain nonracial voting turnout

	data for all of Alabama.
NIH	Map of proportion of non-English speakers in Alabama by county.
Public Libraries	Used to obtain estimates on the number of public libraries in each county in the Black Belt
Rural Institute	2019-2023 5-year American Community Survey, the 2020 Decennial Census, and the 2020 OMB Metropolitan-Micropolitan Statistical Areas delineation files.
State of Alabama Department of Revenue Motor Vehicle Division	Used to count the registered vehicles in each Alabama county
Statistical Atlas	Map showing demographic data per county in Alabama.
The M Transit	Used to check Montgomery county's transit system and verify if it's a fixed system
United States Census Bureau	Table showing demographic data per county in Alabama.
U.S. Census Blocks - Esri Federal Data	A map of all census blocks in Alabama
Rural Urban Continuum Codes	Used to obtain ranks for rurality of counties in the Black Belt
US Religion Census	Used to obtain an estimate of the number of congregations in each county
West Alabama Public Transportation	Used to verify if WAPT transit system was a fixed transit system or not. This system covers multiple Black Belt counties
2020 General Election Participation by Race	Used to obtain election participation by race per county

Community Colleges and Civics Report	Used to establish rationale for considering community colleges as an important indicator for civic participation
Resource Model for Civic Participation	Used to explore subindex groupings for our composite index
Creating Composite Indices Using ArcGIS: Best Practices	Used to inform general index development

Table 1. Sources used in index development. Source: BB-CPRI team, 2026.

From these sources, we were able to create a number of variables we thought would be useful when creating a civic participation risk index.

These variables include:

Name	Description	Importance
County	Name of county.	
Total Active Registered Voters	The total number of voters who are registered in each county.	
Disability	Percentage of disabled people in each county. Found from dividing disabled people in a county by total population in a county.	Disabilities may create barriers to voting, such as difficulty accessing polling locations, transportation challenges, or the need for additional assistance.
Age	Percentage of the population that is 65+.	According to the US Census, higher age is correlated to higher voter turnout due to higher income and access. Source
Education Level	Percentage of the population with no high	Lower educational attainment is typically

	school diplomas in each county.	associated with lower political participation, lower awareness of voting procedures, and greater vulnerability to access barriers
Gender	The distance of the female population percentage below 50% to measure male dominance.	Historically, women are more likely to vote over men, so a lower population of women was mapped to higher risk. .Source
Race Turnout Disparity	Pairwise disparity of turnout rates by race. Absolute values of turnout rate pairs are used to find the average.	If there is a wide gap between the voting turnout of one group vs another, that signifies lack of access and a higher risk in civic nonparticipation. To get our numbers, the turnout rate was calculated for each group by dividing the ballots cast with the population size. Then, the pairwise disparity was calculated, which is the absolute difference between every possible pair. Finally, all pairwise disparities were added together then divided by the number of comparisons.
Language Access	Relative proportion of individuals who are not able to communicate in English.	Used to quantify information access
Broadband Access	Relative proportion of	Used to quantify

	individuals who have access to a broadband internet connection.	information access
Registered Vehicles per County	The source was the Alabama Department of Revenue and their statistics sheet from 2023-2024 about the amount of registered vehicles in 2020 for each county.	Used to quantify mobility access
Fixed Public Transit?	Fixed public transit relies on fixed schedules and/or routes to operate public transit. Y is yes and N is no.	Used to quantify mobility access
Rural-Urban Continuum Ranking	Higher values of rurality means increased risk, meaning less access to public transportation and broadband.	Used to quantify mobility access and information access
Number of Physically Unhealthy Days	Average amount of sick days each month per county.	Used to quantify individual capability to vote
Distance to Nearest Polling Place	Used census blocks as origin points to measure the distance of polling places. Used Near analysis in ArcGIS to find the distance to the closest polling place. The closest distances were aggregated by each county by finding the median.	Used to quantify mobility access
Number of Congregations	Defined by the	Used to quantify

	Association of Statisticians of American Religious Bodies as "churches, mosques, temples, or other meeting places. The number of congregations in each county.	community engagement within counties
Number of Public Libraries	The number of public libraries in each county.	Used to quantify community engagement within counties
Number of Community Colleges	The number of community colleges in each county.	Used to quantify community engagement within counties

Table 2. Variables used in index development. Source: BB-CPRI team, 2026.

To elaborate on the "nearest polling place" variable, the Near tool works by using census blocks as starting points, and the second points that were added were the polling places. After the points are added, the Near tool was executed to find the closest distance for every census block. To make use of the distances found, we analyzed the nearest distance information on a dataset from ArcGIS and filtered it by Black Belt counties. After the filtered dataset was created, we found the median of all distances for each county. Median calculations were selected as the method due to their robustness against outliers, which can cause disproportionate results.

Once we gathered the data, we had to consider the weights of each county and the populations of each county. So we created more variables that would determine the proportion of a specific variable to the population of each county.

- Total Active Registered Voters → County Proportion of Active Registered Voters
- Registered Vehicles per County → Registered Vehicles Proportion of County
- Number of Congregations → Congregation County Proportion
- Number of Public Libraries → Library County Proportion
- Number of Community Colleges → Community College County Proportion

After gathering all data for the variables, we normalized them to put them on a similar scale. We started off with a min-max normalization method in order to build a consistent risk framework, in order to prepare the data for statistical analysis. We then used another normalization method, z-scoring, primarily because we had Montgomery County to account for. As the county holding the capital of Alabama, it had a significantly higher population than other counties in the dataset and served as an outlier. To mitigate the skew of that data point, we used z-scoring to shift the distribution of different county values for a certain category around the mean, equalizing the variance.

Principal Component Analysis

Having a great number of variables, we needed a way to categorize the variables in order to better interpret and understand the data. A PCA was conducted to identify patterns among variables and reduce redundancy among variables. This allowed related indicators to be grouped in a broader dimension of civic participation risk.

Many of the variables in the data set overlapped conceptually in broader categories, for example, transportation, infrastructure, and demographic variables. A PCA allowed us to establish those categories, making it easier to interpret. Instead of treating all variables separately, we were able to reduce the redundancy and combine similar variables into a single index. This was seen very clearly in the transportation variables, where we combined them into a single index to make the model cleaner. The outcome of the PCA defined meaningful categories that become interpretable dimensions, such as infrastructure access, demographic variables, geographic isolation, and individual capability.

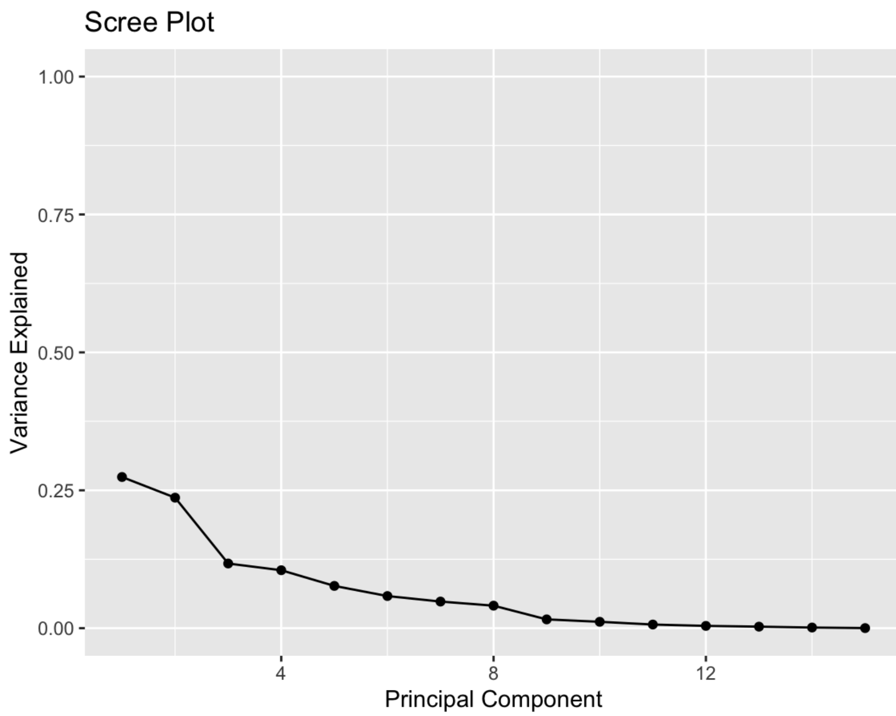


Figure 1. PCA analysis screen plot displaying variance explained by each principal component. Relying on the elbow method, our team decided to retain 4 principal components for our composite index. Source: R Studio, 2026.

Factor Analysis

After our initial data collection, our dataset included 15 variables to account for in the index. To the average person, it can be difficult to identify why each variable is important to the index immediately. Therefore, along with using PCA, our team also conducted a factor analysis (FA) to group variables for subindex creation and easier interpretation of the final composite index.

Before running the factor analysis, the Bartlett's Test of Sphericity and Kaiser-Meyer-Olkin factor adequacy test were run to check that the data met the assumptions required for the factor analysis. The Bartlett's Test was significant, showing that a data reduction technique can actually compress the data in a meaningful way. However, the KMO test gave a measure of sampling adequacy (MSA) score of 0.27, suggesting that only 27% percent of the variance in our variables was caused by underlying factors. Because of the first test's significance, we continued with the factor analysis and took the results with a grain of salt.

Using R and RStudio, a CSV of our raw data was loaded into the workspace and scaled with the scale() function from base R using the z-score technique. The fa() function from the psych package was used to do exploratory factor analysis on the resulting data using the default minimum residuals method. First, the number of factors was set to 15 (the maximum) to determine where the eigenvalues fell below 1. From that test, 11 factors had an eigenvalue of at least 1, but because the goal for the index was interpretability, the results from a factor analysis with 3 factors were ultimately used.

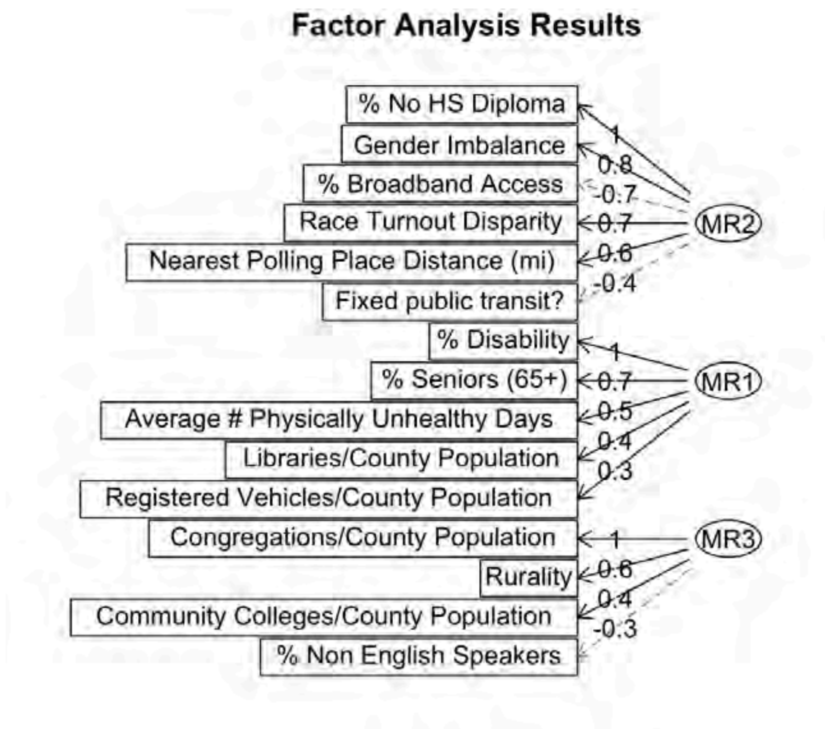


Figure 2. Factor analysis results grouping 15 variables measured across 17 Black Belt counties. Negative associations are shown with red dashed lines, positive associations with black solid lines. Numbers represent the relative importance of each variable for each factor. Source: R Studio, 2026.

From the factor analysis results, these three factors explained 56% of the variance in the dataset. The second factor, MR2, was the most significant factor out of the three, explaining 23% of the variance and including variables such as education level, gender, broadband access, race turnout disparity, nearest polling place distance, and whether the county had fixed public transit. When a county's education level decreases, its gender imbalance tends to increase, its broadband access tends to decrease, the difference in voter turnout between races is greater, polling places are farther away, and fixed public transit is more limited. Overall, these variables suggest that demographic and infrastructure variables influence a county's ability to participate in voting more than the other two factors, which can be defined as a mobility factor and a community access factor.

These factors explained 19% and 14% of the data's variance, respectively. For the mobility factor, when the county's disability population increases, so does the senior population, the average number of self-reported physically unhealthy days, the number of libraries, and the number of vehicles. For the community access factor, when there are more English speakers in a county, the number of congregations, the number of community colleges, and the rurality of that county tend to increase. The factor analysis was used to determine whether our variables could be grouped into broader factors meaningfully, and based on the low correlation between factors and the high regression scores within factors, we can conclude that grouping can be beneficial.

Index Construction

Our final composite index was created in ArcGIS using the Join Features, Model Builder, and Calculate Composite Index tools. Over the course of the development process, we created three versions of the index. While the weighting approach remained the same, changes in variables contributed to differences in the final county rankings.

We conducted a PCA analysis to determine groups of variables for subindices within the composite index. According to the analysis, these were the four factors, and the variables within them had significant correlations with that factor (loadings with an absolute value > 0.3):

- PC1: Community Resources
  - Number of physically unhealthy days: self-reported, per month
  - Proportion of community college campuses to county population
  - Proportion of congregations to county population
  - Broadband Access: percentage of the county with broadband access
  - Education Level: percentage of the county without a high school diploma
- PC2: Socioeconomic Status
  - Language Access: What percent of the county does not speak English
  - Race Turnout Disparity: pairwise disparity of turnout rates by race
  - Disability (%): percentage of the county with disability
  - Gender: whether the county population is male-dominated, female-dominated, or balanced
- PC3: Mobility and access issues
  - Age: percentage of the county population aged 65+

    - Proportion of Vehicles to County Population
    - Fixed public transportation?: binary
    - Number of metro stations
    - Distance to nearest polling place: calculated from county center
- PC4: Rural isolation
    - Rural Ranking: 1-9, 9 means most rural
    - Proportion of libraries to county population

Our reasoning for selecting these four categories pertains to their relevance in the subject of civic participation. Socioeconomic status assesses an individual's access to financial and social services. Community resources evaluate the availability of resources like educational services and technological access. Mobility and access issues identify obstacles pertaining to transportation and physical support. Rural isolation covers the separation of supportive systems like educational institutions, public transit, disability accommodations, etc.

For interpretability, we grouped the rural variable with PC3 and the library variable with PC1 because they had the second-strongest correlations in those categories.

Using ArcGIS, 3 subindices were made for each of the three principal components.

- Some of the variables were reversed in directionality in the platform, so that all of them meant a higher number was better.
- The weights within each of the subindices were calculated according to their loadings. The variable with the lowest loading got a weight of 1, and all other weights were calculated by dividing them by the lowest variable.

The three subindices were combined into one index; the socioeconomic factors were weighted at 0.4, while the community resources and mobility weights were each 0.3. This is because your demographic background plays a big role in your baseline information and access to information, and if people don't recognize a need to vote, there is no point in polling locations being physically accessible or having greater general knowledge due to nearby public resources and the internet (Brady et al., 1995).

For both the subindices and full index, the method to scale and combine variables was to combine scaled values (Mean of scaled values), which creates the index by scaling the input variables between 0 and 1 (minimum-maximum scaling) and calculating the mean of the scaled values. However, this first index iteration had two critical issues: the scaling and combination, as well as some individual variables.

Within our dataset, we created three separate versions of the data: one raw version with differences in units such as percentages, absolute numbers, and proportions, a normalized version using the min-max method, and a normalized version using the z-score method. We input the min-max normalization dataset into ArcGIS for the composite index calculation, but because the combination method also did scaling and normalization, the data was normalized twice, which meant the results were uninterpretable. Additionally, the gender variable was made so that balanced counties (counties with an equal population of male and female residents) had the highest value in the index, meaning that they would be the most likely to have access to vote. However, after further research, the census data about voting and registration for the November 2020 election showed that women registered and voted more than men (US Census Bureau, 2021). Therefore, the variable was edited so that counties with male populations higher than 50% received a numeric score that was their male population percentage - 50 to quantify the difference. Additionally, we changed the metro stations variable from an absolute number in the raw dataset to a proportion by county population in the process, because it was the only absolute count in the dataset, and it could have skewed the index results.

Version 2 of the index rectified these issues by changing the combination method to raw values and using the z-score dataset while reversing some variables to align directionality. By changing the combination to combine raw values (Mean of raw values) for both the subindices and composite index, we only had to reference the dataset to understand how our preprocessing affected the final composite index. Within ArcGIS, we used the built-in Reverse Direction feature for some of the variables so that all the variables signified better civic outcomes when their numbers were higher and aligned directionality according to an Esri technical paper for best practices when creating composite indexes in ArcGIS (Esri, 2024). However, due to a mistake in the metro stations variable, which actually represented water stations, we had to eliminate the variable from our dataset and recalculate the PCA to determine how the weights for the variables and subindices would change.

$$\textbf{Composite Index Score} = 0.38 \times \textsc{Mean}(\mathit{Age} \times 0.24 + \mathit{Broadband\ Access} \times 0.3 + \mathit{Fixed\ public\ transit} \times 0.20 + \mathit{Libraries/County\ Population} \times 0.26) + 0.32 \times \textsc{Mean}(-\mathit{Disability} \times 0.22 - \mathit{Education\ Level} \times 0.24 - \mathit{Gender\ Imbalance} \times 0.3 - \mathit{Race\ Turnout\ Disparity} \times 0.24) + 0.16 \times \textsc{Mean}(-\mathit{Rural\ Ranking} \times 0.34 + \mathit{Congregations/County\ Population} \times 0.28 + \mathit{Community\ Colleges/County\ Population} \times 0.21 - \mathit{Distance\ To\ Nearest\ Polling\ Place} \times 0.17) + 0.14 \times \textsc{Mean}(-\mathit{Language\ Access} \times 0.18 + \mathit{Registered\ Vehicles/County\ Population} \times 0.4 - \mathit{Physically\ Unhealthy\ Days\ Count} \times 0.42)$$

Figure 3. An equation showing the calculation of the civic participation risk composite index that covers the Black Belt counties. Weights were applied according to PCA, and ArcGIS combined raw values using the average within subindices and for the final index. Software may have adjusted final values through multiplication, as the minimum index value was set to 0 and the maximum index value was set to 100 for all steps. Source: ArcGIS, 2026.

Version 3, the final version of the composite index, had four subindices that explained 73% of the variance within the data. These subindices included infrastructure access, demographic variables, geographic isolation, and individual capability.

Infrastructure access (38%)	**Demographic variables (32%)**
- Age (% Seniors (65+)) (24%) - % Broadband Access (30%) - Fixed public transit? (20%) - Libraries/County Population (26%)	- % Disability (22%) - % No HS Diploma (24%) - Gender Imbalance (30%) - Race Turnout Disparity (24%)
Geographic isolation (16%)	**Individual capability (14%)**
- Rurality (34%) - Nearest Polling Place Distance (mi) (17%) - Congregations/County Population (28%) - Community Colleges/County Population (21%)	- % Non-English Speakers (18%) - Registered Vehicles/County Population (40%) - Average # Physically Unhealthy Days (42%)

Table 3. Breakdown of subindices and weight allocations based on PCA. Source: BB-CPRI team, 2026.

Within subindices, weights were calculated by summing the absolute values of the loadings in their principal components and then dividing a loading for an individual variable by that sum to get weights that sum to 1. Between subindices, their weight in the composite index was calculated by the proportion of the variance their principal components explained in the total variance from all principal components to get weights that sum to 1. PC1, the infrastructure access factor, explained 27.4% of the variance, PC2, the demographics factor, explained 23.7%, PC3, the geographic isolation factor, explained 11.7%, and PC4, the individual capability factor, explained 10.5% to total to 73.3% of variance explained. As an example, 27.4/73.3 is around 37.3%, but its weight was rounded up to 38% so that all the weights added to 1 and to account for rounding error. All this information can be found in the pcaRloadings sheet within the Civic Participation Factors Dataset deliverable.

Tools/Technology

Python was used for the principal component analysis(PCA), Excel for data collection, ArcGIS for Near analysis, as well as creating the composite index, and R Studio for factor analysis. Other tools that were used were React and TypeScript for developing the web application. In addition to the web application, GitHub was used as a repository for the web application's files.

Challenges

One of the challenges we encountered was the limitations of our variables. The number of public libraries had uncertainty about what the website considers as a boundary for counties. For the race group, some groups had fewer than ten people, which can cause disproportionate results. The severity of the disability is not accounted for in the disability variable, which can be important for assessing voter access. Also, there were no considerations of accommodations that could reduce the impact of disabilities. The physically unhealthy days variable is self-reported, so the responses are subjective and may vary by person. The values reflect short-term conditions and not long-term conditions.

There was some uncertainty in addressing the geographical portion of the project. At the time, only one member had installed and used the geographic information software ArcGIS, which can take a while to install. This can make it difficult to progress and collaborate since geographical knowledge is dependent on a single member.

How were they addressed?

All limitations from the variables are identified and flagged in the report and the Civic Participation Factors Dataset. The challenge with ArcGIS was addressed by creating an ArcGIS Online group where any user can share findings without installing the app. These groups increased accessibility, facilitated collaboration, and made it easier to share any updates.

Deliverables and Findings

- Civic Participation Dataset
    - **Description**: A spreadsheet table with all the variables we considered in order to assess civic participation in each county. This table meets the project goals because the required composite index that is created is built from data in this table.
    - **Link:** https://docs.google.com/spreadsheets/d/1Xbi_A5L1z0DOj2c9kfynB_2yYOw-ZWb8rhKKwPTJeCk/edit?gid=1305686700#gid=1305686700

- Composite Index
    - **Description**: A map showing the Black Belt counties and the risk involved in civic participation. This index meets the project goal because a composite index is required and is the central point of the project.
    - **Picture:**

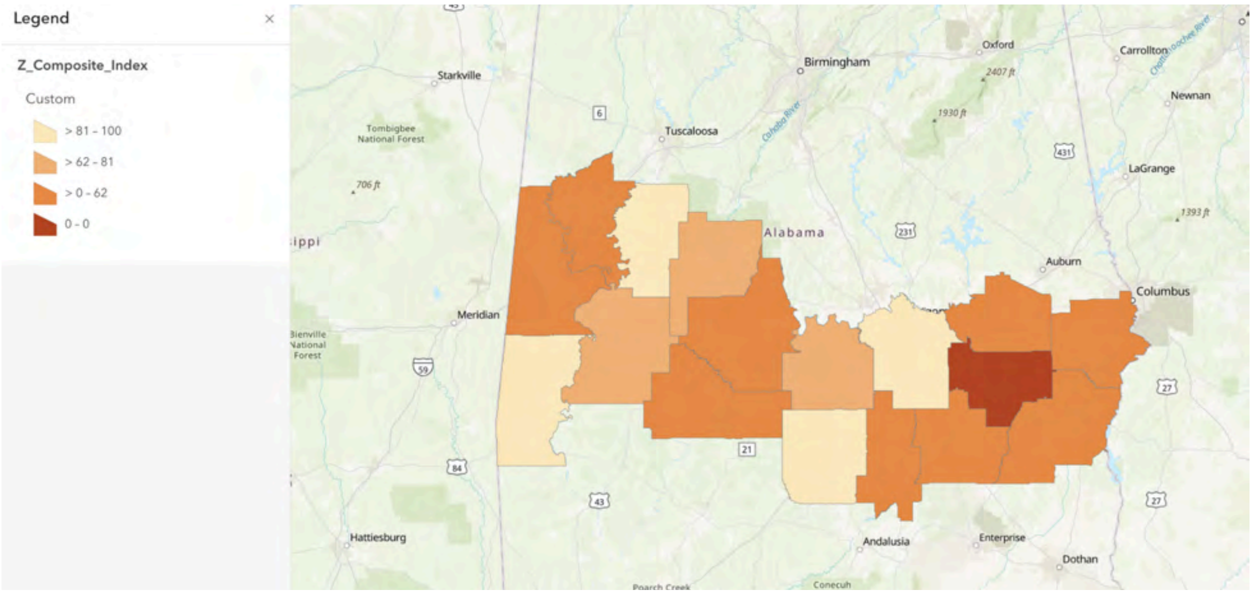


Figure 4. The civic participation risk composite index that covers the Black Belt counties. Counties with shades of green have a higher score and less risk. Counties that have shades of red and orange have lower scores which means higher risk from civic participation Source: ArcGIS, 2026.

- Web application
  - **Description:** An interactive web application connected to the composite risk index. The purpose of this app is to be a front-facing interface that provides an accessible experience for users by working with risk data. This application meets the project goal by being connected to the central composite index and is created for user interaction.
  - **Link:** https://github.com/vsayasa/risk-index

Findings:

From the overall composite index, the standout counties are Bullock, with a low index score of 0.00, signifying the highest risk of not participating in voting. Barbour has the second-lowest score of 46.31. Butler has a score of 100, meaning the least risk, and Montgomery is the second highest with 94.42.

Montgomery County, the largest county by population and home to Alabama's capital, ranked second in civic participation access despite a low subindex 4 ranking, which correlated to individual capability factors such as language access, mobility access through personal vehicles, and health status with self-reported physically unhealthy days. This suggests that while Montgomery's external resources, such as transportation and broadband access, provide good conditions for voting, individual citizens may be hindered by personal issues.

To expand on the subindex, this factor covers the individual capability of the county, which means what a single person can do. Having low risk and low individual capability can lead to many interpretations of a county. One example is that there is a fair amount of individuals from marginalized groups like those with disabilities, low income, ethnicity, etc. To add on to the interpretation, there could be stable support systems to accommodate people who are less equipped. For example, disability services, public transit, and low-income housing.

Looking at individual subindices also informs SPLC where counties are failing in some areas and succeeding in others. For instance, within the geographic isolation subindex that measured factors such as the proportion of community colleges in the county, the proportion of congregations in the county, polling place distance, and the rurality of the county, over half the counties were low risk, while a segment of counties in the southeast corner of the map were 4 out of the 5 highest risk counties in that category. This implies a possible gap in the public resources provided in that region, requiring more hands-on work to connect citizens to information.

Recommendations

Recommendations for SPLC

From the composite civic participation risk index, SPLC should focus its efforts on counties that exhibited high civic participation risks repeatedly throughout the various subindices. Counties that lack broadband access, have fewer transport options, experience longer distances to polling places, and have large racial disparities in voter turnout may face compounding barriers to increasing voter participation opportunities. From the final civic participation risk index, infrastructure and demographic-related barriers play the most crucial roles in decreasing civic participation rates. In turn, counties experiencing infrastructure or demographic deficits might see the greatest benefits from SPLC's interventions.

An essential recommendation for SPLC is that the organization should investigate the issue of Black Belt counties lacking adequate transportation systems. According to the findings, the distances to polling places as well as the lack of fixed public transit systems were two variables highly correlated with civic participation risks. This means that counties lacking adequate transportation might need additional outreach and advocacy efforts, perhaps including mobile civic education programs, partnering with transport companies, or advocating for polling place accessibility issues.

Broadband access and education level were shown to be two variables that play an important role in determining participation risks. Counties that lack sufficient broadband access also had low levels of educational attainment and higher levels of turnout disparities among racial groups. It may be beneficial for SPLC to explore digital civic engagement tools in these areas as part of future initiatives, for instance, voter education on the Internet, multilingual outreach programs, etc.

Another recommendation is that SPLC continue investigating data gaps that exist concerning accessibility to polling places for voters with disabilities and voters who speak languages other than English. Although the database provided estimates for these categories, they lacked information about the severity of disabilities and quality of accommodations, as well as translation services available in different counties. Such limitations affect the reliability of the index. Future efforts at gathering additional localized data may help overcome this problem.

Lastly, SPLC needs to keep in mind that the proposed civic participation risk index is a decision-making tool, not a definitive evaluation of civic participation barriers. According to factor analysis, about 56% of the variance is accounted for by the underlying factors identified in this study. Thus, there is still a need for investigating potential social, economic, or political factors impacting participation results. The index will work best for pattern recognition, county selection for further investigation, and future advocacy planning.

Recommendations for Future Project Teams

Future teams should further refine the civic participation risk index. This can be done through improvements in data collection and selection from more recent years. In the present case, many variables needed significant preprocessing and interpretation to be successfully added to the index. Transportation data, for instance, required extra validation to determine the presence of fixed transit systems in some counties, whereas library numbers were based on how county boundaries were interpreted outside. Additionally, the directionality of the variables in the index was based on research, but a correlation test could be conducted between the variables and the county proportion of active voters to determine whether each variable is positively or negatively correlated. Standardized data processing methods should be used early in the research process to minimize inconsistencies.

Future teams should add more variables, especially those that can increase geographic granularity. The present analysis is based on county-level data, which does not account for variations existing within counties. Census tract-level variables such as transportation accessibility, polling places density, Internet reliability, and more granular disability and language data would be beneficial for making the index more accurate. Additionally, information about voting resources available for non-English speakers could also help measure barriers to civic participation better.

Although the present principal component analysis was effective for constructing subindices, future teams should conduct additional tests involving alternative weighting methods. The sample adequacy test revealed relatively low results, indicating that some correlations between variables cannot be explained by underlying factors only. Thus, future efforts in this area might include using larger datasets, adding more counties, and experimenting with alternative dimensionality reduction techniques.

In regard to workflow, future teams should start data collection and preprocessing early in the project timeline. Much of the timeline of this particular effort was spent looking for reliable public datasets, verifying their accuracy, and preparing for transformations needed for calculating civic participation risk indices. Keeping a data dictionary that documents all data processing steps and decisions would be highly beneficial for new capstone groups. This way, teams would spend less time onboarding.

Last but not least, future teams should update the civic participation risk index periodically. Civic participation barriers keep changing as infrastructure develops, policies change, demographic profiles fluctuate, and transport systems expand. Regularly updating the index would be useful for monitoring changes in civic participation and evaluating how SPLC's interventions impacted Black Belt counties over time.

Conclusion

Through synthesizing data collected across several variables, we were able to compile them into indices, allowing the SPLC to visualize several factors that serve as barriers to civic participation across several Alabama counties, thanks to the successful construction of the Composite Civic Participation Risk Index (BB-CPRI). One of the most significant findings from the index was the strong influence of infrastructure and demographic-related variables on civic participation risk across the Black Belt. The PCA allowed us to analyze variable groups as a whole to identify these influential indicators. Standout variables such as broadband access, educational attainment, transportation accessibility, and racial turnout disparities appeared repeatedly throughout our analysis, indicating larger regional challenges rather than being isolated to an individual county. These findings suggest that allocating regional resources towards elements such as expanding transportation systems, increasing broadband access, and reducing informational barriers can improve civic participation among many counties across the Black Belt. The region's demographics can't be changed, but the index helps inform SPLC what types of information different counties need to fill in their specific gaps. By compiling the collected data into a single index, the BB-CPRI can be presented to the public, allowing them to see how their participation differs from one another.

Contacts

Michelle Akem: michelleakem6@gmail.com

- Ask about ArcGIS index development, factor analysis

Vonn Sayasa: vonnsayas@gmail.com

- Ask about web application development and race turnout custom variable

Noah Lee: noahdlee03@gmail.com

- Ask about civic participation factors dataset, PCA

Eli Jean: daveyjean@icloud.com

- Ask about ArcGIS, polling place distance custom variable, deliverables, and findings

Ethan Schwartzberg: ethanschwartzberg@gmail.com

- Ask about recommendations, web application design

Deliverables Checklist

Deliverable	Status	Link
Civic Participation Factors Dataset	Complete	BB-CPRI-Composite...
Composite Index	Complete	Composite Index
Web Application	Complete	Web Application

References

Brady, H. E., Verba, S., & Schlozman, K. L. (1995). Beyond SES: A Resource Model of Political Participation. *The American Political Science Review*, *89*(2), 271–294. https://doi.org/10.2307/2082425

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US Census Bureau. (2021). *Voting and Registration in the Election of November 2020*. Census.Gov. https://www.census.gov/data/tables/time-series/demo/voting-and-registration/p20-585.html

Voting & Civic Participation

Alabama Black Belt Counties

Team 8 BB-VIZ: Data Visualization for Voting Access and Civic Infrastructure

Prepared for: Southern Poverty Law Center (SPLC)

Team Members: Hawareyawe Ketema, Eyoha Henoke, Getnet Workalemahu, Milca Teklezgi, Michael Melaku

Course: INST490 · Section: 0103

Date: May 18, 2026

Team 8: BB-VIZ

Team Brief

Section Brief

The BB-VIZ team designed and built an interactive, client-facing dashboard that organizes county-level voting access and civic infrastructure data for the Southern Poverty Law Center. Rather than leaving SPLC staff to navigate raw spreadsheets or disconnected data files, the dashboard provides a single interface for exploring election results, turnout patterns, candidate margins, voting methods, and county-level comparisons across the Alabama Black Belt and Gulf Coast regions. The team also developed a consistent design system—colors, typography, layout, and reusable components—so that future teams can extend the dashboard without breaking its visual consistency. This work matters to the overall project because it serves as the public-facing layer through which SPLC can communicate and act on findings produced by all other teams.

Key Findings

- A dashboard format can make voting access and civic infrastructure data substantially easier to understand, compare, and communicate than raw spreadsheets or written summaries alone.
- Data availability and cross-team consistency are the primary constraints on dashboard reliability: sections that depended on finalized datasets from other teams were completed using sample or semi-cleaned data, and those sections should be updated as final data become available.
- Simplicity improves usability for civic data dashboards; a smaller number of clear, accurate visuals outperforms a larger number of cluttered ones for the audiences SPLC serves.
- The dashboard's county comparison functionality—allowing side-by-side analysis of vote composition, margins, and turnout across counties—is its most immediately actionable feature for internal SPLC planning.

Deliverables Produced

- **Interactive BB-VIZ Dashboard:** Live web application displaying county-level election and civic infrastructure data.https://inst490-project-swo8.vercel.app/
- **County-Level Visualizations:** Maps and charts embedded in the dashboard showing vote composition, victory margins, turnout change, and voting methods by county.
- **Data Files/CSVs:** Organized source files used to populate dashboard visuals, drawn from 2024 general election results and BB-VOTE team datasets.
- **Dashboard Design System:** Documented fonts, colors, layout rules, and reusable components for visual consistency and future extensibility.
- **Client Handoff Documentation:** Guide explaining dashboard structure, data sources, known limitations, and continuation instructions for future teams.
- **Source Code:** Full codebase (HTML, CSS, JavaScript, Python) available on GitHub. github.com/eyohah/inst490-project

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Challenges

Data availability was the most significant obstacle: because the dashboard depended on finalized datasets from other teams, some sections were built and tested with sample or semi-cleaned data, and final integration was constrained by what was available at time of deployment. A secondary challenge was scope management—the team had to make deliberate decisions about which indicators to include, prioritizing a stable core dashboard over a more expansive but less reliable one.

Connections to Other Teams

BB-VOTE: Primary data source; BB-VOTE's county-level turnout and race participation datasets were used directly to populate dashboard visualizations.

BB-DIG: Standardized CSV structure and FIPS-coded datasets provided the data foundation the dashboard was designed around.

All BB teams: The dashboard is designed to serve as the visual output layer for findings produced across the full 11-team regional analysis; as other teams' datasets are finalized, they can be integrated into the existing dashboard structure.

Final Report and Deliverables

SPLC BB-VIZ: Voting Access and Civic Infrastructure Dashboard

Team Members

Hawareyawe Ketema, Eyoha Henoke, Getnet Workalemahu, Milca Teklezgi, Michael Melaku

Client and Project Overview

This project was completed for the Southern Poverty Law Center (SPLC) as part of a broader regional analysis focused on voting access, civic infrastructure, and public data in Alabama's Black Belt and Gulf Coast regions. The BB-VIZ team's main goal was to create a client-facing dashboard that organizes county-level information into a clear visual format. Instead of leaving users to interpret raw spreadsheets or disconnected data files, the dashboard provides a more accessible way to explore patterns in voting access, changes in turnout, and civic infrastructure.

Data and Methodology

The team worked with county-level datasets connected to voting access and civic infrastructure, including data from SPLC, BB-Vote, public election sources, and supporting datasets provided or referenced during the capstone project. After gathering the data the team cleaned and organized, then structured it into usable files for dashboard development. We also created a consistent design system, built dashboard components, tested visuals with sample or semi-cleaned data, and deployed the project as an interactive site.

Key Findings

The main finding from this project is that a dashboard format can make voting access and civic infrastructure data easier to understand, compare, and communicate. The project also showed that data availability and consistency are major factors in how reliable and complete the final dashboard can be. While the dashboard helps identify patterns and possible areas for closer attention, it should be treated as a starting point for further investigation rather than a final conclusion. This follows the style of the example report, which combines summary, methods, findings, limitations, and visual deliverables into one client-facing document.

Project Context

The BB-VIZ team contributed to the broader 11-team SPLC regional analysis by focusing on the visual presentation of voting access and civic infrastructure data. While other teams focused on collecting, analyzing, or interpreting different parts of the regional voting landscape, BB-VIZ worked on turning data into a dashboard that could be used by SPLC and future project teams. Our work focused on Alabama's Black Belt and Gulf Coast region, with attention to county-level patterns and access-related indicators.

This project depended on data and findings from SPLC, BB-Vote, public data sources, and other project teams working in the same regional analysis. Because the larger project included multiple teams, our dashboard was designed to support cross-team use by presenting information in a clear, structured, and client-facing format. Other teams may be able to use the dashboard to display their findings, compare county-level trends, or connect their own deliverables to a larger regional picture.

The dashboard is not intended to make partisan claims or provide final explanations for voter behavior. Instead, it supports SPLC's work by organizing voting access and civic infrastructure information in a way that is easier to explore and communicate. The project's value comes from making data more usable, visible, and understandable for future analysis.

Methods

Project Approach

The BB-VIZ team approached the project by focusing on three main areas: data organization, dashboard design, and client-facing usability. Our first step was to understand the project goals and identify what kind of information SPLC and future users would need from the dashboard. From there, we reviewed available datasets, discussed what information could realistically be visualized, and planned a dashboard structure that would support county-level analysis.

The team worked in phases. First, we reviewed the project requirements and available data. Second, we created early dashboard ideas and design concepts. Third, we began developing the dashboard with sample or available data. Fourth, we refined the dashboard's layout, fonts, colors, and visualization structure. Finally, we prepared the final report, documentation, and deliverables folder for client handoff.

Data Sources

The team used the following data sources:

- 2024-General 2nd Congressional District

The dashboard was built around county-level data because county comparisons were most useful for the project's scope. In some cases, final datasets were not fully available at the time of dashboard development, so the team used sample or semi-cleaned data to continue building and testing the dashboard structure.

Tools Used

The team used the following tools during the project:

- Code: HTML, Python, CSS and JavaScript
- Deployment: Vercel and Github
- Dataset: CSV Files
- Design: Canva
- Collaboration: Google Drive and Github
- https://github.com/eyohah/inst490-project

These tools helped the team manage development, design, documentation, and project coordination.

Dashboard Development Process

The dashboard development process began with deciding what information should be displayed and how users should interact with it. The team wanted the dashboard to be simple enough for a client or future student team to understand quickly, but detailed enough to support meaningful analysis.

The dashboard includes sections for Current Leader, Candidate Totals, County Comparison, County Victory Margins, Highest-Volume Precincts, Voting Methods, and County Vote Composition. Each section was designed to answer a specific user need, such as comparing counties, viewing trends, or identifying access-related differences.

The team also worked to create a consistent user experience. This included selecting a color palette, organizing dashboard sections, standardizing chart layouts, and making sure the interface was not overloaded with too much information at once.

Challenges and How They Were Addressed

One major challenge was data availability. Because the project depended on multiple sources and some cross-team deliverables, the team did not always have complete or finalized data at every stage. To address this, we used sample or semi-cleaned data to build the dashboard structure while leaving space for final data to be added later.

Another challenge was scope. Voting access and civic infrastructure are large topics, and it would be easy to include too many variables or dashboard sections. The team addressed this by focusing on a smaller number of clear dashboard elements that could realistically be completed and explained by the end of the semester.

A third challenge was keeping the dashboard neutral and responsible. Since voting access can be politically sensitive, the team avoided partisan framing and focused on documented access-related data. The dashboard is meant to support exploration and decision-making, not make unsupported claims.

A final challenge was team coordination. Different team members were responsible for project management, communication, coding, design, and documentation. The team addressed this by assigning roles, tracking progress, and focusing on deliverables that supported the final client handoff.

Deliverables and Findings

Deliverable 1: Interactive BB-VIZ Dashboard

Description

The main deliverable for this project is the interactive BB-VIZ dashboard. The dashboard organizes county-level voting access and civic infrastructure data into a visual format. It is designed to help SPLC, future capstone teams, and other stakeholders explore regional patterns more easily.

Rationale

This deliverable meets the project goal because it turns complex data into a more usable client-facing tool. Instead of relying only on spreadsheets or written summaries, users can view data through maps, charts, and dashboard sections. This makes it easier to identify possible patterns, compare counties, and decide where further investigation may be needed.

Content

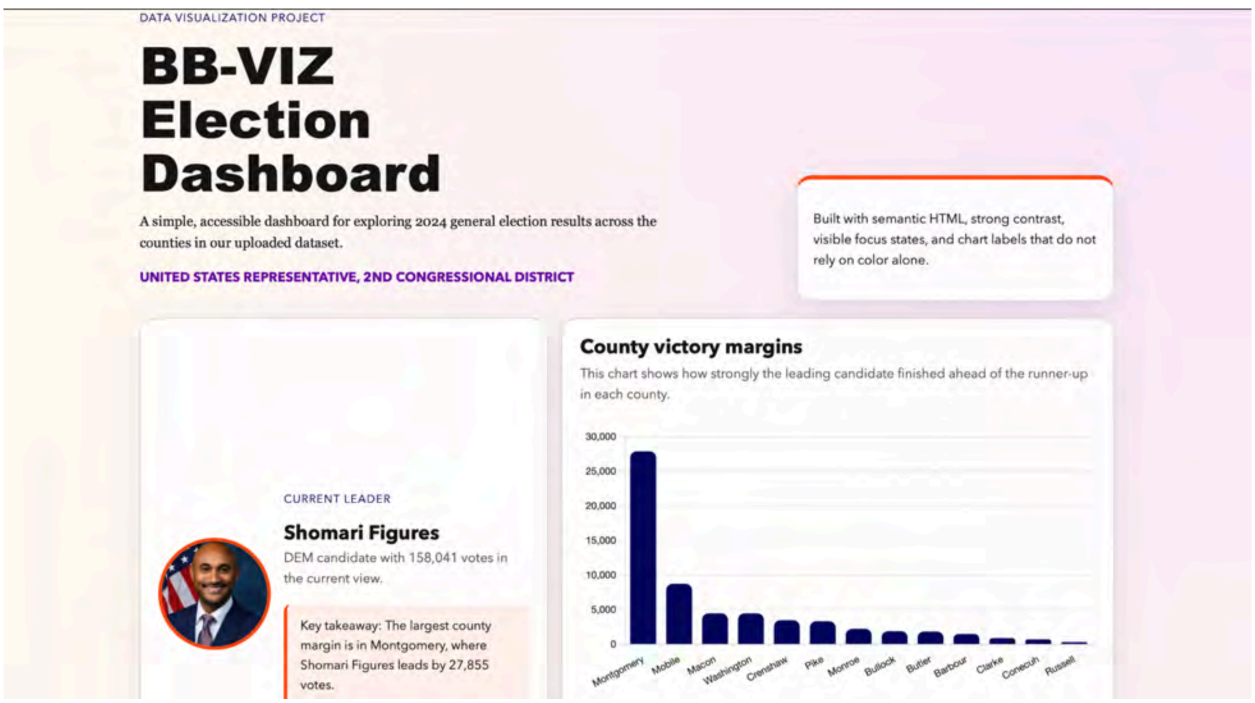

Figure 1. BB-VIZ Dashboard Homepage.
This figure shows the main landing page of the BB-VIZ dashboard. The homepage introduces the project and provides users with a starting point for exploring county-level voting access and civic infrastructure information.

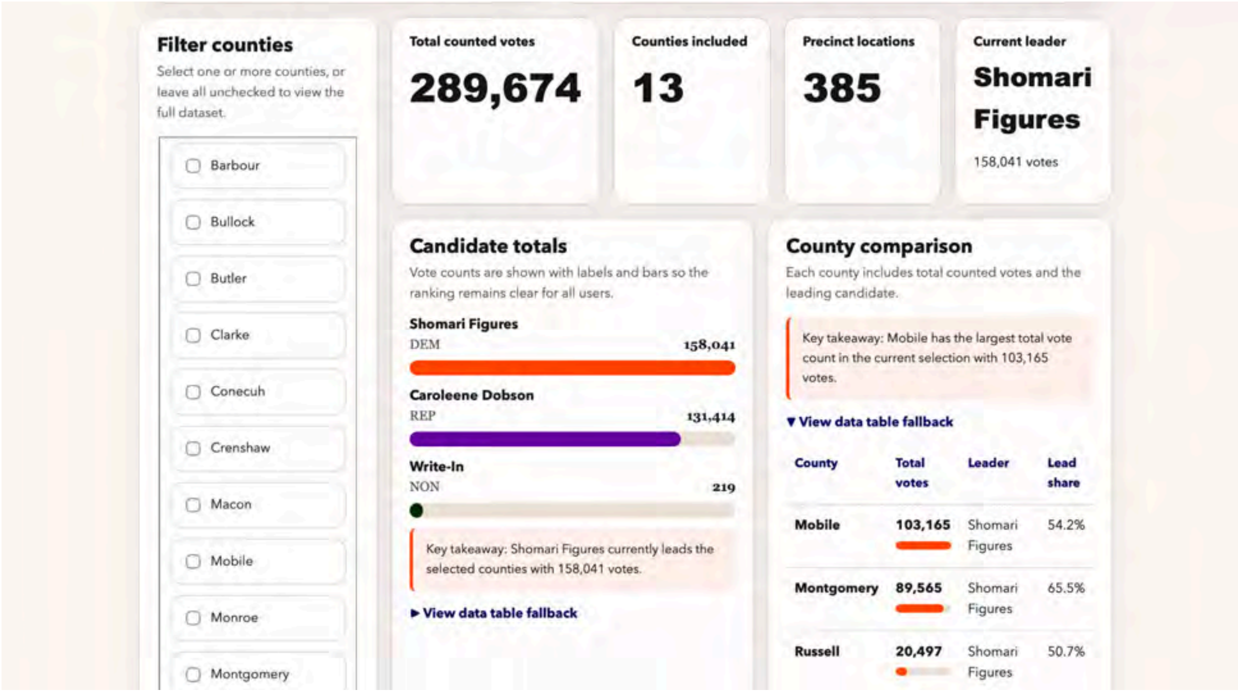

Figure 2. Dashboard Navigation and Layout.
This figure shows how the dashboard is organized so users can move between major sections of the project.

Interactive Dashboard Link: https://inst490-project-swo8.vercel.app/

Deliverable 2: County-Level Data Visualizations

Description

The dashboard includes visualizations that display county-level data related to voting access, turnout change, civic infrastructure, or other relevant indicators. These visuals help users compare counties and recognize patterns that may not be obvious from raw data alone.

Rationale

Visualizations are important because they make public data more understandable. For SPLC, this can support communication, planning, and future research. For future project teams, the visualizations provide a foundation that can be expanded with more complete data or additional variables.

Content

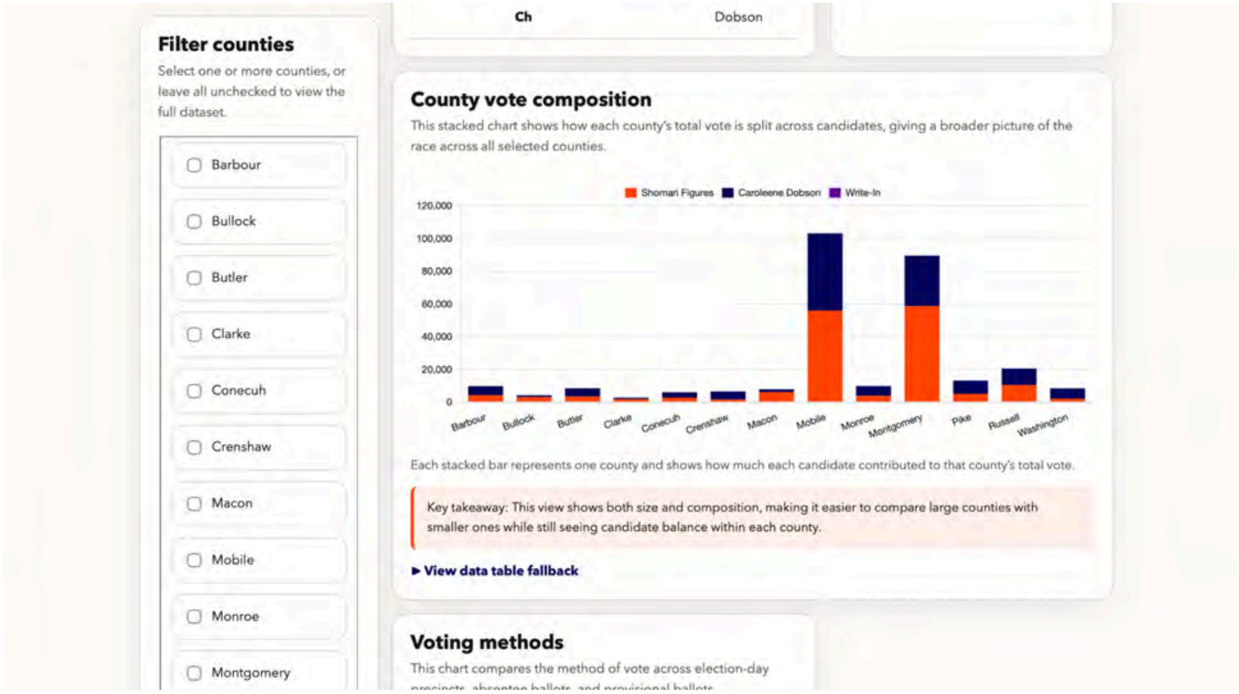

Figure 3. County Vote Composition Visualization.
This figure shows how each county's total vote is divided across candidates, giving users a broader picture of both county size and candidate distribution.

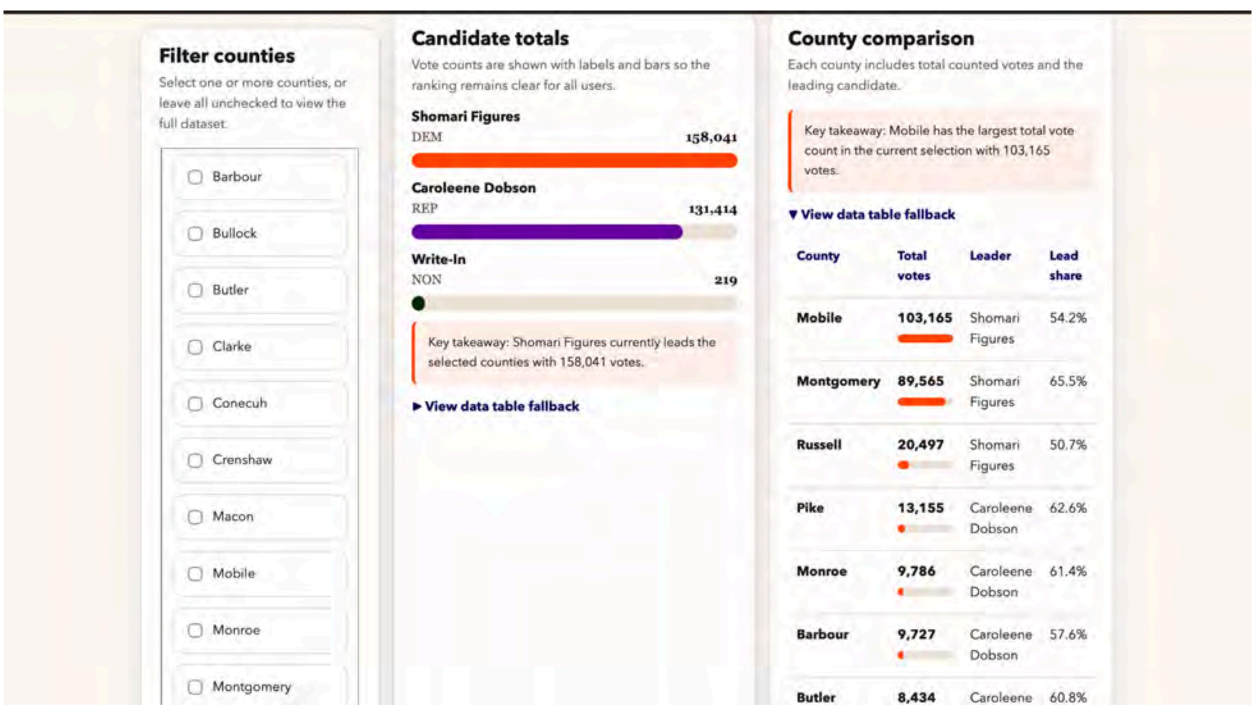

Figure 4. County Comparison Chart.

This figure shows a chart comparing selected counties across one or more access-related measures.

Figure 5. County Victory Margins Chart.

This figure shows the vote margin between the leading candidate and the runner-up in each county, helping users identify where the race was most decisive.

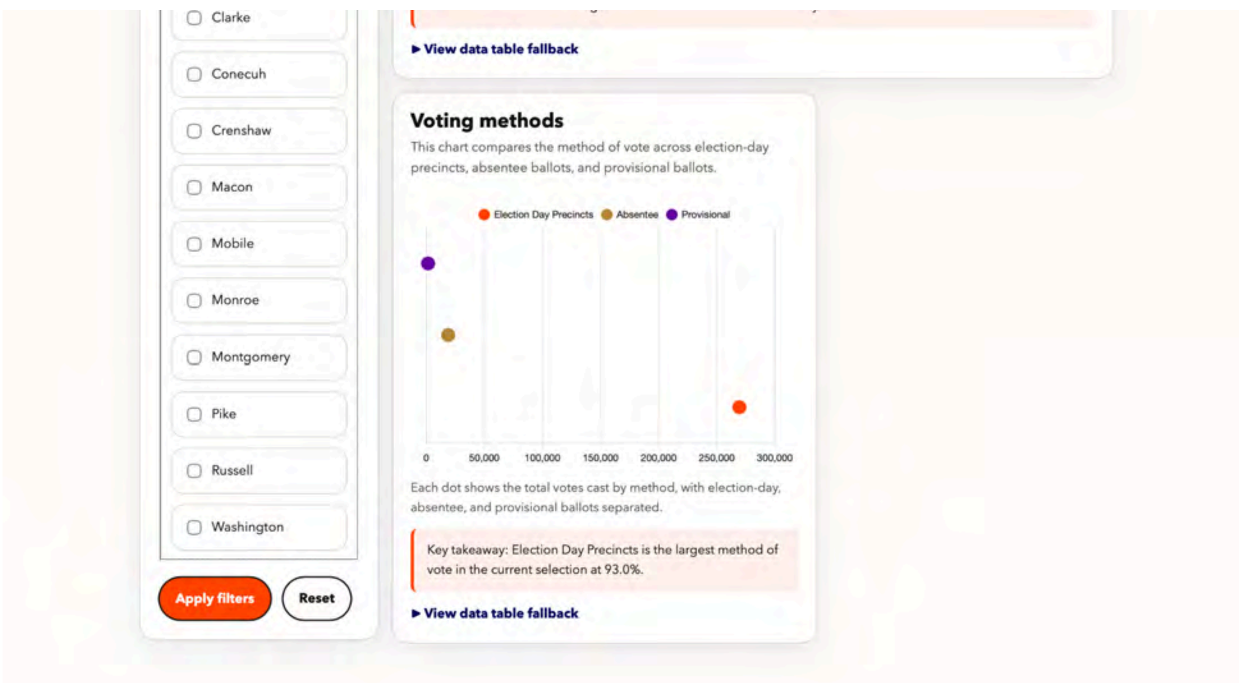

Figure 6. Voting Methods or Civic Infrastructure Visualization.

This figure shows how the dashboard presents one of the project's key indicators.

Findings

The dashboard shows that visual organization can make county-level differences easier to notice. While the dashboard does not prove why certain patterns exist, it helps users identify where closer attention may be needed. For example, users may be able to compare counties based on total votes, victory margin, and candidate vote composition.. These comparisons can help SPLC decide where additional research or outreach may be useful.

Deliverable 3: Data Organization and CSV Structure

Description

The team organized project data into structured files that support the dashboard. These files were used to connect data to dashboard visuals and maintain consistency across the project.

Rationale

A dashboard depends on organized data. If the data is unclear, incomplete, or inconsistent, the dashboard becomes harder to update and less reliable. By organizing the data into structured files, the team created a foundation that future teams can continue building on.

Content

Contest Title	Party	Candidate	101 WILSON COMM & ATHLETIC CTR	102 VAUGHN PARK CH	103 MUSEUM OF FINE ARTS	104 WHITFIELD METHODIST	105 ALDERSGATE METHODIST	106 CITY OF REFUGE CH	107 TRENHOLM COMM COLLEGE	201 ST PAUL AME CH	202 BEULAH BAPT CH	203 HAYNEVILLE RD COMM CTR	204 REBIRTH CHRISTIAN MINISTRI
UNITED STATES REPRESENTATIVE, 2ND CONGRESSIONAL DISTRICT	DEM	Shomari Figures	1385	2484	1697	1350	2376	1967	1010	1804	1875	709	2792
UNITED STATES REPRESENTATIVE, 2ND CONGRESSIONAL DISTRICT	REP	Caroleene Dobson	1809	1233	644	232	393	83	46	434	67	114	148
UNITED STATES REPRESENTATIVE, 2ND CONGRESSIONAL DISTRICT	NON	Write-in	7	7	0	0	3	5	1	3	2	1	1
UNITED STATES REPRESENTATIVE, 2ND CONGRESSIONAL DISTRICT		Over Votes	0	1	0	1	0	1	0	0	0	4	0

Contest Title	Party	Candidate	205 SOUTHLAWN BAPT	206 TWIN GATES COMM CTR	207 HUNTER STATION CC	208 STONETANK ANTIOCH BAPT	209 1ST SOUTHERN BAPT	210 PINTLALA FIRE DEPT	211 RUFUS LEWIS LIBRARY	212 MACEDONIA BAPT CH	801 DALRAIDA CH CHRIST	802 EASTERN HILLS BAPT
UNITED STATES REPRESENTATIVE, 2ND CONGRESSIONAL DISTRICT	DEM	Shomari Figures	1359	1064	204	327	89	401	616	1609	1394	922
UNITED STATES REPRESENTATIVE, 2ND CONGRESSIONAL DISTRICT	REP	Caroleene Dobson	82	102	88	508	192	789	26	207	1500	741
UNITED STATES REPRESENTATIVE, 2ND CONGRESSIONAL DISTRICT	NON	Write-in	2	1	0	2	0	0	0	2	4	1
UNITED STATES REPRESENTATIVE, 2ND CONGRESSIONAL DISTRICT		Over Votes	0	0	0	0	0	0	0	0	0	1

Montgomery General CD-2

Figure 7. Sample Data File Structure.
This figure shows an example of the data structure used to support the dashboard. Organized data files make it easier to update and maintain the project.

Field Name	Description	Source	Notes
Contest Title	Name of the election contest being reported in the workbook	County-level election result workbook / public election source	In this project, the files show the United States Representative, 2nd Congressional District race
Party	Party affiliation associated with the candidate or result row	County-level election result workbook	Examples include DEM, REP, and NON; helps group and interpret candidate records
Candidate	Candidate name or ballot result category	County-level election result workbook	Includes named candidates such as Shomari Figures and Caroleene Dobson, as well as Write-In, Over Votes, and Under Votes
Precinct / Voting Location Columns	Vote totals reported for each precinct, polling place, or ballot category	County-level election result workbook	Each column after Candidate contains vote counts by location or category; these columns were used to build county comparisons, voting method summaries, and composition charts

Findings

The data organization process showed that future teams will need clear field definitions and source tracking. Some datasets may use different formats, naming conventions, or geographic boundaries. A shared data dictionary would make the project easier to continue and reduce confusion during future dashboard updates.

Deliverable 4: Dashboard Design System

Description

The team created a consistent design style for the dashboard. This included choices related to layout, fonts, colors, spacing, icons, and chart presentation.

Rationale

The dashboard needed to look professional and be easy to understand. Since the project is client-facing, the design had to support trust, readability, and accessibility. A consistent design system also makes the dashboard easier for future teams to expand without making it look disconnected.

Content

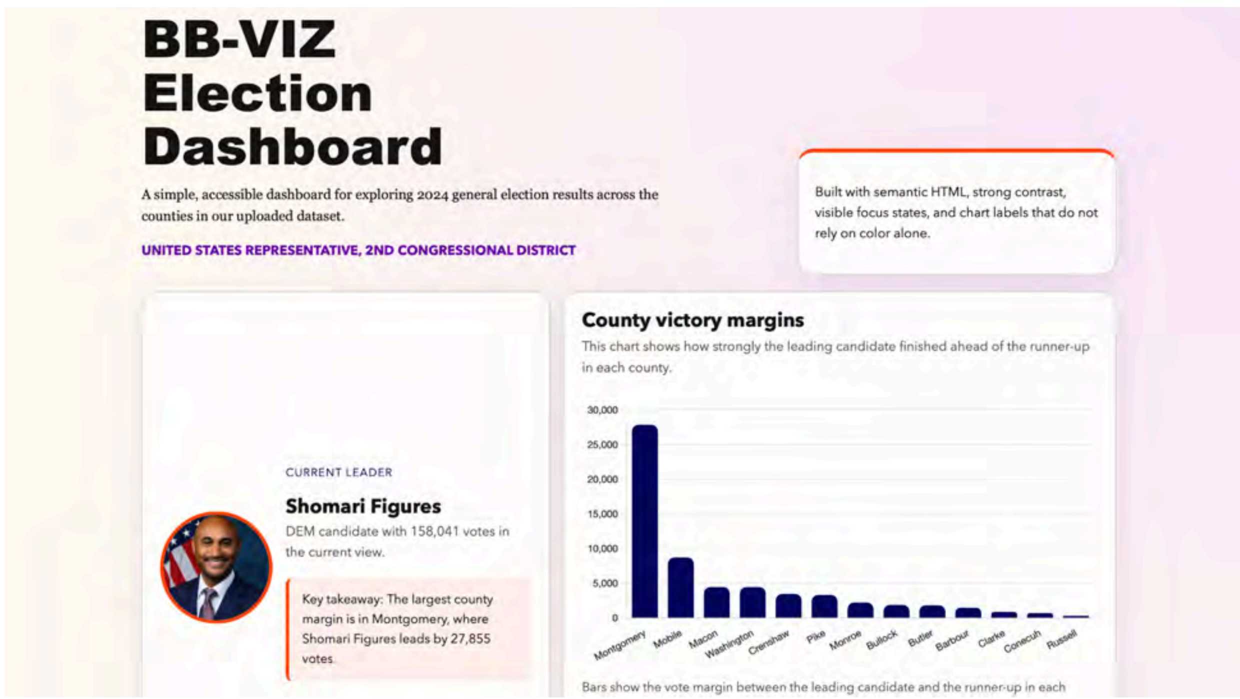

Figure 8. BB-VIZ Design Style.
This figure shows the dashboard's visual design choices, including colors, typography, and layout structure.

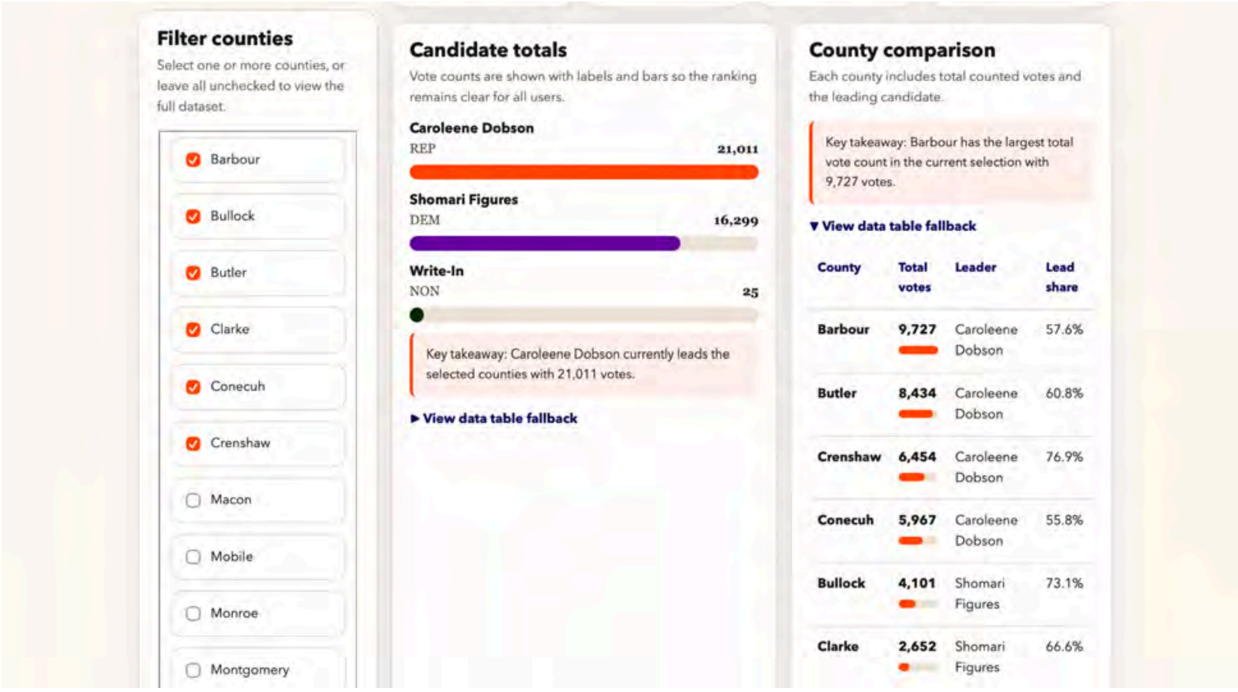

Figure 9. Reusable Dashboard Component.
This figure shows an example of a reusable visual component designed for consistency across the dashboard.

Findings

The design process showed that simplicity is important for civic data dashboards. Too many visuals, colors, or sections can make the dashboard harder to use. A clean layout helps users focus on the data and reduces the chance of misinterpretation.

Deliverable 5: Client Handoff Documentation

Description

The team prepared documentation to explain the dashboard, data structure, and future continuation of the project. This documentation is meant to help SPLC and future teams understand how the dashboard works and what still needs to be improved.

Rationale

Client handoff documentation is important because the project should remain useful after the semester ends. Without documentation, future teams may have to spend extra time figuring out how the dashboard was built, where the data comes from, and what limitations exist.

Content

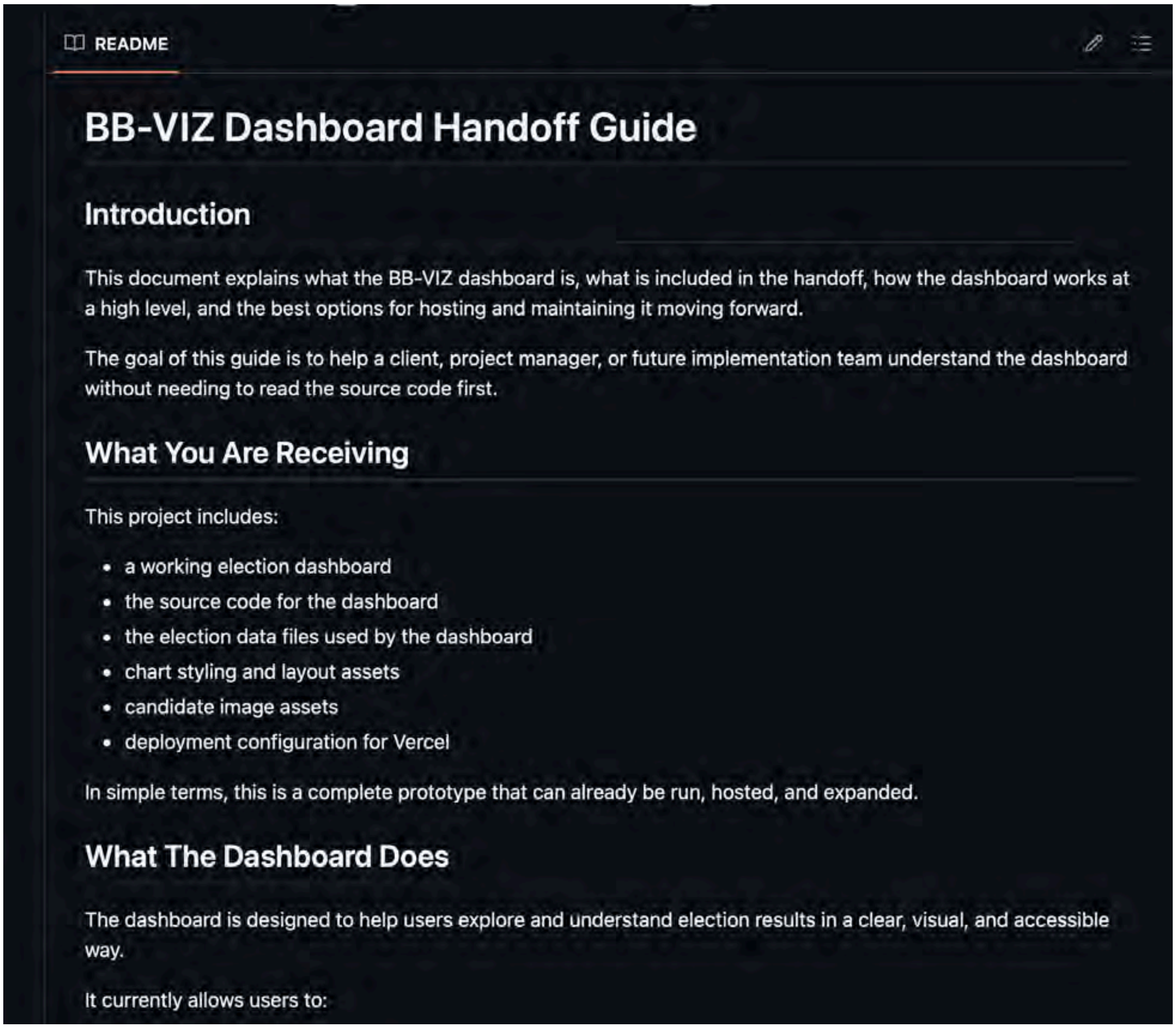

Figure 10. Client Handoff Documentation Example.
This figure shows part of the documentation prepared to support future use of the BB-VIZ dashboard.

Findings

The documentation process showed that future teams need clear instructions, file organization, and notes about limitations. The more clearly the current team documents the project, the faster future teams can continue the work.

Limitations

One major limitation was time. Like the example report explains for its own team, capstone teams have to balance project goals with course deadlines, client needs, and other responsibilities. Our team had to make decisions about what could realistically be completed within the semester.

Another limitation was data availability. Some datasets were incomplete, still being developed, or dependent on other teams. This affected how much the dashboard could show in its final version. Because of this, some dashboard sections may use sample, semi-cleaned, or placeholder data until final datasets are confirmed.

A third limitation was data consistency. Different sources may use different formats, field names, geographic definitions, or reporting methods. This can make it difficult to compare counties directly without additional cleaning and documentation.

A fourth limitation was interpretation. The dashboard can help identify patterns, but it does not explain every cause behind those patterns. For example, if one county appears to have a different access-related measure than another, additional research would be needed to understand why.

Finally, the dashboard should be viewed as a foundation rather than a finished long-term product. It provides a useful structure for SPLC and future teams, but it will need continued updates, stronger data validation, and additional user testing to become more complete.

Recommendations

Recommendations for SPLC

SPLC should use the BB-VIZ dashboard as a starting point for identifying counties or access-related issues that may need closer attention. The dashboard can help show where patterns appear across counties, but it should be paired with local knowledge, additional research, and validated data before major conclusions are made.

SPLC should also prioritize consistent data standards across future regional analysis projects. A shared data dictionary, clear source list, and version history would make it easier for teams to build accurate dashboards and reports. This would also reduce the amount of time future teams spend cleaning and interpreting data.

SPLC should consider continuing the use of dashboards and visual tools for civic infrastructure work. Visual tools can make complex public data easier to communicate to staff, partners, and community stakeholders. The BB-VIZ dashboard shows how regional data can be organized visually, but future versions could become stronger with more complete data and additional interactive features.

SPLC should also consider identifying which dashboard indicators are most useful for its work. For example, the client may want to prioritize turnout-related county differences, voting method distribution, or high-volume precinct activity. Choosing priority indicators would help future teams focus on what matters most instead of adding too many unrelated features.

Recommendations for Future Project Teams

Future teams should begin by confirming data sources as early as possible. Before building visuals, the team should know which datasets are final, which are still drafts, and which fields need cleaning. This would prevent major changes later in the semester.

Future teams should also create a data dictionary at the beginning of the project. The data dictionary should define each field, explain where the data came from, and note any limitations. This would make the project easier to understand and continue.

Future teams should keep the dashboard simple before expanding it. A smaller number of clear and accurate visuals is better than a large number of confusing visuals. Once the core dashboard is stable, future teams can add more counties, more indicators, or more advanced filters.

Future teams should also test the dashboard with real users. This could include SPLC staff, course instructors, classmates, or community partners. User feedback would help identify confusing sections, missing explanations, or accessibility issues.

The natural continuation of this project would be to expand the dashboard with finalized data, connect it more directly to other teams' deliverables, improve accessibility, and add stronger documentation for long-term maintenance.

Conclusion

The BB-VIZ team completed a client-facing dashboard and supporting deliverables focused on voting access, civic infrastructure, and public data in Alabama's Black Belt and Gulf Coast region. The project helped turn complex county-level information into a clearer visual format that SPLC and future teams can build on.

The team's work included data organization, dashboard development, visual design, documentation, and project reporting. While the project faced challenges related to time, data availability, and scope, the final deliverables provide a strong foundation for continued analysis.

The dashboard should not be treated as a final answer to every question about voting access. Instead, it should be used as a tool for exploration, communication, and future research. With more complete data and continued development, the BB-VIZ dashboard can become an even more useful resource for SPLC and its partners.

Team Contact Information

Name	Role	Email
Hawareyawe Ketema	Project Manager	hketema1@terpmail.umd.edu
Eyoha Henoke	Deputy Project Manager / Client Liaison	Eyoha Henoke
Getnet Workalemahu	Lead Coder / Integration	gworkale@terpmail.umd.edu
Milca Teklezgi	Lead Designer	mteklezg@terpmail.umd.edu
Michael Melaku	Documentation / QA	mmelaku1@terpmail.umd.edu

Deliverables Checklist

Deliverable	Description	Status	Link
Interactive BB-VIZ Dashboard	Client-facing dashboard for county-level voting access and civic infrastructure data	Completed	LINK
Dashboard Screenshots	Screenshots of major dashboard pages, maps, charts, and interface sections	Completed	github and v…
County-Level Visualizations	Maps, charts, or dashboard components showing access-related patterns	Completed	Included in Dashboard
Data Files / CSVs	Organized data files used to support dashboard visuals	Completed	2024-Gener…
Dashboard Design System	Fonts, colors, layout, and visual design decisions used in the dashboard	Completed	Design Visu…
Client Handoff Documentation	Guide explaining how the dashboard works and what future teams should know	Completed	PDF BB-VIZ_Clien…
Final Project Report	Full report explaining context, methods, deliverables, findings, recommendations, and conclusion	Completed	Final Project …

Glossary

SPLC: Southern Poverty Law Center, the client organization for this project.

BB-VIZ: The team identifier for the dashboard and visualization team.

Civic Infrastructure: The systems, places, processes, and resources that support public participation in civic life.

Voting Access: The ability of eligible voters to participate in elections without unnecessary barriers.

County-Level Data: Data organized by county, allowing users to compare patterns across geographic areas.

Dashboard: An interactive visual tool that displays data through charts, maps, tables, and other interface elements.

CSV: A comma-separated values file used to store structured data.

Data Dictionary: A document that defines each data field, explains its source, and describes how it should be interpreted.

Client Handoff: The final set of files, documentation, and instructions given to the client so the project can be used or continued after the semester.

Voting & Civic Participation

Alabama Black Belt Counties

Team 9 BB-CCP: Community Context & Personas

Prepared for: Southern Poverty Law Center (SPLC)

Team Members: Alison Wu, Alejandra Saravia, Kyrene Jamero, Yafet Tedros

Course: INST490 · Section: 0103

Date: May 18, 2026

Table of Contents

Team 9: BB-CCP

Team Brief

Key Findings

- Voting barriers in Alabama are cumulative rather than isolated — voters frequently encounter multiple overlapping obstacles simultaneously, including logistical constraints, administrative uncertainty, transportation limitations, caregiving responsibilities, and psychological deterrents.
- Structural and policy barriers represent some of the most significant participation obstacles: the absence of early voting compresses access into a single day, strict voter ID requirements create administrative friction, and voter roll maintenance practices generate confusion and anxiety among otherwise eligible voters.
- Time scarcity is one of the strongest determinants of voting access, particularly for hourly workers, caregivers, agricultural laborers, and physically demanding occupations who cannot freely manage their schedules around a single Election Day.
- Geographic and transportation barriers disproportionately affect rural Alabama residents, who face long distances to polling locations with limited or no public transit infrastructure, compounding economic and scheduling constraints.
- Psychological and contextual factors — including fear of procedural mistakes, low perceived electoral impact, and lack of civic encouragement in social networks — meaningfully reduce participation even among voters who are motivated in principle, suggesting that access barriers alone do not fully explain nonvoting behavior.

Deliverables Produced

Voter Barrier Persona Cards (Figure 1) — Six illustrated persona cards depicting overlapping civic participation barriers across distinct Alabama voter profiles. [INST490 Final Personas]

Alabama Voting Barrier Ecosystem Concept Map (Figure 2) — A visual concept map illustrating the relationships among structural, logistical, and psychological barriers to civic participation. [INST490 Final Personas]

Categorized Voting Barrier Framework — A written analytical framework organizing Alabama's key voting barriers into six thematic categories with supporting evidence. [INST490 Final Personas]

Full Final Report (BB-CCP) — Comprehensive project report including methods, findings, personas, and recommendations submitted to SPLC. [Final Presentation Contribution Brief - BB-CCP]

Challenges

Localized qualitative data on lived voter experiences was more limited than anticipated — while structural barriers such as voter ID laws and absentee voting rules were well documented, psychological barriers like voter anxiety and feelings of political disengagement were harder to measure consistently and required triangulation across multiple source types. The inherent limitations of persona-based research also posed a challenge, as personas necessarily simplify complex lived experiences and cannot be treated as statistically representative of entire demographic groups; the team addressed this by grounding every persona attribute in documented patterns across multiple sources rather than relying on assumption or generalization.

Connections to Other Teams

Team 3 (BB-PAD) and Team 2 (BB-VOTE) — Their findings (identifying where turnout disparities exist by county or precinct) provided context that informed which barrier categories and persona demographics the BB-CCP team prioritized; in turn, the personas and barrier framework offer a qualitative explanation layer for patterns those teams surface in the data.

Team 6 (BB-BAND) and Team 10 (BB-PORT) helped develop the six personas and barrier framework were directly designed to support their work by identifying which populations face which specific barriers, enabling outreach messaging and materials to be tailored rather than generalized (e.g., Spanish-language materials for Carlos-type voters, scheduling-focused messaging for Danielle-type voters).

Team 5 (BB-AACS) helped identify the transportation and rural distance as recurring barrier themes overlapped thematically with any team conducting spatial or transit analysis; those teams' county-level data could validate or sharpen the geographic claims made in the persona profiles.

Final Report and Deliverables

Team Members

Project Manager - Alison Wu

Alison was responsible for coordinating and delegating responsibilities and tasks amongst members to maintain collaboration and understanding. Her role involved acting as the communications liaison in order to ensure understanding and clarity amongst team members and the client.

Data Researcher - Alejandra Saravia

Alejandra was responsible for conducting and gathering secondary research from reliable sources, as well as analyzing information related to the civil participation and barriers in the Black Belt counties.

Quality Assurance - Yafet Tedros

Yafet was responsible for evaluating the quality of team deliverables to ensure client satisfaction. His role also involved conducting secondary research and gathering data from reliable sources to support the team's analysis of civic participation barriers in Black Belt counties.

Visual Designer - Kyrene Jamero

Kyrene was responsible for creating the six illustrated persona cards to help visualize the different barriers citizens of Black Belt counties in Alabama and creating the concept map that showcased the voting barrier ecosystem.

Project Context

This project was conducted as part of the INST490 Capstone course in partnership with the Southern Poverty Law Center (SPLC). Across eleven student teams, the broader initiative aimed to better understand barriers to civic engagement and voting access throughout Alabama. Each team contributed specialized research and deliverables that supported a larger regional analysis intended to assist SPLC in identifying opportunities for advocacy, outreach, and voter support efforts. While some teams focused on quantitative analyses such as demographic trends, turnout data, or geographic disparities, the Community Context and Personas (BB-CCP) team focused on understanding how voting barriers are experienced at the individual and community level. The purpose of the BB-CCP team was to contextualize voter participation challenges through research-driven voter personas and an analysis of community-specific barriers. Rather than solely identifying where turnout disparities exist, this team sought to understand why eligible voters may encounter difficulties participating in elections. Through the development of representative voter profiles, the team translated broader structural and behavioral trends into lived experiences that demonstrate how barriers intersect in practice. Findings generated by this team were designed to complement the work of other capstone groups by adding a qualitative layer of understanding to quantitative turnout analyses. For example, geographic or demographic findings produced by other teams may identify regions of lower turnout, while BB-CCP personas provide insight into the real-world circumstances contributing to those outcomes. Similarly, communication and outreach teams may use these personas to better understand how voter education and engagement strategies can be tailored to different populations.

Methods

The Community Context and Personas team used a mixed-methods research approach to identify major voting barriers in Alabama and develop representative voter personas grounded in documented patterns of voter participation. The team primarily relied on secondary research, combining policy analysis, academic literature, demographic data, and reports from civic engagement organizations to better understand the structural, social, and behavioral factors that influence voter turnout. Rather than relying on a single data source, the team triangulated information across multiple forms of evidence in order to identify recurring themes and barriers affecting different populations across Alabama.

The research process began with an examination of Alabama's election laws and voting policies. Particular attention was given to policies that may unintentionally increase participation barriers, including the absence of early voting, strict voter identification requirements, excuse-required absentee voting, and voter registration maintenance practices such as inactive voter status and voter roll purges. These structural barriers were reviewed because they can create additional logistical and administrative burdens for eligible voters, particularly among populations facing economic or geographic constraints.

In addition to policy analysis, the team reviewed demographic and behavioral research related to voter participation. Academic studies, census data, public voting reports, and nonprofit advocacy materials were used to identify patterns associated with lower turnout. Through this process, barriers emerged across several recurring categories, including structural and policy barriers, accessibility and health barriers, socioeconomic and geographic barriers, information and language barriers, historical and racial disparities, and psychological or contextual barriers. The team found that voting obstacles rarely occur independently; instead, barriers frequently overlap and compound one another. For example, a rural voter may simultaneously experience transportation limitations, restrictive work schedules, and uncertainty regarding election requirements.

Following the identification of recurring voting barriers, the team developed six voter personas intended to represent common voter experiences in Alabama. These personas were not designed to represent real individuals or entire demographic groups. Rather, they serve as research-informed archetypes intended to synthesize documented voting barriers into relatable and realistic profiles. Characteristics such as age, occupation, race, geographic location, caregiving responsibilities, and religious affiliation were included when supported by broader contextual research, as these factors often shape civic participation experiences. By translating statistical trends into individual narratives, the personas provide a human-centered lens through which SPLC and future project teams may better understand barriers to civil participation.

Challanges

Throughout the research process, the team encountered several methodological challenges. One significant limitation involved the availability of localized qualitative data regarding voter experiences. While structural barriers such as voter identification laws and absentee voting requirements were well documented, psychological barriers—including voter anxiety, confusion, and feelings of political disengagement—were more difficult to measure consistently. Additionally, the team recognized the inherent limitations of persona-based research. Personas simplify complex lived experiences and therefore should not be interpreted as representative of every voter within a demographic category. To address these limitations, the team relied on multiple sources of evidence and emphasized overlapping barriers rather than singular explanations for voter behavior.

Deliverables and Findings

Deliverables

The primary deliverables produced by the Community Context and Personas team included a categorized framework of key voting barriers in Alabama and a set of six evidence-based voter personas. Together, these deliverables were intended to provide SPLC with both a systems-level understanding of participation barriers and a human-centered interpretation of how those barriers affect individuals in practice.

Figure 1. Voter Barrier Persona Cards

The Overworked Caregiver
Danielle

Gender: Female
Race: Black
Occupation: Nursing Assistant
Location: Wilcox County, Alabama

Barriers to Voting/Voter Registration

- Works long shifts and has no time to vote on Election Day
- No option to vote early
- Responsible for childcare and an elderly parent
- Confuion on registration status

Behavior towards Voting:

- Intends to vote by often misses the window due to time constraints

The Disengaged Young Adult
Marcus

Gender: Male
Race: Black
Occupation: Community College Student
Location: Russell County, Alabama

Barriers to Voting/Voter Registration

- Low perseived impact of vote
- Not fully registered to vote
- Weather sensitivity (won't go out in bad conditions)

Behavior towards Voting:

- Says he'll vote but minor inconveniences (ex. rain, long lines) deter him

The First-Generation Voter
Carlos

Gender: Male
Ethnicity: Latino
Occupation: Manufacturing
Location: Montgomery County, Alabama

Barriers to Voting/Voter Registration

- Limited Spanish-language voting information
- Unsure about ID requirements
- Irregular work hours

Behavior towards Voting:

- Politically interested but hesitates due to uncertainty and lack of clear guidance

The Bureaucratically Blocked Voter
Linda

Gender: Female
Race: White
Occupation: Retail Worker
Location: Dallas County, Alabama

Barriers to Voting/Voter Registration
- Recently moved which causes an ID address mismatch
- Flagged as an “inactive voter”
- Fear of being turned away

Behavior towards Voting:
- Showed up once but had issues and is now hesitant to try again

The Weather-Sensitive Laborer
James

Gender: Male
Race: Latino
Occupation: Construction Worker
Location: Dallas County, Alabama

Barriers to Voting/Voter Registration
- Physically exhausted after work
- Extreme heat or rain discourages turnout
- No flexibility to vote earlier

Behavior towards Voting:
- Voting depends heavily on conditions and feelings that day (ex. weather, energy level)

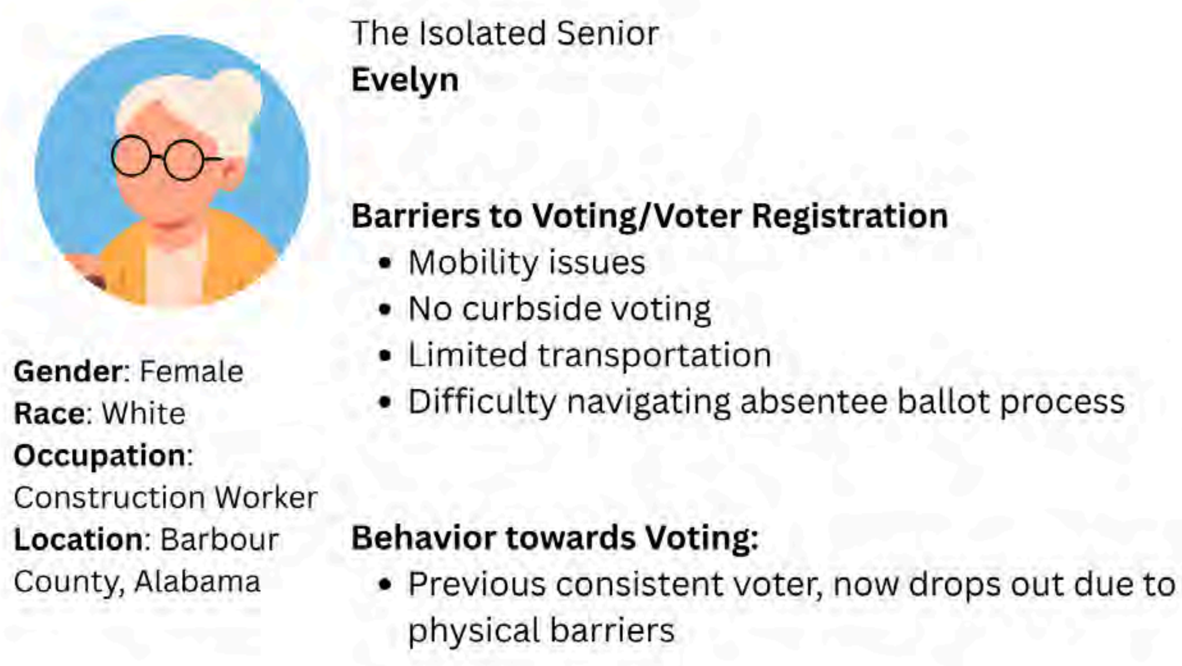

Figure 1. Voter barrier persona cards illustrating overlapping civic participation barriers across six Alabama voter profiles. Source: USCCR, 2020; Brennan Center for Justice, 2015–2023; Alabama Election Protection Coalition, 2024

Figure 2. Concept map of the Alabama voting barrier ecosystem

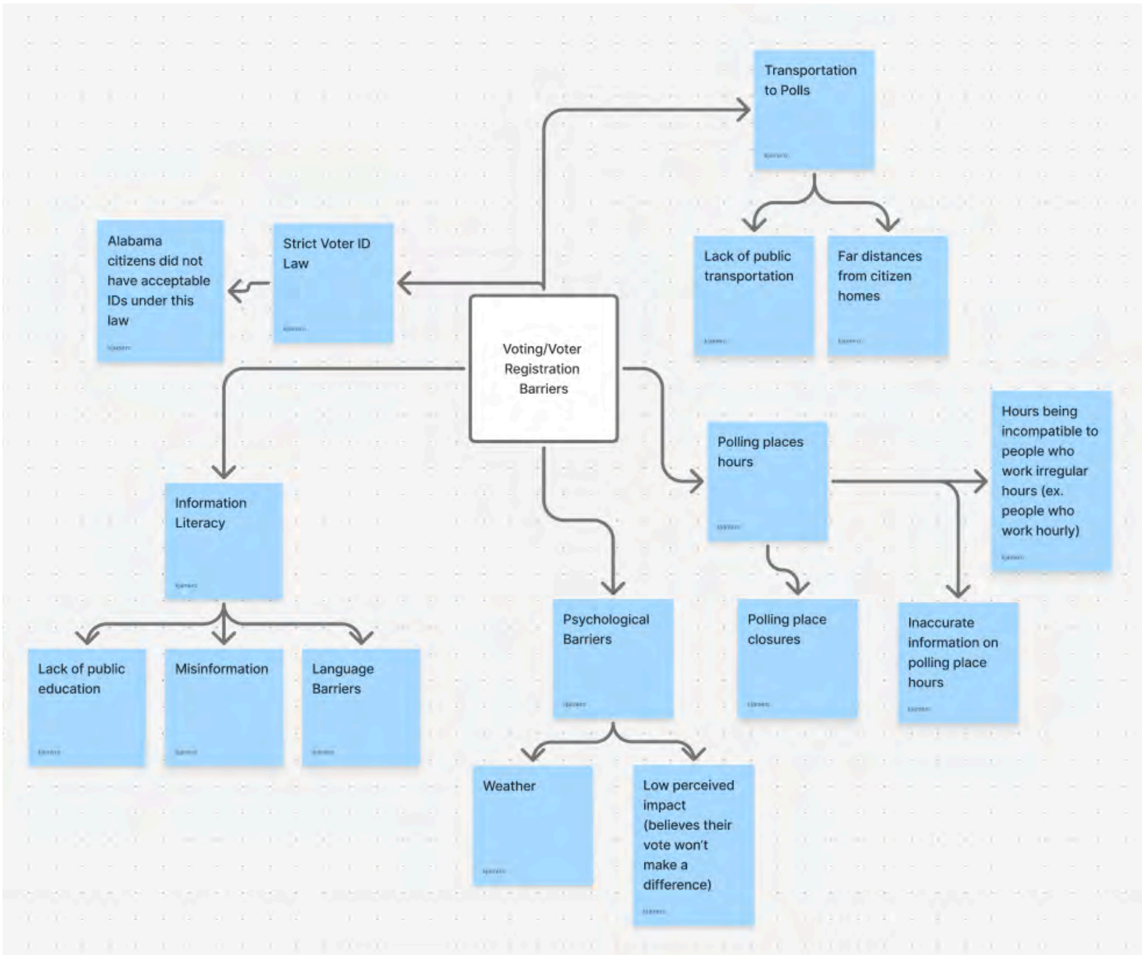

Figure 2. Concept map of the Alabama voting barrier ecosystem showing relationships among structural, logistical, and psychological barriers to civic participation. Source: USCCR, 2020; Brennan Center for Justice, 2018

Findings

One of the team's major findings was that voting barriers in Alabama extend beyond political preference or civic motivation and are frequently shaped by structural and logistical constraints. Structural and policy barriers emerged as some of the most significant obstacles to participation.

Alabama's lack of early voting places greater pressure on voters to participate during a single election day, increasing the importance of transportation access, work flexibility, and caregiving arrangements. Strict voter identification laws and excuse-required absentee voting may further complicate participation, particularly for individuals with limited resources or irregular schedules. In addition, voter roll maintenance procedures may create confusion regarding registration status, requiring some voters to complete additional administrative steps at the polls.

Accessibility and health-related barriers also emerged as a significant concern. Individuals with disabilities, mobility limitations, chronic illness, or caregiving obligations may experience increased difficulty accessing polling locations or navigating absentee voting processes. Research indicates that illness and caregiving responsibilities contribute substantially to nonvoting behavior, highlighting the role of physical accessibility in shaping turnout outcomes. These barriers suggest that participation challenges are often less about political disengagement and more closely linked to practical limitations.

Socioeconomic and geographic barriers were especially prominent among rural populations. Many Alabama residents face long travel distances to polling locations and limited access to reliable transportation. For hourly workers and individuals in physically demanding jobs, voting may require sacrificing wages or energy after long shifts. Because Alabama does not offer early voting, participation often depends on whether voters can navigate competing obligations during a narrow timeframe.

Information and language barriers also affected voter accessibility. Limited multilingual election resources may disproportionately impact Spanish-speaking and bilingual households, creating confusion regarding registration procedures, eligibility, and voter identification requirements. Even relatively minor uncertainties about voting processes can discourage turnout, particularly among first-time voters or individuals with limited prior engagement in elections.

The team additionally identified psychological and contextual factors that shape civil participation. Weather conditions, such as heavy rain or extreme heat, were found to slightly reduce turnout among less-motivated voters. Feelings of political disengagement, fear of making procedural mistakes, and low perceptions of electoral impact also emerged as meaningful barriers. These findings suggest that voting participation is often influenced by accumulated friction rather than a single decisive obstacle.

To further contextualize these findings, the team developed six voter personas reflecting recurring patterns identified through research. For example, "Danielle," an overworked caregiver living in rural Alabama, illustrates how long work shifts, caregiving responsibilities, and a lack of voting flexibility can prevent otherwise motivated voters from participating. Similarly, "Carlos," a first-generation voter, represents the challenges faced by bilingual households navigating limited Spanish-language election information and uncertainty surrounding identification requirements. Other personas—including "Evelyn," an isolated senior, "Marcus," a disengaged young adult, "Linda," a bureaucratically blocked voter, and "James," a weather-sensitive laborer—demonstrate how barriers frequently intersect and reinforce one another.

Across all personas, the team identified several recurring themes. First, time consistently emerged as one of the strongest determinants of voting access, particularly for working adults and caregivers. Second, uncertainty regarding administrative procedures frequently discouraged participation. Third, barriers were rarely experienced independently; instead, voters often faced multiple forms of friction simultaneously. Finally, environmental and social contexts—including transportation, weather, and civic encouragement—meaningfully shaped turnout behavior.

Recommendations

Recommendations for the Southern Poverty Law Center (SPLC)

The findings generated through the Community Context and Personas (BB-CCP) research suggest that barriers to civil participation in Alabama are often cumulative rather than isolated. Rather than identifying a single dominant obstacle, this project found that voters frequently encounter multiple overlapping barriers, including logistical constraints, administrative uncertainty, transportation limitations, caregiving responsibilities, and psychological deterrents. Based on these findings, the following recommendations are intended to support SPLC's voter advocacy, outreach, and future research efforts.

One of the most consistent findings across both the barrier analysis and voter personas was uncertainty surrounding election procedures. Confusion regarding voter identification requirements, registration status, inactive voter designations, absentee voting eligibility, and polling procedures repeatedly emerged as barriers that may discourage otherwise eligible voters from participating. This uncertainty was especially evident in personas such as "Carlos," the first-generation voter, and "Linda," the bureaucratically blocked voter, both of whom demonstrate how fear of procedural mistakes can reduce turnout motivation.

As a result, SPLC should consider prioritizing voter education initiatives that focus specifically on reducing administrative confusion. Rather than relying on broad voter mobilization messaging, outreach efforts may be more effective when centered on practical procedural guidance. For example, clear educational materials explaining acceptable forms of voter identification, how to verify voter registration status, what "inactive voter" designation means, and how absentee voting eligibility functions in Alabama may help reduce anxiety and improve voter confidence.

Additionally, the findings suggest that educational resources should be designed for accessibility and clarity. Because even small uncertainties can discourage turnout, particularly among infrequent or first-time voters, simplified materials using visual aids, multilingual content, and step-by-step guidance may help lower participation friction.

A major pattern identified throughout the research was the role of time scarcity as a barrier to participation. Alabama's lack of early voting compresses voting access into a single day, making election participation significantly more difficult for individuals with inflexible schedules. Personas such as "Danielle," the overworked caregiver, and "James," the weather-sensitive laborer, highlight how work obligations, exhaustion, and caregiving responsibilities often interfere with voting intentions.

SPLC may benefit from prioritizing outreach efforts toward populations most likely to experience scheduling constraints, including healthcare workers, hourly employees, agricultural laborers, retail workers, caregivers, and single-parent households. While this research cannot definitively conclude which intervention strategies would be most effective, the findings suggest that practical reminders, workplace-friendly election resources, transportation assistance information, and scheduling-focused messaging may improve turnout among these groups.

Additionally, future research may investigate whether particular counties with large rural workforces or limited transportation infrastructure experience compounded time-related participation barriers.

The research identified geography as a recurring structural issue, particularly among rural Alabama residents. Long travel distances to polling locations, limited public transportation infrastructure, and inconsistent access to reliable vehicles may create meaningful participation challenges. Because transportation barriers often overlap with economic and scheduling constraints, voters in rural communities may experience multiple forms of participation friction simultaneously. As a result, SPLC should consider prioritizing further investigation into county-level geographic access disparities. While this project identified rural access as a meaningful barrier, limitations in available county-specific transportation data prevented deeper analysis regarding which areas may be disproportionately affected. Future investigation into counties characterized by low population density, transportation scarcity, or historically lower turnout may help identify opportunities for targeted intervention and advocacy. The findings do not establish causation between transportation barriers and turnout declines; however, they suggest sufficient evidence to warrant closer examination.

Accessibility concerns emerged as another major area requiring further attention. Personas such as "Evelyn," the isolated senior, illustrate how mobility limitations, transportation barriers, and absentee ballot complexity may reduce participation among otherwise civically engaged individuals. Alabama's limited accommodations for some disabled or mobility-constrained voters may increase participation costs and discourage turnout. SPLC should consider further research into accessibility-related barriers, particularly among elderly voters, disabled populations, and caregivers. This may include investigating the usability of absentee ballot systems, transportation availability to polling sites, and barriers related to physical polling place accessibility.Importantly, this project cannot determine the prevalence of accessibility barriers across all Alabama counties due to limited localized data. However, the consistency of these concerns throughout broader voter research suggests that accessibility remains an important area for future advocacy.

The research also identified social context and civic encouragement as meaningful influences on participation. Several personas demonstrated that voting behavior is often shaped by family expectations, community engagement, and social reinforcement. For younger or less politically engaged individuals—such as "Marcus," the disengaged young adult—low perceived impact and limited civic encouragement contributed to decreased motivation. As a result, SPLC may benefit from expanding community-centered voter engagement efforts that emphasize trusted local relationships and social reinforcement. Civic participation messaging delivered through churches, schools, neighborhood organizations, or culturally relevant community groups may be more effective than generalized outreach alone. Because trust and social influence frequently shape participation decisions, leveraging existing community networks may reduce barriers related to disengagement and skepticism. At the same time, this project recognizes that motivational barriers are more difficult to quantify than structural barriers. Therefore, additional qualitative research may help clarify which forms of civic encouragement are most effective within different Alabama communities.

Recommendations for Future Project Teams

Future capstone teams building upon this work should recognize that researching voting barriers requires balancing quantitative evidence with qualitative interpretation. While structural barriers such as voter identification laws, absentee voting restrictions, and transportation limitations were relatively easy to document, behavioral and psychological barriers—including anxiety, motivation, and perceptions of political efficacy—proved significantly more difficult to measure consistently. Teams should expect ambiguity in these areas and avoid over-interpreting limited data.

One of the strongest methodological approaches used in this project involved organizing barriers into thematic categories before developing personas. Categorizing barriers into structural, accessibility-related, socioeconomic, informational, historical, and psychological dimensions helped simplify an otherwise overwhelming research process and made recurring patterns easier to identify. Future teams would likely benefit from beginning with a barrier framework before attempting persona creation, as it provides a stronger evidentiary foundation for later analysis.

Additionally, persona development proved particularly useful for translating complex findings into client-facing deliverables. Rather than presenting voting barriers solely through statistics, personas helped contextualize how barriers overlap within real-life circumstances. However, future teams should approach personas carefully and avoid treating them as representative demographic profiles. Personas function best when framed as evidence-informed archetypes designed to illustrate patterns, not stereotypes or predictions about individual behavior.

Regarding data sources, some materials proved substantially more useful than others. Election policy information from the Alabama Secretary of State and advocacy organizations provided strong context regarding structural barriers, while academic literature and public health research were valuable for understanding turnout behaviors and caregiving-related participation challenges. However, county-level qualitative data regarding lived voter experiences was more limited than anticipated. Future teams may significantly strengthen this work by incorporating interviews, focus groups, or surveys with Alabama residents to validate or refine persona assumptions.

If this project were continued, a natural next step would involve geographic validation of personas. Future teams could explore whether specific personas are more likely to emerge in certain counties based on demographic, transportation, economic, or turnout patterns. Geographic mapping or county-level clustering may allow SPLC to better target outreach and interventions. Similarly, future teams may investigate multilingual voting access, transportation deserts, or accessibility barriers with greater precision.

Finally, future teams should remain transparent regarding the limitations of the research. This project identified recurring patterns and plausible barriers to participation but cannot definitively determine causation. The findings illustrate how barriers may shape turnout behavior, not whether a particular factor directly causes nonparticipation. Being clear about these limits strengthens the integrity of the analysis and ensures that recommendations remain grounded in available evidence rather than overstated conclusions.

Conclusion

The Community Context and Personas (BB-CCP) team sought to better understand how structural, social, and behavioral barriers influence voter participation in Alabama as part of the broader INST490 partnership with the Southern Poverty Law Center. Through policy analysis, demographic research, academic literature review, and behavioral synthesis, the team developed a framework of key voting barriers and a series of evidence-based voter personas designed to contextualize how participation challenges are experienced in everyday life. By translating broader trends into human-centered narratives, the team aimed to complement quantitative findings produced by other capstone groups and contribute a more holistic understanding of civic participation barriers.

The findings demonstrate that voting barriers in Alabama are frequently interconnected rather than isolated. Structural policies such as strict voter identification laws, the absence of early voting, and absentee ballot restrictions often intersect with socioeconomic, geographic, informational, and psychological challenges. Across the six voter personas developed through this project, recurring themes included limited time availability, transportation barriers, administrative confusion, caregiving responsibilities, accessibility concerns, and low perceptions of political efficacy. These findings suggest that voter participation is often shaped not by a lack of civic interest, but by the cumulative effect of logistical and contextual obstacles.

While this research does not establish causal relationships between specific barriers and voter turnout outcomes, it highlights recurring areas of friction that warrant additional investigation and support. The personas and barrier framework developed through this project provide SPLC with a foundation for future outreach, voter education, and advocacy efforts while also offering future capstone teams a starting point for deeper community-based analysis. Continued research involving county-level investigation, community interviews, and targeted accessibility studies may further strengthen understanding of how barriers affect participation across Alabama's diverse communities.

Contacts

Alison Wu: foralisonwu@gmail.com

Alejandra Saravia: alisaravia19@gmail.com

Yafet Tedros: yafettedros4@gmail.com

Kyrene Jamero: kyrenejamero@gmail.com

Deliverables Checklist

BB_CCP - Community Context & Personas-Assignments

BB_CCP - Community Context & Personas-Final Personas

BB_CCP - Community Context & Personas-Contribution Brief

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COLLEGE OF INFORMATION

Voting & Civic Participation

Alabama Black Belt Counties

Team 10 BB-PORT: Policy and Organizational Translation

Prepared for: Southern Poverty Law Center (SPLC)

Team Members: Ryan Mitchell, Nayab Safdar, Sudenur Yavez, Zohair Farouk

Course: INST490 · Section: 0103

Date: May 18, 2026

Team 10: BB-PORT

Team Brief

Key Findings

- Finding 1: Many of the rural areas in the Black Belt region suffer from the limitation of broadband access, meaning that they deal with difficulty when it comes to online registration of voting. There are also factors of information that get limited due to this issue including access to information that relates to polling, being able to verify one's status relating to voting registration, and other things that would be deemed important to certain voters.
- Finding 2: There are multiple socioeconomic factors that heavily impact multiple communities within the Black Belt Counties that contribute to lower participation and voter turnout within the county. Lower income communities have to deal with factors that are talked about frequently in this report such as limited access to transportation, less knowledge and access to digital technology and usage, and reliance on employment.
- Finding 3: Due to the lowered civic engagement and voter registration, this is something that affects the communities directly and heavily. The reduced participation of voting leads to the limited representation of these communities and decreases any sort of policy attention towards those who are underserved, and may reduce any investment that could assist in healthcare, education, transportation and infrastructure.
- Finding 4: One of the most monumental issues is the transportation, or lack thereof, as well as accessibility to these locations geographically. Many of the Black Belt counties are very well spread out from each other and are considered to be rural, which means that residents of these counties have no choice but to travel far distances to vote.

Deliverables Produced

Policy Translation Report: Policy Report - An organized list of important laws, policies, court cases, amendments and organizations related to voting and barriers surrounding voting for minority communities.

ABWRT: ABWRT Deliverable - Informative document to help support ABWRT in making informed decisions about advocacy, programming, and partnerships.

SRBWI : SRBWI Deliverable - Informative document to help support SRBWI in making informed decisions about advocacy, programming, and partnerships.

Challenges

Our group struggled to find a good medium when recommending ideas for our clients and organizations involved in the Alabama voting process. We just needed to discuss it with our client to get a better understanding of how we should be targeting these stakeholders.

Connections to Other Teams

Team 8: BB-Viz - We interacted with this group to determine whether certain details and information were worth mentioning in our report, but they were already covering it so we decided to remove it.

Final Report and Deliverables

Project Context

Our class' collaboration with the Southern Poverty Law Center (SPLC) gave us the opportunity to analyze voting barriers and civic participation in the Black Belt region of Alabama. Our group, team 10 (BB-PORT), was responsible for translating historical and current laws and policies that may influence voting accessibility, discrimination, and turnout. Specifically, we focused on communities that faced struggles with being eligible to vote at all. Our team's research consisted of searching court cases and relevant policies which were specific to Alabama voters. Our deliverables are meant to serve as a useful guide to translate and summarize important information regarding voting accessibility and discrimination.

Methods

To fulfill our section of the project, our group worked to find and translate to plain language as many laws, policies, cases, or barriers related to voting in minority communities in Alabama. To start, we each did individual research and put everything we could find into a shared Notion page for reference. We collected sources, as well as descriptions so we could easily go back and know each link was for. During our research, we also familiarized ourselves with the Alabama voting data from the 2020-2024 general elections, which gave us more insight into the voting turnout ratings and potential barriers certain communities face when voting. Our group used sources such as the Alabama Secretary of State website, as well as other informational sites such as Vote, Ballotpedia, NAACP, the Brennan Center, and more. These sites were greatly helpful since they had the majority of the information we needed regarding how voting has developed and changed over time. A more in-depth list of our sources can be found below.

Once we felt comfortable with our research, we started to create a rough draft document which would contain an organized list of all laws, policies, important court cases, key organizations, and relevant amendments. For each topic, we tried to explain what each law/policy does, how it directly applies to Alabama voters, and other implications we found to be relevant. Additionally, we tried to apply our research with previous elections (before 2020), and how civic engagement in minority communities have decreased since then. However, since other groups had already included this in their research, we decided to leave it out. On top of laws and policies in place, we also included a case which just finished, explaining how it will affect the voting process from here forward. As well as the SAVE act, which could have a major impact for minority communities if it were passed.

Once we had an organized draft from our research, Dr. Klein suggested we create informative reports and guides for specific stakeholders. After consulting with the clients, they suggested we target certain key organizations, like The Alabama Black Women's Roundtable and Southern Poverty Black Women's Initiative. The organizations strive to expand, educate, and strengthen underserved communities to give everyone proper resources to use their right to vote. For each organization, we translated key policies and cases, structural barriers, community and economic implications, and recommended next steps for the organization. We hope these deliverables can support these groups in making informed decisions about advocacy, programming, and partnerships.

Provided below is a more detailed list of our sources we used for research.

Source Name:	Description:	Link:
NAACP Legal Defence Fund	This source gave us	https://www.naacpldf.org/shel

	information regarding Shelby v Holder, which marked a landmark decision for the state of Alabama, eliminating the preclearance requirement of the Voting Rights Act of 1965.	by-county-v-holder-impact/
Brennan Center	This report gave us great background information on the current voting racial gap being the highest it's been in years in the state.	https://www.brennancenter.org/our-work/analysis-opinion/alabamas-racial-turnout-gap-hit-16-year-high-2024
Ballotpedia	This served as a great source for all general voting policies and laws in Alabama. Gave us great insight on absentee voting laws, same day voting registration policies, and photo ID laws.	https://ballotpedia.org/Election_policy_in_Alabama
Alabama Secretary of State Website	The SoS website also provided us with great insight on Alabama specific policies, such as their absentee voting and same day registration policies.	https://www.sos.alabama.gov/alabama-votes/voter/absentee-voting

Brennan Center	We used this site again to research and track current cases and acts, which could greatly impact voters of Alabama.	https://www.brennancenter.org/our-work/research-reports/louisiana-v-callais & https://www.brennancenter.org/our-work/analysis-opinion/save-act-reaches-senate
US Congress	This sight gave us good information on the Voting Accessibility for the Elderly and Handicapped Act, which led to better accessibility for elderly and handicapped voters.	https://www.congress.gov/bill/98th-congress/house-bill/5762

Deliverables and Findings

Deliverables

1. Final Policy Translation Report

Description

The Final Policy Translation Report is the primary deliverable created for the Southern Poverty Law Center (SPLC). It provides a nonpartisan overview of the federal laws, Alabama state policies, court cases, constitutional amendments, and organizations that shape voting rights and civic engagement in Alabama, with a focus on Black Belt and LatinX communities.

The report includes:

- Simplified explanations of laws and court decisions
- Alabama-specific policy impacts
- Community-specific implications
- Supporting data and source references

Rationale / How It Meets Project Goals

This deliverable supports the project goal of translating complex legal and policy information into accessible language for organizational use. By consolidating key voting-rights information into one reference document, the report improves usability during internal communications, outreach efforts, and conversations with policymakers.

The report also highlights how structural barriers disproportionately affect Black Belt and LatinX communities, aligning with the project's focus on voting access and equity.

Content Included

The report covers:

- The Voting Rights Act of 1965
- Shelby County v. Holder
- HAVA and NVRA requirements
- Alabama voter ID and absentee voting laws
- Felony disenfranchisement policies
- Key court cases such as Allen v. Milligan
- Relevant constitutional amendments

- Key organizations involved in Alabama voting-rights work

Link

SPLC Policy Translation Report

2. Stakeholder-Specific Policy Translation Documents

Description

Two stakeholder-specific policy translation documents were created as shorter, more targeted versions of the full policy translation report. These briefs were tailored to the goals, priorities, and operational contexts of specific organizations connected to voting-rights and community advocacy work in Alabama, including the Southern Rural Black Women's Initiative (SRBWI) and the Alabama Black Women's Roundtable (ABWRT).

Each document translated key voting laws, court decisions, and structural barriers into organization-specific implications and potential action areas relevant to their community engagement and advocacy work.

Rationale / How It Meets Project Goals

These deliverables support the project goal of improving accessibility and usability of policy information by adapting the broader research into concise, stakeholder-centered resources. Rather than providing only general policy summaries, the documents connect legal and structural developments directly to the organizations' missions, outreach efforts, and community priorities.

The stakeholder briefs were designed to:

- Simplify complex voting-rights information
- Highlight community-specific impacts
- Connect policy developments to organizational strategy
- Support advocacy, outreach, and civic engagement planning
- Provide actionable reference materials for staff and leadership

This approach made the information more practical and immediately applicable for organizations working directly with underserved communities.

Content Included

The stakeholder-specific documents included:

- Simplified explanations of major voting laws and court decisions
- Analysis of structural barriers affecting Black Belt and LatinX communities
- Equity and economic implications of voting-related policies
- Rural and community-specific considerations
- Organizational action implications tied to advocacy and outreach work
- Recommended next steps and engagement strategies

Link

SRBWI and ABWRT

3. Findings and Data Visualization Graphics

Description

Two informational graphics were created to visually summarize major findings from the policy analysis process. The graphics focus on:

1. The impact of Shelby County v. Holder on voting policies in Alabama
2. Felony disenfranchisement and voting barriers in Alabama

Rationale / How It Meets Project Goals

The graphics make complex policy information easier to understand and communicate by presenting major findings in a visual format. They are designed to support presentations, outreach, and internal discussions while improving accessibility for non-legal audiences.

Content Included

The graphics highlight:

- Polling location closures and voter ID changes
- Disproportionate impacts on minority communities
- Felony disenfranchisement statistics
- Legal, financial, and administrative voting barriers

Graphics

Figure 1. The Shelby County v. Holder decision removed federal oversight, leading to stricter voter ID laws and polling location closures that mainly affect low income and minority voters (Source: Brennan Center for Justice, 2018 & U.S. Supreme Court. Shelby County v. Holder, 570 U.S. 529, 2013).

Figure 2. Alabama has one of the highest rates of felony disenfranchisement, with about 227,000 people unable to vote because of legal and financial barriers (Source: Encyclopaedia Britannica, 2026 & The Sentencing Project, 2022).

4. Stakeholder Map Visualization

Description

A stakeholder map visualization was created to show organizations and resources connected to voting-rights work in Alabama and their relationship to SPLC.

The map includes organizations such as:

- NAACP Legal Defense Fund
- ACLU of Alabama
- StandUp Mobile
- Redistricting Data Hub

Rationale / How It Meets Project Goals

This deliverable supports the organizational analysis component of the project by identifying collaboration networks, advocacy partners, and research resources relevant to voting-rights work in Alabama.

Content Included

The stakeholder map categorizes organizations by their primary role, including:

- Legal advocacy and litigation
- Community engagement and outreach
- Research and election data analysis

It also demonstrates how these organizations contribute to voting-rights efforts affecting Black Belt and LatinX communities.

Graphic

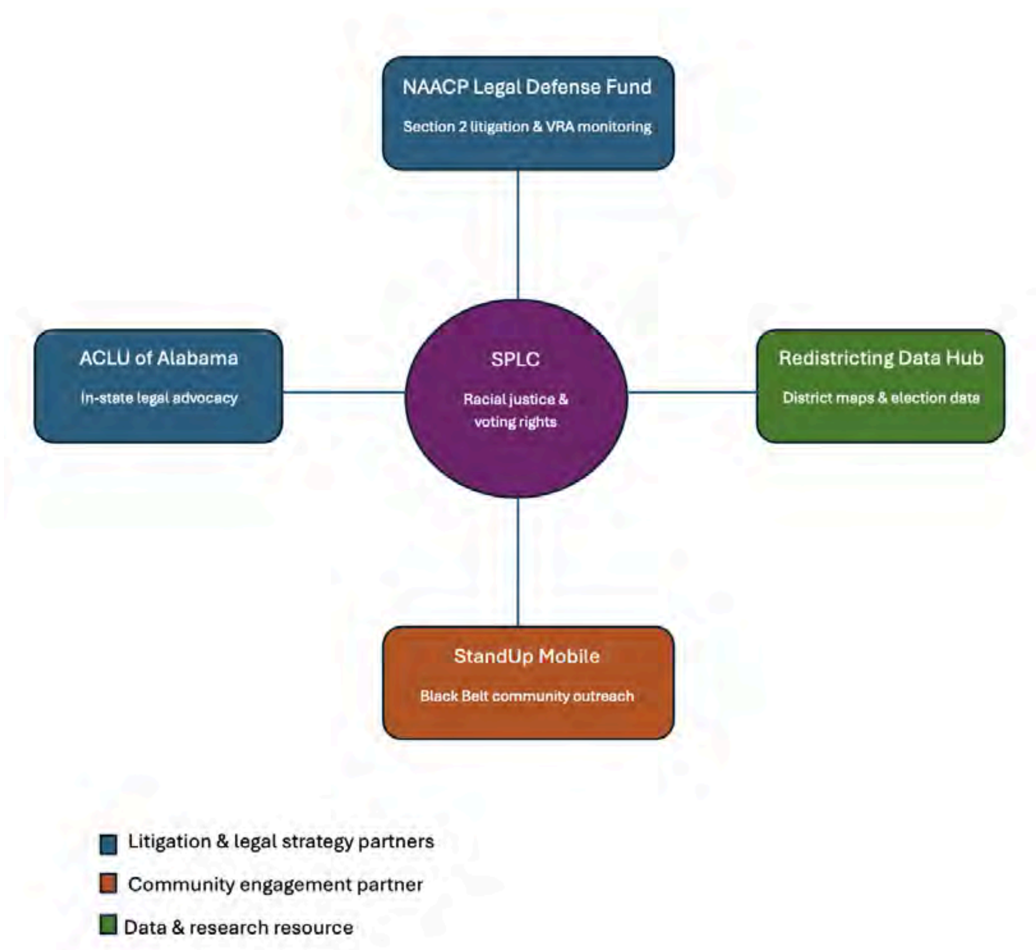

Figure 3. Stakeholder map of key organizations that SPLC can collaborate with since they share similar missions.

Findings

Our research identified several interconnected legal, structural, and socioeconomic barriers that contribute to lower civic engagement in Alabama's Black Belt communities. Rather than stemming from a single issue, the strongest pattern throughout our findings was that multiple overlapping barriers disproportionately affect low-income, minority, and rural populations, making civic participation significantly more difficult for these communities.

One major finding involved the long-term impact of Shelby County v. Holder. The removal of federal preclearance protections allowed Alabama to implement stricter voter ID policies and make polling location changes without prior federal review. Research showed that polling location closures and stricter identification requirements have had a greater impact on underserved communities, particularly residents in rural Black Belt counties who already face transportation and accessibility challenges.

Transportation and geographic accessibility were recurring barriers throughout the research. Many Black Belt counties are rural and geographically spread out, requiring residents to travel long distances to polling locations, registration offices, or ID-issuing locations. These barriers disproportionately affect elderly residents, low-income individuals, voters without reliable transportation, and workers with inflexible schedules. Limited public transportation infrastructure further increases these difficulties and contributes to lower participation rates.

Another major trend identified was the impact of strict voter identification requirements and administrative voting procedures. Alabama's voter ID laws create additional obstacles for residents who may struggle to obtain qualifying identification documents such as birth certificates or passports. These barriers were found to especially affect low-income residents, elderly populations, and formerly incarcerated individuals attempting to restore voting rights.

The research also highlighted felony disenfranchisement as a significant factor affecting civic engagement in Alabama. Alabama remains among the states with the highest rates of voting ineligibility due to felony convictions, with more than 227,000 residents disenfranchised. Legal fines, administrative requirements, and unclear eligibility rules create additional challenges for individuals seeking to restore voting rights, limiting re-registration and participation within affected communities.

Digital inequality and broadband access also emerged as important structural barriers. Many rural Black Belt communities experience limited broadband infrastructure, reducing access to online voter registration, election information, polling updates, and voter verification tools. As election systems and civic engagement resources become increasingly digital, limited internet access creates additional disadvantages for rural and low-income populations, as well as elderly residents with lower levels of digital accessibility and literacy.

Socioeconomic pressures were another consistent pattern across the findings. Many residents in underserved communities must balance employment demands, caregiving responsibilities, transportation coordination, and household management, leaving limited time and flexibility for civic participation. The research found that these burdens particularly affect Black women in rural communities, who often experience overlapping responsibilities related to employment, caregiving, and household support.

Overall, our findings demonstrate that reduced civic engagement is not solely a political or legal issue, but also one closely connected to broader economic, infrastructural, and social inequalities. Lower participation rates can contribute to reduced political representation and decreased policy attention toward underserved communities, potentially limiting investment in healthcare, education, transportation, and infrastructure throughout Alabama’s Black Belt region.

These findings reinforced the importance of creating clear, evidence-based deliverables that stakeholders can easily reference and apply in outreach, advocacy, and policy discussions. By presenting these barriers in accessible language and connecting them to real-world community impacts, the project aimed to make complex voting-rights issues more understandable, actionable, and relevant for organizations working directly with affected populations.

Recommendations for SPLC

Transportation Support and Election Accessibility:

As mentioned earlier, we found that one of the biggest barriers behind this issue is lack or limited access to transportation across rural areas in Black Belt Counties. Many citizens have difficulties due to many factors including long travel distances, unreliable infrastructure of roads, no or limited access to personal transportation, and lack of public transportation. We find that SPLC should consider supporting and partnering with organizations with the intent of providing Election Day transportation by supplying shuttle services, initiating ride-sharing services community wise, and other forms of transportation assistance programs that SPLC deems reasonable. These efforts are relatively small and would benefit much of the communities that are affected by transportation factors, including the elderly population, those with limited vehicle access, and low-income citizens.

Elderly and Handicapped Support with Election Participation:

In our draft document, we highlighted 'The Voting Accessibility for the Elderly and Handicapped Act of 1984', which served to be a landmark act for this discriminated group. This act forced all governments running elections to accommodate all handicapped and elderly voters, so that everyone can safely access the polling locations. However, despite there being some provided accommodations, there are still other barriers elderly and handicapped people may face if they want to vote. There are other impacting factors, such as obtaining a valid photo ID, getting reliable transportation that accommodates the handicapped and/or elderly. To help prevent this barrier, SPLC may look to target this group of people, in order to provide a resource in order to make the process smoother and more accessible. Whether it be a representative to help answer questions and guide voters through the process, or giving transportation to those in need as mentioned above. This could also be a good opportunity to inform the public on this issue, which could raise awareness and lead to increased funding to help this underserved group of people.

Recommendations for future project teams

For future teams working with this topic, there are a few things we deem to be important knowledge going into this subject. First, I want to explain how our groups' ideas and mindset changed throughout the first few weeks of starting. We were a bit confused at first on what was expected from us in terms of deliverables. At first, we were trying to think of ways we could make a change and improve what was currently in place. We were taking the entire project into account, and didn't really understand how our group played a factor in the overall objective. Our mission was simply to find as much information on the topic, and create clear concise translations of these policies, and direct them to stakeholders who are involved in the process. This confusion caused us to start a little slower until we got clarification from the clients. So for future groups with this task, I would start by giving them a clear idea of what is expected with some examples to guide them from the start.

When we started our research, we knew little about voting laws, especially in Alabama. We didn't really know where to start, so we tried just writing down every little thing we could find. It was hard to split up tasks for research, considering we didn't know what specifically to find. By the time we were done with our research, we had multiple people researching the same things and it really hurt our efficiency. For future groups researching this topic, we would recommend familiarizing themselves with the most basic policies, such as VRA and other nationally followed voting laws. Once, they get an idea for the national voting laws, then look into how Alabama differs from other state policies. We recommend this because our group dove straight into researching everything, and it proved to be a bit confusing later on trying to go back and figure out how to apply our findings.

They should take it slowly and meaningfully, instead of overwhelming themselves, which would lead to a smoother and more efficient process.

Another recommendation to the next group working on this project would be to ask the clients for stakeholders you should be targeting as early as possible. The clients provided us very useful insight, along with a few groups to research. This led our group to choose our targeted stakeholders for our deliverables, being Southern Rural Black Women's Initiative and Alabama Black Women's Roundtable. While we only focused on the two groups, the client also suggested we research and target Stand Up Mobile. The next group working on this project could target this organization, as well as use the other two groups mentioned as resources along the way.

As for the research itself, the information and sources are easy to find. Our group starts by using ChatGPT to find us a few useful sources from all different topics of voting policies. From absentee voting, to handicapped and elderly policies, and popular court cases to look into. However I think the most reliable sources are the Alabama Secretary of State site, along with informational sites like Ballotpedia, Vote, and the Brennan Center for Justice. The only thing we recommend is not copying any information directly from AI, because it can make mistakes and the information is so publicly accessible it's not working getting the information wrong.

When deciding your final deliverables, one thing we struggled with was finding a way to provide the most information in the smallest amount of words. We had too much information and we had to cut it down and shorten it. It was really hard trying to decipher what was worth sharing or deleting. So, we recommend keeping your research in short straight-forward bullet points, instead of long blocky paragraphs. It makes your contents so much easier to read and follow along with when going back to apply it.

Overall, the part we could have improved on the most was getting off to a good start. It can be very overwhelming trying to learn about all voting policies and trying to get as much information as you can. We recommend researching together at the beginning so you can collectively gain a grasp on the voting process and potential barriers for different communities. Once you have some general research, they can split off and do individual research on more specific cases and Alabama specific policies.

Conclusion

This project was a joint effort in order to research and analyze potential barriers for minority communities in our voting system. In particular, our group was in charge of policy and organizational translation. Our extensive research helped us gather plenty of context into the history of voting discrimination, including how there are still barriers today. We used this context to create a draft of important cases and policies, along with a couple of deliverables to important organizations. We were also shocked to learn of the low civic engagement of minority communities, leading to the highest racial voting gap in many years. This is why groups like SRBWI and ABWRT support and educate these communities to strengthen their influence in our elections, and our group was happy to be able to potentially support them with our deliverables. Our groups' emails are listed below.

Ryan: rymit9@terpmail.umd.edu

Nayab: nsafdar@terpmail.umd.edu

Sudenur: syavuz@terpmail.umd.edu

Zohair: zfarouk@terpmail.umd.edu

Deliverables Checklist

Policy Translation Report: Policy Report

ABWRT: ABWRT Deliverable

SRBWI : SRBWI Deliverable

Voting & Civic Participation

Alabama Black Belt Counties

Team 11 BB-CPAD: Civic Participation Process & Access Design

Prepared for: Southern Poverty Law Center (SPLC)

Team Members: Lakshya Sajal Kumar, Uziel Saldivar, Amari Harrington, Matthew Romalis

Course: INST490 · Section: 0103

Date: May 18, 2026

Table of Contents

Team 11: BB-CPAD

Team Brief

Section Brief

Our team studied the barriers that may affect voter participation in Alabama Black Belt communities, particularly among younger and first-time voters, and explored how digital tools and modern outreach methods could improve civic engagement, specifically encouraging voter turnout. This research matters to the overall project because it helped identify gaps in accessibility, communication, and voter understanding that may discourage participation in the democratic process. By analyzing these challenges, the team contributed ideas and solutions to make voting information more accessible, engaging, and easier to understand for underrepresented communities. These solutions include an interactive Alabama voter guide and social media outreach concepts

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Key Findings

- **Finding 1**: Voters aged 18-39 have significantly lower turnout rates compared to older age groups in previous general elections, despite making up a similar share of the population, highlighting a major participation gap among younger voters.
- **Finding 2:** The voter turnout rate of young adults ages 18 to 39 in the 2022 midterm elections in Alabama was only 15%, one of the worst records of youth voter participation nationwide. The voter participation rates for those 65 years and above were more than 75% nationwide in 2024, which is 28% greater than those aged 18 to 39.
- **Finding 3:** Low youth voter participation in Alabama is influenced more by accessibility and engagement barriers than by a lack of interest in civic issues. Many younger voters face challenges such as limited civic information, confusion about the voting process, and reduced exposure to traditional outreach methods, making clear, digital, and user-friendly civic resources especially important for increasing participation.

Deliverables Produced

- **Deliverable 1 - Civic Participation Process Model:**
  An interactive civic participation guide that simplifies Alabama's voting process for first-time and underserved voters. Website | Source Code

- **Deliverable 2 - User Research:**
  A research analysis examining the barriers, challenges, and factors affecting civic participation and voter engagement in Alabama's Black Belt counties. Document

- **Deliverable 3 - Social Media Campaign:**
  A nonpartisan civic engagement guide designed to increase voter awareness and participation through clear, accessible, and easy-to-understand voting information.Canva

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Challenges

One limitation of this research was the inability to directly gather feedback from constituents living in Alabama regarding the factors that may contribute to lower voter turnout among younger voters compared to older populations. While a firsthand perspective could have provided valuable qualitative insight, opportunities to conduct surveys or informal outreach were limited. In particular, many online communities and social platforms, such as Reddit, maintain strict policies regarding surveys and participant solicitation, which restrict the team's ability to engage directly with potential respondents. As a result, the project relied primarily on existing research, publicly available voter participation data, and secondary sources to identify trends and potential barriers to civic engagement.

Connections to Other Teams

BB-VOTE overlaps and reinforces our finding that voter turnout was higher for the 2020 general election than compared to general elections of 2024

Final Report and Deliverables

Project Context

Voter participation plays a major role in ensuring that communities are represented in the democratic process. However, barriers such as limited access to information, confusion about voting procedures, and a lack of engagement through modern communication methods may affect participation among certain groups of voters. These challenges can be especially important to address in communities where civic engagement and voter outreach efforts are critical to increasing accessibility and awareness.

The Southern Poverty Law Center (SPLC) is interested in exploring ways to improve voter engagement and accessibility within Alabama Black Belt communities, with a particular focus on younger and first-time voters. Traditional outreach methods may not always align with the ways younger populations consume information and interact with civic resources. As a result, there is a growing need for accessible, user-friendly, and digital approaches that simplify the voting process and encourage participation.

Our project focused on identifying trends in voter participation by age group and examining how digital tools and outreach strategies could help improve engagement among underrepresented voters. To support this effort, our team developed concepts including an interactive Alabama voter guide and a social media outreach campaign designed to make voting information easier to access and understand. These resources were intended to reduce confusion surrounding the voting process while presenting information in a format that is more engaging and accessible to younger audiences.

Methods

Voter turnout analysis

Our team conducted an analysis of voter turnout data from the three most recent general elections within Alabama's Black Belt counties. The purpose of this analysis was to identify demographic groups with lower levels of electoral participation and use those findings to guide the design priorities of the project.

The analysis focused specifically on age demographics because differences in participation across age groups could influence how voting information should be presented and delivered. Understanding which age groups experienced lower turnout helped inform decisions regarding the platform's format, accessibility, and communication approach.

Analytical Approach

Our approach for this analysis involved collecting and examining ballot count datasets from the 2018, 2020, and 2024 general elections. These datasets included the total number of ballots cast for each county and categorized voter participation by age demographic. From each dataset, only the counties classified within Alabama's Black Belt region were extracted for analysis.

The original election datasets categorized voter participation using the following age groups:

- 18–29
- 30–39
- 40–49
- 50–59
- 60–69
- 70–79
- 80–89
- 90–99
- 100+

To calculate voter turnout percentages, population data by county and age demographic were also collected. The primary population dataset used for this analysis was sourced from the National Institutes of Health (NIH), which provided county-level population estimates organized by age group. Similar to the election datasets, only Black Belt counties relevant to the project scope were extracted from the population dataset.

The population dataset categorized residents into the following age demographics:

- 18–39
- 40–64
- 65 and over

To align the election ballot count data with the population dataset categories, the original election age groups were consolidated into broader demographic ranges. Ballot counts for the

18–29 and 30–39 categories were combined to create the 18–39 demographic group. Ballot counts for the 40–49 and 50–59 categories, along with a proportional allocation of the 60–69 category, were combined to create the 40–64 demographic group. The remaining proportion of the 60–69 category was combined with the 70–79, 80–89, 90–99, and 100+ categories to create the 65 and over demographic group.

Once the datasets were standardized across election years and demographic categories, voter turnout rates were calculated using the following formula:

$$Turnout\ Rate(\%)\ =\ \left(\frac{Number\ of\ Ballots\ Cast}{Eligible\ Voting\ Population}\right)\ \times\ 100$$

Disclaimer/Challenges

Methodological Considerations

This analysis used general population estimates rather than eligible voter population data to calculate voter turnout percentages by age group. While this approach does not produce exact turnout rates among eligible voters, it still provides a consistent framework for identifying comparative participation patterns across election cycles and demographic groups. The methodology was applied uniformly throughout the analysis to support reliable trend observation and demographic comparison.

Data Limitations

One challenge encountered during this analysis involved the availability and consistency of county-level population datasets organized by age demographic. The population dataset sourced from the National Institutes of Health, while the closest match for the needs of this project, reported population estimates covering the years 2019 through 2023 rather than aligning exactly with each election year analyzed. Despite this limitation, the dataset still provided a stable demographic baseline that allowed for meaningful comparison of participation trends across the 2018, 2020, and 2024 election cycles.

Data Used

Data source used for this analysis includes:

- Alabama’s Official 2024 General Election Participation by Age Source
- Alabama’s Official 2020 General Election Participation by Age Source
- Alabama’s Official 2018 General Election Participation by Age Source
- National Institute of Health Alabama Population - Table Source

Tools Used

The tools used in this analysis were

- Excel for using formulas and displaying data
- Canva for displaying the website and making the flyers

Deliverables and Findings

Civic Education Information Materials

Description

To support voter education and civic literacy efforts in Alabama Black Belt communities, we created a series of public-facing informational materials that provide accessible, nonpartisan guidance on the voting process. These materials focused on simplifying key voting information for first-time voters and rural residents through visually engaging, easy-to-understand formats. The goal of these deliverables was to reduce confusion surrounding voter registration, voting requirements, and civic participation while encouraging greater engagement among younger audiences.

Rationale

1. Figure 1 shows an infographic explaining voter eligibility requirements in Alabama. The infographic outlines the qualifications required to register to vote, including citizenship, residency, age requirements, and voting eligibility restrictions. The design uses concise language and visual elements to improve readability and accessibility for first-time voters.

Figure 1: Alabama voter eligibility infographic.

2. Figure 2 displays a social media infographic focused on voter registration methods in Alabama. The material explains different ways individuals can register to vote, including downloading and mailing registration forms, requesting forms by email, and visiting local County Boards of Registrars. The infographic also identifies locations where voter registration resources are available, such as DMVs, libraries, universities, and public assistance offices. A QR code linking directly to voter registration resources was incorporated to improve accessibility and simplify access to information. If viewers want to access voter registration resources directly, they can scan the QR code included within the flyer.

Figure 2: Alabama voter registration information graphic

3. Figure 3 presents an informational graphic explaining acceptable forms of identification required at polling locations in Alabama. The material highlights different forms of valid photo identification, including driver's licenses, passports, student IDs, military IDs, and government-issued identification cards. Additional accepted forms of identification were also included in order to clarify voting requirements and reduce uncertainty surrounding what voters need to bring to the polls.

Figure-3: Accepted voter ID requirements infographic.

4. Figure 4 shows a civic engagement-focused social media graphic encouraging voter participation. The graphic emphasizes the importance of voting by highlighting themes such as being heard, encouraging positive change in communities, and exercising constitutional rights. The design incorporated recognizable Alabama imagery and Black Belt regional mapping to create a stronger sense of community connection and civic identity. A source link was also included at the bottom of the graphic to allow viewers to access additional voting information and FAQs directly from the Alabama Secretary of State website.

Figure-4: Civic engagement and voter participation graphic.

5. These informational materials were specifically designed within social media dimensions to improve accessibility and engagement among younger audiences who are more likely to consume civic information digitally. The graphics emphasized readability through concise wording, organized layouts, visual icons, and mobile-friendly formatting appropriate for platforms such as Instagram.

The overall purpose of these deliverables was to provide SPLC with accessible outreach resources that could be used to support voter education and increase civic participation within Alabama Black Belt communities. By presenting voting information in a simplified and visually engaging format, the materials aimed to reduce informational barriers and encourage greater participation among first-time voters and historically underrepresented populations.

Civic Process Model

Description

The Civic Participation Process Model is an interactive digital resource designed to help Alabama residents navigate the voting process by providing clear, accessible, and organized guidance on voter registration, election preparation, voting procedures, and voter rights.

Findings

During the research phase of this project, one of the most consistent patterns observed across Black Belt counties was lower voter turnout among individuals between the ages of 18 and 39 compared to older age groups. Research conducted for this project identified several factors that may contribute to lower participation rates among younger voters, including limited familiarity with voting procedures, difficulty locating reliable election information, and the complexity of navigating multiple voting requirements and deadlines.

A major issue identified throughout the research process was the volume and fragmentation of voting-related information available to voters. Information regarding voter registration, absentee voting requirements, identification requirements, polling locations, election dates, and voting rights is often spread across multiple sources and agencies. For first-time voters or individuals who have not participated in recent elections, locating and understanding this information can become difficult and time-consuming.

This challenge aligns with the concept of information overload, which occurs when individuals are presented with more information, steps, or choices than they can reasonably process efficiently. Research on information overload suggests that excessive complexity can reduce engagement, increase procrastination, and discourage participation in procedural tasks. In the context of voting, this may create additional barriers for individuals attempting to understand election procedures or determine eligibility requirements.

These barriers may be further amplified for younger adults balancing work schedules, school, childcare responsibilities, or multiple jobs. Alabama's voting structure includes limited in-person voting availability, no same-day voter registration, and restricted absentee voting eligibility outside specific qualifying circumstances. These requirements increase the importance of advance planning and procedural awareness for voters attempting to participate in elections.

The relationship between procedural complexity and voter participation became particularly noticeable when comparing turnout trends across election cycles. Figure 1 compares voter turnout rates among three age demographics: ages 18—39, 40—64, and 65+ within Alabama Black Belt counties. The data show that voter turnout among individuals ages 18—39 was significantly higher during the 2020 election cycle than during the 2018 and 2024 election cycles.

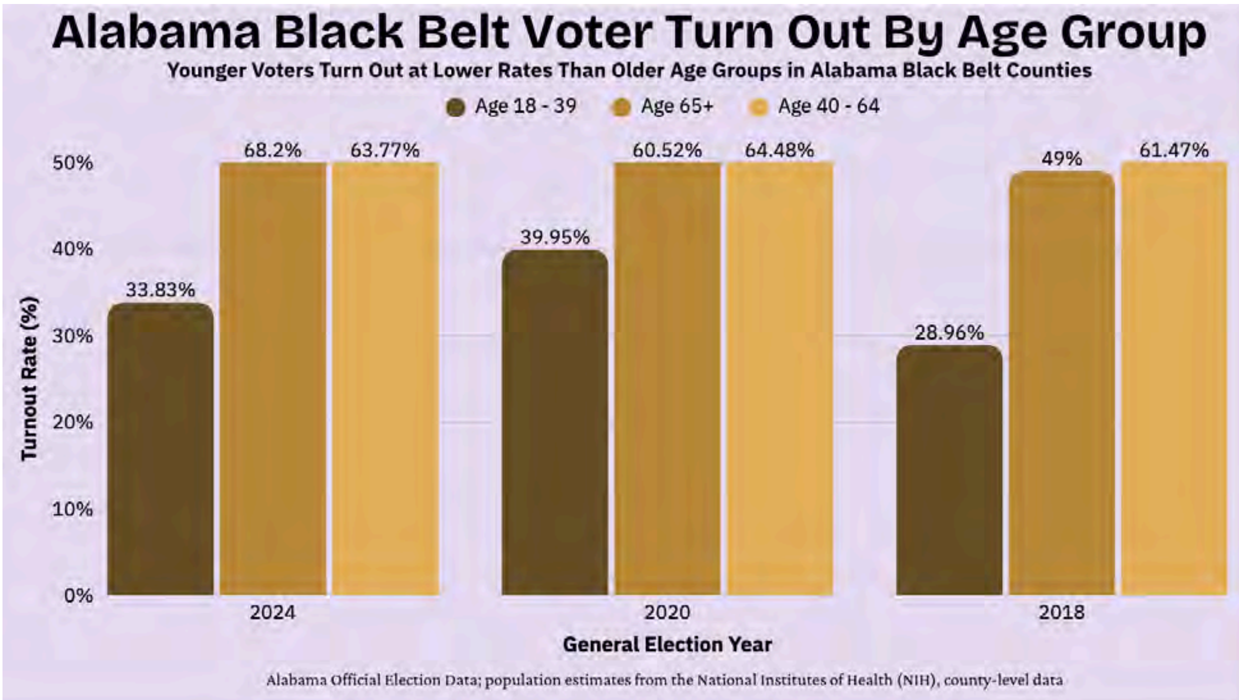

Figure 1. Voter turnout rates among Alabama Black Belt constituents ages 18–39, 40–64, and 65+ across the 2018, 2020, and 2024 election cycles. Source: Alabama Secretary of State voter turnout data, compiled by project team.

One notable distinction during the 2020 election cycle was the temporary expansion of absentee voting eligibility related to COVID-19 accommodations. During this period, absentee voting became more broadly accessible than in previous or subsequent election cycles. By the 2024 election cycle, these temporary accommodations had been removed, and turnout among younger voters returned to levels more consistent with earlier election cycles. While this project does not establish causation, the observed relationship between expanded voting accessibility and increased turnout among younger voters suggests that procedural accessibility may influence participation rates within this demographic.

Rationale Behind the Design

These findings significantly influenced the design strategy for the project deliverable. The research suggested that younger voters may benefit from resources that reduce procedural confusion, simplify access to information, and provide clearer guidance throughout the voting process.

Based on these findings, the project team identified several core objectives that the solution needed to achieve:

- Provide information in a digital and mobile-friendly format
- Reduce procedural confusion by organizing information into clear and manageable steps
- Support both first-time and returning voters
- Accommodate users with different levels of familiarity with voting procedures
- Provide quick-access pathways for users seeking immediate information
- Help users prepare for important election deadlines and requirements in advance

To address these objectives, the team developed a mobile-friendly interactive website focused on guiding users through the voting process in Alabama. The website was designed to function as both an instructional resource for first-time voters and a reference tool for returning voters seeking specific election-related information.

The interactive website format was selected because it allows information to be organized dynamically based on user needs. Rather than requiring users to navigate large amounts of information at once, the interface guides users toward relevant topics and resources depending on their circumstances. The platform also supports accessibility across different devices, languages, and learning preferences.

Additional design considerations included:

- Compatibility with mobile devices
- Minimal technical requirements for users
- Support for visual and text-based learning
- Integration of reminders and election planning tools
- Simplified navigation between related voting topics
- Access to source material without interrupting the user experience

Determining Included Topics

Determining which topics to include in the interactive website required extensive review of Alabama voting procedures and official election resources. Because the voting process contains multiple pathways, requirements, and exceptions, the team approached the research process from the perspective of a first-time voter navigating the system for the first time.

The team documented the procedural steps required for different voting activities, including voter registration, absentee voting, polling location verification, identification requirements, and ballot tracking. Research was conducted using official Alabama election resources and publicly available government materials.

For each process, the team identified:

- Required steps
- Available methods for completing the task
- Potential barriers users could encounter
- Alternative pathways available to voters

For example, research regarding voter registration identified multiple registration methods, including online registration and registration by mail. Additional review revealed that online registration requires a valid driver's license or state-issued identification card. This requirement was identified as a potential barrier for some voters, leading the team to include additional guidance related to obtaining acceptable voter identification.

The research process also extended beyond standard voting procedures. The team examined scenarios involving voters without identification, voters requiring accessibility accommodations, absentee voters, and individuals seeking information regarding voting rights restoration. This approach resulted in the development of a broader journey map outlining the different paths a voter may take while attempting to participate in an election.

The final website structure condensed this research into simplified, user-oriented pathways organized into three primary sections:

1. Registering to Vote
2. Preparing for Election Day
3. Additional Voting Information and Special Circumstances

Figure 2. Voter Registration menu within the interactive civic participation Source: Project team website prototype.

Figure 3. Election Day Preparation menu within the interactive civic participation website Source: Project team website prototype.

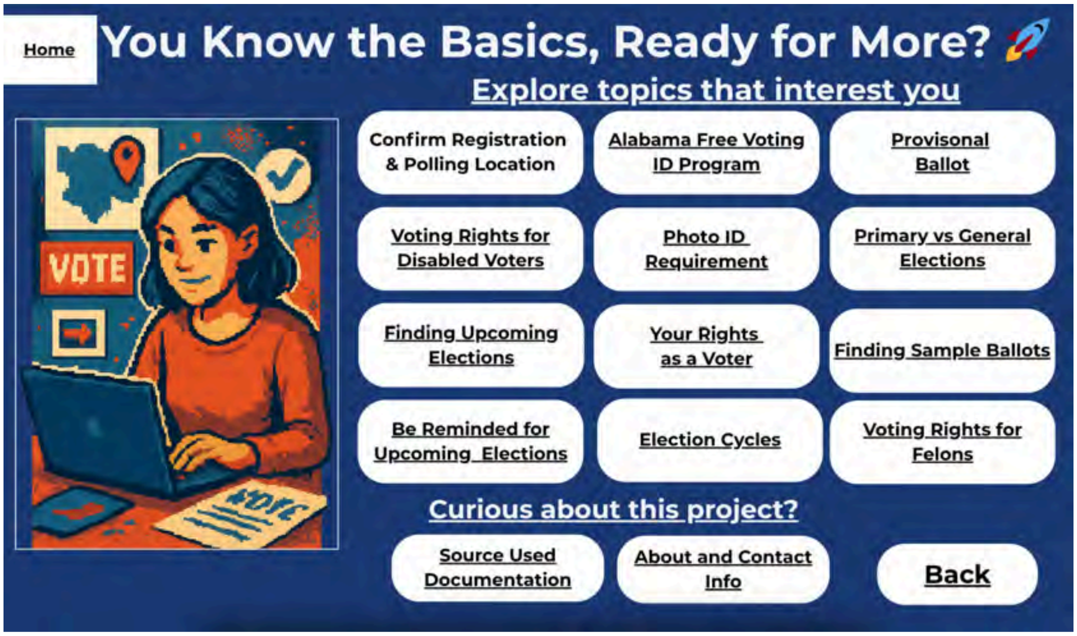

Figure 4. Additional Voting Information and Special Circumstances menu within the interactive civic participation website Source: Project team website prototype.

User Experience and Design Approach

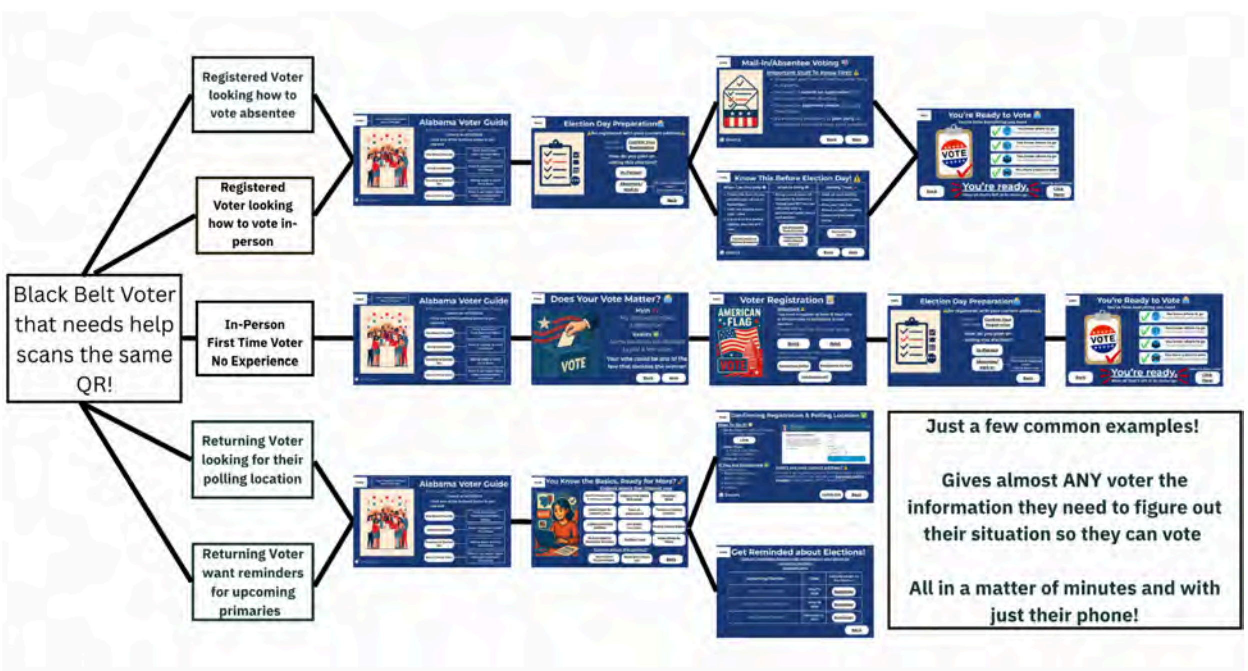

Figure 5. User navigation flow demonstrating multiple possible pathways through the interactive civic participation website, depending on voter needs. Source: Project team interface design.

The user experience flow of the website was designed to adapt to different voter needs while minimizing unnecessary navigation. Depending on the user's situation, the website presents different pathways that guide individuals toward relevant voting information and procedures.

The interface itself was developed using presentation slides created in Canva and integrated into a custom web-based framework optimized for both mobile and desktop viewing. Slides were exported as image assets and paired with interactive navigation elements programmed within the website.

Figure 6. Example Canva slide assets integrated into the interactive website framework for mobile and desktop viewing. Source: Project team materials.

Figure 7. Mobile-responsive implementation of slide-based interface components within the interactive platform. Source: Project team website prototype.

This design approach was selected to prioritize:

- Ease of maintenance
- Low technical overhead
- Compatibility across devices
- Fast loading speeds
- Accessibility in areas with limited internet connectivity

Because the platform relies primarily on lightweight static assets, the website can function efficiently without requiring advanced infrastructure or server-side systems.

Language Accessibility

Users can switch between English and Spanish versions of the website at the beginning of the experience. This feature was included to improve accessibility for users who may prefer to navigate election information in Spanish. Due to project time constraints, broader multilingual support was not implemented during the current development cycle.

Figure 8. English and Spanish language versions of the interactive civic participation website. Source: Project team website prototype.

Visual Instructional Content

Slides involving procedural tasks include short visual demonstrations or animated clips designed to supplement written instructions. These visuals provide users with examples of how to complete specific voting-related tasks and support individuals who prefer visual learning formats.

Figure 9. Example instructional slide containing embedded visual demonstrations to support procedural learning. Source: Project team website prototype.

Source Transparency

Slides containing election-related information include expandable source tabs that provide citations and direct links to official Alabama government resources. This feature allows users to verify information while maintaining a streamlined user interface that minimizes visual clutter during navigation.

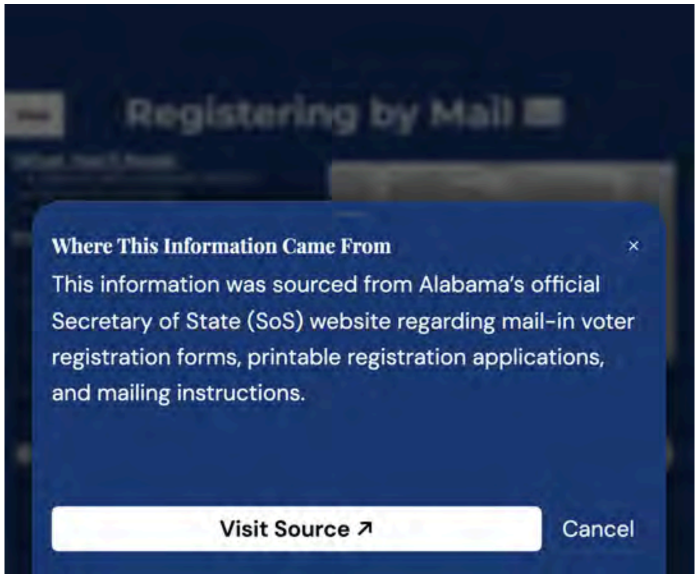

Figure 10. Expandable source tab providing citations and direct access to official Alabama election resources within the website interface. Source: Project team website prototype.

Sustainability and Project Handoff

Upon project completion, all source code, assets, and documentation will be transferred to the client through GitHub. The project is hosted separately from personal repositories to ensure long-term ownership and maintainability by the client organization.

The handoff package will include:

- Complete HTML, CSS, and JavaScript source files
- English and Spanish slide assets
- A maintainer guide documenting update procedures and deployment instructions
- Documentation explaining content management and button configuration workflows

The website was intentionally designed to require minimal technical expertise for future maintenance. Most content updates involve replacing slide image assets and modifying a centralized configuration file rather than altering application logic.

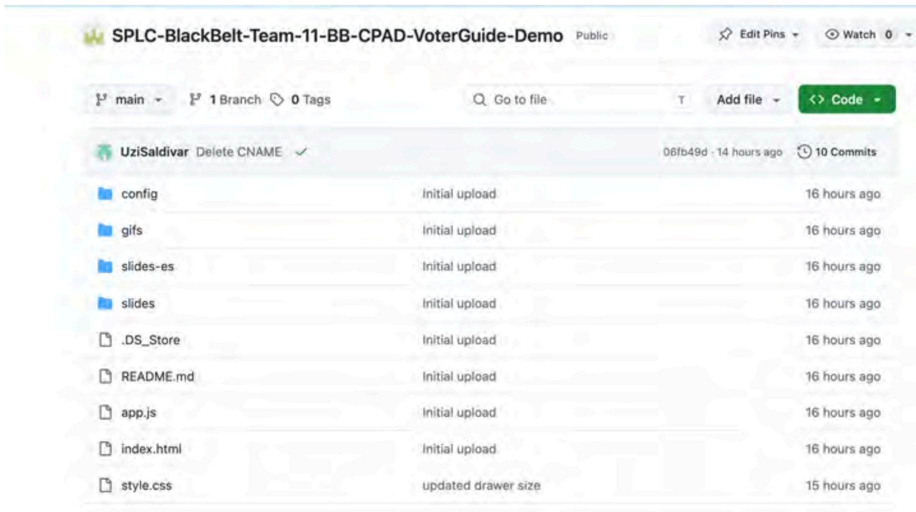

Figure 11. GitHub repository structure and documentation prepared for project handoff and long-term maintenance. Source: Project team repository documentation.

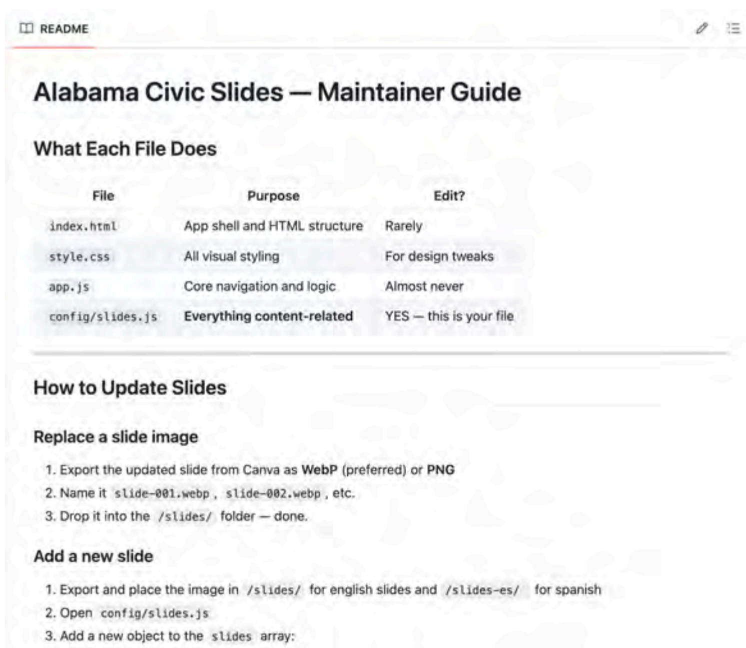

Figure 12. Example maintainer documentation describing slide updates, deployment procedures, and configuration management. Source: Project team README documentation.

The project is deployed through GitHub Pages, which eliminates server maintenance requirements and provides a cost-effective long-term hosting solution.

Final Project Presentation

To support the final reporting phase of the BB CPAD project, Team 11 compiled its research findings, supporting visuals, outreach recommendations, and civic education materials into contributions for the final presentation and written report delivered to SPLC. This deliverable focused on documenting the team's research regarding youth voter participation barriers, accessibility challenges, and digital civic engagement within Alabama Black Belt communities. The deliverable also highlighted how research findings were translated into practical communication materials and outreach recommendations designed to improve civic literacy and voter participation among younger and first-time voters.

The purpose of this deliverable was to ensure that SPLC could review and reuse the project findings, educational graphics, and outreach concepts developed throughout the semester for future civic engagement initiatives. Team 11's contributions were intended to support long-term voter education efforts by presenting information in a clear, research-based, and visually accessible format appropriate for community outreach, educational programming, and public communication efforts. The final reporting deliverable also served as documentation of how the project findings connected directly to the educational materials and digital outreach strategies developed during the project.

The findings, visual materials, and communication recommendations developed by Team 11 were incorporated by the project working group into the final presentation slides and overall reporting structure. The working group combined deliverables and research contributions from multiple project teams into a unified final presentation that connected identified civic participation barriers with proposed outreach solutions and educational strategies. Team 11's materials specifically contributed to the sections discussing youth voter participation trends, accessibility focused civic engagement strategies, and the role of digital communication tools in reducing informational barriers for younger audiences.

Figure 1 shows a presentation slide created by the working group using Team 11's research findings regarding voter turnout disparities across age groups in Alabama Black Belt counties. The slide highlights that younger voters consistently participate at significantly lower rates compared to older age groups and emphasizes that Alabama youth voter turnout during the 2022 midterm elections was only 15%, one of the lowest youth turnout rates in the country. The slide also incorporates comparative turnout visualizations and supporting analysis demonstrating that these participation gaps reflect structural and informational barriers rather than a lack of civic interest among younger populations. The presentation framing emphasized accessibility, institutional support, civic awareness, and outreach limitations as contributing factors affecting youth participation.

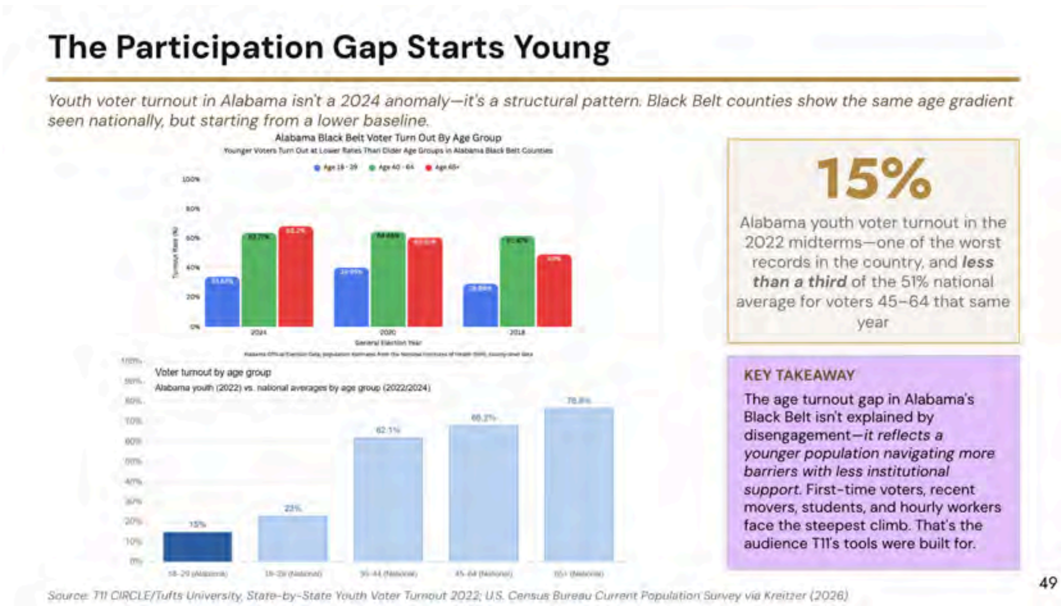

Figure 1: Working group presentation slide summarizing Team 11's findings on youth voter participation disparities and accessibility barriers.

Figure 2 presents another slide created by the working group that incorporated Team 11's community and educational communication materials developed throughout the semester. The slide combines research findings with examples of outreach-focused deliverables, including informational graphics, social media campaign concepts, and the interactive Alabama Voter Guide homepage created to support first-time and younger voters. These materials were designed to simplify the voting process by presenting voter eligibility information, registration guidance, election preparation resources, and civic participation information in a visually engaging and mobile-friendly format. The slide also demonstrates how these communication materials can be adapted for use by voters, educators, community organizations, and outreach groups working within Alabama Black Belt communities.

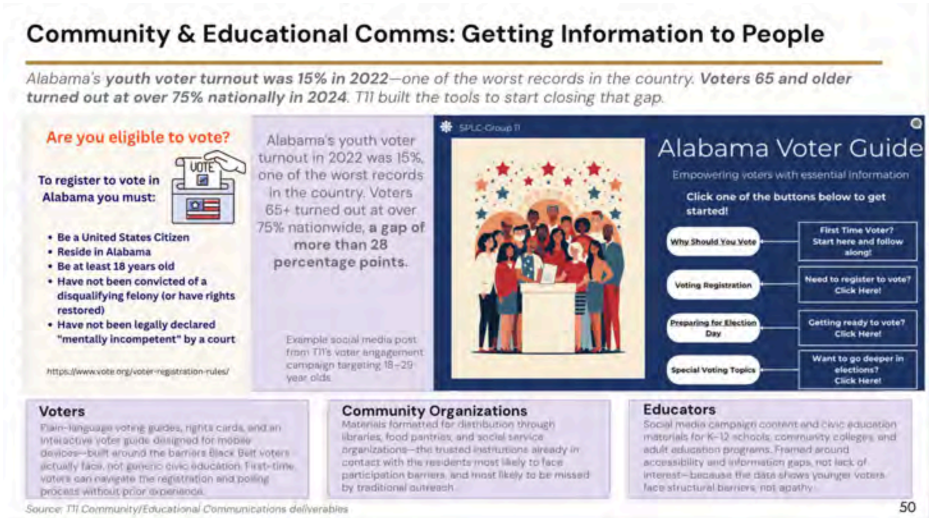

Figure 2: Working group presentation slide featuring Team 11's civic education materials, outreach graphics, and interactive voter engagement resources.

The final project reporting deliverable directly supported SPLC's broader project goals by ensuring that research findings, process documentation, educational resources, and outreach recommendations were consolidated into a reusable and accessible format. By incorporating Team 11's findings into the final presentation and written report, the working group created a comprehensive resource that documents both the identified civic participation barriers and the strategies developed to address them. The deliverable allows SPLC staff to reference research findings, reuse outreach materials, and build upon the project's recommendations for future civic education and voter engagement efforts within Alabama Black Belt communities.

Additionally, the final reporting deliverable demonstrated the relationship between research-based findings and practical community outreach implementation. Team 11's work helped show how data regarding youth voter participation and civic accessibility could be translated into communication strategies and educational tools aimed at reducing informational barriers and increasing voter confidence among historically underrepresented populations. Through the integration of research, visual communication, and educational outreach materials, the deliverable contributed to the overall objective of improving civic participation accessibility and supporting informed voter engagement throughout the region.

User Research

Description

The user research deliverable looked into the main barriers that impact voter engagement in the Alabama Black Belt region. The objective of this research was to look into the obstacles and issues that prevent or turn people away from voting, especially first-time and young voters. This deliverable used data from publicly available sources on voter turnout statistics, research papers, non-profit organizations, and the official website of elections in Alabama.

Rationale Behind Research

This deliverable directly supported the project goal of improving civic participation accessibility within the Alabama Black Belt region. This was done by identifying the barriers most likely to affect voter engagement. The findings helped guide the development of the team's deliverables and the following tasks within our project. The deliverables that benefited from this research include our interactive civic participation website, educational graphics, and social media outreach materials by highlighting the importance of simplified voting information, digital accessibility, and mobile-friendly civic resources throughout the research. The research also helped the team better understand how younger voters consume civic information and what communication methods may be more effective for increasing engagement among underrepresented populations.

Research Process

Secondary-source research was done by the team using available data on elections, reports of non-profit organizations, government resources on elections, and academic literature about civic participation and voter turnout.

In particular, research was focused primarily on barriers that may prevent or hinder Alabama Black Belt community residents, young voters, rural populations, and new voters. The research process included finding out themes and patterns concerning barriers to civic participation, as well as finding new data that contributed to all deliverables. Sources discussing voter turnout dynamics, voter registration access, transportation barriers, digital divide, voter ID laws, and civic information access were analyzed by the team. The results of this analysis were grouped into major barrier categories that would be helpful for further project deliverables.

Research conducted by the team during earlier project phases and voter turnout dynamics studies, especially those concerning low voter participation rates for 18–39 age category voters, was also used in this user research.

Introduction

The problem of civic participation and its structural and informational barriers continues to persist in the Alabama Black Belt region. Previous phases of this project discovered that civic participation difficulties did not stem from any single reason but rather represented an intersection of several issues, among which were problems with transportation, voter information, digital access, and low participation of younger voters. The goal of this deliverable is the synthesis of the existing body of research and publicly available statistics on barriers impacting civic participation and voter registration in Alabama Black Belt counties to create civic participation materials that will be used by SPLC to assist civic engagement in the area.

Barrier 1: Lower Civic Participation of Younger Voters

According to our previous findings, the participation rate among younger voters in Alabama is relatively low. Thus, in the 2022 election, voter participation of Alabamians under the age of 30 accounted for only 15% of the total number of young people eligible to cast their votes, making it one of the lowest rates in the nation. On average, people older than 65 tend to vote in far greater numbers than those belonging to younger demographics.

Tufts University researchers note that the problem of civic participation does not lie with a lack of interest in civic issues but rather manifests itself in insufficient civic knowledge, registration awareness, and inadequate outreach efforts. Many newly registered voters are unaware of how to perform various operations, such as registration and voting. In the case of residents of rural Black Belt areas, these problems are likely compounded due to poor geographical location and poor civic outreach efforts.

As findings revealed, younger audiences prefer to use digital media to find out more information. Therefore, providing these individuals with digital resources will help solve the identified problem.

Barrier 2: Transportation and Geographic Challenges in Access to Voting Infrastructure

Another problem faced by residents of Alabama Black Belt counties is associated with poor access to voting infrastructure and general geographical difficulties. Specifically, due to the rural character of Alabama's Black Belt area, traveling significant distances becomes necessary when trying to access such civic structures as county offices, libraries, polling places, etc. Moreover, residents in this area do not necessarily possess personal transportation, which may create additional obstacles. For instance, it may be difficult for elderly voters, residents with disabilities, and low-income earners to move around and reach such facilities as DMVs or voter registration places, which require in-person visits.

In some cases, it might even become challenging to gather the necessary documentation for voting because of transportation problems. As the League of Women Voters suggests, document verification might create additional barriers for voters lacking passports or birth certificates. The Alabama law mandates that all in-person voters have to produce government-issued photo identification for authentication purposes. However, if the identification document was lost, the State of Alabama allows residents to apply for a voter ID card at no cost. Nevertheless, in terms of the geographical context, finding such cards may prove troublesome.

Barrier 3: Poor Civic Information Accessibility Due to Digital Disparities

Poor civic information accessibility represents yet another problem faced by the target audience. As per our previous findings, many residents are likely to have unreliable access to the internet, mailing address, and timely civic information. Participating in civic activities entails being aware of several procedural aspects, including voter registration period, location, polling place, absentee balloting requirements, identification needed, etc. Lack of consistency in information availability and its fragmented distribution via different platforms may further add to the confusion and hinder civic participation.

In a rapidly changing legal environment, voters become confused due to conflicting information about civic laws. Despite the fact that in most cases the actual law did not change, misunderstanding and confusion may prevent certain voters from casting their ballots. Digital disparities are another obstacle that might impede civic information dissemination and acquisition. Due to poor internet access or unreliable connections, some residents rely primarily on information provided by community organizations, local libraries, churches, and other sources.

Our research shows that civic information centers in the Alabama Black Belt include community organizations and libraries. Thus, nonpartisan civic participation materials and civic process models will serve in the best interest of the residents.

Recommendations for the client (SPLC)

The research indicates that younger voters (18 - 39) in Alabama Black Belt counties have significantly lower voter turnout rates compared to older voters. A major contributing factor found through our research is the sheer volume of information spread across various locations needed to vote. Election deadlines, polling locations, registration requirements, election candidates, etc., are often spread across multiple locations, making it difficult and overwhelming for younger voters.

Based on these findings, SPLC should prioritize reducing information overload by centralizing voter resources into a single, accessible platform tailored to younger voters. Similar to the website we produced, an easily accessible platform that provides county-specific information on voter ID requirements, registration deadlines, polling locations, election cycles, and absentee voting would directly address the gap between younger and older voters. To support this effort, SPLC should enhance outreach efforts through social media commonly used by younger people, such as TikTok and Instagram. Social media efforts should be creative and concise through engaging infographics and videos. Additionally, linking the platform to social media accounts and posts would provide easy accessibility for younger users.

Furthermore, SPLC should expand on community outreach efforts that promote the centralized platform and social media accounts. Partnering with local high schools, colleges, and community organizations across the Alabama Black Belt region to distribute flyers and QR codes would enhance younger voter awareness and reduce information overload. Continued engagement should also be a priority, such as through email and text message services that keep voters updated on civic participation processes. Overall, SPLC should leverage technology and community outreach efforts to engage younger voters.

Recommendations for future project teams

Several methods from this topic were effective and should be continued. A clear target population was found through user research, and the civic participation guide provides a consolidated resource for voters. Data showed that younger voters, aged 18 - 39, had much lower voter turnout rates compared to older voters. Although our user research demonstrated clear barriers for Black Belt residents, such as transportation and digital access, more research can be conducted on the most effective ways to break these barriers. Our efforts involved creating a civic participation guide and a social media campaign, but there may be other efforts that could potentially be more effective in reaching younger voters.

Additionally, future work on this topic should utilize the voter guides and social media campaigns to measure the success that these efforts have on younger voters. Ultimately, an increase in younger voter participation in Black Belt counties during election cycles would indicate a success in outreach efforts. However, analyzing election results may not be feasible if there are no recent elections on which to build upon this topic. Metrics can be analyzed in other ways, such as engagement in social media posts, QR code usage, and website visits.

Conclusions

This project examined barriers affecting civic participation and voter engagement within Alabama Black Belt communities, with a particular focus on younger and first-time voters. Through voter turnout analysis, secondary-source research, and the development of digital civic engagement materials, the project identified several key challenges impacting participation, including information overload, procedural confusion, transportation limitations, and barriers related to digital accessibility.

To address these challenges, Team 11 developed multiple deliverables designed to support civic education and voter accessibility. These included an interactive civic participation website, social media outreach materials, and research-based recommendations intended to simplify the voting process and improve access to election information. The project emphasized mobile-friendly, visually accessible, and nonpartisan communication strategies aimed at reducing confusion surrounding voter registration, election preparation, absentee voting, and voter rights.

The findings from this research suggest that improving accessibility to clear and centralized voting information may help reduce barriers that disproportionately affect younger voters and underrepresented communities. Additionally, the project demonstrated how research findings can be translated into practical outreach tools and educational resources that organizations such as the Southern Poverty Law Center may continue to adapt and expand for future civic engagement efforts.

Overall, this project contributed research, educational materials, and digital engagement concepts intended to support broader efforts focused on increasing civic participation accessibility within Alabama Black Belt communities. Future work may continue building upon these findings by evaluating the effectiveness of outreach materials, expanding multilingual accessibility, and exploring additional strategies for increasing engagement among younger voters.

Lakshya Sajal Kumar: lakshya.sajalk@gmail.com

Uziel Saldivar: uzielsaldivar10@gmail.com

Amari Harrington: amarivashon@gmail.com

Matthew Romalis: romalismatt@gmail.com

Deliverables Checklist

Social Media Campaign Part 1: Link

Social Media Campaign Part 2: Link

Deliverable 2 User Research: Link

Presentation Slide: Link

Civic Process Model GitHub: Source | Website